

Unleashing the Power of IBM Storage FlashSystem Grid

A Comprehensive Guide

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**Unleashing the Power of IBM Storage FlashSystem
Grid: A Comprehensive Guide**

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Note: Before using this information and the product it supports, read the information in “Notices” on page vii.

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Preface

The IBM FlashSystem® grid is an innovative and efficient solution for overcoming data storage management challenges. This federated, scalable storage platform enables unified management of multiple IBM FlashSystem and IBM Storage Virtualize (IBM SAN Volume Controller (SVC)) controllers. Its key strengths include unmatched scale-out capability and streamlined, non-disruptive management across federated IBM FlashSystem and SVC environments, which enhances both agility and resilience.

This IBM Redbooks® publication provides a practical guide to deploying and managing the IBM FlashSystem grid. The intended audience includes storage administrators and storage architects who need practical guidance for IBM FlashSystem grid implementation and ongoing management.

Note: We used a pre-GA version of the product for the scenarios in this book, so some screen captures might differ slightly from the ones that are in the GA release.

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Throughout his career, Digambar has contributed to several core features across the IBM Storage Virtualize product family, including IBM Storage FlashSystem, SVC, and Storwize systems. His areas of expertise include clustering technologies (FC, iSER/RDMA), iSCSI performance optimization, remote copy over IPsec, and grid infrastructure reliability. As a technical leader, he has driven feature development, led customer beta engagements, and collaborated with global teams to deliver enterprise-class solutions. Digambar has also authored internal white papers, presented at IBM TechXchange on storage virtualization and connectivity performance and filed multiple US patents in enterprise storage technologies.

Ashwin Joshi is a technical leader for IBM Storage Virtualize product in Pune, India. Since 2013, he has contributed to numerous IBM Storage FlashSystem features. His expertise includes AIOPS and IBM FlashSystem grid technologies. He has led design and development teams in areas such as secured data replication over IPsec tunnels, quality of service with I/O throttling, FC and NVMe over FC protocol development, seamless FC adapter error recovery, data encryption and key management in Transparent Cloud Tiering (TCT) environments, and support for Microsoft ODX (Offloaded Data Transfers). He has also authored open-source code for automatic storage discovery in Linux environments. Before joining IBM, he worked on storage virtualization, volume management software, file backup and recovery software, embedded Linux kernel, boot loaders, and Linux device driver development. He has represented IBM at events and written white papers on product features.

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Introduction

Business services and application environments in organizations are increasingly complex, with evolving requirements for storage. These requirements include business continuity, data resiliency, performance, scalability, security, compliance, and cyber resiliency.

Effective storage management must address these needs and control storage resources and costs. A fragmented approach to data storage creates application and organizational silos, which increases complexity and manual effort.

The IBM FlashSystem grid provides an innovative and efficient solution to these challenges. It is a federated, scalable storage platform that manages a grid that is composed of multiple IBM FlashSystem and IBM Storage Virtualize (IBM SAN Volume Controller (SVC)) controllers.

This chapter introduces the IBM FlashSystem grid concept and outlines its architecture.

This chapter describes the following topics:

- ▶ “IBM FlashSystem grid overview” on page 2
- ▶ “IBM FlashSystem grid architecture and terminology” on page 2
- ▶ “Business value of an IBM FlashSystem grid” on page 8
- ▶ “Main features” on page 9
- ▶ “Key use cases” on page 12
- ▶ “Supported hardware platforms and software releases” on page 14

1.1 IBM FlashSystem grid overview

An IBM FlashSystem grid is a federated storage platform that unifies multiple IBM FlashSystem or SVC systems into a single storage solution. It provides a single management interface for configuration, monitoring, and administration. With IBM FlashSystem grid, you can migrate storage partitions between systems within the grid without disruption. This capability provides private cloud functions and helps maximize storage infrastructure usage.

Each grid includes a coordinator system (either an IBM FlashSystem or SVC) that handles configuration and management operations. The remaining systems act as members. You can view the grid from any system; however, the coordinator runs all management commands.

An IBM FlashSystem grid supports up to 32 storage systems (as of IBM Storage Virtualize 9.1).

1.2 IBM FlashSystem grid architecture and terminology

The IBM FlashSystem grid architecture uses a layered design and introduces new concepts for managing storage at scale. This section explains the key terms, components, and architectural layers.

Table 1-1 explains the key IBM FlashSystem grid terminology.

Table 1-1 IBM FlashSystem grid terminology

Terminology	Description
Coordinator system	The designated system that creates and manages the IBM FlashSystem grid. It also automatically becomes the first member of the grid. You can add or remove members only through the coordinator system.
Member system	An IBM FlashSystem or SVC controller system that the IBM FlashSystem grid manages.
Physical storage controller	An individual IBM FlashSystem or SVC unit that acts as a building block of the grid.
Virtual storage controller	A software-defined virtualization layer that unifies the underlying physical storage controllers into a single, manageable system.
Storage partition	A higher-level object introduced with IBM FlashSystem grid that encapsulates application data and all logical configurations that are required to make that data accessible to servers, such as volumes, volume groups, host objects, mappings, and replication policies-required to make that data accessible to servers.
Volume group	A logical grouping of volumes that serves as the unit of application-level cross-volume consistency. Multiple volume groups can exist within a storage partition.

1.2.1 Grid layered architecture

The IBM FlashSystem grid consists of four layers:

- ▶ Artificial intelligence (AI)-powered grid intelligence layer
- ▶ Grid management layer
- ▶ Virtual storage partitions layer
- ▶ Physical layer

Figure 1-1 shows the layered view of the grid architecture.

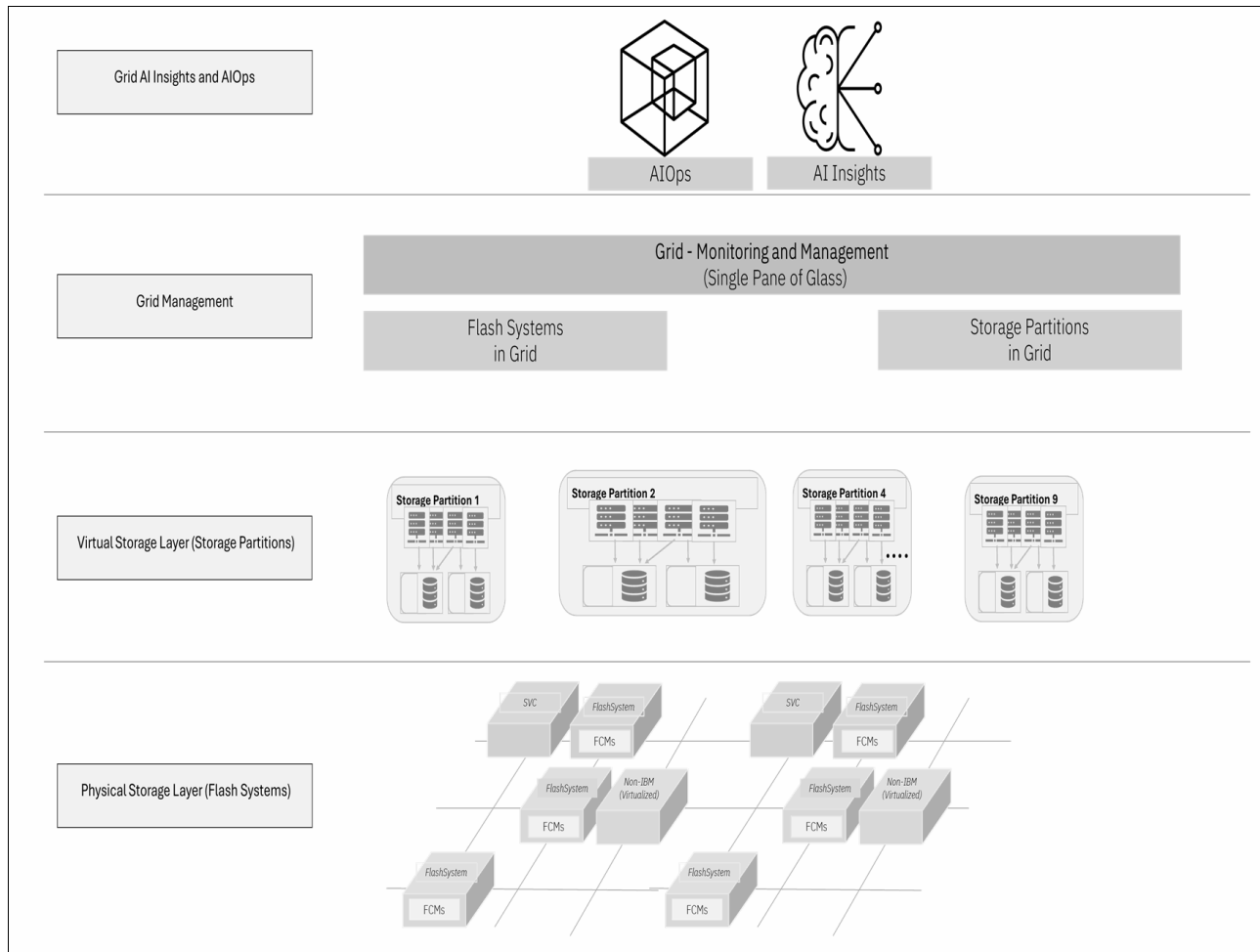


Figure 1-1 IBM FlashSystem and SVC grid-layered architecture

Each layer plays a specific role in delivering scalability, automation, and resiliency across the grid.

Physical storage layer

At the physical layer, an IBM FlashSystem grid includes multiple IBM FlashSystem and SVC controllers. These systems form a federated storage environment. I/O groups serve as the fundamental building blocks. This architecture supports up to 32 systems per grid, delivering linear scalability in performance, capacity, and the number of volumes, hosts, and snapshots.

IBM FlashSystem grid abstracts the physical layer by introducing a virtual storage layer called storage partitions. This design enables seamless data mobility across the grid.

Figure 1-2 shows how the IBM FlashSystem grid scales by adding systems to create a federated cluster.

Note: Systems with multiple I/O groups cannot participate in the grid.

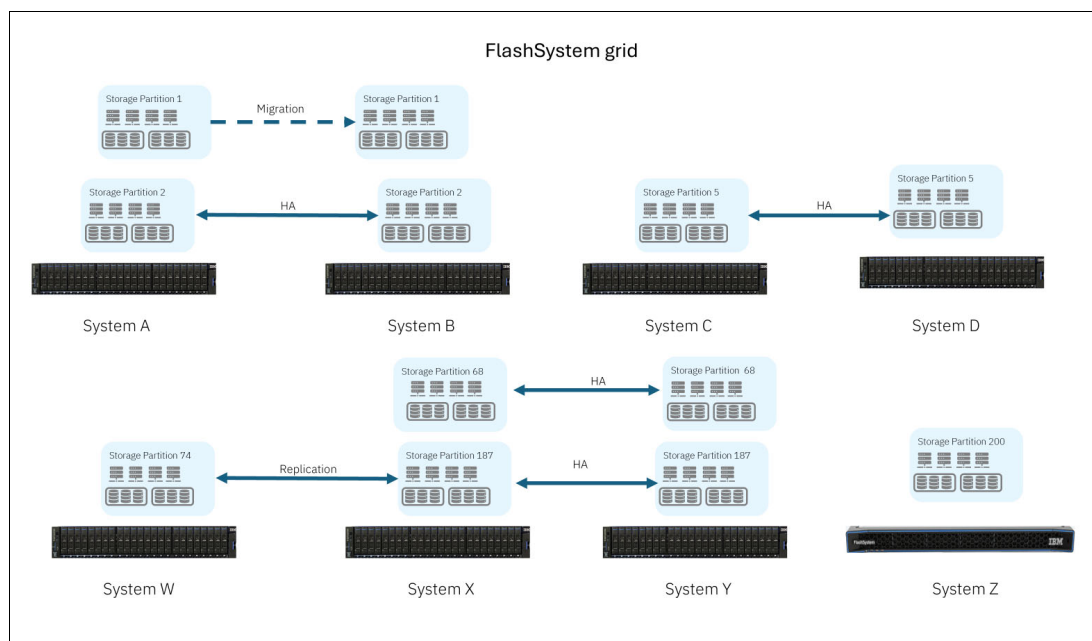


Figure 1-2 IBM FlashSystem grid

Virtual storage layer

IBM FlashSystem grid introduces storage partitions as a new virtual storage layer. Hosts and applications access storage through these partitions. Each partition bundles application data with all logical configurations that are required for access, such as volumes, host mappings, and replication policies. This design simplifies management and enables seamless data mobility across the grid.

Storage partitions and mobility

A storage partition acts as a dedicated, virtual storage controller for a specific workload. It keeps all related storage elements together, which simplifies management and can ensure consistency.

You can use an IBM FlashSystem grid to move storage partitions between systems in the grid without disruption. AI-driven insights suggest optimal placement and migration targets to maintain performance and balance capacity across the grid. The grid integrates with host environments to automate rescans and multi-path health checks, which supports non-disruptive migration with minimal user intervention.

Figure 1-3 shows how storage partitions move across the grid without disruption.

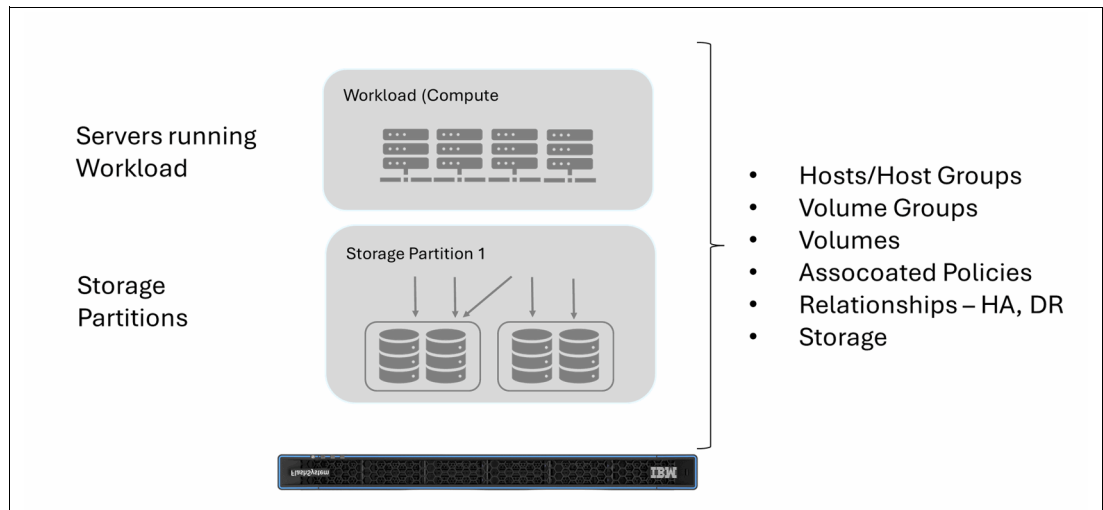


Figure 1-3 Storage partitions

Note: You can have up to 32 storage partitions per grid.

1.2.2 Grid management layer

This grid management layer provides automation and centralized control for the entire grid. It delivers a single management interface for monitoring, provisioning, and policy enforcement. You can manage the grid from any connected system, and configuration updates automatically propagate to all members, as shown in Figure 1-4.

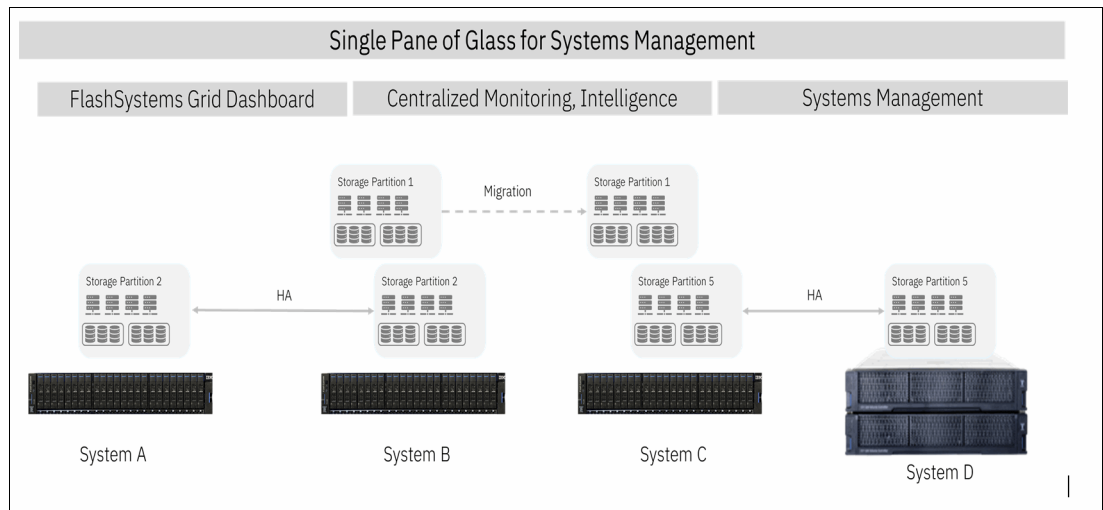


Figure 1-4 Federated management

The grid architecture incorporates a control plane that is designed for federated management of interconnected physical systems and their associated virtualized storage. This plane provides a single pane of glass for monitoring and administration, consolidating storage resources and partitions into a unified view.

IBM FlashSystem grids offer flexible updates and hardware compatibility. You can update systems independently, and a single grid can include diverse hardware models and generations.

Note: You can manage the grid from any connected storage system. If your system is not directly connected to the grid coordinator, it automatically receives configuration updates from all other grid members.

Separate update domain

IBM FlashSystem grids offer flexible updates and hardware compatibility. You can update systems independently. A single grid can include diverse hardware models and generations. This design eliminates the need for uniform software levels and allows I/O group-by-I/O group updates. Older hardware can also participate in the grid and be managed centrally through the same interface.

Grid AI layer

This AI-driven layer provides insights through IBM Storage Insights, a cloud-based management tool. AI-generated capacity and performance recommendations help you optimize storage placement within the grid and select suitable migration targets.

Note: If your system is not directly connected to the coordinator, it automatically receives configuration updates from other grid members. For more information about IBM Storage Insights, see *Maximize Your Storage Investment: Use IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, SG24-8584.

1.2.3 Grid networking

IBM FlashSystem grid uses multiple networking types to support different functions. This section describes the control plane, intra-grid communication, data transport, host connectivity, and trust domain.

Control plane

This control plane manages configuration and communication across the grid by using IP networking. Systems form a loosely coupled cluster that operates within a unified configuration domain.

Intra-grid communication

All grid members exchange information to support federated views and centralized management. Each system communicates over IP connectivity. Choose one of the following options for the grid management IP address:

- ▶ IPv4
- ▶ IPv6
- ▶ IPv4 with fully qualified domain names (FQDNs) and a configured Domain Name System (DNS) server
- ▶ IPv6 with FQDNs and a configured DNS server
- ▶ A combination of IPv4 and FQDN with a configured DNS server

Best practice: To help ensure reliable communication, configure a VLAN for grid management and connect all storage system management ports to this VLAN to isolate grid traffic. This setup helps isolate grid traffic from broader network issues.

Data plane

The data plane transports data between systems for high-availability replication. Use one of the following networking types:

- ▶ Fibre Channel (FC). Used for replication between storage systems
- ▶ High-speed Ethernet (HSE). Used for fast data transport
- ▶ IP networking. Used for replication between storage systems across data centers

Host connectivity (host-to-partition)

This plane provides host connectivity to the grid. Hosts access their data through the host connectivity plane. Supported networking types include the following options:

- ▶ FC. Uses either the SCSI or Non-Volatile Memory Express over Fabrics (NVMe-oF) protocols.
- ▶ HSE. Uses the iSCSI protocol.

Trust domain

IBM FlashSystem grid implements a trust domain to secure communication between systems. When you add a storage system to the grid, you must verify authorization. After you add the storage system, the grid stores the attributes of the new member system. During the add process, systems exchange trust certificates. After trust is established, it extends to all grid members to enable secure communication across the grid. By default, the IBM FlashSystem grid uses self-signed certificates.

For more information about certificate management, see the [System Certificates](#).

1.2.4 Grid storage

IBM FlashSystem grid supports traditional volumes and modern methods for presenting storage to hosts.

- ▶ In NVMe over Fibre Channel environments, you continue to use traditional volumes and volume groups, consistent with earlier IBM FlashSystem configurations.
- ▶ For hosts that connect by using SCSI over FC or iSCSI, the IBM FlashSystem grid introduces storage partitions as logical management entities.

Storage partitions encapsulate volume groups, volumes, host ports, and host-to-volume mappings. They enable Policy-based High Availability (PBHA) and replace configurations such as HyperSwap and stretched clusters.

A key advantage of storage partitions is the ability to encapsulate both data and host-facing configurations. You can migrate these partitions seamlessly across systems in the grid without disrupting applications. This capability simplifies system upgrades, host migrations, and infrastructure changes by eliminating manual reconfiguration steps.

1.2.5 I/O groups in IBM FlashSystem grid

In an IBM FlashSystem grid, the fundamental building block is a single I/O group. Each participating IBM FlashSystem or SVC unit must be configured as a stand-alone I/O group to help ensure compatibility and optimal integration.

Eligibility requirements

To join an IBM FlashSystem grid, systems must meet the following criteria:

- ▶ Use a single I/O Group (disable multi-I/O group configurations).
- ▶ Use the standard topology setting.
- ▶ Enable the replication layer.

For SVC clusters, configure each cluster as a separate 2-node cluster, forming a single I/O group. For example, a 6-node SVC deployment must be split into three 2-node clusters before it can join the grid.

Advantages over traditional clustering

This architecture offers several benefits compared to traditional N-way clusters:

- ▶ Scalability. Traditional clusters are limited in volume count regardless of node count (for example, 16,000 volumes for both 2-node and 8-node clusters). A grid of eight 2-node systems supports up to 64,000 volumes (16,000 per system).
- ▶ Simplified code and management. Each system operates independently, which reduces complexity and eases development and support.
- ▶ Improved reliability. The federated design minimizes interdependencies, which enhancing fault isolation and system resilience.

Streamlined maintenance

You can upgrade systems independently, which eliminates long, sequential upgrade cycles. Each system can be updated on its own schedule, reducing maintenance windows and minimizing operational risk. You can migrate storage partitions to another system in the grid before an upgrade to maintain application availability and avoid downtime.

1.3 Business value of an IBM FlashSystem grid

An IBM FlashSystem grid transforms traditional storage into a modern, intelligent, and resilient infrastructure. The following capabilities help you address evolving IT demands, improve operational efficiency, and reduce risk:

- ▶ Elastic storage infrastructure with insights. The *grid control plane* creates an elastic storage infrastructure and provides insights into resource usage across the grid.
- ▶ Workload-centric storage management. Storage partitions act as the unit of storage management, streamlining operations and aligning storage with application needs.
- ▶ Enhanced business continuity and high availability (HA). Built-in features protect data from cyberattacks, hardware failures, and software issues, and enable fast recovery.
- ▶ Cloud-like user experience and automation. The grid delivers a consistent, automated experience for on-premises storage, reducing complexity and skill gaps.
- ▶ AI-powered decision making. Provides AI-driven recommendations that improve planning, accelerate decisions, and reduce management.

- ▶ Non-disruptive partition mobility. You can move partitions without downtime, improving agility and eliminating data silos across systems without downtime.
- ▶ Simplified operations through automation. Automated processes reduce complexity and operational costs for network, storage, and security tasks.
- ▶ Adaptive and scalable storage. The grid scales to meet changing business needs, including hybrid deployments and cloud repatriation.
- ▶ Optimized resource usage. Intelligent data mobility helps ensure efficient use of capacity and performance resources.
- ▶ Elimination of data silos. AI-driven placement and mobility keep data accessible across the grid.

1.4 Main features

IBM FlashSystem grid provides a modern approach to managing and scaling enterprise storage through a unified, virtualized platform. The design supports automation, scalability, and resiliency. The IBM FlashSystem grid transforms traditional storage architectures into agile, federated environments that are simpler to manage and adapt.

This section describes the platform's most significant technical capabilities, focusing on federated storage, centralized management, and data mobility.

1.4.1 Federated storage in grid

Traditional storage systems operate in silos, and each requires separate management and configuration. The IBM FlashSystem grid addresses this challenge with a federated architecture that unifies multiple IBM FlashSystem and SVC systems into a single logical entity. This approach provides the following benefits:

- ▶ Centralized control. Manage all systems through a single pane of glass that is powered by IBM Storage Virtualize and IBM Storage Insights.
- ▶ AI-driven workload placement. Optimize performance and resource usage with intelligent recommendations.
- ▶ Policy-based automation. Apply consistent policies for HA, disaster recovery (DR), and replication.
- ▶ Fleet-wide monitoring. Monitor diverse models and locations from one interface.
- ▶ Scalable infrastructure. Support up to 32 systems, more than 1 million volumes, and over 1 exabyte of usable block storage.

IBM FlashSystem grid supports hybrid environments by allowing different IBM FlashSystem models (such as IBM FlashSystem C200, IBM FlashSystem 5300, IBM FlashSystem 7300, IBM FlashSystem 9500, and SVC) to coexist. This flexibility makes it ideal for organizations with varied performance and cost requirements.

1.4.2 Single pane for grid storage management

IBM FlashSystem grid simplifies traditional storage management by consolidating multiple interfaces into a single unified dashboard. This centralized view reduces complexity, eliminates redundant tasks, and minimizes the risk of human error.

You gain comprehensive control over storage resources across diverse locations and hardware platforms. From a single intuitive interface, you can efficiently manage provisioning, resource allocation, capacity planning, policy enforcement, performance monitoring, and troubleshooting.

IBM FlashSystem grid dashboards: Key areas and user benefits

You gain unified visibility and control over storage resources across diverse locations and hardware platforms. Through a single, intuitive interface, you can efficiently manage provisioning, resource allocation, capacity planning, policy enforcement, performance monitoring, and troubleshooting.

The dashboard provides a centralized view of system status and performance to help you monitor and manage your IBM FlashSystem grid effectively. The dashboard includes the following key areas:

- ▶ System health overview
 - What it shows. Real-time status of all nodes, alerts, and hardware health.
 - User benefit. Identifies and resolves issues before they affect workloads.
- ▶ Performance metrics
 - What it shows. IOPS, latency, throughput, and resource usage across the grid.
 - User benefit. Enables performance tuning and proactive capacity planning.
- ▶ Capacity insights
 - What it shows. Storage usage trends, compression savings, and available capacity.
 - User benefit. Helps you optimize storage efficiency and forecast future needs.
- ▶ Data protection status
 - What it shows. Snapshot schedules, replication status, and DR readiness.
 - User benefit. Helps ensure compliance with protection policies and simplifies audits.
- ▶ AI-Powered recommendations
 - What it shows. Suggestions for workload placement, tiering, and anomaly detection.
 - User benefit. Automates decision-making to improve performance and reduce costs.
- ▶ Migration and mobility tools
 - What it shows. Active migrations, source and target systems, and policy adherence.
 - User benefit. Simplifies and accelerates data movement with full visibility and control.

1.4.3 Data mobility in the grid

IBM FlashSystem grid introduces *advanced data mobility*, enabling non-disruptive migration and dynamic rebalancing of data across storage resources. This capability is especially valuable in environments with changing workload demands or during routine hardware upgrades.

By using storage partitions, you can migrate both logical configurations and physical data seamlessly across systems within the grid, without application downtime or host involvement. This approach contrasts with traditional migrations, which often require extensive coordination and planned outages.

The automated data mobility feature helps ensure business continuity by dynamically adjusting resource allocation and balancing workloads. This capability enhances resilience, improves resource usage, and enables organizations to respond quickly to evolving business needs.

The migration process includes the following steps:

1. Initiate the migration request through the management interface:
 - Select the source and target systems.
 - Identify the partitions or volumes to migrate.
 - Choose the scheduling options (immediate or deferred).
 - Validate the prerequisites, such as compatibility and capacity.
2. Mirror the partition's data and configuration to the target system:
 - Copy the data by using block-level replication or snapshot-based methods.
 - Include configuration details such as volume metadata, access controls, and performance settings.
 - Maintain synchronization until cutover to help ensure minimal downtime.
3. Present the hosts with new paths to the volumes:
 - Update the zoning and masking in storage-attached network (SAN) environments.
 - Reconfigure the iSCSI targets or NVMe namespaces.
 - Detect and add the new paths through multi-path software.
4. Switch traffic to the new system after verifying multi-path connectivity:
 - Test the paths for latency, throughput, and failover.
 - Perform an automatic or manual switchover of I/O traffic.
 - Confirm a successful transition by using monitoring tools.
5. Retire the original source data:
 - Unmount and remove volumes from host access.
 - Securely wipe or archive data.
 - Release resources on the source system for reuse.

Key advantages

This model offers several key advantages that enhance operational efficiency and data protection:

- ▶ Zero downtime migrations. These migrations occur without requiring host reconfiguration or causing application downtime.
- ▶ Policy continuity. Snapshot and DR policies remain intact throughout the migration process, helping ensure uninterrupted data protection.
- ▶ Flexible use cases. The model supports scenarios such as load-balancing, hardware refreshes, and site failovers without disrupting application performance.
- ▶ AI-driven optimization. AI guides storage movement decisions based on policies, performance metrics, and user-defined criteria to optimize resource usage.

1.5 Key use cases

IBM FlashSystem grid delivers advanced storage management capabilities that meet diverse business requirements. It enables seamless workload mobility across storage arrays with varying performance tiers, provides centralized single-pane-of-glass management, and supports data movement. With its intelligent, automated architecture, the IBM FlashSystem grid optimizes performance, maximizes capacity usage, and enhances operational efficiency. It helps ensure that data is placed intelligently for optimal results across the entire infrastructure.

This section describes the key use cases.

1.5.1 Moving workload to the correct performance class

An IBM FlashSystem grid can contain many different systems with different capabilities, capacities, and workload requirements. You can define custom performance classes based on host count, workload type, and each storage system's performance characteristics. This approach helps ensure that workloads run on the most suitable hardware.

Example use case

A grid includes two IBM FlashSystem 9500 systems, one IBM FlashSystem 7300 system, and one IBM FlashSystem C200 system:

- ▶ First IBM FlashSystem 9500. Reserved for mission-critical workloads that use small FC fan-in ratios (2:1) to help ensure maximum performance and minimal latency. Of the available 48 FC ports, eight are reserved for grid communications, with 40 ports for host connections. At a 2:1 fan-in ratio, this configuration supports up to 80 high-performance hosts.
- ▶ Second IBM FlashSystem 9500. Used for mixed workloads with higher FC fan-in ratios (8:1). Eight ports are reserved for grid communication, and the remaining 40 ports support up to 320 hosts, which are given the 8:1 fan-in ratio.
- ▶ IBM FlashSystem 7300. Optimized for VMware environments, providing balanced performance tailored for virtualization.
- ▶ IBM FlashSystem C200. Designed for capacity-optimized Flash Storage, ideal for backup, archive, and cold data storage, significantly reducing storage costs for large-scale archival workloads and infrequently accessed data.

You can manually migrate workloads between IBM FlashSystem 9500 systems without disruption. For example, if you detect increased latency on the second IBM FlashSystem 9500, because of a higher FC fan-in ratio, move the affected workload to the first IBM FlashSystem 9500 without disruption. This approach helps prevent noisy neighbor effects and helps ensure consistent performance.

For more information, see Chapter 5, “Configuring and managing storage partition mobility” on page 103.

1.5.2 AI-driven workload placement optimization

As storage environments grow, managing workload placement becomes complex. An IBM FlashSystem grid integrates with IBM Storage Insights to provide automated placement recommendations based on capacity trends, central processing unit (CPU) load, IOPS behavior, latency forecasts, and power efficiency.

Example use case

A latency-sensitive workload on an IBM FlashSystem 7300 shows sporadic increases in response time. IBM Storage Insights detects increased throughput on another server in the same system, which causes contention. Using historical telemetry and predictive modeling, The AI engine recommends moving the workload to an IBM FlashSystem 9500 with more bandwidth and reduced projected latency.

You can trigger a non-disruptive migration from IBM Storage Insights. Host mappings, volume groups, and high availability and disaster recovery (HADR) policies move automatically without downtime.

Key capabilities

The following capabilities enhance workload placement and migration efficiency:

- ▶ AI-driven fitment scores that are based on forecasted latency, throughput, and capacity
- ▶ Visual impact analysis with and without the new workload
- ▶ One-click migration from the IBM Storage Insights interface
- ▶ Policy-aligned placement based on performance or power targets

For more information, see Chapter 6, “IBM FlashSystem grid monitoring with IBM Storage Insights” on page 113.

1.5.3 Grid-wide data mobility for hardware refreshes

Refreshing hardware without downtime is a major challenge. Traditional storage platforms require planned outages, complex host coordination, and manual data migrations, which increase risk, cost, and operational complexity. IBM FlashSystem grid eliminates this issue with non-disruptive, grid-wide data mobility.

Storage partitions encapsulate application data, metadata, host mappings, and replication policies. These partitions act as virtual storage controllers with predefined HADR policies.

When you replace hardware, such as upgrading an IBM FlashSystem 5300 to a new IBM FlashSystem 9500 you can move all associated partitions without disruption.

Migration process

The following steps describe how to migrate data and configuration between systems in the IBM FlashSystem grid. Each step helps ensure minimal disruption and maintains data integrity throughout the process:

1. Initiate the migration from the management interface:
 - a. Select the source and target systems.
 - b. Identify the partitions or volumes to migrate.
 - c. Choose the scheduling options (immediate or deferred).
 - d. Validate the prerequisites (compatibility and capacity).
2. Mirror the data and configuration to the target system:
 - a. Copy the data by using block-level replication or snapshots.
 - b. Include configuration details such as volume metadata and access controls.
 - c. Maintain synchronization until the cutover.

3. Present the hosts with new paths:
 - a. Update the zoning and masking in SAN environments.
 - b. Reconfigure the iSCSI targets or NVMe namespaces.
 - c. Detect and add the new paths through multi-path software.
4. Switch traffic to the new system after you verify connectivity:
 - a. Test the paths for latency, throughput, and failover.
 - b. Perform an automatic or manual switchover.
 - c. Confirm a success by using the monitoring tools.
5. Removing the old configuration:
 - a. Unmount and remove the volumes from host access.
 - b. Securely wipe or archive the data.
 - c. Make resources available for reuse.

Key advantages

- ▶ Zero downtime migrations
- ▶ Policy continuity for snapshots and DR
- ▶ Flexible use cases, including load-balancing, hardware refresh, and site failover
- ▶ AI-driven optimization for resource usage

Strategic benefits

This approach helps ensure zero application downtime, avoids risky reconfiguration steps, and simplifies lifecycle management. By combining policy-based automation and grid-level orchestration, IBM FlashSystem grid transforms storage refreshes from disruptive projects into routine operations.

With IBM FlashSystem grid, you can modernize continuously. Perform hardware upgrades incrementally and intelligently without using forklifts or requiring host reconfiguration. When combined with [IBM Storage Assurance](#), this approach helps you maintain a modern infrastructure with predictable costs.

Data mobility is no longer just a feature. It is a strategy. The IBM FlashSystem grid helps ensure that your workloads always run on the most optimal platform across your storage estate.

1.6 Supported hardware platforms and software releases

This section lists supported hardware platforms and software releases for IBM FlashSystem grid and compares features across releases. This information helps you decide the best code level for your systems.

1.6.1 Supported platforms

In release 9.1.x, up to 32 systems can join an IBM FlashSystem grid. You can mix hardware models and generations; however all systems must support storage partitions, except for IBM FlashSystem 5045.

The following products are supported:

- ▶ IBM FlashSystem 5200 and IBM FlashSystem 5300
- ▶ IBM FlashSystem 7200 and IBM FlashSystem 7300
- ▶ IBM FlashSystem 9200 and IBM FlashSystem 9500
- ▶ IBM FlashSystem C200
- ▶ IBM SVC SA2, IBM SVC SV2, and IBM SVC SV3

IBM FlashSystem grid is not supported on IBM FlashSystem 50x5 platforms or IBM Storage Virtualize for Public Cloud. For more information, see the “Software feature limits” section in the relevant Hardware Guide.

For example, Figure 1-5 shows the [Software feature limits for IBM Storage FlashSystem 5045](#). The table for IBM FlashSystem 5045 shows that the IBM FlashSystem grid cannot be configured on this hardware.

Software feature limits

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IBM® Storage FlashSystem 5000 runs the IBM Storage Virtualize software stack to provide its rich storage functionality. Due to limitations of hardware resource, the following software feature limits should be noted.

Table 1. IBM Storage FlashSystem 5045 Software feature limits

Software function	Hardware-derived limit
Embedded VASA Provider/vVols	Cannot be configured on this hardware
Deduplication	Requires 32 GB memory per node. Supported if policy-based replication or policy-based high availability is not configured
Policy-based replication	Requires 32GB memory per node. Supported if Data Reduction Pools (DRP) are not used
Policy-based high availability (HA)	Requires 32GB memory per node. Supported if Data Reduction Pools (DRP) are not used
Storage Insights Integration (Cloud Call-home)	Cannot be configured on this hardware
FlashSystem grid	Cannot be configured on this hardware
Partition Migration	Cannot be configured on this hardware

Figure 1-5 Software feature limits for IBM Storage FlashSystem 5045 in IBM Documentation

1.6.2 Supported software levels

IBM FlashSystem grid was introduced in release 8.7.0.0 with command-line interface (CLI)-only support. In subsequent releases, GUI capabilities and scalability increased.

The following release summary shows the supported software levels and key capabilities:

- ▶ 8.7.0.0
 - CLI-based grid creation and management
 - A maximum of eight systems per grid, which is limited by the truststore size (seven entries plus the coordinator)
 - No GUI management for the grid
- ▶ 8.7.2.0
 - Partial GUI support to create or join an IBM FlashSystem grid in addition to using the `manageflashgrid` command
 - A grid dashboard view to manage multiple independent systems from a single pane of glass
- ▶ 8.7.3.0
 - Full GUI support for visualization and management
 - GUI-driven storage partition migration between grid members
 - Automated truststore provisioning and storage-pool linking
- ▶ 9.1.x
 - Updated CLI and GUI terms (for example, `managegrid`, `lsgrid`, `lsgridpartition`, `lsgridsystem`, and `lsgridmembers`)
 - Increased maximum grid size from 8 to 32 systems

Mixed-version orchestration and troubleshooting

You can include multiple members with systems that run different code levels in the same grid (for example, 9.1.0.0 and 8.7.3.2). However, to support IBM FlashSystem grid orchestration, each member system must run 8.7.3.0 or later. If you attempt to manage a system that runs an earlier code level under a coordinator at 8.7.3.0 or later, the IBM FlashSystem grid dashboard displays the following error:

Grid orchestration is paused due to configuration error.

In addition, the affected system logs an unfixed event code 3930 in the management GUI (to see this error, select **Monitoring** → **Events**). To fix this error, upgrade the member to 8.7.3.0 or later. After the upgrade, the event is marked as fixed automatically.

Best practice: Run 8.7.3.x or later to use IBM FlashSystem grid effectively.

IBM FlashSystem grid features comparison across code levels

Table 1-2 shows a comparison of IBM FlashSystem grid features across code levels. Select the code level that best meets your requirements.

Table 1-2 Comparing the IBM FlashSystem grid between different code releases

Function	8.7.0	8.7.2	8.7.3	9.1.0
CLI support	Yes The manageflashgrid command creates a grid, and adds and removes members. The lsysteminfo command is used to display a summary of system information that can be used by an IBM FlashSystem grid. It provides key attributes about the system and its health status.	Yes The same commands that are used in 8.7.0, plus the following commands: ► lsflashgrid ► lsflashgridpartition ► lsflashgridsystem These commands provide more information about the IBM FlashSystem grid.	Yes The same commands that are used in 8.7.2, plus the lsflashgridmembers command.	Yes Use managegrid to create a grid, and add and remove members. The following commands provide more information about the IBM FlashSystem grid: ► lsgrid ► lsgridpartition ► lsgridsystem ► lsgridmembers
Management GUI support	No	Partly Use the GUI to create IBM FlashSystem grids or join an existing grid.	Full Use the GUI to create an IBM FlashSystem grid, join an existing grid, or perform storage partition migration for IBM FlashSystem grid members.	Full
IBM FlashSystem grid dashboard	No	Partly Use the dashboard to add members to a grid, see the grid and members on the coordinator system, and see storage partitions on the coordinator system.	Full Use the dashboard to see the IBM FlashSystem grid, monitor all systems, and enable error reporting.	Full
Certificate exchange between systems	Use to manually create truststores for member and coordinator systems.	Use to manually create truststores for member and coordinator systems.	Use to automatically provision truststores between the member systems, and enable error reporting for trust provisioning.	Automatic
Pool linking	Manual	Manual	Use to automatically link storage pools for IBM FlashSystem grid members.	Automatic
Number of systems in a grid	8	8	8	32



IBM FlashSystem grid construction configuration

This chapter provides steps to set up an IBM FlashSystem grid. IBM FlashSystem grid is a method to scale out IBM FlashSystem and IBM SAN Volume Controller (SVC) systems by adding independent systems. It explains how to perform the following tasks:

- ▶ Add and remove systems from a grid.
- ▶ Review system requirements.
- ▶ Upgrade existing IBM FlashSystem grid configurations to newer code levels.

This chapter describes the following topics:

- ▶ “Prerequisites for setting up an IBM FlashSystem grid” on page 20
- ▶ “Creating an IBM FlashSystem grid” on page 36
- ▶ “Adding and removing systems” on page 38
- ▶ “Upgrading existing IBM FlashSystem grid configurations to release 8.7.3.x or later” on page 52

2.1 Prerequisites for setting up an IBM FlashSystem grid

Your system must meet specific requirements to be used in an IBM FlashSystem grid. This section explains each requirement and provides steps to make the necessary adjustments.

If your system does not meet the prerequisites, the following error message might appear:

CMMVC1270E The command failed as the system does not meet the requirements to be a member of a grid

2.1.1 System layer and topology

The IBM FlashSystem grid creates a federated cluster of multiple independent IBM FlashSystem or SVC systems, allowing them to be managed as a single entity. This section describes considerations for the system layer and topology.

System layer

The system layer influences the interaction between systems that run IBM Storage Virtualize software.

- ▶ IBM Storage Virtualize systems operate in the *storage layer* or the *replication layer*.
- ▶ To create or join an IBM FlashSystem grid, set the system layer to replication.
- ▶ Starting with code level 8.7.0, the default setting for new systems is the replication layer.

A system can create partnerships with systems in the same layer.

Table 2-1 lists the differences between systems in the storage layer and systems in the replication layer.

Table 2-1 Differences between storage and replication layer

Storage layer	Replication layer
The system can participate in replication with other storage-layer systems.	The system can participate in replication with other replication-layer systems.
The system can provide external storage for replication-layer systems. This means it can be used as a back-end controller.	The system cannot provide external storage for other IBM Storage Virtualize systems. This means it cannot be added as a back-end controller to another system.
The system cannot use another IBM Storage Virtualize system as external storage.	The system can use another IBM Storage Virtualize system that is configured with storage layer as external storage.

Differences between layers (summary)

The following list summarizes the key differences between the storage layer and the replication layer:

► **Storage layer**

The storage layer manages physical and virtualized storage resources and provides storage services to other systems. It can also act as a back-end controller for replication-layer systems.

Capabilities

- Participates in replication with other storage-layer systems.
- Provides external storage to replication layer systems (acts as a back-end controller).
- Does not use another IBM Storage Virtualize system as external storage.

► **Replication layer**

The replication layer focuses on data movement and disaster recovery (DR). Systems in this layer manage replication relationships and depend on storage-layer systems for back-end storage.

Capabilities

- Participates in replication with other replication-layer systems.
- Provides external storage to other systems.
- Does not use a storage-layer system as external storage.

Checking and changing the layer of the system

Run the following command on the system that you plan to add to the grid:

```
> svcinfo lssystem | grep layer  
layer replication
```

To change the system layer from storage to replication, use the following command:

```
> svctask chsystem -layer replication
```

Note: The system layer can be changed during normal host I/O.

Important: When you run **chsystem -layer replication**, ensure that no other IBM Storage Virtualize systems are visible through the storage-attached network (SAN) fabric.

Remove any zoning connections to other IBM Storage Virtualize systems. Ensure that no host objects are configured with worldwide port names (WWPNs) from another IBM Storage Virtualize system and no other partnerships exist.

► If these conditions are not met, you receive a CMMVC error message:

CMMVC7143E The command cannot be initiated because nodes from another cluster are visible.

► If required, run **svctask detectmdisk** to rescan the SAN after zoning changes.

► You can change the system layer during normal host I/O. you can Restore the SAN configuration after the change.

System topology

To participate in an IBM FlashSystem grid, the system must be configured with a *standard topology*. In a standard topology, all nodes within the system are at the same physical site.

Before you join an IBM FlashSystem grid, reconfigure the system from an *enhanced stretched* or *HyperSwap topology* to a standard topology.

In a stretched topology, the nodes within an I/O group are physically distributed across different sites. Each component is assigned to a specific site, and the system topology is set to stretched. This configuration is supported on SVC systems.

To check the current system topology, use the command **svcinfo lssystem**:

```
> svcinfo lssystem | grep topology
```

If the output shows the following entries, no change is required because the system topology is already set to standard:

```
topology standard
topology_status
```

For a stretched system, the output includes the following entries:

```
topology stretched
topology_status dual_site
```

To configure the system topology, complete the following steps:

1. Change the topology from stretched to standard. Add zoning from all nodes within the system to the back-end storage controllers.
 - In enhanced stretched system configurations, zoning is only required for nodes that are at the same site as the controllers.
 - In a standard topology configuration, all nodes must have back-end storage connectivity.
2. Change the topology to standard:

```
> svctask chsystem -topology standard
```
3. Remove site awareness:
 - a. Remove the site assignment from the host objects:

```
> svctask chhost -nosite <host-id>
```
 - b. Remove the site assignment from the controller objects:

```
> svctask chcontroller -nosite <controller-id>
```
 - c. Remove the site assignment from the nodes:

```
> svctask chnode -nosite <node-id>
```
4. Confirm that the topology_status is empty:

```
> svcinfo lssystem | grep topology
```

Note: You can convert the system non-disruptively. However, the behavior changes during a fail-over scenario.

Converting from HyperSwap to standard topology

To convert the topology from HyperSwap to standard, complete all prior steps for converting from stretched to standard. Then complete the following tasks before you change the topology:

Convert existing HyperSwap volumes to standard volumes by using the following steps:

1. HyperSwap volumes have two active copies (one at each site) for high availability (HA). To convert the volumes, remove one copy so that each volume becomes a single-copy standard volume by running the following command:

```
svctask rmvolume copy -pool <pool-name> <volume-name>
```

2. Use the **-pool** parameter and specify the name or ID of the storage pool that is at site 2, and specify the volume name.

Instead of using the pool, specify the site directly by using the following command:

```
svctask rmvolume copy -site <site-id-or-name> <volume-name>
```

3. After removing the extra copy, confirm that no HyperSwap volumes remain by running the following command:

```
svcinfolsvdisk -delim : -gui -filtervalue 'ha_type=hyperswap'
```

If no output is displayed, the command was successful.

4. After converting the volumes, remove any extra I/O groups from the system. IBM FlashSystem grid supports only one I/O group per system. For more information, see 2.1.2, “Single I/O group” on page 23.

2.1.2 Single I/O group

Before you can add a system to an IBM FlashSystem grid, it must meet specific architectural requirements. One of the most important is the I/O group configuration.

An IBM FlashSystem grid supports only one I/O group per system and the following configurations apply:

- For enclosure-based products, this configuration means a single control enclosure.
- For SVC systems, this configuration means a 2-node system.

If your system has multiple I/O groups, you must remove the additional I/O groups before joining the grid. To do so, migrate all volumes from the extra I/O group to the remaining group to preserve data.

Migrating volumes from an I/O group

For volumes in standard pools, you can migrate them using the following command:

```
svctask movevdisk -iogrp <destination-iogrp> <vdisk-name>
```

For more information about this command and possible restrictions, see [IBM Documentation](#).

After migration, verify that the VDisk is accessible through the newly assigned I/O group. If it is not, add access manually by running the following command:

```
svctask addvdiskaccess-iogrp <destination-iogrp> <vdisk-name>
```

Tip: To perform this action without disrupting I/O, ensure that hosts support Non-Disruptive Volume Movement (NDVM). Supported hosts include the following operating systems and platforms:

- ▶ IBM AIX®
- ▶ Windows
- ▶ Red Hat Enterprise Linux (RHEL)
- ▶ SUSE Linux Enterprise Server
- ▶ VMware

Alternative migration method (adding a mirrored copy)

If you cannot use the standard migration method or need a non-disruptive approach, you can migrate data by creating a mirrored copy of the volume. This method uses the **addvdiskcopy** command to create a second copy of the VDisk in the same or a different storage pool. After synchronization, the original copy is deleted automatically if you specify the **-autodelete** option:

```
svctask addvdiskcopy <vdisk-name> -pool <pool-name> -autodelete
```

This method works with Data Reduction Pools (DRP). For the available command options, see [IBM Documentation](#). The original copy is deleted automatically after synchronization if you specify **-autodelete**.

To check the synchronization progress, run the following command:

```
svcinfolsvdisksyncprogress
```

Benefits of using addvdiskcopy

Compared to other migration methods, using the **addvdiskcopy** provides the following benefits:

- ▶ Adjust the synchronization rate to optimize migration performance:
 - Increase the synchronization rate to accelerate migration.
 - Decrease the synchronization rate if application performance is affected.
- ▶ Stop migration if needed by deleting the new copy by running the following command:

```
svctask rmvdiskcopy <vdisk-name> <copy-id>
```

Note: Using the **addvdiskcopy** command for migrating data is the preferred option.

Removing extra I/O groups

After all volumes are migrated, remove the additional I/O group, as described in [Removing a control enclosure from a system](#). If needed, configure the removed I/O group as a stand-alone system by initializing it, as described in [initializing a system](#).

Disabling multiple I/O group compatibility mode

The compatibility mode for multiple I/O groups must be disabled before adding a system to an IBM FlashSystem grid. You can check and change this setting by using the CLI to run the following commands:

- ▶ Check the current setting by running the following command:

```
svcinfolscompatibiltymode -delim :
```
- ▶ The following example shows the output of the command:

```
feature:enabled:in_use  
multiple_io_groups:no:no
```

This output shows that the `multiple_io_groups` feature is disabled (no).
- ▶ If the feature is enabled, you can disable it by running the following command:

```
svctask chcompatibiltymode -disable -feature multiple_io_groups
```

2.1.3 Network connectivity

IBM FlashSystem grid requires IP connectivity for secure communication, certificate exchange, and management operations. Related features, such as replication and partition migration, also require FC connectivity between the systems. SAN connectivity is not required for IBM FlashSystem grid.

Ethernet network connectivity

All systems in an IBM FlashSystem grid must be able to communicate over an Ethernet network.

Each system requires a management IP address, which can be IPv4 or IPv6. You can also use a fully qualified domain name (FQDN) if a Domain Name System (DNS) server is configured.

Systems must allow HTTPS connectivity and REST API access between each other. The same IP connection is also used to exchange certificates automatically between the systems.

Ensure that TCP ports 443, 6443, and 7443 are open for inbound and outbound traffic on the management IP address.

Verifying the IP connectivity and related settings

To help ensure communication between systems in an IBM FlashSystem grid, verify the IP connectivity, DNS configuration, and time synchronization by completing the following steps:

1. Check the IP connectivity:
 - a. Run the following **ping** command:

```
svctask ping -srcip4 <source-system-ip> <target-system-ip>
```
 - b. If ping is disabled, run the following **traceroute** command:

```
sainfo traceroute -ip_or_name <target-system-ip>
```

Run the command on the configuration node by using SSH.

Note: If you are using a DNS server, ensure that it is configured to include a Pointer (PTR) Resource Record for the system's management IP address.

2. Verify the DNS configuration:
 - a. Ensure that the DNS server includes a PTR response record for the system's management IP address. The IBM Storage Virtualize system uses reverse DNS lookup to validate connectivity. If the reverse lookup fails, Error Code 1585 is logged.
 - b. Check the DNS server access:


```
svcinfolsdnsserver
```
 - c. Ensure that the status is active.
3. Synchronize time:

Time must be synchronized between the systems to prevent issues during TLS certificate validation. Use a Network Time Protocol (NTP) server for consistency.

 - a. Verify the IP address of the configured NTP server by running the following command:


```
svcinfolssystem | grep ntp
```

This value shows the IP address in the `cluster_ntp_IP_address` field.

Test the connectivity to the NTP server by running the following command:

```
ping <ntp-server-ip>
```
 - b. Check the current date and time by running the following command:


```
svqueryclock
```

If the time is not synchronized on both systems, TLS handshake validation might fail when it checks remote system certificates.

2.1.4 Authorization requirements

To enable secure communication between systems in an IBM FlashSystem grid, each system must exchange TLS certificates.

These certificates provide mutual authentication and protect management operations during communication:

- ▶ The REST apiserver presents its system certificate. The client verifies the certificate by using its truststore.
- ▶ The client presents its system certificate. The server verifies the certificate by using its own truststore.

This mutual verification helps ensure that both systems are trusted and that communication is securely established. This subsection explains certificate requirements, scope values, certificate chains, and truststore configuration.

Certificate requirements for IBM FlashSystem grid

Starting with release 9.1.x, IBM FlashSystem grid supports multiple SSL certificates. Each certificate can be assigned to a specific feature or use case. For more information, see "Multiple SSL certificates" on page 34.

With this update, the system uses separate internal certificates for all communication between IBM FlashSystem grid members. By default, new installations or systems that are upgraded to release 9.1.0 or later, the system continues to use the main system certificate until you configure a dedicated certificate. As a best practice, use the *IBM FlashSystem grid system certificate*, which is designed for grid communication.

If your system is part of an IBM FlashSystem grid and you upgrade to release 9.1.0.0 or later, the system automatically generates this certificate. It is internally signed and named the *internal communication certificate*, helping ensure uninterrupted communication between grid members.

Certificate scope values

Starting with release 9.1.0.0, system certificates include a *scope value* that defines their intended use:

- Scope 0. Default scope. Used for web browser access, key servers, and general purposes.
- Scope 1. Key server scope. The certificate is presented to key servers.
- Scope 3. Internal communication scope. Used for secure communication between member systems within an IBM FlashSystem grid.

Each scope helps ensure that the certificate is applied correctly in the context of the operation.

Figure 2-1 shows the GUI Certificates view for a system that upgraded to release 9.1.0.0. It displays the main system certificate and the internally signed certificate that is used for grid communication. This certificate is used by the IBM FlashSystem grid.

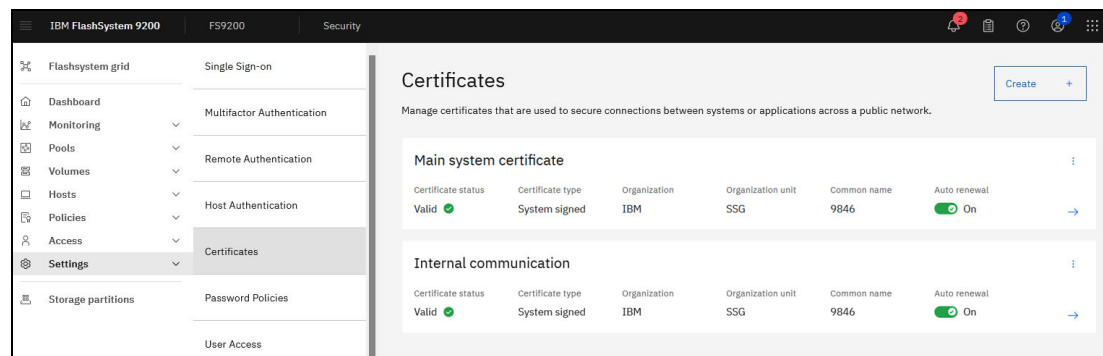


Figure 2-1 Main system certificate and internal communication certificate in the release 9.1.0.0 GUI

Switching from externally signed certificates

If your system previously used an externally signed certificate and upgraded to release 9.1.x, you can switch to the new internally signed certificate by completing the following steps:

1. Remove the system from the IBM FlashSystem grid, which clears the existing certificate and trust relationship.
2. Add the system back to the grid, which automatically applies the new internal certificate.

The following sections explain how certificates are used and managed within IBM FlashSystem environments, helping you understand their role in secure communication.

System certificates

Your IBM Storage Virtualize system supports multiple methods for signing system certificates. By default, your system uses internally signed certificates. The internal root certificate authority (CA) signs these certificates. IBM introduced support for this internal CA in release 8.5.3.

Alternatively, you can use certificates that are signed by a third-party provider (external CA) or by your organization's private CA.

If your system currently uses a self-signed certificate, it continues to function until it expires. However, newer code releases support only internally or externally signed certificates.

As a best practice, update your system certificate before the self-signed certificate expires.

Best practice: Use internally signed certificates because they support *automatic renewal* by default. Your system automatically generates a new certificate 30 days before expiration, simplifying renewal and helping prevent security warnings due to expired certificates.

In contrast, you must renew externally signed certificates manually.

To view the system certificate and its expiration date, run the following command:

```
svcinfo lssystemcert
```

If automatic renewal is enabled, the output includes `automatic_renewal:yes`.

The command also displays the certificate chain. Each certificate begins with `-----BEGIN CERTIFICATE-----` and ends with `-----END CERTIFICATE-----`.

The `certificate_type` is listed as `self_signed`, `system_signed` or `externally_signed`.

The GUI certificate window was updated in code release 9.1.0.0. You might notice differences in the layout and menu options compared to earlier versions.

To update an externally signed certificate, create a Certificate Signing Request (CSR) in the 9.1 GUI by selecting **Settings** → **Security** → **Certificates** → **Replace**. The system saves the CSR file as `certificate.csr`, in the `/dumps` directory. Submit this CSR to a CA for signing. After the CA returns the signed certificate, install it on the system.

Note: Using self-signed certificates is not a best practice.

Installing a signed certificate and completing the CSR form

To install the signed certificate in the 9.1 GUI, select **Settings** → **Security** → **Certificates** → **Install certificate**, as shown in Figure 2-2.

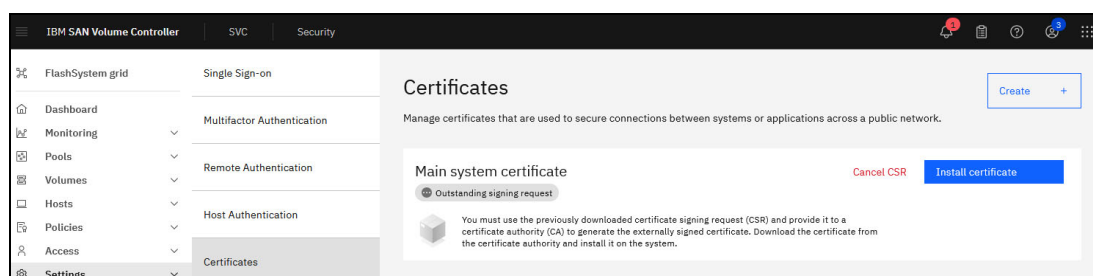


Figure 2-2 Outstanding Certificate Signing Request in the 9.1.0.0 GUI Certificates window

When you create a CSR, complete the form and include the following required details:

- ▶ In the certificate Subject Alternative Name field, include the system management IP address, the service IP addresses for each node, and any IP addresses that are configured for IP replication.
- ▶ Use domain names, email addresses, Uniform Resource Identifiers (URIs), or a combination of them.
- ▶ At a minimum, include the system management IP address or DNS name.

The following list includes other important fields:

- ▶ Organization name (O)
- ▶ Organizational unit (OU)
- ▶ Common name (CN)

The internal root CA uses the following values:

- ▶ Organization name. International Business Machines Corporation
- ▶ OU. Spectrum Virtualize Root CA
- ▶ CN. The serial number

Those root certificates are valid for 20 years.

Certificate file requirements

Ensure that the following requirements are met:

- ▶ Do not include blank lines in the certificate file.
- ▶ Ensure that each certificate ends with a newline character.
- ▶ Use Linux line endings and do not use Carriage Return Line Feed (CRLF) at the end of the line.

You can use a certificate decoder to verify validity.

Starting with release 9.1.0.0, the system also supports .crt and .cer file types.

Key usage requirements

Ensure that the following requirements are met:

- ▶ The signed certificate must contain the Digital Signature X509v3 Key Usage extension.
- ▶ Each externally signed certificate must also include these Extended Key Usage extensions:
 - TLS Web Client Authentication
 - TLS Web Server Authentication
 - Any Extended Key Usage

Figure 2-3 shows the tip that the GUI displays in the Manage main system certificate window when you click **Externally Signed Certificate**.

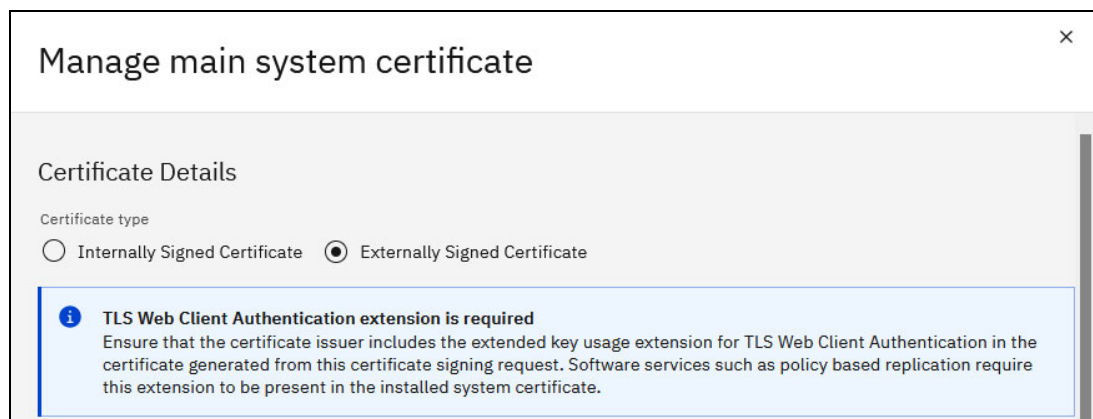


Figure 2-3 TLS Web Client Authentication extension is required

The GUI displays the following warning in the Install Signed Certificate wizard if any required extensions are missing:

The extended key usage extension for TLS Web Client Authentication is missing from the uploaded certificate. Software services such as policy-based replication require this extension to be present in the installed system certificate.

This issue can occur if the CA removes required extensions from the certificate.

Use OpenSSL to inspect and manage certificates. With OpenSSL, you can perform the following tasks:

- ▶ View the complete contents of a certificate, including subject, issuer, and validity details.
- ▶ Extract specific details, such as X.509 certificate extensions.
- ▶ Convert certificates between different formats, for example, PEM and DER, which is helpful if your certificate is not in PEM format, which is required for many systems.

Example 2-1 shows how to display the contents of a certificate that uses the **openssl** tool.

Example 2-1 Using the openssl tool to interact with certificates

```
> openssl s_client -showcerts -connect <system-ip>:<port>
```

Certificate chain

Either a root CA or an intermediate CA signs a system certificate. The root CA is a trusted third-party authority at the top of the certificate hierarchy. It issues certificates to intermediate CAs or directly to servers. Because the root CA verifies its own identity, the subject and issuer distinguished names in a root certificate are identical.

An intermediate CA is subordinate to the root CA. It issues certificates to organizations or devices further down the trust chain. Either a root CA or another intermediate CA signs an intermediate certificate.

A *certificate chain* is a file that contains a sequence of certificates. The chain typically starts with the system certificate, continues with any intermediate certificates, and optionally ends with the root certificate. This chain establishes a path of trust from the system certificate back to a trusted root.

When you use externally signed certificates, create a single file that includes the following item:

- ▶ The signed system certificate
- ▶ Any intermediate CA certificates
- ▶ The root CA certificate (optional)

For internally signed certificates, the system automatically includes the internal root CA certificate in the system certificate file.

Note: A certificate chain might include intermediate certificates, depending on the configuration and the CA structure.

TLS handshake validation

Figure 2-4 illustrates the certificate chain for an externally signed certificate.

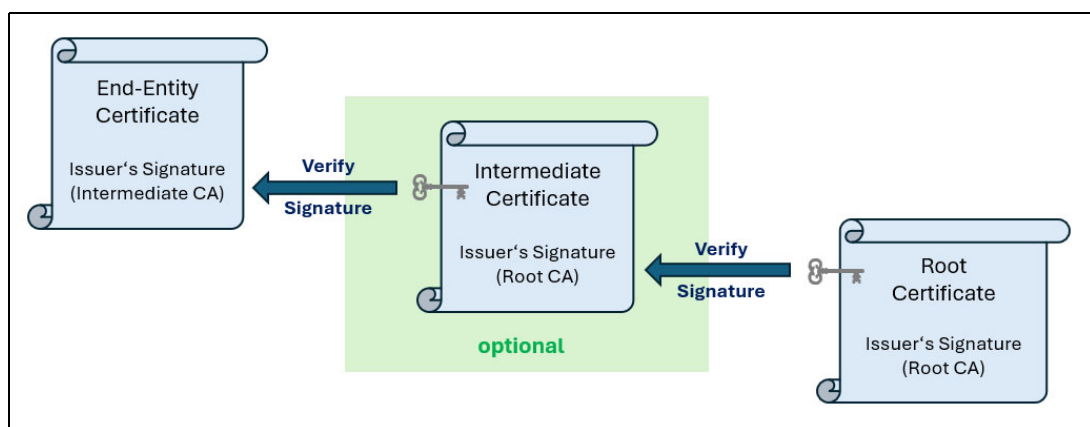


Figure 2-4 Chain of trust

During the TLS handshake, the server sends its certificate to the client. The client validates the certificate by performing the following steps:

1. Verify the signature of the end-entity (server) certificate by using the public key of the intermediate CA.
2. Verify the signature of the intermediate certificate that uses the public key of the root CA.

If the entire chain of trust is valid and traces back to a trusted root certificate, the client considers the server certificate as authentic and establishes a secure connection.

Upgrade requirement for release 8.7.3.0 or later

If you upgrade to release 8.7.3.0 or later, the software upgrade test utility might display the following error when the system certificate does not contain the full certificate chain:

From release 8.7.3.0 onwards, the IBM FlashSystem with IBM FlashSystem grid enabled requires a full certificate chain, including a root certificate, to be installed as the system certificate. Use the `chsystemcert` command to install the full certificate chain and rerun the CCU checker.

Starting with release 8.7.3.0, IBM FlashSystem grid requires a full certificate chain, including the root certificate, to be installed as the system certificate.

Installing the full certificate chain

To install the full certificate chain in the GUI, complete the following steps:

1. Select **Settings** → **Security** → **Certificates**.
2. Click **Install certificate**.
3. Upload the `certificate.pem` file that includes the full certificate chain (system certificate, intermediate certificate, and root certificate).

Note: The browser connection might drop briefly during installation (approximately 1 - 2 minutes).

Figure 2-5 shows the error message that appears when you try to join an IBM FlashSystem grid without installing the full certificate chain.

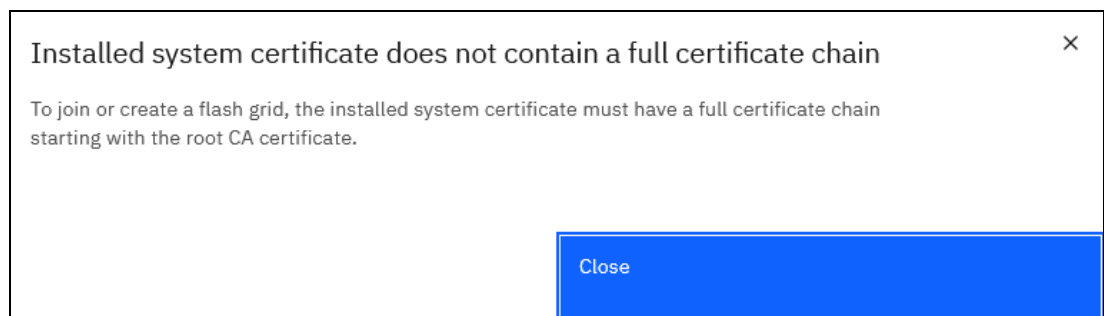


Figure 2-5 System certificate does not contain a full certificate chain

In this example, the system was using a self-signed certificate with an outstanding CSR. After canceling the request and replacing the certificate to an internally signed certificate, the system successfully joined the IBM FlashSystem grid.

Exporting certificates by using the GUI and CLI

You can export certificates by using the GUI or CLI.

Exporting certificates by using the GUI

To export certificates by using the GUI, complete the following steps:

- ▶ Open the GUI and select **Settings** → **Security** → **Certificates** → **Export**.
- ▶ To export the system certificate, click **End point certificate**.
- ▶ To export the root certificate, click **Root CA Certificate**.
- ▶ Choose **Export** to download the certificate, or **Copy to clipboard** to copy its contents.

Exporting certificates by using the CLI

To export certificates by using the CLI, complete the following steps:

1. Export the default system certificate by running the following command:

```
svctask chsystemcertstore -export -scope default
```

The system saves the file as `certificate.pem` in the `/dumps` directory.
2. Export the internal communication certificate (used by IBM FlashSystem grid) by running the following command:

```
svctask chsystemcertstore -export -scope internal_communication
```

The system saves the file as `system_certificate_slot_3.pem`, where 3 is the scope value for internal communication. For more information about the scope values, see “Certificate requirements for IBM FlashSystem grid” on page 26.

3. Export the default root CA certificate by running the following command:

```
svctask chsystemcert -exportrootcert
```

The system saves the file as `rootcertificate.pem` in the `/dumps` folder.

4. Export the root CA certificate for internal communication by running the following command:

```
svctask chsystemcertstore -exportrootca -scope internal_communication
```

The system saves the file as `system_rootcertificate_slot_3.pem`.

5. Use `svcinfo lsdumps | grep “.pem”` to display exported files.
6. Use Secure Copy (SCP) or Secure File Transfer Protocol (SFTP) to copy the files from the `/dumps` directory to your local system.
7. Upload the exported files to the target system in the `/tmp` directory and add them to the truststore.

You can also add certificates to browsers or other devices (such as key servers) to establish trust.

Adding certificates to the truststore

IBM FlashSystem grid systems add certificates to each other’s truststore to enable secure communication.

To add a certificate manually, use the **-grid** parameter with the value *on* when adding certificates to the truststore. Specify the syntax in the following format:

```
-grid on | off
```

This optional parameter specifies that the certificate in the truststore is for IBM FlashSystem grid use.

This parameter also sets **-restapi on**, which is required for the IBM FlashSystem grid use case. It cannot be used with any other parameter.

A truststore is a CA store that contains certificates from external devices. The system uses those certificates to establish trust with those devices.

For IBM FlashSystem grid, add only the root certificate to the truststore because the system supports chain-of-trust validation.

Important: Starting with code release 8.7.3.0, if you attempt to add the system certificate along with a full chain to the truststore, the system might display the following error:

```
CMMVC1373E The command failed as the certificate provided as trust doesn't meet the required criteria of trust in the grid.
```

When you add a system to an IBM FlashSystem grid, complete the following steps:

1. Download the root certificate of the coordinator system and copy it to the new system.
2. Add the certificate to the truststore by running the **svctask mktruststore** command.
3. Add the new system’s root certificate to the coordinator system’s truststore by performing the previous steps in reverse order.

After the certificates are exchanged, verify the system certificate and corresponding truststore by running the following commands:

```
svcinfo lssystemcert  
svcinfo lstruststore
```

Example 2-2 demonstrates how to add the coordinator system's certificate to another system's truststore.

Use the following naming convention for the truststore entries followed by the management IP address of the coordinator system:

TS	The truststore
FG	IBM FlashSystem grid

This consistent naming helps ensure clarity and alignment across systems in the IBM FlashSystem grid.

Example 2-2 Adding a partner certificate to the internal truststore

```
> svctask mktruststore -file /tmp/root_certificate.pem -grid on -name  
TS_FG_9.80.82.80
```

Starting with release 8.7.3, the GUI procedure for adding a system to an IBM FlashSystem grid automatically manages certificates. This process adds the required certificates to the truststore, removing the need for manual steps. Manually creating truststore entries can cause conflicts. To avoid issues, use either the GUI or the CLI consistently when adding or removing systems.

If a remote GUI response does not include the issuer certificate, the GUI prompts you to upload a file that contains the complete certificate chain. This file must be a PEM-encoded combination of certificates. After you upload the file, the system adds the full certificate chain to the truststore with the REST API flag enabled, helping ensure proper trust configuration.

Multiple SSL certificates

Traditionally, IBM Storage Virtualize systems used a single SSL certificate for all secure communications. This certificate is shared across the following features:

- ▶ Browser access to the Management and Service GUI
- ▶ Communication with key servers
- ▶ REST API access
- ▶ IP Quorum application communication
- ▶ Other certificate-dependent functions

Starting with release 9.1.0.0, IBM Storage Virtualize supports multiple SSL certificates. Administrators can install and assign different certificates to specific features, enhancing both security and flexibility.

One key change in this release is the introduction of a dedicated IBM FlashSystem grid system certificate. If your system is part of an IBM FlashSystem grid and is upgraded to release 9.1.0.0 or later, the system automatically generates this new certificate. This certificate replaces the use of the default system certificate for grid-related communications.

Managing SSL certificates for IBM FlashSystem grid

With the new system certificate store, IBM Storage Virtualize systems manage SSL certificates for individual use cases more effectively. One key enhancement is the IBM FlashSystem grid system certificate, which secures communication for the following features:

- ▶ Peer-to-peer communication between IBM FlashSystem grid members
- ▶ Policy-based Replication (PBR)
- ▶ Policy-based High Availability (PBHA)
- ▶ Partition migration

When you add a system to an IBM FlashSystem grid, the system automatically generates and installs a dedicated, system-signed certificate for grid-related operations. If you remove the system from the grid, the certificate is automatically uninstalled and deleted to maintain security.

Creating a certificate for a specific use case

To create a certificate for a specific use case, complete the following steps:

1. Select **Settings** → **Security** → **Certificates**.
2. Click **Create**.
3. Select one of the following certificate scopes:
 - Key server
 - Partition
4. If you select **Partition**, choose a partition from the drop-down list. Only partitions with an assigned IP address are eligible for certificate use.
5. Complete the required fields, and click **Create and Install**.

For partition-scoped certificates, you can choose between a system-signed or an externally signed certificate.

In IBM FlashSystem grid environments, each partition requires its own certificate. These certificates are managed in the partition certificate store, which supports multiple IBM FlashSystem grid partition certificates. They enable secure authentication of storage partitions and support components such as VMware plug-ins.

CLI commands for Certificate Management

Here are new commands for Certificate Management:

- ▶ `svcinfo mkssystemcertstore`
- ▶ `svcinfo lssystemcertstore`
- ▶ `svctask chssystemcertstore`
- ▶ `svcinfo mkpartitioncertstore`
- ▶ `svcinfo lspartitioncertstore`
- ▶ `svctask chpartitioncertstore`

Example 2-3 demonstrates how to create a new externally signed certificate for a specific use case.

Example 2-3 Creating an externally signed certificate for a specific use case

```
> svctask mkssystemcertstore -externalsigned -scope default
```

Example 2-4 shows the command to create a partition certificate.

Example 2-4 Creating a partition certificate for the IBM FlashSystem grid use case

```
> svctask mkpartitioncertstore -systemsIGNED -partition <partition_id_or_name>
```

Specify either **-systemsIGNED** or **-externalsIGNED**. Other optional parameters can also be provided.

Benefits of multiple SSL certificates

With support for multiple SSL certificates, you can now configure dedicated certificates for individual use cases, improving both security and manageability. Although the system can still use the default system certificate across multiple features, you can also do the following tasks:

- ▶ Create and assign separate certificates for specific use cases, such as key server communication or partition-level operations.
- ▶ Assign individual certificates to each partition for granular control and isolation.

This flexibility enables you to customize your certificate strategy to align with specific security and operational requirements.

2.2 Creating an IBM FlashSystem grid

This section guides you through creating a new IBM FlashSystem grid by using the GUI or the CLI. It also explains the roles of coordinator and member systems.

2.2.1 Coordinator and member systems

In an IBM FlashSystem grid, one system acts as the coordinator, whereas the others serve as member systems. The coordinator system creates and manages the grid. It adds and removes member systems and is always the first system you add to the grid.

The coordinator function is not currently highly available. If the coordinator system becomes unavailable due to a failure or network issue, grid management operations stop. You cannot add or remove systems until you restore the coordinator or create a new grid with a new coordinator and member systems.

You can add member systems to the grid. When you add a system as a member of an IBM FlashSystem grid, the grid manages that system.

You can display the member systems of a grid by running the following command:

```
svcinfo lsgridmembers
```

You can also use the **svcinfo lsgridsystem** command.

To list all systems in an IBM FlashSystem grid, run the following command:

```
svcinfo lsgridsystem
```

The GUI login window for a member system shows the message **Managed by IBM FlashSystem grid**. If the system is part of an IBM FlashSystem grid, run the following command. The output includes the grid's Universally Unique Identifier (UUID):

```
lssysteminfo
```

If the system is not part of a grid, the `grid_uuid` field is blank.

2.2.2 Creating an IBM FlashSystem grid and adding the first system

You can create an IBM FlashSystem grid from the dashboard. If the system supports IBM FlashSystem grid, the dashboard displays an informational banner with a **Learn more** link, such as the one shown in Figure 2-6. Click **Learn more** to open the IBM FlashSystem grid view, where you can join an existing IBM FlashSystem grid or create a new one.

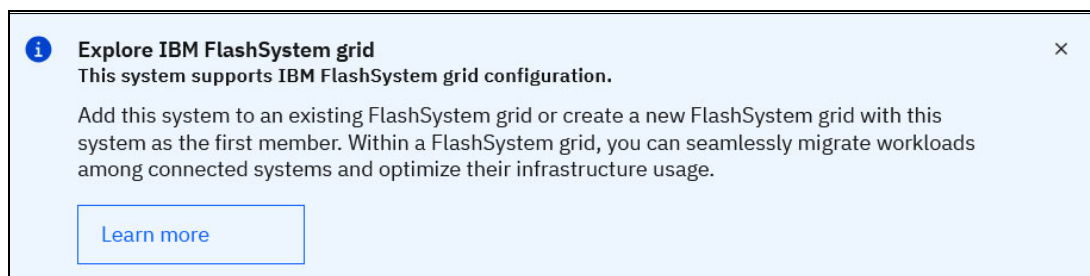


Figure 2-6 IBM FlashSystem grid supported

Creating an IBM FlashSystem grid in the GUI

Complete the following steps:

1. Select **Settings** → **System** → **IBM FlashSystem grid**.
2. Click **Create a new IBM FlashSystem grid**.
3. Enter a name for the IBM FlashSystem grid, and click **Create**, as shown in Figure 2-7.

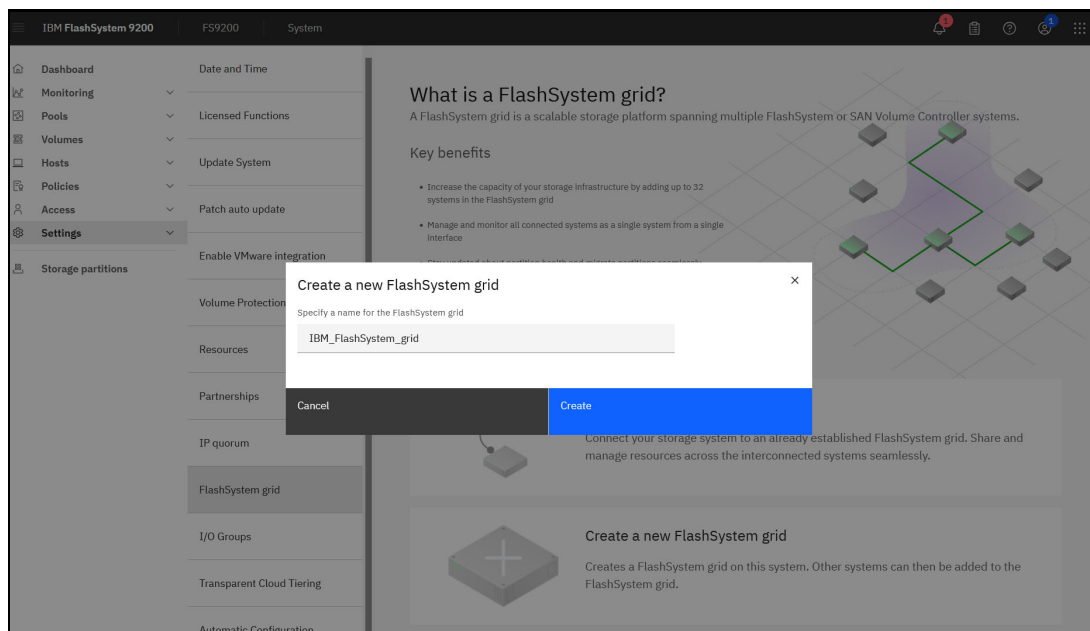


Figure 2-7 Creating an IBM FlashSystem grid by using the GUI

After the task completes, close the dialog box. The system redirects you to the IBM FlashSystem grid dashboard. From this dashboard, you can do the following tasks:

- ▶ Add member systems to expand your grid.
- ▶ View system and partition details.
- ▶ Monitor grid health and activity.

For more information about dashboard features, see Chapter 3, “Managing and monitoring the IBM FlashSystem grid” on page 59.

Creating an IBM FlashSystem grid in the CLI

To create an IBM FlashSystem grid in the CLI, complete the following steps:

1. Connect to the coordinator system by using SSH.
2. To create the system grid and add the current system as the coordinator, run the following command:

```
svctask managegrid -create -name <flashsystem-grid-name>
```

3. Run the following command to verify that the grid was created:

```
svcinfolsssysteminfo
```

4. Check the `grid_uuid` field for the new grid ID.

- a. To verify the grid details, run the following command:

```
svcinfolsgrid
```

This command displays the grid name, UUID of the grid, number of systems, the number of storage partitions, and capacity information.

- b. Review 2.1, “Prerequisites for setting up an IBM FlashSystem grid” on page 20 to prepare your system for use with IBM FlashSystem grid if you see the following error message:

```
CMMVC1270E The command failed as the system does not meet the requirements  
to be a member of a grid
```

If the system is already a member of an IBM FlashSystem grid, the command fails with the following message:

```
CMMVC1265E The command failed as this system is already a member of a grid
```

2.3 Adding and removing systems

You can add systems to an IBM FlashSystem grid to increase capacity, improve performance, or replace older hardware. By adding a system, you can scale the number of supported volumes, hosts, and snapshots. You can also remove systems when they are no longer needed. These actions can only be performed from the coordinator system.

2.3.1 Adding a system to an IBM FlashSystem grid

You can add a system to an IBM FlashSystem grid by using the GUI or CLI.

Adding a system by using the GUI

1. In your browser, connect to the IBM FlashSystem grid coordinator system and log in.
2. From the left navigation pane, click **Storage systems**. The table lists all current members of the IBM FlashSystem grid.
3. Click **Add a system** (shown in Figure 2-8 on page 39) to begin adding a member system to the IBM FlashSystem grid.

Storage systems	Product name	Code level	Last contacted	Connectivity health	Hardware health	Physical capacity
FS9200 coordinator	IBM FlashSystem 9200	9.1.0.0	now	Unhealthy	Unhealthy	49.43 GiB / 37.00 TiB (0.13%)
SVC	IBM SAN Volume Controller	9.1.0.0	now	Healthy	Healthy	0 bytes / 0 bytes (0.00%)

Figure 2-8 Storage systems view in the IBM FlashSystem grid dashboard

- Enter the IP address or FQDN of the new member system that you want to add, as illustrated in Figure 2-9. In this example, the IP address of the IBM FlashSystem 7300 system is 9.80.82.60.

Add a system

FQDN or IP address of the system

Cancel
Login to the system

Figure 2-9 Entering the IP address or FQDN of the new member system

- Click **Login to the system**. You are redirected to the management GUI of the new member system, where you can complete the process of adding it to the IBM FlashSystem grid.

6. In the GUI of the new member system, click **Verify certificate**, as shown in Figure 2-10, to retrieve the coordinator system's certificate.

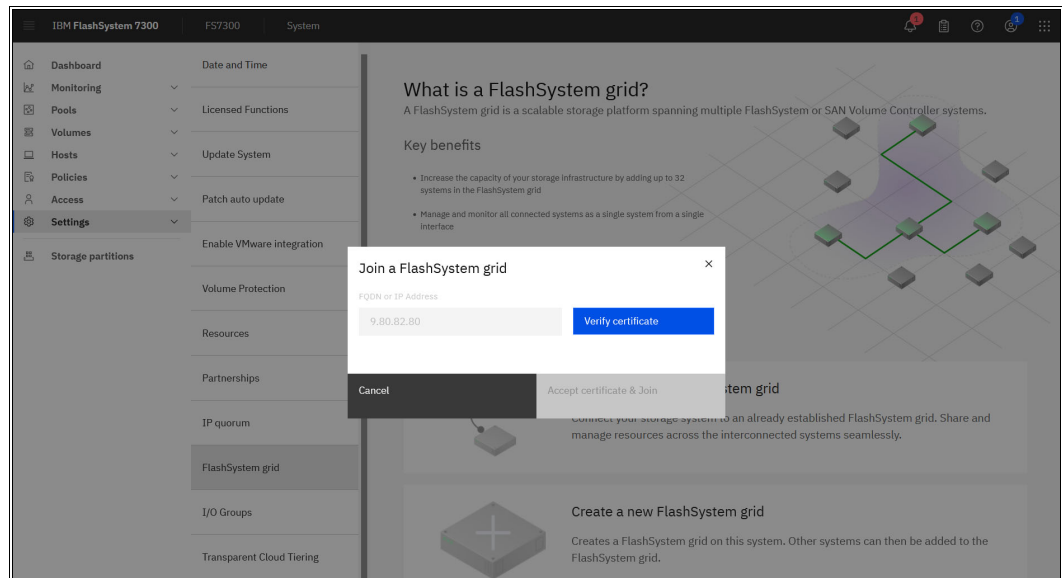


Figure 2-10 Joining an IBM FlashSystem grid

7. After all volumes are migrated, remove the additional I/O group, which is described in [Removing a control enclosure from a system](#).

This action automatically performs the following steps:

- Adds the coordinator system's certificate to the member truststore
- Sends a join request to the coordinator

After the process finishes, click **Close** to return to the main interface.

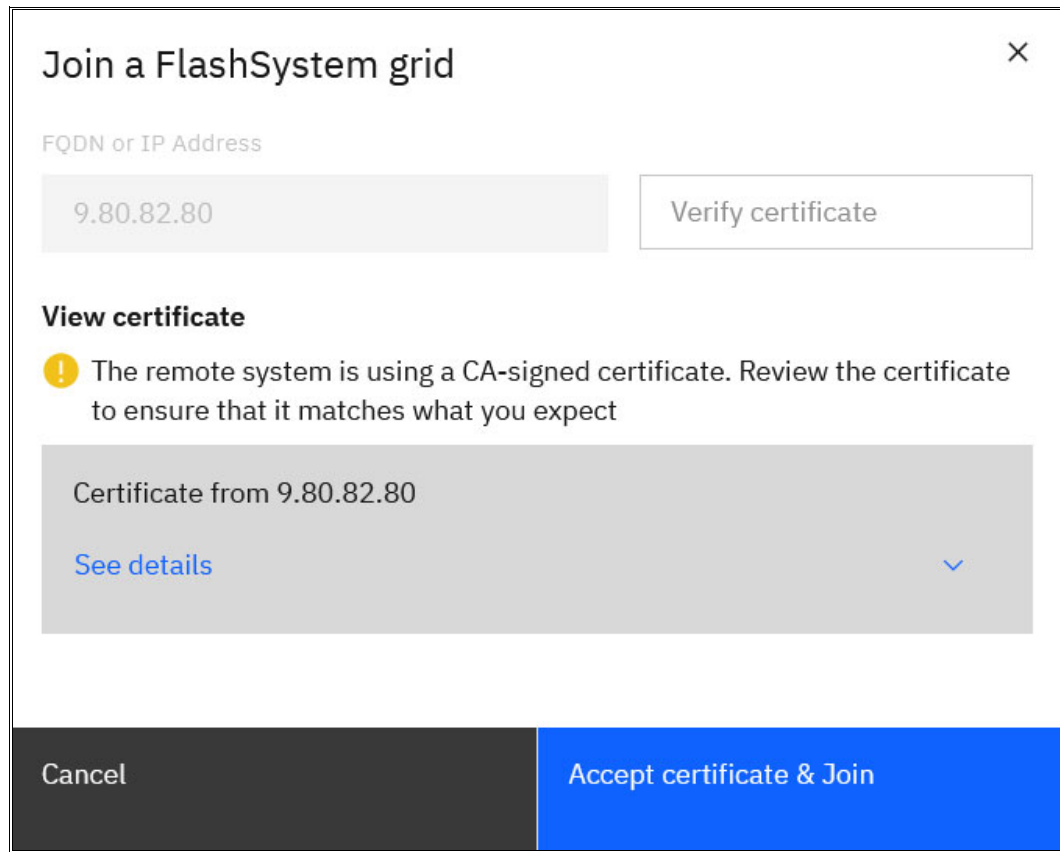


Figure 2-11 Verifying the certificate of the coordinator system

8. Click **Continue**, as shown in Figure 2-12. You are automatically redirected back to the coordinator system.

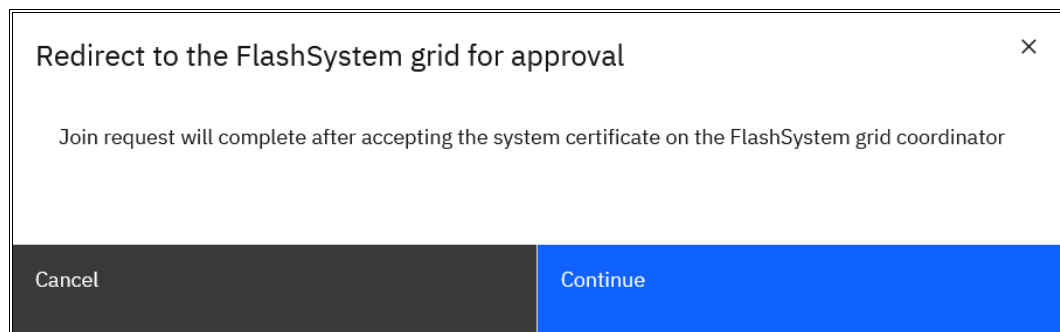


Figure 2-12 Proceeding with redirection to the coordinator system for approval

- In the coordinator's grid dashboard, review the member's certificate, and click **Approve join request**, as shown in Figure 2-13.

This action performs the following steps:

- Adds the new member system's certificate to the coordinator's truststore.
- Accepts the join request.
- Adds the system to the IBM FlashSystem grid as a new member.

After the task finishes successfully, click **Close** to finish the process.

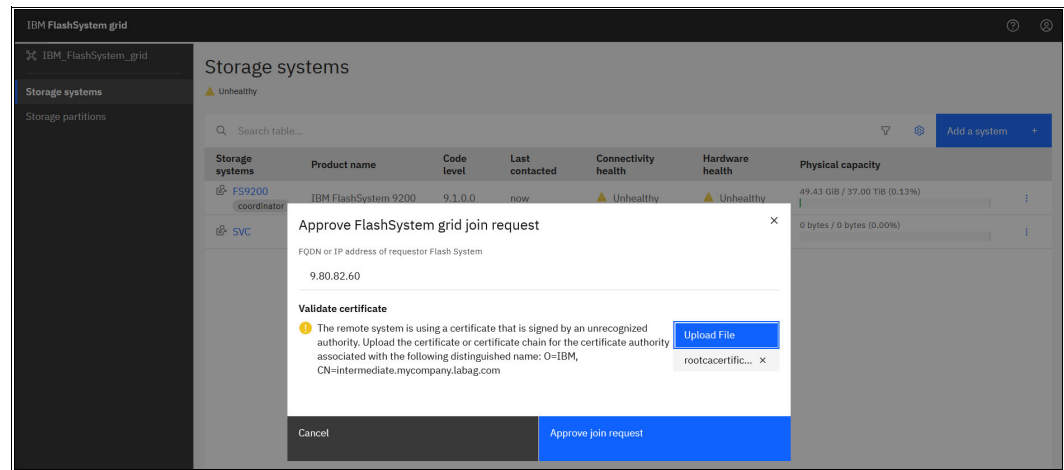


Figure 2-13 Approving the join request to add the system to the IBM FlashSystem grid

After you finish all the steps, the Storage systems window displays the newly added member system in the table (Figure 2-14). In this example, the IBM FlashSystem 7300 is successfully added as a new member of the IBM FlashSystem grid.

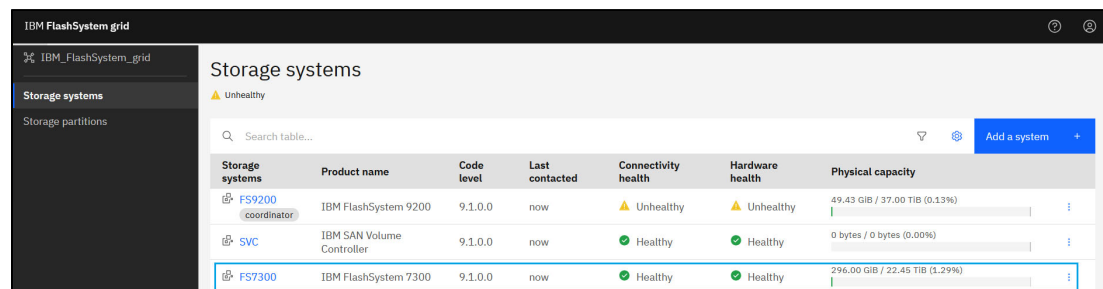


Figure 2-14 The Storage systems window now shows three member systems

10. Alternatively, initiate the join request directly from the management GUI of the system that you intend to add as a member. Select **Settings** → **System** → **IBM FlashSystem grid** and click **Join an existing IBM FlashSystem grid**, as shown in Figure 2-15.

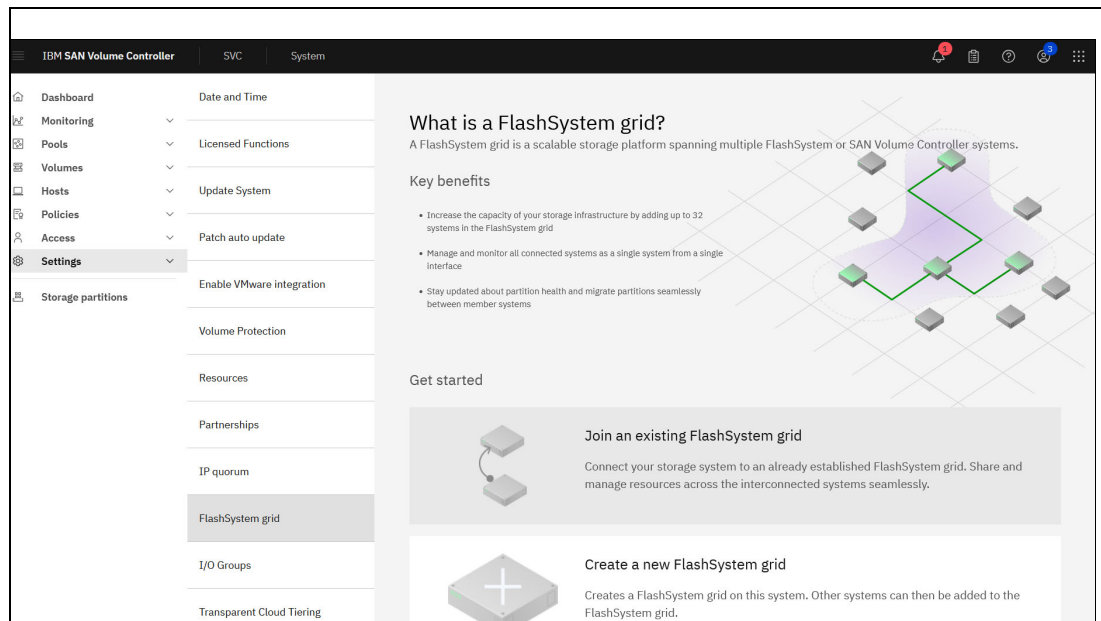


Figure 2-15 Join an existing IBM FlashSystem grid

11. Then, enter the IP address or FQDN of the IBM FlashSystem grid coordinator system. Click **Verify certificate**, and then click **Accept certificate & Join**, as shown in Figure 2-16.

Join a FlashSystem grid

FQDN or IP Address

9.80.82.80

Verify certificate

View certificate

! The remote system is using a CA-signed certificate. Review the certificate to ensure that it matches what you expect

Certificate from 9.80.82.80

[See details](#)

Cancel

Accept certificate & Join

Figure 2-16 Accepting the certificate of the coordinator system and sending the join request

12. Click **Continue** to redirect to the IBM FlashSystem grid coordinator system for approval. After you confirm On the coordinator, review the join request and click **Approve join request**. The system joins the grid.

Note: If the new member system uses an externally signed certificate, the coordinator prompts you to upload the CA certificate or certificate chain that signed the remote system's certificate, as shown in Figure 2-13 on page 42.

Click **Upload File** to upload the corresponding root CA certificate file that you previously exported from the new member system. This step helps ensure that the coordinator system can validate and trust the certificate that is presented by the new member system.

Adding a system by using the CLI

To add a system by using the CLI, exchange root certificates and create truststore entries on both systems, then submit and approve the join. Complete the following steps:

1. Export the root certificate by using either the GUI or the following CLI command:

```
svctask chsystemcertstore -exportrootcert -scope internal_communication
```

For more information about how to export certificates, see “Exporting certificates by using the GUI and CLI” on page 32.

Download the certificates by using the GUI or use SCP from the /dumps directory. Then, transfer the file to the /tmp directory on the partner system.

2. Add the coordinator root CA to the member's truststore:

```
svctask mktruststore -file /tmp/system_rootcacertificate_slot_3.pem -grid on  
-name <truststore-name>
```

3. Similarly, add the member's root CA to the coordinator's truststore.

4. From the member, submit the join request by running the following command:

```
svctask managegrid -join -ip <ip-address-coordinator-system> -truststore  
<truststore-id-or-name>
```

Ensure that you specify the correct IP address for the IBM FlashSystem grid. The truststore entry must correspond to the correct system:

- When you join the grid, use the coordinator system's truststore entry.
- When you accept a join request, use the new member system's truststore entry.

5. From the coordinator system, accept the join request:

```
svctask managegrid -accept -ip <ip-address-new-member-system> -truststore  
<truststore-id-or-name>
```

The truststore entry that you refer to must correspond to the new member system. Ensure that you specify the correct IP address of the new member system.

If you specify the wrong IP address or the member is not requesting to join, request, the coordinator system reports the following error:

```
CMMVC1150E The command failed because an internal command failed with  
[CMMVC1268E] on a remote system.
```

Explanation of CMMVC1268E: The system is not requesting to join this grid.

For the join and accept commands to succeed, both systems must communicate over the Ethernet management network. You can use the FQDN instead of the IP address in the commands. If the accept command completes successfully, the system is added as a new member of the IBM FlashSystem grid.

6. To confirm that the system is listed:

```
svcinfolsgridmembers -delim
```

Adding two members

Example 2-5 shows how to add two new member systems, an IBM FlashSystem 7300 (IP address 9.80.82.60) and an SVC (IP address 9.80.82.74), to an existing IBM FlashSystem grid with an IBM FlashSystem 9200 as the coordinator (IP address 9.80.82.80).

Example 2-5 Adding member systems to an existing IBM FlashSystem grid

```
#Export the root certificates from each system
IBM_FlashSystem:FS7300:superuser>svctask chsystemcertstore -exportrootcacert
-scope internal_communication
IBM_FlashSystem:FS7300:superuser>svcinfolddumps | grep "root"
92 system_rootcacertificate_slot_3.pem
```

```
IBM_2145:SVC:superuser>svctask chsystemcertstore -exportrootcacert -scope
internal_communication
IBM_2145:SVC:superuser>svcinfolddumps | grep "root"
89 system_rootcacertificate_slot_3.pem
```

```
IBM_FlashSystem:FS9200:superuser>svctask chsystemcertstore -exportrootcacert
-scope internal_communication
IBM_FlashSystem:FS9200:superuser>svcinfolddumps | grep "root"
99 system_rootcacertificate_slot_3.pem
```

```
#Copy the certificates to your local workstation by using Putty Secure Copy
C:\>cd PuTTY
C:\PuTTY>PSCP.EXE -unsafe
superuser@9.80.82.60:/dumps/system_rootcacertificate_slot_3.pem
FS7300-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:
```

```
End of keyboard-interactive prompts from server
FS7300-rootcacertificate.pem | 1 kB | 1.3 kB/s | ETA: 00:00:00 | 100%
C:\PuTTY>PSCP.EXE -unsafe
superuser@9.80.82.74:/dumps/system_rootcacertificate_slot_3.pem
SVC-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:
```

```
End of keyboard-interactive prompts from server
SVC-rootcacertificate.pem | 1 kB | 1.3 kB/s | ETA: 00:00:00 | 100%
C:\PuTTY>PSCP.EXE -unsafe
superuser@9.80.82.80:/dumps/system_rootcacertificate_slot_3.pem
FS9200-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:
```

```
End of keyboard-interactive prompts from server
FS9200-rootcacertificate.pem | 2 kB | 2.0 kB/s | ETA: 00:00:00 | 100%
C:\PuTTY>
```

```
#Transfer the coordinator system certificate to the member systems
C:\PuTTY>PSCP.EXE FS9200-rootcacertificate.pem
superuser@9.80.82.74:/tmp/FS9200-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:
```



```

End of keyboard-interactive prompts from server
FS9200-rootcacertificate.pem | 2 kB | 2.0 kB/s | ETA: 00:00:00 | 100%
C:\Putty>PSCP.EXE FS9200-rootcacertificate.pem
superuser@9.80.82.60:/tmp/FS9200-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:

```

```

End of keyboard-interactive prompts from server
FS9200-rootcacertificate.pem | 2 kB | 2.0 kB/s | ETA: 00:00:00 | 100%
C:\Putty>

```

```

#Transfer the member system certificates to the coordinator system
C:\Putty>PSCP.EXE SVC-rootcacertificate.pem
superuser@9.80.82.80:/tmp/SVC-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:

```

```

End of keyboard-interactive prompts from server
SVC-rootcacertificate.pem | 1 kB | 1.3 kB/s | ETA: 00:00:00 | 100%
C:\Putty>PSCP.EXE FS7300-rootcacertificate.pem
superuser@9.80.82.80:/tmp/FS7300-rootcacertificate.pem
Keyboard-interactive authentication prompts from server:
| Password:

```

```

End of keyboard-interactive prompts from server
FS7300-rootcacertificate.pem | 1 kB | 1.3 kB/s | ETA: 00:00:00 | 100%
C:\Putty>

```

```

#Add the member system certificates to the truststore of the coordinator system
IBM_FlashSystem:FS9200:superuser>svctask mktruststore -file
/tmp/SVC-rootcacertificate.pem -grid on -name TS_FG_9.80.82.74
Truststore, id [1], successfully created
IBM_FlashSystem:FS9200:superuser>svctask mktruststore -file
/tmp/FS7300-rootcacertificate.pem -grid on -name TS_FG_9.80.82.60
Truststore, id [2], successfully created
IBM_FlashSystem:FS9200:superuser>svcinfoltruststore -delim :
id:name:percent_used:space:restapi:ipsec:vasa:email:snmp:syslog:grid_address:grid_
references
0:common_CA_database:100:0:off:off:off:off:off:off:off:
1:TS_FG_9.80.82.74:16:10.0KB:on:off:off:off:off:off:off:0
2:TS_FG_9.80.82.60:11:10.7KB:on:off:off:off:off:off:off:off:0

```

```

#Add the coordinator system certificate to the truststore of each new member
IBM_2145:SVC:superuser>svctask mktruststore -file
/tmp/FS9200-rootcacertificate.pem -grid on -name TS_FG_9.80.82.80
Truststore, id [1], successfully created
IBM_2145:SVC:superuser>svcinfoltruststore -delim :
id:name:percent_used:space:restapi:ipsec:vasa:email:snmp:syslog:flash_grid_address
:flash_grid_references
0:common_CA_database:100:0:off:off:off:off:off:off:off:
1:TS_FG_9.80.82.80:16:10.0KB:on:off:off:off:off:off:off:0

```

```

IBM_FlashSystem:FS7300:superuser>svctask mktruststore -file
/tmp/FS9200-rootcacertificate.pem -grid on -name TS_FG_9.80.82.80

```

```

Truststore, id [1], successfully created
IBM_FlashSystem:FS7300:superuser>svcinfolstruststore -delim :
id:name:percent_used:space:restapi:ipsec:vasa:email:snmp:syslog:flash_grid_address
:flash_grid_references
0:common_CA_database:100:0:off:off:off:off:off:off::
1:TS_FG_9.80.82.80:16:10.0KB:on:off:off:off:off:off::0

#Initiate the join request from each member system
IBM_FlashSystem:FS7300:superuser>svctask managegrid -join -ip 9.80.82.80
-truststore TS_FG_9.80.82.80

IBM_2145:SVC:superuser>svctask managegrid -join -ip 9.80.82.80 -truststore
TS_FG_9.80.82.80

#Accept the join requests on the coordinator system
IBM_FlashSystem:FS9200:superuser>svctask managegrid -accept -ip 9.80.82.74
-truststore TS_FG_9.80.82.74
IBM_FlashSystem:FS9200:superuser>svctask managegrid -accept -ip 9.80.82.60
-truststore TS_FG_9.80.82.60

#Confirm that the new systems are listed as member
IBM_FlashSystem:FS9200:superuser>svcinfolsgridmembers -delim :
id:system_uuid:member_address:truststore_id:role
0:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:::coordinator
1:1B6BCA40-2C4E-5845-80D6-53E4297132F9:9.80.82.74:1:member
2:64E27F48-7B7C-5CA1-B93E-FE20BEA29651:9.80.82.60:2:member

IBM_FlashSystem:FS9200:superuser>svcinfolsgrid | grep systems
number_of_systems 3
number_of_reachable_systems 3
number_of_systems_reported 3

```

A system can belong to only one IBM FlashSystem grid. If a system is already a member of an IBM FlashSystem grid, you cannot add it to another grid.

2.3.2 Removing a system from an IBM FlashSystem grid

You can remove a system from an IBM FlashSystem grid by using the GUI or CLI.

Removing a system by using the GUI

Complete the following steps:

1. From the IBM FlashSystem grid dashboard, go to the Storage systems view.
2. From the overflow menu (three vertical dots) next to the system you plan to remove, click **Remove system**, as shown in Figure 2-17 on page 49. In this example, the SVC system is removed.

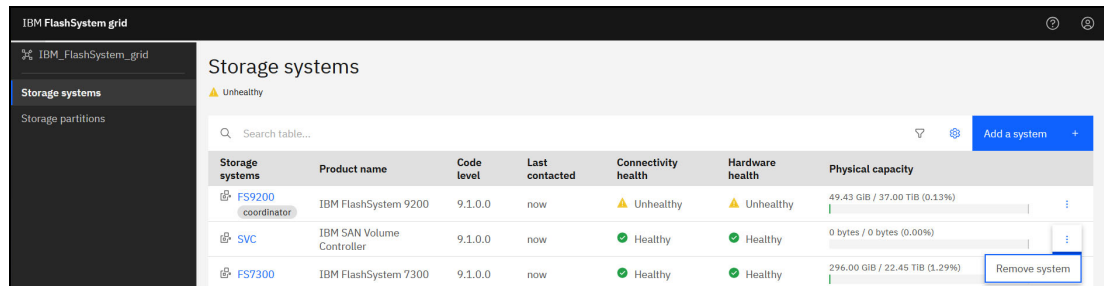


Figure 2-17 Removing a member system from the IBM FlashSystem grid

3. The system displays the following warning message:

Removing system ‘<system-name>’ from IBM FlashSystem grid ‘<grid-name>’ results in data accessibility issues, including loss of shared resources. Do you want to remove the system?”

4. Confirm the removal by clicking **Remove system** again. After the task completes, click **Close**. The system is removed from the table of storage systems.

Important: If you remove the last remaining system from the IBM FlashSystem grid (the coordinator) the IBM FlashSystem is automatically deleted. Before proceeding, the system displays a warning message, as shown in Figure 2-18.

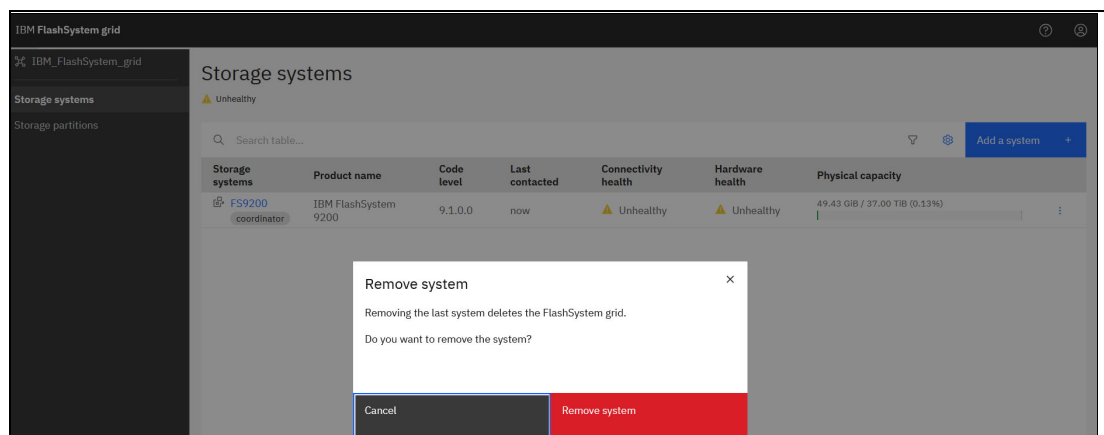


Figure 2-18 Removing the coordinator system results after deleting the IBM FlashSystem grid

5. Click **Remove system**. The system runs the leave request and redirects you to the default system management GUI, as shown in Figure 2-19.

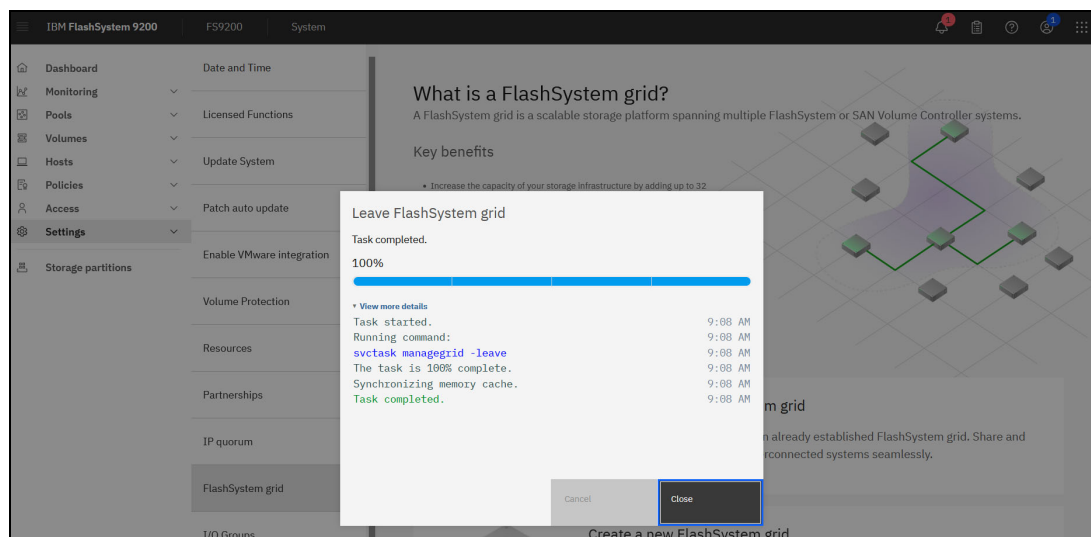


Figure 2-19 Leave request completed

You can remove the coordinator only after you remove all other member systems from the IBM FlashSystem grid. If you attempt to remove the last system (the coordinator) when other member systems are still part of the grid, the system displays an error message, as shown in Figure 2-20.

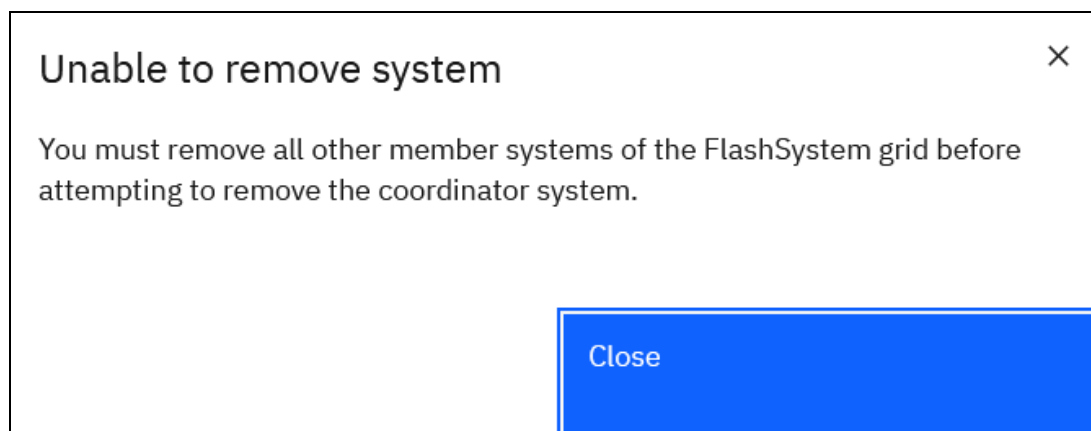


Figure 2-20 Unable to remove system

Removing a system by using the CLI

To remove a member system, run the `svctask managegrid -remove` command from the coordinator system. Specify the IP address or the FQDN of the system that you want to remove. The command also removes the related truststore entry automatically.

Example 2-6 on page 51 shows how to remove a member system with IP address 9.80.82.60.

Example 2-6 Removing a member system from an IBM FlashSystem grid

```
IBM_FlashSystem:FS9200:superuser>svcinfolsgridmembers -delim :
id:system_uuid:member_address:truststore_id:role
0:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:::coordinator
1:1B6BCA40-2C4E-5845-80D6-53E4297132F9:9.80.82.74:1:member
2:64E27F48-7B7C-5CA1-B93E-FE20BEA29651:9.80.82.60:2:member

IBM_FlashSystem:FS9200:superuser>svctask managegrid -remove 9.80.82.60
IBM_FlashSystem:FS9200:superuser>svcinfolstruststore | grep TS_FG_9.80.82.60
IBM_FlashSystem:FS9200:superuser>

IBM_FlashSystem:FS9200:superuser>svcinfolsgrid | grep systems
number_of_systems 2
number_of_reachable_systems 2
number_of_systems_reported 2

IBM_FlashSystem:FS9200:superuser>svcinfolsgridmembers -delim :
id:system_uuid:member_address:truststore_id:role
0:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:::coordinator
1:1B6BCA40-2C4E-5845-80D6-53E4297132F9:9.80.82.74:1:member
```

In the latest release, running the **svctask managegrid -remove** command from the coordinator system automatically removes all objects that are associated with the member system. However, if connectivity issues occur during the removal, you might need to manually delete the related truststore entry or run the **svctask managegrid -leave** command on the member system.

For systems that use a code level earlier than 8.7.3, the remove action does not clean up the configuration on the member system. In this case, you must manually run the **svctask managegrid -leave** command to complete the cleanup.

This step prevents future issues if you add the system to an IBM FlashSystem grid again or create a new IBM FlashSystem grid with this system as the coordinator.

Example 2-7 shows how to run the **svctask managegrid -leave** command on a member system and verify that the system is no longer part of any IBM FlashSystem grid.

Example 2-7 Running the managegrid -leave command

```
IBM_FlashSystem:FS7300:superuser>svctask managegrid -leave
IBM_FlashSystem:FS7300:superuser>svcinfolssysteminfo | grep uuid
grid_system_uuid 64E27F48-7B7C-5CA1-B93E-FE20BEA29651
grid_uuid
```

The `grid_uuid` field is blank, which indicates that the system is not part of any IBM FlashSystem grid.

Note: The recommended way to remove a system from an IBM FlashSystem grid is to run the `svctask managegrid -remove` command from the coordinator (owner) system.

This method helps ensure that the coordinator system performs all necessary cleanup tasks. The following tasks are included in this process:

- ▶ Removing the system from the grid
- ▶ Deleting associated truststore entries
- ▶ Updating grid information across all remaining member systems

If you run only the `managegrid -leave` command on the system that you want to remove, the coordinator is not notified. This action can leave stale entries on the coordinator and other member systems, which might cause inconsistencies in the grid configuration.

You cannot remove the coordinator system when other members are still part of the IBM FlashSystem grid.

Removing the last system by running the `managegrid -leave` command on the coordinator system deletes the entire IBM FlashSystem grid. If any other member systems remain, the command fails with the following error:

CMMVC1392E. The command failed because the owner system can't leave grid while member systems are still present.

2.4 Upgrading existing IBM FlashSystem grid configurations to release 8.7.3.x or later

Before release 8.7.3.0, the GUI did not provide full IBM FlashSystem grid visualization and monitoring capabilities on member systems. Starting with 8.7.3.x, IBM introduced changes that are related to certificate management and grid prerequisites.

Beginning with release 8.7.3.0, the truststore entry for the IBM FlashSystem grid feature must contain only the root certificate. If your systems are running software Version 8.7.2.x or earlier, you might need to follow a special procedure when upgrading to 8.7.3.0 or later.

The software upgrade test utility notifies you if this procedure is required, and it provides a link to [Upgrading a system within an IBM FlashSystem grid to 8.7.3 or later](#).

Here are the high-level upgrade steps:

1. Prepare the system for the update.
2. Remove member systems from the IBM FlashSystem grid.
3. Run the software upgrade test utility again.
4. Update the systems.
5. Add the updated systems back to the IBM FlashSystem grid.

Preparation

Run the software upgrade test utility several days before the planned upgrade and review any reported errors.

Testing the GUI

To test in the GUI, complete the following steps:

1. Select **Settings** → **System** → **Update System** → **Alternate update options** → **Provide a package** → **Provide the package manually** → **Upload**.
2. Upload the software upgrade test utility and the code package.
3. After the upload completes, click **Close**.
4. Click **Test only**.
5. Enter the target code level (for example, 8.7.3.0) and click **Test**.
6. Use the most recent software upgrade test utility.

Note: Always use the latest available software upgrade test utility.

Figure 2-21 shows the Update System view in the GUI after the software upgrade test utility and the 8.7.3.0 code package are uploaded.

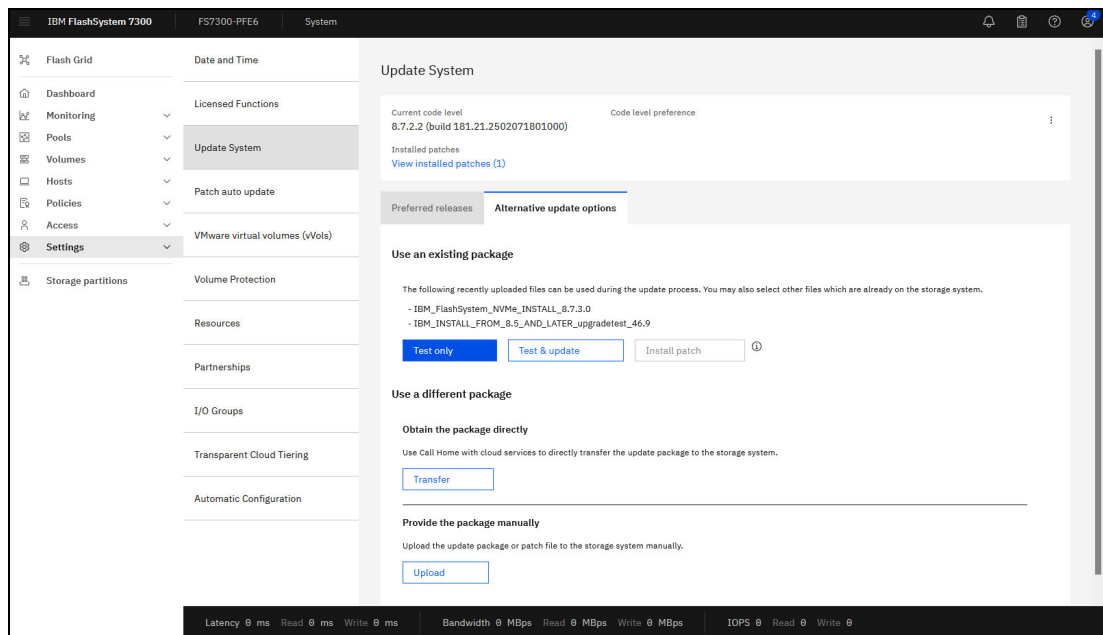


Figure 2-21 Test for update

Figure 2-22 shows the window that appears after you click **Test only** and enter the target code level. In this example, the code level is 8.7.3.0. Click **Test** to run the software upgrade test utility.

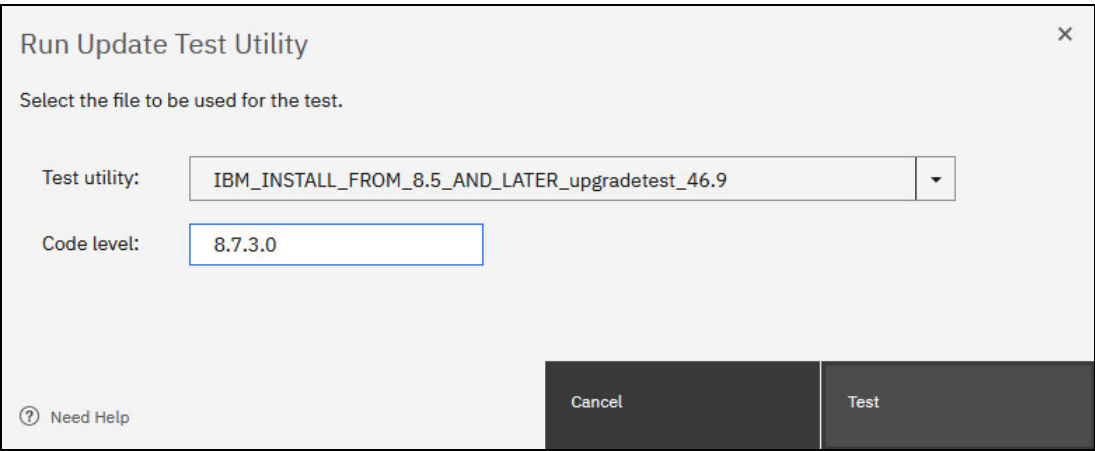


Figure 2-22 Running the software upgrade test utility with target release 8.7.3.0

Figure 2-23 shows the error message. If the truststore entry contains more than one certificate (root plus intermediate or full chain).

If the software upgrade test utility reports this condition, complete the following steps in the Special procedure for upgrading from 8.7.2.x to 8.7.3.x subsection.

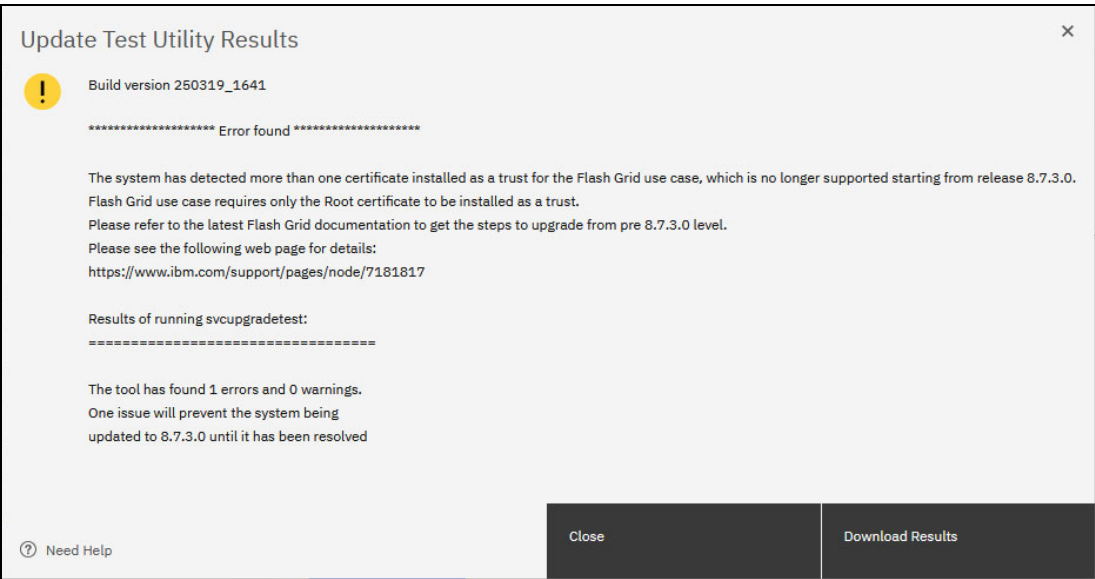


Figure 2-23 The software upgrade test utility detects an error during an upgrade to code release 8.7.3.0

Special procedure for upgrading from 8.7.2.x to 8.7.3.x

If the software upgrade test utility does not return any errors, you can start the update by selecting **Settings** → **System** → **Update System** → **Alternate update options** → **Use an existing package** → **Test and update**.

This option reuses the uploaded files for the software upgrade test utility and the code package. If the test utility returns the error message that is shown in Figure 2-23 on page 54, follow the special procedure instead.

If the software upgrade test utility reports that a truststore entry contains more than one certificate (root plus intermediate or full chain), complete the following steps:

1. Remove each member system from the IBM FlashSystem grid before you update.

Important: Wait for any partition migrations to complete before you remove a system.

You cannot remove a member system by using the GUI if migrations are in progress. Although you can remove a system by using the CLI during migration, doing so causes an authentication error that interrupts migration management. The data migration continues.

2. Check the migration status before you remove a system.

In the GUI, click **Storage partitions** in the IBM FlashSystem grid dashboard. The Migration status column shows a dash (-) when a migration is not in progress. Use the following commands as needed:

- Use **svcinfo lspartition** for a quick check. A blank migration_status field indicates that the migration is not in progress.
- Use **svcinfo lsvolumegroupreplication <volumegroup-id-or-name>** to display the detailed progress, including the following information:
 - Amount of data to copy (locationX_sync_required and locationX_sync_remaining)
 - Estimated completion time (locationX_sync_estimated_completion_time)

3. Log in to the CLI of the IBM FlashSystem grid coordinator system by using SSH. Then run the following command to remove the member system:

```
svctask manageflashgrid -remove <IP-address-or-FQDN>
```

Specify the IP address or FQDN of the member system that you want to remove.

Example 2-8 shows how to remove an IBM FlashSystem 7300 member system (IP address 9.80.82.60) by using the CLI of the IBM FlashSystem 9200 coordinator system (IP address 9.80.82.80).

Example 2-8 Removing a member system by using the manageflashgrid -remove command

```
IBM_FlashSystem:FS9200:superuser>svcinfo lsflashgridmembers -delim :
id:system_uuid:member_address:truststore_id:role
0:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:::coordinator
1:64E27F48-7B7C-5CA1-B93E-FE20BEA29651:9.80.82.60:1:member
IBM_FlashSystem:FS9200:superuser>svctask manageflashgrid -remove 9.80.82.60
IBM_FlashSystem:FS9200:superuser>svcinfo lsflashgridmembers -delim :
id:system_uuid:member_address:truststore_id:role
0:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:::coordinator
IBM_FlashSystem:FS9200:superuser>
```

4. Log in to the CLI of the member system and run the following command:

```
svctask manageflashgrid -leave
```

Example 2-9 shows how you can remove the IBM FlashSystem 7300 member system from the IBM FlashSystem grid.

Example 2-9 Running the manageflashgrid -leave command from the CLI of the member system

```
IBM_FlashSystem:FS7300:superuser>svcinflsflashgrid
flash_grid_name New_FlashSystem_grid
flash_grid_uuid 239D6448-DF7E-5300-8311-7F91F13A4AC2
flash_grid_owner_uuid 9C83CD3A-A4CB-550E-9CC5-8C6C4639B440
flash_grid_ip_address 9.80.82.80
last_contact_time
number_of_storage_partitions 0
number_of_storage_partition_copies 0
total_physical_capacity 17.29TB
used_physical_capacity 0.00MB
number_of_systems 2
number_of_reachable_systems 2
number_of_systems_reported 2
IBM_FlashSystem:FS7300:superuser>svctask manageflashgrid -leave
IBM_FlashSystem:FS7300:superuser>svcinflsflashgrid
IBM_FlashSystem:FS7300:superuser>
```

Repeat this process for all member systems that require an update. Do not remove the coordinator system because doing so deletes the IBM FlashSystem grid. Alternatively, you can also use the GUI procedure instead, which is described in “Removing a system by using the GUI” on page 48.

5. After the member systems are removed from the grid, delete the related truststore entries by running the following command:

```
svctask rmtruststore <truststore-id>
```

Then, confirm that the entries were deleted by running the following command:

```
svcinflstruststore
```

Upgrading and rejoining systems

As the procedure finishes, the next time you run the software upgrade test utility, the following message appears:

The update test utility completed and found no issues.

Now, you are ready to upgrade your system by completing the following steps:

1. Upgrade the coordinator system and all member systems of the IBM FlashSystem grid to software Version 8.7.3.0 or later.
2. After the code upgrade completes, add each system back to the IBM FlashSystem grid.
3. Add the root certificate to the truststore of each other and run the following commands:

```
svctask manageflashgrid -join
svctask manageflashgrid -join -accept
```

Note: In release 9.1.x, the command is **managegrid** instead of **manageflashgrid**.

Alternatively, follow the GUI procedure, which is described in “Adding a system by using the GUI” on page 38.

Exporting certificates

You can export certificates by using the GUI or CLI. For more information about release 9.1.0.0, see “Exporting certificates by using the GUI and CLI” on page 32.

On a system that is running code level 8.7.3.x, either run the commands through the GUI or by using the CLI.

To export the root certificate in the GUI, run the following commands:

1. Select **Settings** → **Security** → **System Certificates** → **Export System Certificate**.
2. To export the root certificate, select **GUI** → **Settings** → **Security** → **System Certificates**, and select one of the following options:
 - a. **Export Internal Root Certificate** for internally signed certificates
 - b. **Export External Root Certificate** for certificates signed by an external root CA
3. Use the **Export Certificate** option to download the certificate directly through the browser.

To export certificate by using the CLI, run the following commands:

1. Run the following command to export the system certificate:

```
svctask chsystemcert -export
```

The command creates a file that is called `certificate.pem` that includes the full certificate chain.
2. To export the root certificate, run the following command:

```
satask exportrootcertificate
```

The resulting file is named `root_certificate.pem`.
3. In 8.7.3.x, you can also use the new command:

```
svctask chsystemcert -exportrootcacert
```

The resulting file is called `rootcacertificate.pem`.

Important: An upgrade to 8.7.3.0 for systems with DRP is not supported because of issue SVAPAR-156976. This restriction will be removed in a future PTF. Standard pools are not affected.



Managing and monitoring the IBM FlashSystem grid

In an IBM FlashSystem grid, you can manage multiple IBM FlashSystem systems as a single, unified entity. By creating a grid, you can efficiently manage storage resources, monitor system health, and perform data migrations across systems. The grid architecture consists of a coordinator system and multiple member systems.

This chapter guides you through monitoring and managing an IBM FlashSystem grid by using the IBM FlashSystem grid dashboard and its associated subdashboards. These dashboards provide detailed insights into the systems within the grid and the storage partitions they own. You can use the dashboard to initiate migrations, view key system details, and manage storage partitions. This chapter helps you in setting up and maintaining a scalable, resilient, and efficient storage environment.

This chapter describes the following topics:

- ▶ “IBM FlashSystem grid dashboard ” on page 60
- ▶ “Systems window under IBM FlashSystem grid dashboard” on page 61
- ▶ “Partitions window under IBM FlashSystem grid dashboard” on page 63

3.1 IBM FlashSystem grid dashboard

The IBM FlashSystem grid dashboard appears when you log in to the Management GUI of any system within the grid. It provides a centralized interface to monitor and manage the entire IBM FlashSystem grid.

From the dashboard, you can view all systems and storage partitions in the grid, along with their health status and connectivity. The dashboard also shows consolidated capacity information across all systems, giving you a comprehensive view of storage usage and system health at a glance.

The IBM FlashSystem grid dashboard includes the following key components (see Figure 3-1):

- ▶ System overview. Displays the total number of systems and storage partitions in the grid.
- ▶ Health monitoring. Indicates the health status of hardware and connectivity for each system.
- ▶ System reachability. Indicates whether the coordinator system can connect with each member system in the grid.
- ▶ Capacity usage. Provides an aggregated view of physical and logical storage capacities across the grid.
- ▶ Migration status. Shows the count and status of active storage partition migrations in the grid.

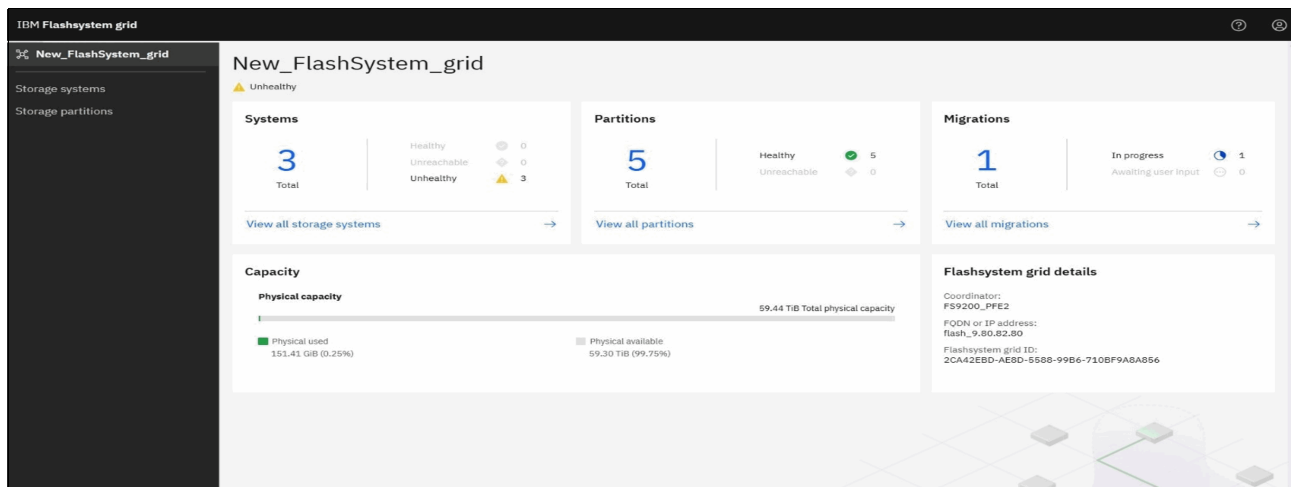


Figure 3-1 IBM FlashSystem grid dashboard

To view the IBM FlashSystem grid information from the command-line interface (CLI), enter the command `svcinfo lsgrid` (see Example 3-1 on page 61).

Example 3-1 The svcinfo lsgrid command output

```
>svcinfo lsgrid
grid_name New_FlashSystem_grid
grid_uuid 2CA42EBD-AE8D-5588-99B6-710BF9A8A856
grid_owner_uuid 9C83CD3A-A4CB-550E-9CC5-8C6C4639B440
grid_ip_address 9.80.82.80
last_contact_time
number_of_storage_partitions 5
number_of_storage_partition_copies 7
total_physical_capacity 59.44TB
used_physical_capacity 247.41GB
warning_exceeded no
total_logical_capacity
total_allocated
total_written
total_written_snapshots
total_provisioned_host
total_reserved_host_volume
total_reserved_contingency
total_reserved_childpools
number_of_systems 3
number_of_reachable_systems 3
number_of_systems_reported 3
```

The IBM FlashSystem grid dashboard contains different windows that provide system-specific data.

Table 3-1 shows the types of available windows and their purpose.

Table 3-1 Types of windows under IBM FlashSystem grid dashboard

Window type	Purpose
Storage systems	Shows a list of storage systems that are part of the IBM FlashSystem grid along with details.
Storage partitions	Shows a list of storage partitions that are part of the IBM FlashSystem grid along with details.

3.2 Systems window under IBM FlashSystem grid dashboard

You can use the Storage systems window to view a comprehensive overview of all storage systems in the IBM FlashSystem grid. On this page, you can monitor key attributes and operational statuses for each system.

The following information is available for each system:

- ▶ Storage system. Provides a link to open that system's Management GUI.
- ▶ Product name. Identifies the specific model or family of the IBM FlashSystem storage device.
- ▶ System code level. Indicates the current software or firmware version that is installed on the system.
- ▶ Last contacted time. Shows the most recent time that the system successfully communicated with the grid.

- **Connectivity health.** Reflects the status of system connectivity and reports any connectivity-related errors or alerts.
- **Hardware health.** Indicates the operational state of hardware components and highlights any detected faults or alerts.
- **Physical capacity.** Displays the total physical storage capacity available on the system.
- **Total number of storage partitions.** Displays the total number of storage partitions that are owned by that system.
- **Configuration problem.** Displays an alert when a configuration error occurs during the process of adding member systems to the grid. In such cases, the system redirects you to the appropriate Device Management Page (DMP) on the coordinator system for further action.

You can use the search bar to find a system by name or attributes in the system table.

To filter results, click the **Filter** icon on the right side of the search bar. You can filter by the following system attributes:

- Storage system
- Product name
- Connectivity health
- Hardware health

The Storage systems window also displays error data that indicates the type of error that is encountered when displaying other systems in the IBM FlashSystem grid. Icons identify errors in applicable fields and suggests actions to resolve them.

Figure 3-2 shows the Storage systems window under the IBM FlashSystem grid dashboard.

ID	Storage systems	Product name	Code level	Last contacted	Connectivity health	Hardware health	Physical capacity	Total number of partitions	Configuration problem
0	FS9200-PFE2 coordinator	IBM FlashSystem 9200	9.1.0.0	now	Unhealthy	Unhealthy	21.41 GiB / 37.00 TiB (0.06%)	3	0
1	SVC-PFE3	IBM SAN Volume Controller	9.1.0.0	now	Unhealthy	Healthy	0 bytes / 0 bytes (0%)	-	0
2	FS7300-PFE6	IBM FlashSystem 7300	9.1.0.0	now	Unhealthy	Healthy	130.00 GiB / 22.45 TiB (0.57%)	3	0

Figure 3-2 Storage systems window

To view member storage systems and related attributes from the CLI, run the following command:

```
svcinfo lsgridsystem -delim :
```


Example 3-2 shows the output of the command.

Example 3-2 The svcinfo lsgridsystem command output

```
>svcinfo lsgridsystem -delim :
id:name:system_id:ip_address:grid_system_uuid:partition_count:total_physical_capacity:used_physical_capacity:last_contact_time:code_level:product_name:upgrade_status:hardware_health:connectivity_health:warning_exceeded:total_logical_capacity:allocated:written:written_snapshots:provisioned_host:reserved_host_volume:reserved_contingency:reserved_childpools:error_data
0:FS9200_PFE2:000002042240B714:9.80.82.80:9C83CD3A-A4CB-550E-9CC5-8C6C4639B440:3:37.00TB:21.41GB::9.1.0.0 (build 187.20.2505131102000):IBM FlashSystem
9200:success:unhealthy:unhealthy:no::::::::::
1:SVC_PFE3:00000204E060837A:9.80.82.74:1B6BCA40-2C4E-5845-80D6-53E4297132F9:0:0.00MB:0.00MB::9.1.0.0 (build 187.20.2505131102000):IBM SAN Volume
Controller:success:healthy:unhealthy:no:0.00MB:0.00MB:0.00MB:0.00MB:0.00MB:0.00MB:0.00MB:0.00MB:0.00MB:
2:FS7300-PFE6:0000020420E142D8:9.80.82.60:64E27F48-7B7C-5CA1-B93E-FE20BEA29651:4:22.45TB:226.00GB::9.1.0.0 (build 187.20.2505131102000):IBM FlashSystem
7300:success:healthy:unhealthy:no::::::::::
```

3.3 Partitions window under IBM FlashSystem grid dashboard

The Storage partitions window provides an overview of all configured storage partitions in the IBM FlashSystem grid. This window displays details such as the storage partition location, replication topology and status, associated volume groups, migration progress, and Evaluation Placement status.

Each storage partition that is listed on the Storage partitions window includes the following details:

- ▶ Storage partition name. Provides a link to open the storage partition's Management GUI.
- ▶ Systems. Indicates the system that owns the storage partition.
- ▶ Replication topology. Describes the replication configuration, such as high availability (HA), DR, or high availability and disaster recovery (HADR).
- ▶ Replication status. Displays the state of HA or DR replication that is associated with the storage partition.
- ▶ Volume groups. Lists the number of volume groups that are owned by the storage partition.
- ▶ Migration status. Displays the migration status if a migration is in progress, or remains blank if none.
- ▶ Evaluation placement status. Displays the artificial intelligence (AI)-based evaluation status from IBM Storage Insights for optimal storage partition placement by using migration.

You can use the search bar to find a storage partition in the storage partition table by name or attribute.

To filter results, click the **Filter** icon at the right side of the search bar. You can filter the results by using the following attributes:

- ▶ Storage partition name
- ▶ Systems
- ▶ Replication topology
- ▶ HA status
- ▶ DR status
- ▶ Migration status

Figure 3-3 shows the Storage partitions window under the IBM FlashSystem grid dashboard.

Partition name	Systems	Replication topology	HA status	DR status	Migration status	Placement evaluation status	Volume groups	Last updated
Storage_Insights_Migration_Partition	FS9200_PFE2	Local	-	-	In progress	-	1	now
FS7300_storage_partition1	FS9200_PFE2	Local	-	-	-	-	1	now
FS9200_partition1	FS9200_PFE2 FS7300-PFE6	HA	HA established	-	-	-	1	now
test3	FS7300-PFE6	Local	-	-	-	-	0	now
FS7300_storage_partition2	FS7300-PFE6	Local	-	-	-	-	3	now

Figure 3-3 Storage partitions window

To view the storage partitions and related attributes from the CLI, run the following command:

```
svcinfo lsgridpartition -delim :
```

Example 3-3 shows the output of the command.

Example 3-3 The svcinfo lsgridpartition command output

```
>svcinfo lsgridpartition -delim :
system_index:id:name:preferred_management_system_name:active_management_system_name:
replication_policy_id:replication_policy_name:location1_system_name:location1_status:
location2_system_name:location2_status:host_count:volume_group_count:default_volume_group_id:
default_volume_group_name:ha_status:link_status:desired_location_system_name:migration_status:
draft:draft_volume_group_count:draft_host_count:ownership_group_id:ownership_group_name:
management_portset_id:management_portset_name:host_cluster_count:draft_host_cluster_count
0:0:FS9200_partition1:FS9200_PFE2:FS9200_PFE2:0:ha-policy-0:FS9200_PFE2:healthy:FS7300-PFE6:
healthy:0:1:::established:synchronized:::no:0:0:::0:0
```

```

0:1:FS7300_storage_partition1::::FS9200_PFE2:healthy::0:1::::no:0:0::4:ports
et4:0:0
0:2:Storage_Insights_Migration_Partition:FS7300-PFE6:FS7300-PFE6:1:StoragePartitio
nMO_9918D13A744CA5835F11F34977DDC3D5:FS9200_PFE2:healthy:FS7300-PFE6:healthy:0:1::
:established:synchronized:FS7300-PFE6:in_progress:no:0:0::::0:0
2:0:FS9200_partition1:FS9200_PFE2:FS9200_PFE2:0:ha-policy-0:FS9200_PFE2:healthy:FS
7300-PFE6:healthy:0:1::established:synchronized::no:0:0::::0:0
2:1:test3::::FS7300-PFE6::::0:0::::yes:0:0::::0:0
2:2:FS7300_storage_partition2::::FS7300-PFE6:healthy::0:3::::no:0:0::::0:0
2:3:Storage_Insights_Migration_Partition:FS7300-PFE6:FS7300-PFE6:1:StoragePartitio
nMO_9918D13A744CA5835F11F34977DDC3D5:FS9200_PFE2:healthy:FS7300-PFE6:healthy:0:1::
:established:synchronized:FS7300-PFE6:awaiting_user_input:no:0:0::::0:0

```



Configuring and managing storage partitions

This chapter describes how you configure and manage storage partitions in an IBM FlashSystem grid. You learn core concepts, create and manage partitions by using the GUI and command-line interface (CLI), and configure replication topologies for continuous data availability.

This chapter describes the following sections:

- ▶ “What is a storage partition ” on page 68
- ▶ “Step-by-step configuration to create an IBM FlashSystem grid storage partition” on page 70
- ▶ “Continuous data availability by using replication” on page 89

4.1 What is a storage partition

A storage partition is a logical configuration entity. It defines and controls access between one or more hosts and one or more volumes, also known as logical unit numbers (LUNs). This access occurs across the virtualized storage fabric that the system manages.

A storage partition is central to host access control within the IBM Storage Virtualize ecosystem. You use storage partitions to implement LUN masking, coordinate access for multiple hosts, and enforce isolation and security policies at the storage-system level independent of fabric zoning.

4.1.1 Storage partition as a storage management construct

A storage partition groups the configuration for one or more applications that you manage together. It simplifies resource management within a storage system or across a broader IBM FlashSystem grid. Conceptually, a storage partition acts like a virtual storage system that focuses on the logical resources an application uses.

4.1.2 Benefits of using storage partitions

Storage partitions provide a structured approach to managing resources in virtualized environments. They help improve security, simplify administration, and support scalability for enterprise workloads.

The following benefits highlight why storage partitions are essential in modern storage management:

- ▶ **Security.** Define explicit host-to-volume mappings to prevent unauthorized access.
- ▶ **Manageability.** Group-related resources to simplify tasks such as monitoring, provisioning, and troubleshooting.
- ▶ **Scalability.** Adjust a storage partition without disrupting existing configurations.
- ▶ **Isolation.** Separate applications so that dedicated resources within each storage partition do not interfere with another storage partition.

Note: In IBM Storage Virtualize, a storage partition groups volume groups that contain volumes. You map these volumes to hosts through host mappings and apply policies to govern their behavior and protection. For more information, see 4.1.3, “Elements of a storage partition” on page 68.

4.1.3 Elements of a storage partition

A storage partition organizes resources for workloads and hosts. The following components are included in a storage partition:

- ▶ **Volumes or LUNs**
LUNs or virtual disks that you present to hosts.
- ▶ **Volume groups**
Collections of volumes that you manage such as a single unit, enabling streamlined operations such as consistent snapshots and replication across all member volumes.

- ▶ **Hosts**
Server or initiators (worldwide port names (WWPNs) for FC or IQNs for iSCSI) that require access to the volumes.
- ▶ **Host clusters**
Groups of logical host objects that you manage together. For example, you can create a volume mapping that every host in the cluster shares.
- ▶ **Policies**
Sets of rules that simplify storage management. The system supports several types of policies, including provisioning policies, snapshot policies, and replication policies.

4.1.4 Limits of storage partitions

Table 4-1 shows the storage partition limits. Storage partitions were first introduced in code release 8.6.1.0. These limits remained the same in releases 8.6.1.0, 8.7.0.x, 8.7.2.x, and 8.7.3.x.

Table 4-1 Storage partition limits across versions

Version	Limit
8.6.1.0 - 8.7.3.x	4 storage partitions per I/O group
9.1.0.0	32 storage partitions per I/O group

4.1.5 Drafting storage partitions

You can create hosts, host clusters, volumes, and volume groups within a storage partition. However, you cannot directly add existing storage resources to a storage partition. To work around this limitation, use a draft storage partition to build a partition by using existing objects that meet the requirements. Some requirements are not strictly enforced during the draft stage. This flexibility enables incremental resource additions while you identify and resolve application dependencies.

A draft storage partition cannot be migrated or made highly available. It becomes operational only after you publish it.

Requirements for publishing a draft storage partition

Before you can publish (activate) a draft storage partition, the partition must satisfy all conditions that govern storage partitions. Meeting these conditions helps ensure that the partition is valid and prepared for activation.

The requirements for publishing a draft partition include the following conditions:

- ▶ All volumes are part of volume groups.
- ▶ Hosts are mapped only to volumes within the same partition.
- ▶ SCSI hosts only, connected through FC or iSCSI.
- ▶ No cross-system volume mappings to other IBM Storage Virtualize systems.
- ▶ Validation of dependencies (performed by the GUI).

4.2 Step-by-step configuration to create an IBM FlashSystem grid storage partition

This section describes how to create a storage partition on an IBM FlashSystem grid by using the GUI and CLI. It also explains the storage partition GUI.

4.2.1 Storage partition GUI

From the IBM FlashSystem grid GUI, click **View all partitions** or click **Storage partitions** from the left pane (see Figure 4-1).

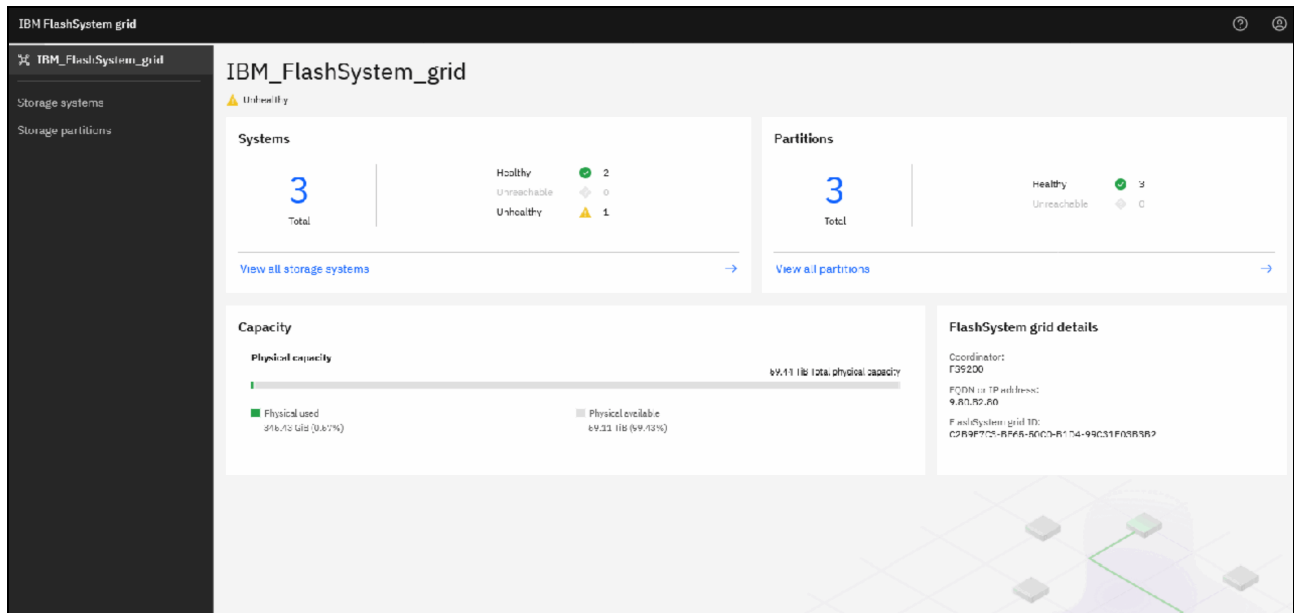


Figure 4-1 IBM FlashSystem grid dashboard

The storage partition GUI view lists all existing partitions in the IBM FlashSystem grid. It shows the status for high availability (HA), disaster recovery (DR), migration, the number of volume groups per partition, and the placement evaluation status (see Figure 4-2 on page 71).

Partition name	Systems	Replication topology	HA status	DR status	Migration status	Placement evaluation status	Volume groups
Fs7300_migration	FS9200	Local	-	-	-	-	0
Storage_Insights_Migration_Partition	FS7300 FS9200	HA	HA established	-	-	-	2
Fs7300_partition1	FS7300 FS9200	HA	HA established	-	-	-	1

Figure 4-2 Storage partitions overall view

From the storage partitions view, you can open the Create partition dialog. Use the overflow menus on the right side to migrate a partition or obtain artificial intelligence (AI) recommendations for storage partition placement. IBM Storage Insights analyzes the environment to identify the best target systems for the storage partition. Partition migration is described in 4.2.4, “Migrating a partition through the IBM FlashSystem grid GUI” on page 76, and AI-driven placement is described in Chapter 6, “IBM FlashSystem grid monitoring with IBM Storage Insights” on page 113.

4.2.2 Creating a storage partition in an IBM FlashSystem grid through GUI

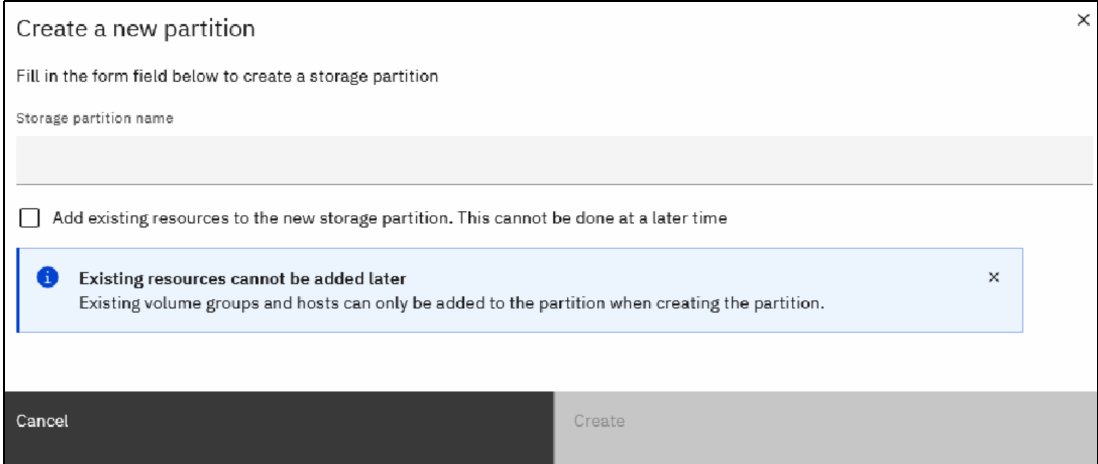
This section describes how to create a storage partition in an IBM FlashSystem grid by using the GUI.

Complete the following steps to create a storage partition in an IBM FlashSystem grid:

1. Select the system where you want to create the partition. The system redirects you to its GUI.

2. In the Create a new partition dialog, which is shown in Figure 4-3, specify a name for the new storage partition. To assign existing volume groups and hosts to this storage partition, select the checkbox. This action creates a draft storage partition. You can assign new volume groups later (see 4.2.5, “Exploring the IBM FlashSystem grid partition GUI” on page 79).

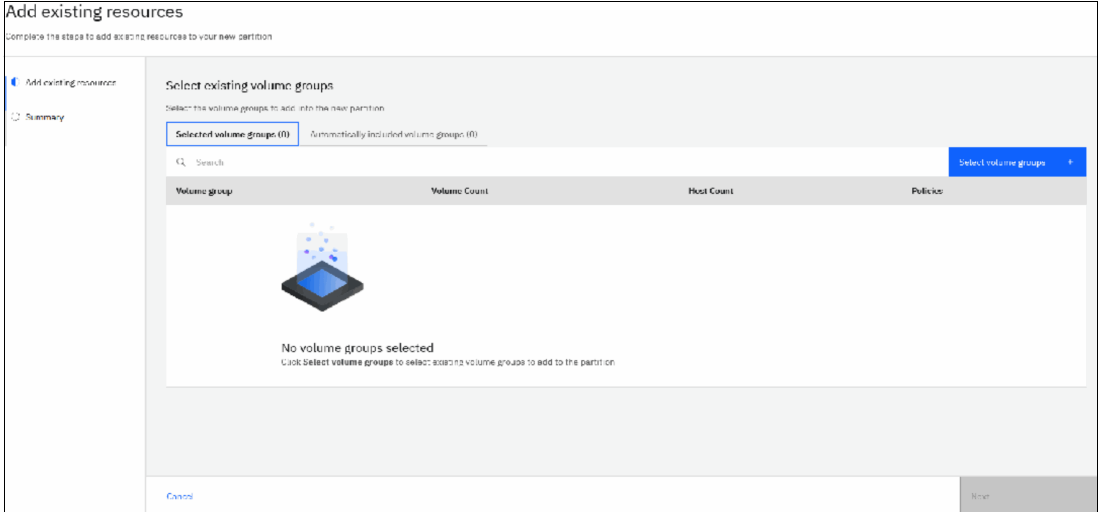
Note: Add existing volume groups and hosts only during the creation of the storage partition.



The dialog box is titled "Create a new partition" and includes a close button (X) in the top right corner. Below the title, it says "Fill in the form field below to create a storage partition". There is a text input field for "Storage partition name". Below this is a checkbox labeled "Add existing resources to the new storage partition. This cannot be done at a later time". A blue information box with an 'i' icon contains the text: "Existing resources cannot be added later. Existing volume groups and hosts can only be added to the partition when creating the partition." At the bottom, there are two buttons: "Cancel" and "Create".

Figure 4-3 Create a new partition dialog

3. If you choose to add existing resources, the dialog continues with more options (see Figure 4-4).



The dialog box is titled "Add existing resources" and includes a close button (X) in the top right corner. Below the title, it says "Complete the steps to add existing resources to your new partition". On the left, there is a sidebar with two options: "Add existing resources" (selected) and "Summary". The main area is titled "Select existing volume groups" and includes the instruction "Select the volume groups to add into the new partition". There are two input fields: "Selected volume groups (0)" and "Automatically included volume groups (0)". Below these is a search bar with a magnifying glass icon and the word "Search". A table with the following columns is shown: "Volume group", "Volume Count", "Host Count", and "Policies". The table is currently empty. Below the table, there is a message: "No volume groups selected. Click Select volume groups to select existing volume groups to add to the partition." At the bottom, there are two buttons: "Cancel" and "Next".

Figure 4-4 Add existing resources

4. Review the summary and close the dialog (see Figure 4-5).

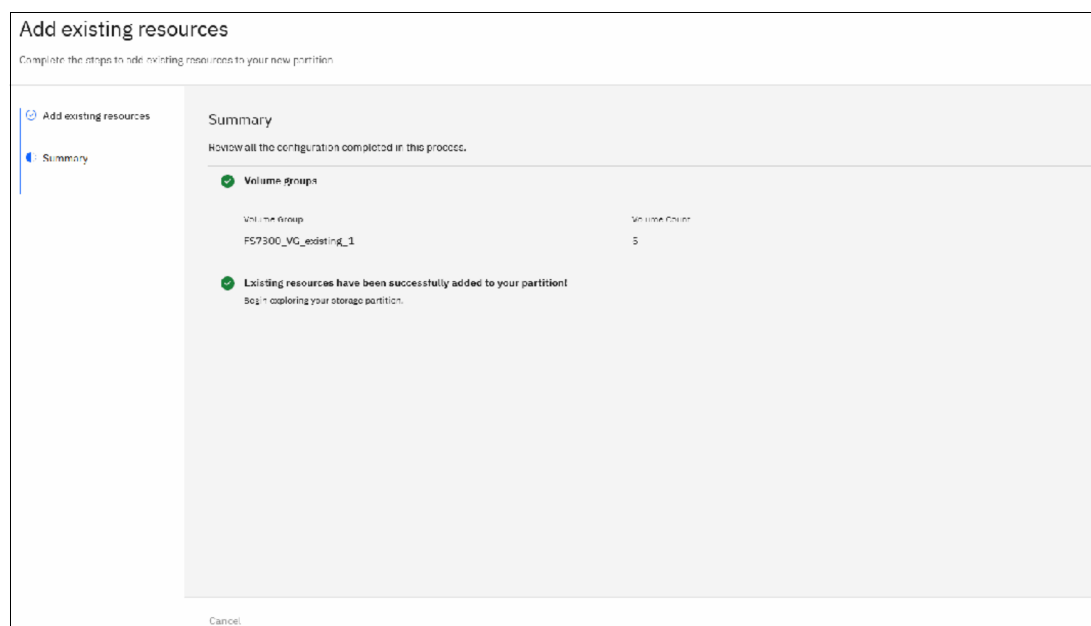


Figure 4-5 Summary of Add existing resources dialog

5. Next, configure replication on your storage or continue exploring the storage partition GUI (see section 4.2.5, “Exploring the IBM FlashSystem grid partition GUI” on page 79).

The following replication types are available and are described later in this chapter:

- HA is described in “High availability” on page 90.
- DR is described in “Disaster recovery” on page 90.
- High availability and disaster recovery (HADR) is described in “High availability and disaster recovery (3-site replication)” on page 90.

4.2.3 Creating and managing a storage partition in an IBM FlashSystem grid by using a CLI

This section explains how to create and manage a storage partition in an IBM FlashSystem grid by using the CLI. A storage partition organizes resources for specific workloads and applications. The process begins by creating a draft partition, which provides flexibility to define the partition with available objects and add resources incrementally as requirements evolve.

Creating a draft storage partition

With a draft storage partition, you can create a storage partition by using existing objects, even if it does not immediately meet all requirements. With this flexibility, you can add resources incrementally as you identify application dependencies.

A draft partition has no effect until you publish it, which you can do only after it meets all partition requirements.

To create a draft storage partition within an IBM FlashSystem grid, connect to the CLI of the storage system and run the **mkpartition** command (see Example 4-1).

Example 4-1 Creating a draft partition

```
IBM_FlashSystem:FS7300:superuser>mkpartition -name Fs7300_for_later_migration
-draft
Storage partition, id [4], successfully created
```

To add an existing volume group (ID 4) to partition ID 4, use the **chvolume** command (see Example 4-2).

Example 4-2 Adding an existing volume group to a draft partition

```
IBM_FlashSystem:FS7300:superuser>svctask chvolume -draftpartition 4 4
```

Listing storage partitions

Use the **lspartition** command to display all existing partitions (see Figure 4-6).

```
IBM_FlashSystem:FS7300:superuser>lspartition
```

id	name	preferred_management_system_name	active_management_system_name	replication_policy_id	replication_policy_name	location1_system_name	location1_status	location2_system_name	location2_status	host_count
portset_id	management_portset_name	host_cluster_count	draft_host_cluster_count							
0	Fs7300_partition1	1	0	no	0	FS7300	0	healthy	0	0
1	FS7300_partition2	0	0	no	0	FS7300	0	healthy	0	0
2	Fs7300_migration	1	0	no	0	FS7300	0	healthy	0	0
3	Storage_Insights_Migration_Partition	2	0	no	0	FS7300	0	healthy	0	0

Figure 4-6 Listing partitions

Publishing a storage partition

To publish a draft storage partition, use the **chpartition - publish** command (see Example 4-3).

Example 4-3 Publishing a partition

```
IBM_FlashSystem:FS7300:superuser>svctask chpartition -publish 4
```

Migrating a partition (CLI)

Migration moves a storage partition from a source system to a target system. This process includes the following steps:

1. The system creates all resources that are associated with the storage partition on the target system, and begins copying data from the source to the target.
2. When synchronization completes, the target system begins serving host I/O.
3. At the end of migration, the system removes the partition and its resources from the source system.

To migrate a partition to another target location, use **chpartition -location** (see Example 4-4). A draft partition cannot be migrated.

Example 4-4 Migrating a partition

```
IBM_FlashSystem:FS7300:superuser>svctask chpartition -location FS7300
```

Check the migration status by running **lspartition <object_id/partition_id>**. The `migration_status` field displays the real-time migration status. One of the following statuses is displayed:

- ▶ `queued`
- ▶ `in progress`
- ▶ `user action required`
- ▶ `aborting`
- ▶ `blank`
- ▶ `suspended`

You can monitor the progress of a storage partition migration by using the **lsvolumegroupreplication** command. This command displays the following details:

- ▶ Amount of data remaining to be copied
- ▶ Estimated time of completion

After data and configuration are migrated to the target storage system, verify the following items:

- ▶ Host path discovery. Affected hosts discover paths to the volumes on the target system.
- ▶ Multipath configuration. Host multipath settings are correct.
- ▶ Path redundancy. Multiple paths are available and functioning as expected.
- ▶ I/O performance. I/O performance meets operational requirements.

To complete the migration, commit it. Committing removes the copy of the storage partition data and configuration from the source storage system.

Rollback migration

If issues are detected before committing, cancel the migration and revert to the original source system. When you start a rollback operation, the source system requires confirmation to proceed with the rollback or cancel it. Continue the migration after you resolve the issues at the target system.

To continue the migration without finishing the rollback, run the following command on the source system:

```
chpartition -migrationaction fixeventwithchecks <partition_id/name>
```

To proceed with the rollback and cancel the partition migration, run the following command on the source system:

```
chpartition -override -location <source_system> <partition_id>
```

Rollback is required if the migrated storage partition does not perform as expected on the target system. The rollback operation switches I/O operations back to the storage partition at the source location.

Merging a partition

Use **mergepartition** to merge a source storage partition into a target storage partition. After the merge completes, the system deletes the source storage partition. This command supports both local-only and HA storage partitions. For HA configurations, changes apply to both systems automatically (see Example 4-5).

Example 4-5 Merging partitions

```
IBM_FlashSystem:FS7300:superuser>mergepartition -targetpartition 1 2
```

Deleting a storage partition

Deleting a storage partition does not remove the objects in the storage partition. If a replication policy is assigned, a copy of the objects exists at each policy location. In this case, you must specify `-deletenonpreferredmanagementobjects` or `-deletepreferredmanagementobjects` to identify the location where the associated objects must be removed. These flags identify the location from which you remove the associated objects. These flags are not allowed if the partition does not have a replication policy (see Example 4-6).

Example 4-6 Deleting a partition

```
IBM_FlashSystem:FS7300:superuser>rmpartition 0
IBM_FlashSystem:FS7300:superuser>
```

4.2.4 Migrating a partition through the IBM FlashSystem grid GUI

Note: Deletion is irreversible, so ensure that all validations are complete before you proceed.

You can migrate storage partitions between systems within the same IBM FlashSystem grid.

In the storage partitions GUI, click **Migrate partition** from the overflow menu. The Migration partition wizard opens with an overview.

When you migrate a storage partition to another system, the operation copies the configuration and data non-disruptively. The original configuration and data remain on the current system until migration completes (see Figure 4-7).

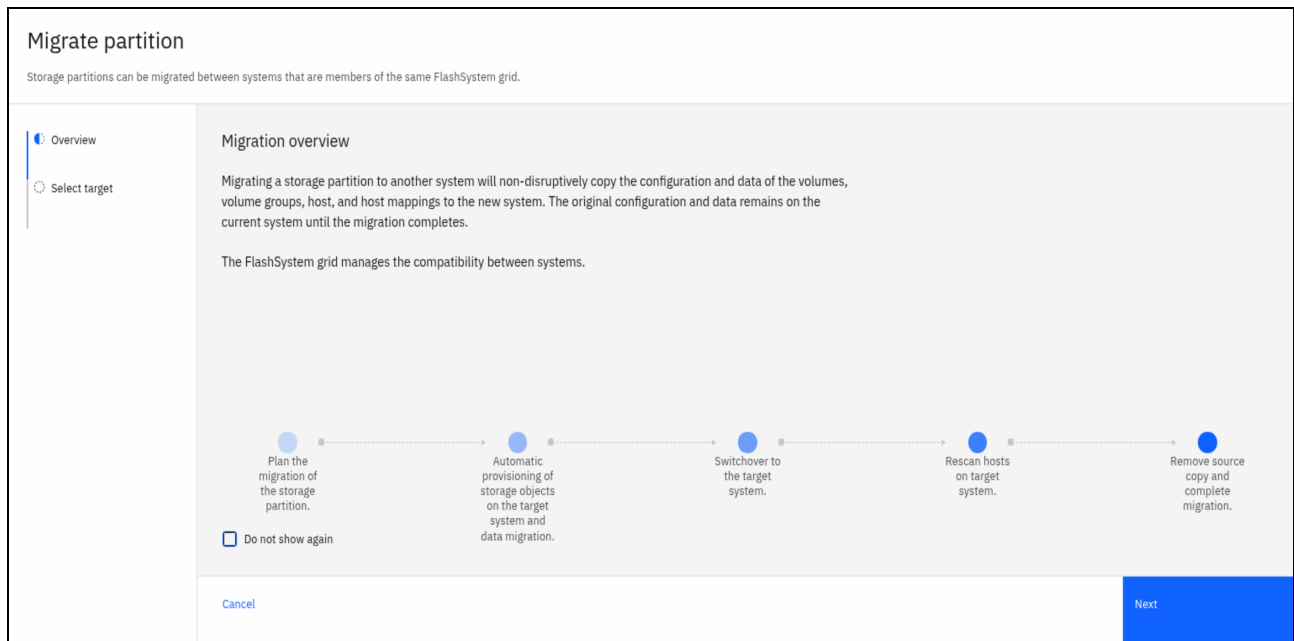


Figure 4-7 Migrate partition wizard overview

To migrate a partition by using the GUI, complete the following steps:

1. Select the target system from the list of suitable systems (see Figure 4-8).

Migrate partition

Storage partitions can be migrated between systems that are members of the same FlashSystem grid.

Select target

Select a target system from the list below that is part of the FlashSystem grid. Based on the chosen partition, the systems are listed in the order of suitability.

Partition details

Storage partition name
Fs7300_migration

Storage partition capacity details
Provisioned capacity 0 bytes

Source system
FS7300
IBM FlashSystem 7300

Select a target system

FS9200
IBM FlashSystem 9200

Used capacity after migration 49.43 GiB / 37.00 TiB

Quick status

Firmware matches Pools linked Portsets linked

SVC
IBM SAN Volume Controller

Used capacity after migration 0 bytes / 0 bytes

Cancel Back Start migration

Figure 4-8 Migrate partition wizard: Selecting a target

2. Confirm to begin copying the configuration and data by clicking **Start** migration (see Figure 4-9).

Confirm

The partition "Fs7300_migration" is about to be migrated from FS7300 to FS9200. This may take some time to complete. Do you want to proceed?

Cancel Start migration

Figure 4-9 Confirming the migration

3. Monitor the migration status in the Storage partitions view (see Figure 4-10).

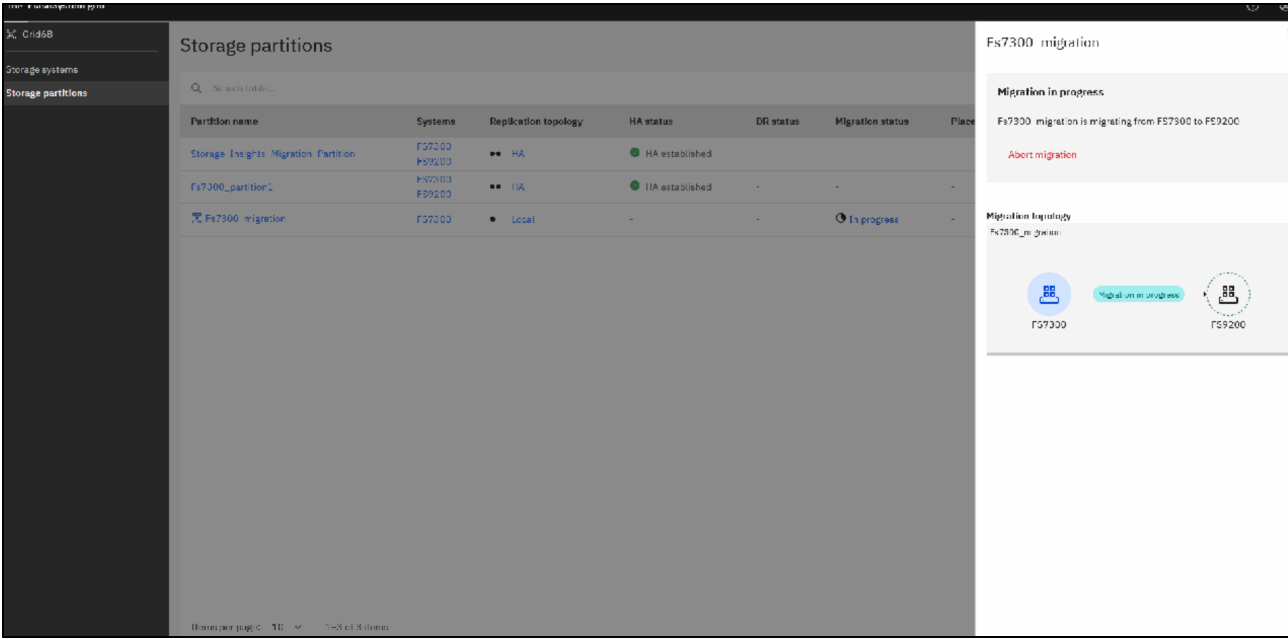


Figure 4-10 Migration in progress

4. When the migration completes and the status changes to Awaiting user input, verify that the partition operates as expected (see Figure 4-11).

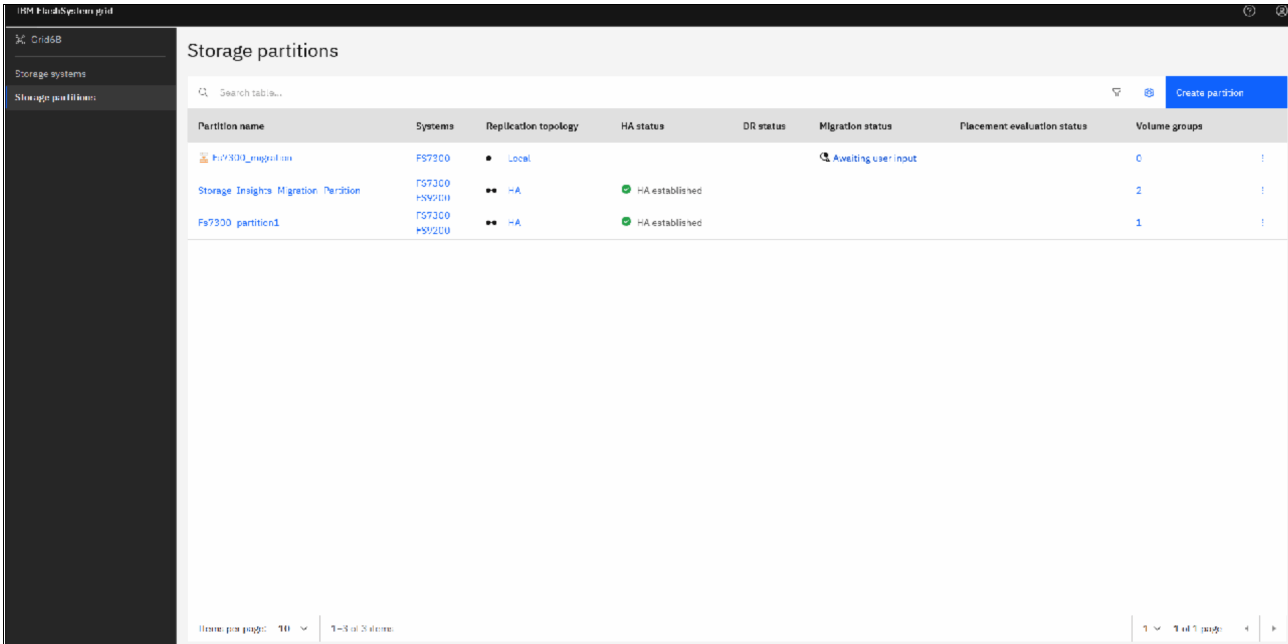


Figure 4-11 Awaiting user input

5. Click **Commit** to finalize the migration. If issues occur, click **Rollback** to revert to the source system (see Figure 4-12).

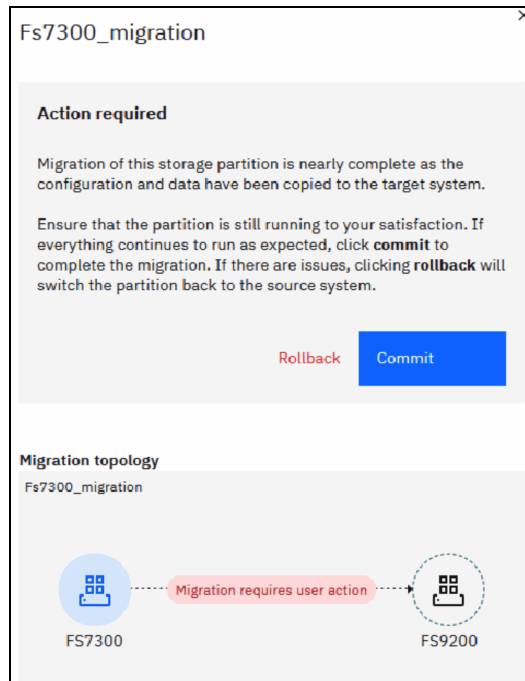


Figure 4-12 Commit or rollback migration

6. After you click **Commit**, the migration is irreversible.

4.2.5 Exploring the IBM FlashSystem grid partition GUI

This section describes how to explore the IBM FlashSystem grid partition GUI. Use the individual storage partition GUI to perform tasks that configure, manage, and maintain partition resources. These tasks include assigning IP addresses, creating volume groups and volumes, defining mappings and hosts. They also involve managing replication and snapshot policies, and other administrative actions.

Use the individual storage partition GUI to perform the following actions:

- ▶ Assign or manage a partition's IP address.
- ▶ Create volume groups.
- ▶ Create volumes.
- ▶ List and define volume mappings.
- ▶ List and define hosts.
- ▶ List and define host clusters.
- ▶ List and create replication policies.
- ▶ List and define snapshot policies.
- ▶ Merge or delete the partition.
- ▶ Download the IP quorum application.
- ▶ Manage partnerships.
- ▶ Exit the individual storage partition GUI and return to the IBM FlashSystem storage grid GUI.

Figure 4-13 shows the individual storage partition GUI.

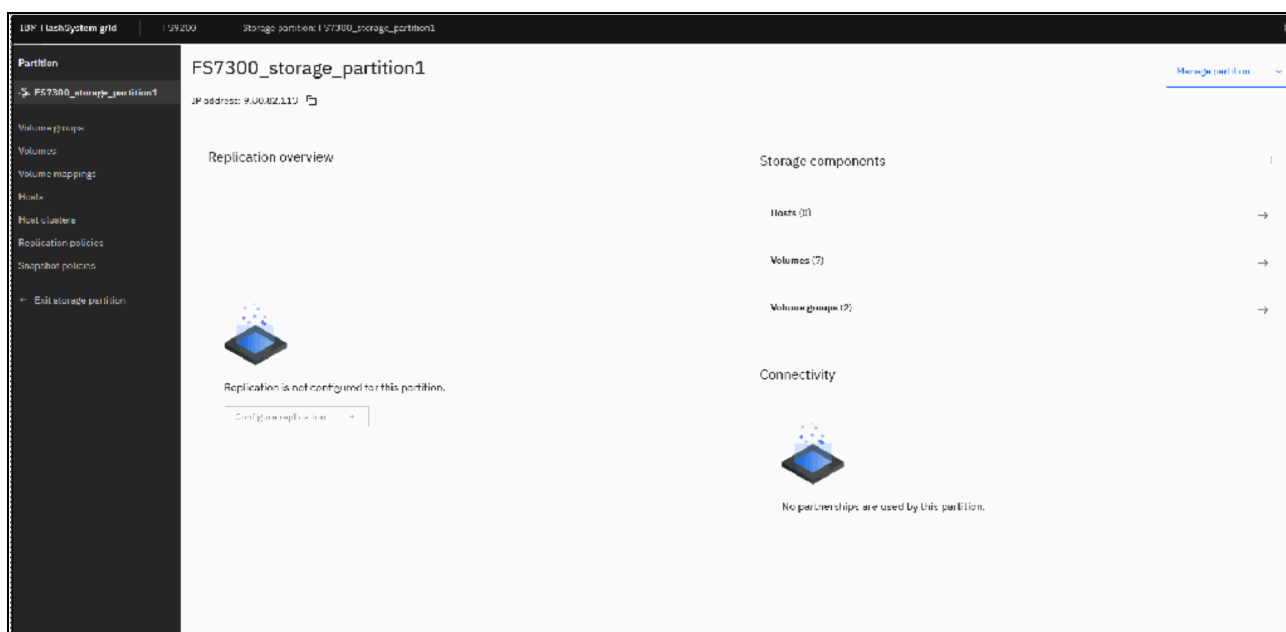


Figure 4-13 Individual storage partition GUI

Assigning a partition IP address

A partition IP address enables remote services to access resources within the storage partition. Currently, direct access to this IP is limited to the CLI. This IP serves as a consistent management endpoint, which is useful for automation during storage partition migration or failover event see Figure 4-14).

Assign IP address

This will add an IP address to the partition.

Partition IP address

☒ IPv4
☐ IPv6

IP address

Subnet Mask

VLAN ID (optional)

Gateway (Optional)

Cancel

Save

Figure 4-14 Assigning a partition IP address

The following restrictions apply:

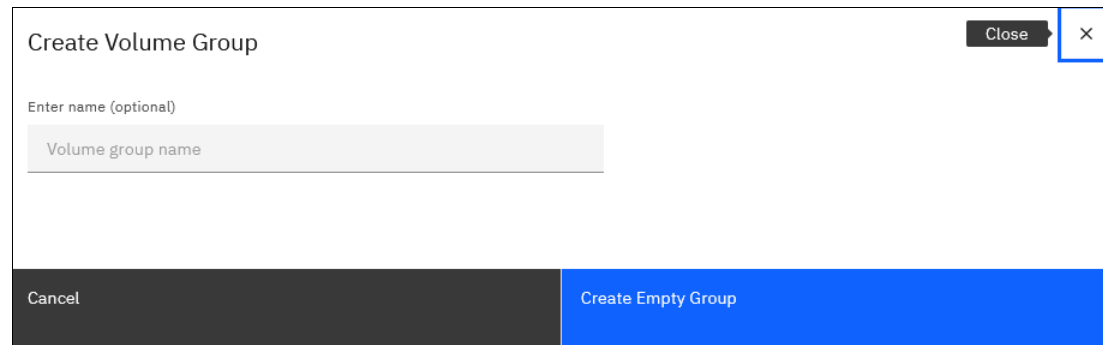
- ▶ Each partition can have one management IP address per management portset.
- ▶ After you map a management portset to a storage partition, its associated management IP address cannot be modified or removed directly. To make changes, unmap the portset, update or delete the IP address, then remap the portset to the storage partition.
- ▶ The GUI automates these steps so that you can change or remove the IP address in the partition settings.

Creating a volume group

Within a storage partition, all volumes must belong to a volume group, and hosts can map only to volumes in the same partition.

To create a volume group, complete the following steps:

1. Click **Create Volume Group** to create an empty volume group.
2. Specify a name for the new volume group (see Figure 4-15).



The image shows a 'Create Volume Group' dialog box. It has a title bar with 'Close' and 'X' buttons. The main area contains a label 'Enter name (optional)' above a text input field labeled 'Volume group name'. At the bottom, there are two buttons: 'Cancel' on the left and 'Create Empty Group' on the right.

Figure 4-15 Creating an empty volume group

3. Save the configuration (see Figure 4-16).

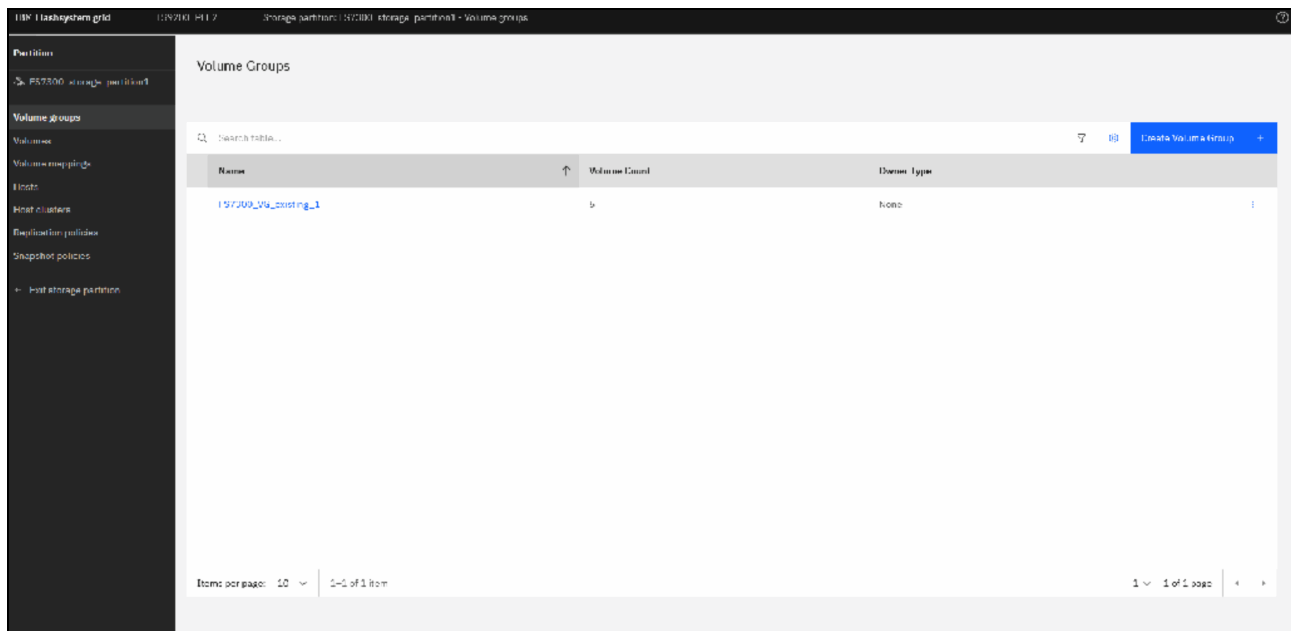


Figure 4-16 Volume group view inside the partition GUI

Notes:

- ▶ You can rename or delete a volume group.
- ▶ You cannot delete a volume group that contains active volumes.
- ▶ If a volume group is used for Policy-based Replication (PBR), remove it from the production system, not the recovery system.

Creating and listing volumes in the storage partition GUI

From the partition GUI, click **Volumes** to view all volumes. The Volumes view can display up to 34 columns, ranging from ID to virtualization type.

To create volumes, complete the following steps:

1. Click **Create volumes** to start the wizard for creating volumes and host mappings (see Figure 4-17).

Create Volumes

Create new volumes here. You can map the volumes to host now, or just create the volumes and map them to hosts later.

Volume group
VG_9200_HA

Pool
database_pool (Base capacity savings)

Provisioning Policy
Volumes created in this pool must follow the provisioning policy assigned to the pool.

Additional Capacity Savings
None

Quantity: 5 Name: Prod Starting suffix: 0 (Prod0 To Prod4)

Capacity: 1 Unit: GiB

☐ Additional settings

Summary

Pool name: database_pool

Available Capacity: 98.85 TiB / 99.08 TiB

Volume group: VG_9200_HA

Volumes

Quantity: 5

Capacity used: 20.00 GiB

Provisioned volume capacity: 5.00 GiB

Buttons: Cancel, Create and map, Create volumes

Figure 4-17 Create volumes wizard

2. Follow the prompts to create volumes and define host mappings:

The overflow menu in the Volumes view provides the following options (see Figure 4-18 on page 83):

- **Create host mappings.** Creates mappings between hosts and volumes.
- **Modify capacity.** Changes the size of an existing volume.
- **Rename volumes.** Renames selected volumes.
- **View snapshots.** Displays available snapshots for a volume.
- **Migrate volumes to another pool.** Moves volumes to a different storage pool.
- **Remove private mappings.** Deletes private host mappings for a volume.

- **Delete volumes.** Removes selected volumes from the system.
- **List properties.** Shows detailed properties of a volume.

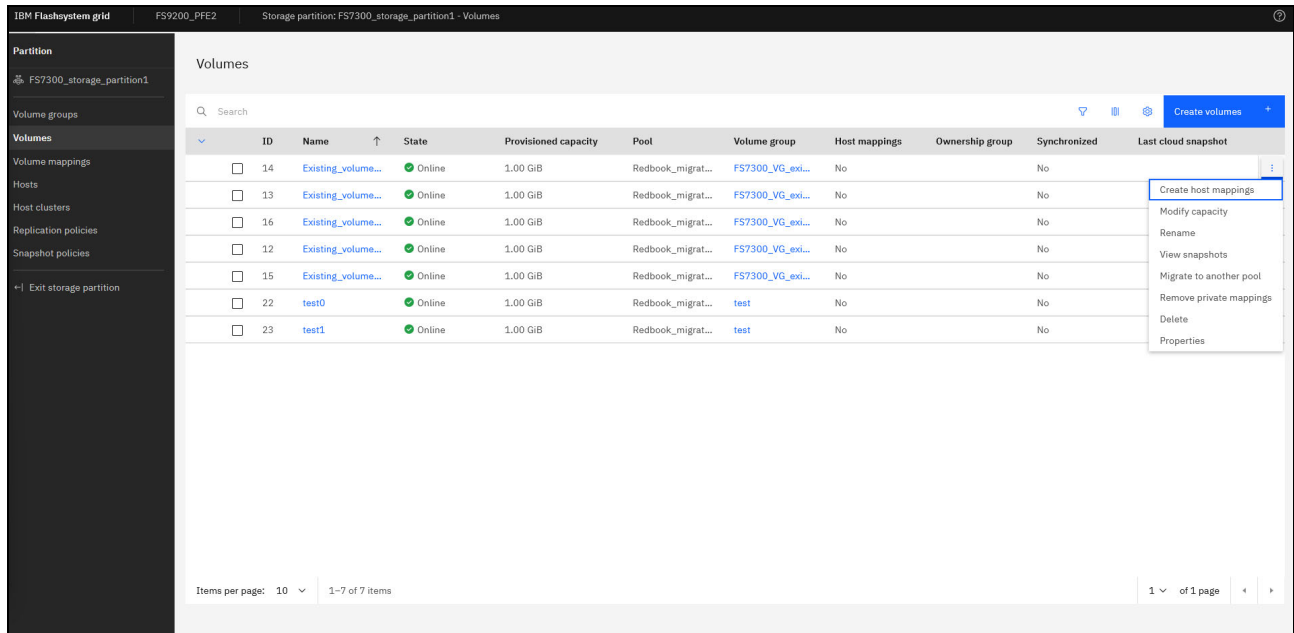


Figure 4-18 Volumes view in the partition GUI

Modifying a volume from the properties view

You can modify a volume from the properties view. For example, you can move the volume to another volume group (see Figure 4-19).

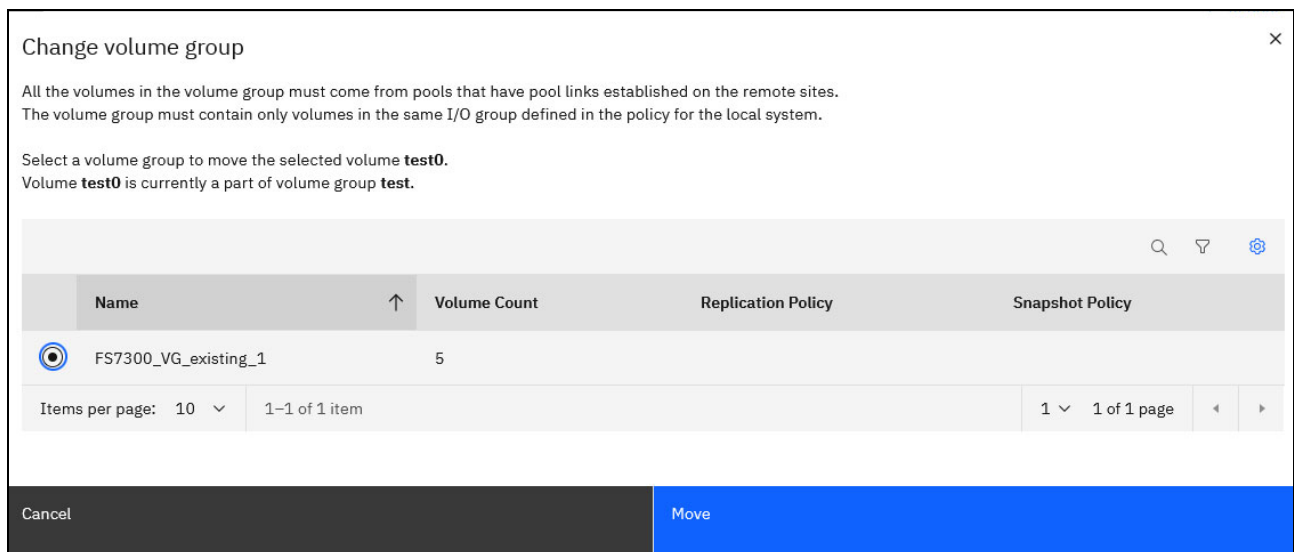


Figure 4-19 Moving a volume to other volume group

Volume mappings

You can view the volume mappings for each host. From the **Host Actions** drop-down menu, you can modify, duplicate, or remove existing mappings under (see Figure 4-20).

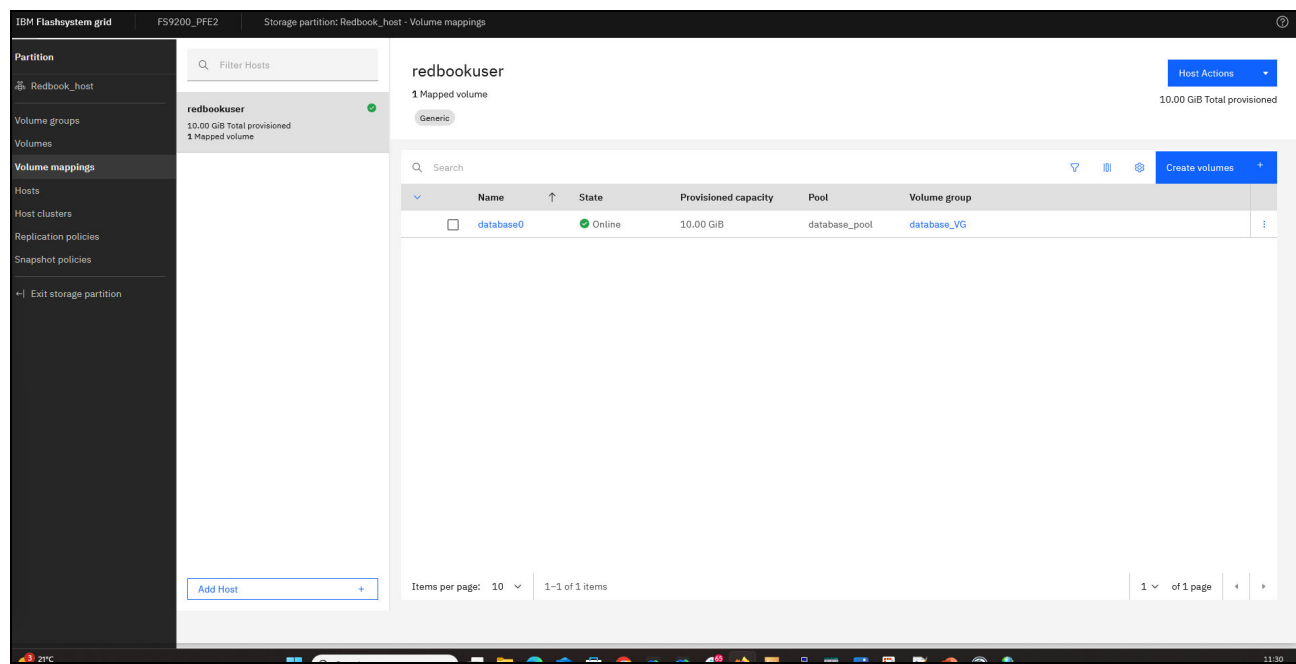


Figure 4-20 Volume mappings in the storage partition GUI

Listing and defining hosts

You can view existing host descriptions and use the wizard to add more hosts. Hosts are either Fibre Channel (FC) attached or iSCSI-attached (see Figure 4-21).

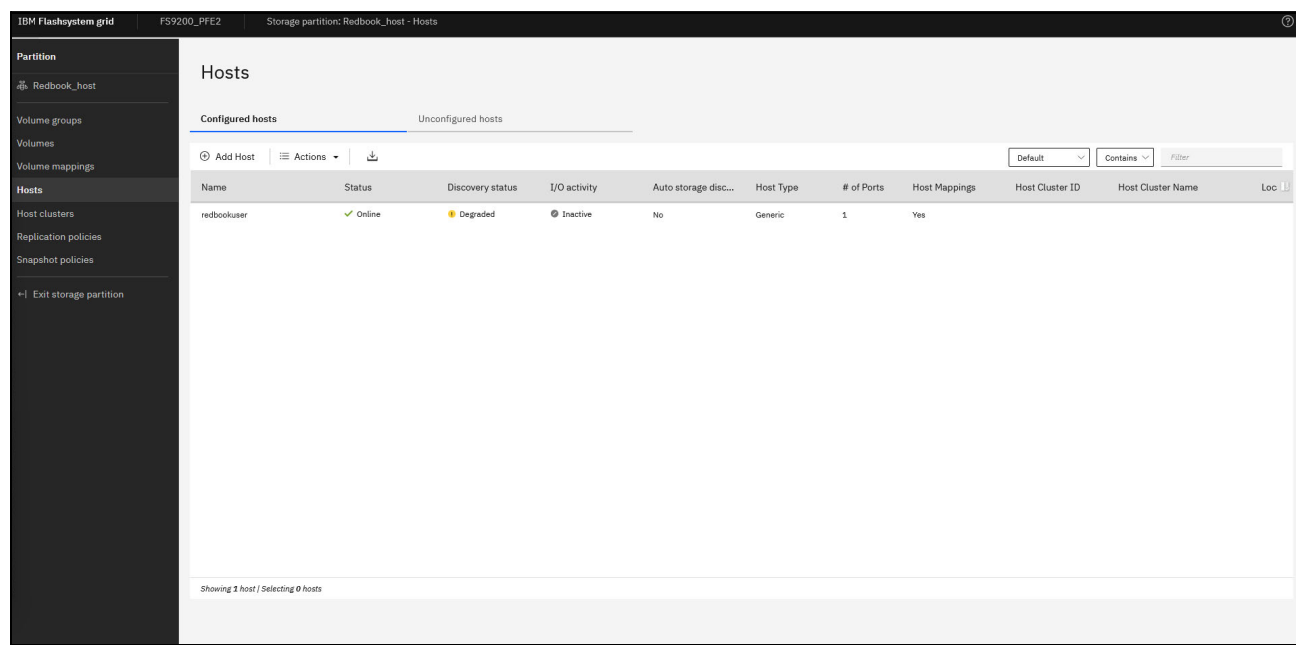


Figure 4-21 Host view in the storage partition GUI

Listing and creating host clusters

View existing host clusters and perform the following actions:

- ▶ Add and remove hosts from a selected cluster.
- ▶ Modify or remove shared volume mappings.
- ▶ Delete the host cluster.

Listing and creating replication policies

You can view existing replication policies (see Figure 4-22).

The screenshot shows the IBM Flashsystem grid web interface. The breadcrumb trail at the top reads: IBM Flashsystem grid > FS9200_PFE2 > Storage partition: FS9200_partition1 - Replication policies. The left sidebar contains a navigation menu with the following items: Partition (selected), FS9200_partition1, Volume groups, Volumes, Volume mappings, Hosts, Host clusters, Replication policies (highlighted), Snapshot policies, IP quorum, and Partnerships. At the bottom of the sidebar is a link: Exit storage partition.

The main content area is titled 'Replication Policies'. It features a search bar with the placeholder text 'Search table...'. To the right of the search bar are icons for a dropdown menu, a settings gear, and a button labeled 'Create replication policy' with a plus icon. Below the search bar is a table with the following columns: Name, Topology, RPO Alert, and Volume group count. The table contains one row with the following data: Name: ha-policy-0, Topology: High availability, RPO Alert: -, and Volume group count: -. To the right of the table is a vertical ellipsis menu icon.

Below the table, the 'Policy Information' section displays the following details: Name: ha-policy-0, ID: 0, HA snapshots: Enabled, and Topology: High availability. The 'Replication Information' section displays two locations: Location 1 (System Name: FS9200_PFE2, System ID: 000002042240B714) and Location 2 (System Name: FS7300-PFE6, System ID: 0000020420E142D8).

At the bottom of the page, there is a pagination bar showing 'Items per page: 10' and '1-1 of 1 item'. On the right side of the pagination bar, there is a page indicator showing '1' and '1 of 1 page' with navigation arrows.

Figure 4-22 Storage partition replication policies

Listing and defining snapshot policies

You can automate snapshot creation and retention by creating snapshot policies (see Figure 4-23).

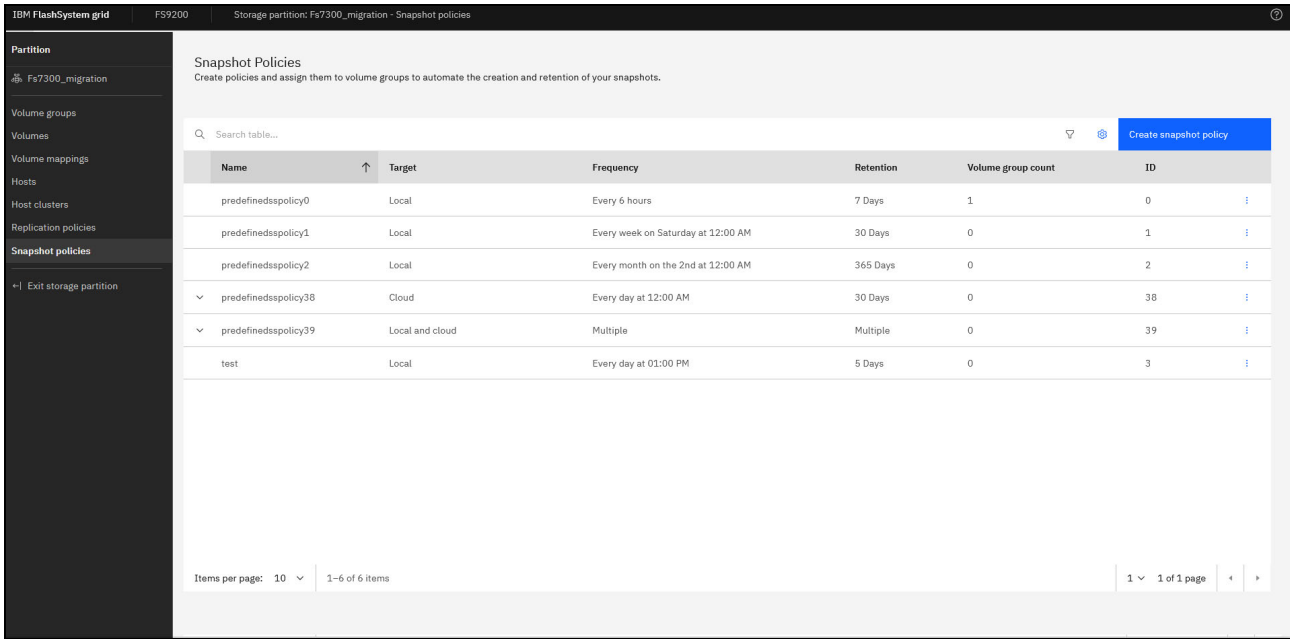


Figure 4-23 Snapshot policy view

Use the wizard to create policies to match your requirements. Specify the policy and the frequency, including the time, day of the week, day of the month, and retention period for snapshots (see Figure 4-24).

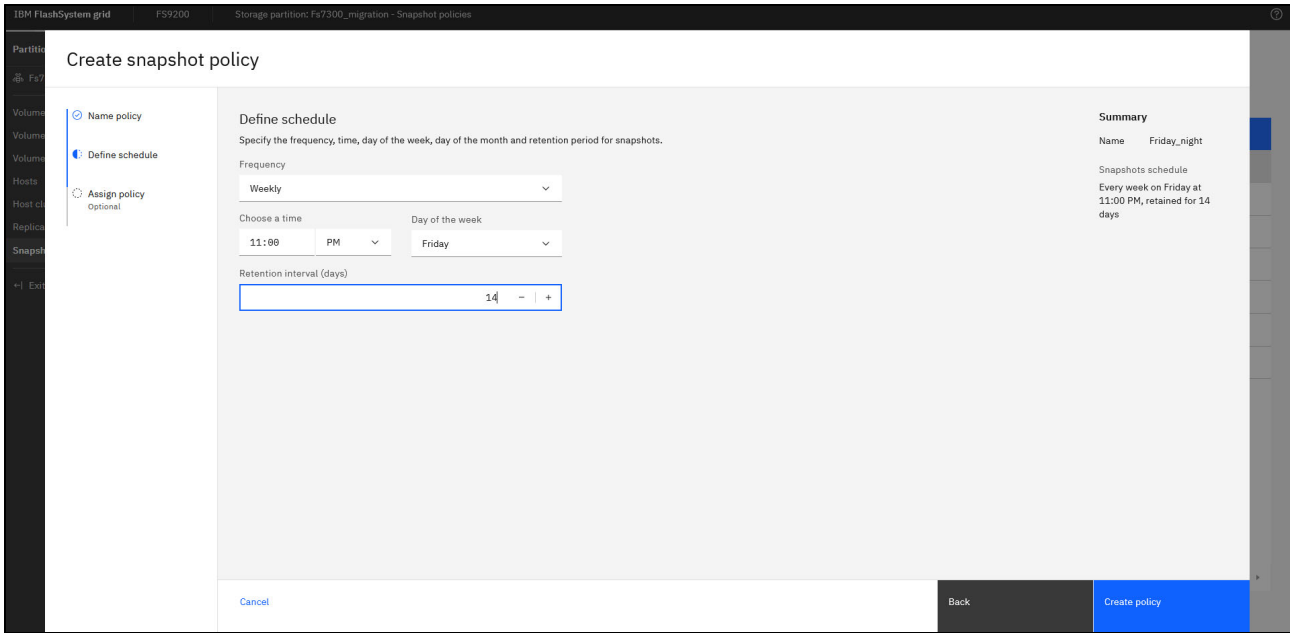


Figure 4-24 Defining a schedule

In the next step of the wizard, assign a volume group or skip this step to do it later (see Figure 4-25 on page 87).

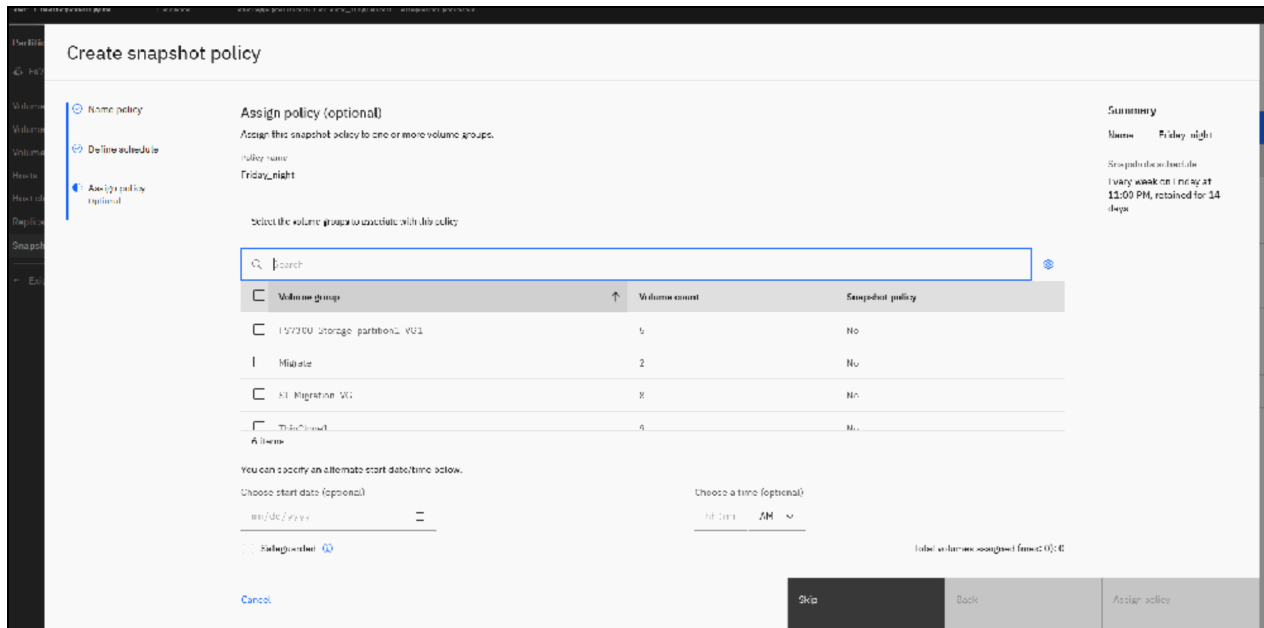


Figure 4-25 Assigning a policy to a volume group

Merging and deleting a partition

Select the partition for the merge. The system moves the objects from the current partition into the target partition and deletes the current partition (see Figure 4-26). Partitions that do not meet the requirements display an exclamation mark.

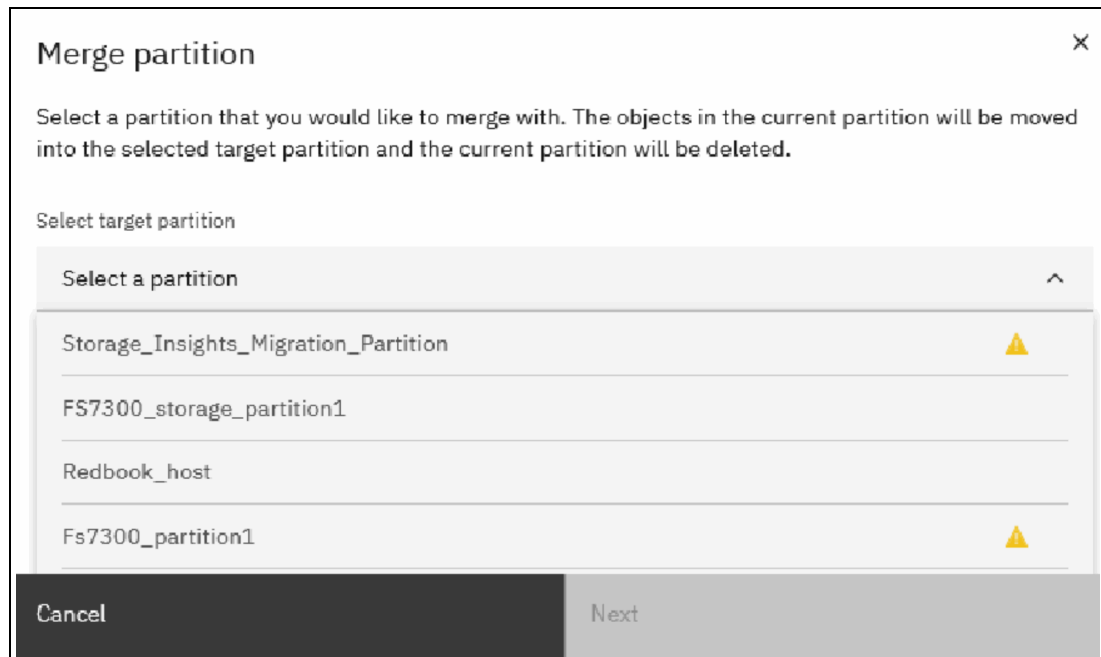


Figure 4-26 Merging and deleting a partition

Review the merged resources (see Figure 4-27).

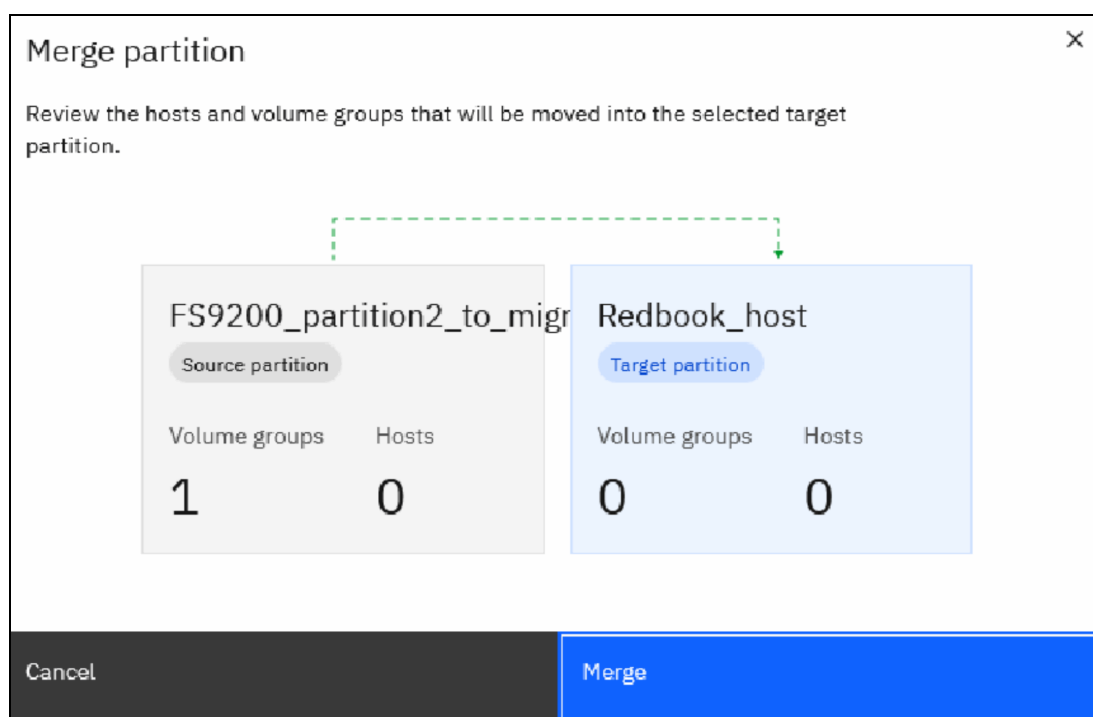


Figure 4-27 Merge partition summary

Deleting a partition

From the Storage Partition GUI, select the storage partition that you want to delete. Then from the **individual storage partition** drop-down menu, click **Delete Storage Partition**. The defined resources (such as hosts and volumes) remain available on the original owning system (see Figure 4-28).

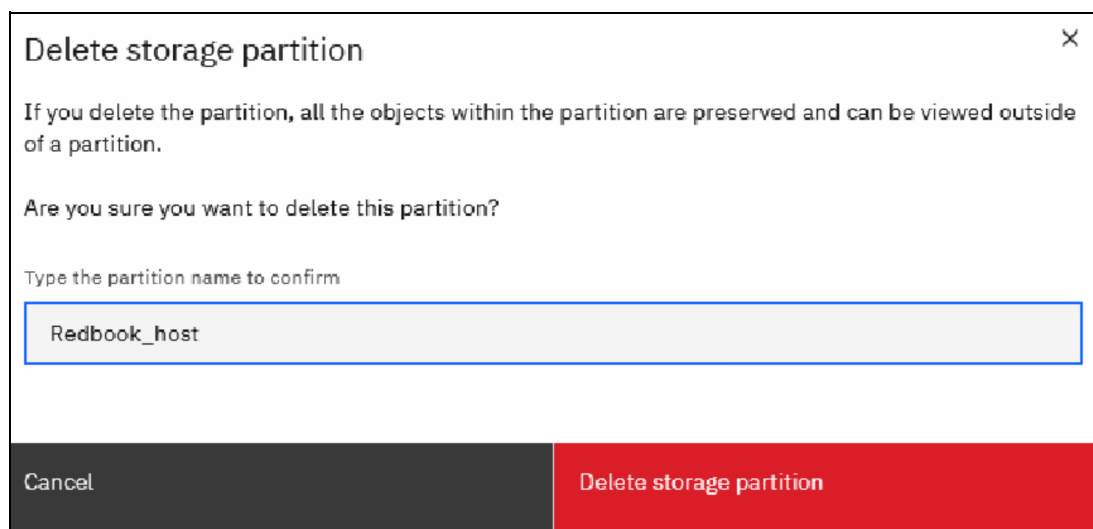


Figure 4-28 Delete storage partition

Partnerships

You can view the status of existing partnerships. From the **Overflow** menu, you can start or stop a partnership, delete a partnership, or modify the CHAP configuration. You can also view partnership details, including the remote system certificate, and change properties such as link bandwidth (see Figure 4-29).

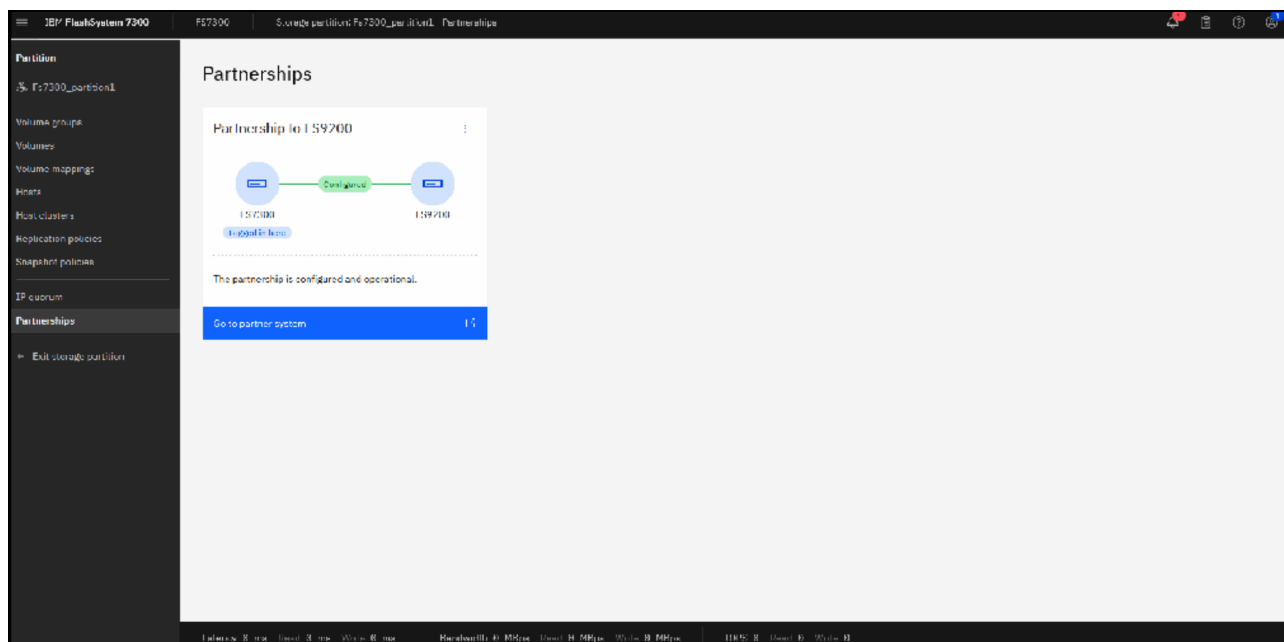


Figure 4-29 Partnership details in the storage partition GUI

4.3 Continuous data availability by using replication

Enterprises require data services that are high-performing and resilient. IBM FlashSystem grid with storage partitions provides a modern approach to data availability through PBR and Policy-based High Availability (PBHA).

Define application-specific protection policies, securely isolate workloads, and achieve low Recovery Point Objective (RPO) and low Recovery Time Objective (RTO) across distributed environments.

4.3.1 Replication

Table 4-2 shows the different types of replication approaches that you can use.

Table 4-2 Types of replication configurations

Type	Purpose
HA	Active-active HA between two independent systems
DR	Asynchronous replication to another system for DR
HADR	Active-active HA between two independent systems, with asynchronous replication to a third system for DR (3-site replication)

You can use PBR to implement HADR solutions between systems in different locations. For more information about implementing PBR, see Chapter 4, “Implementing policy-based replication”, of *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, SG24-8569.

High availability

PBHA introduces storage partitions to manage your HA storage. Partitions provide a modular approach. For more information about implementing PBHA, see Chapter 6, “Implementing policy-based HA”, of *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, SG24-8569.

Disaster recovery

You can use PBR to implement asynchronous replication between systems for DR.

If a disaster prevents access to the production storage, you can enable host access to the recovery copy. When the original production system becomes available again, you can restart replication in either direction to restore redundancy.

You can configure replication for DR with or without storage partitions. When you use storage partitions, you can add HA later and enable IBM FlashSystem grid features. Use storage partitions where the host environment meets the supported configurations.

You can migrate DR configurations without storage partitions into a storage partition by using the CLI. With this migration, you can extend an existing 2-site DR configuration to a 3-site HADR configuration without resynchronizing the recovery copy.

Replication for DR provides flexibility to use different replication policies for different volume groups. You can match business requirements for different recovery point objectives, even on the same partnership or within the same storage partition.

Apply a replication policy to volume groups

To apply a replication policy, complete the following steps:

1. In the management GUI, click **Replication**.
2. Select the partnership that you want to use.
3. Select the replication policy.
4. Select the volume groups where you want to apply the replication policy.
5. Confirm the settings, and then click **Apply**.

For details about implementing DR, see Chapter 2, “Policy-based replication”, of *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, SG24-8569.

High availability and disaster recovery (3-site replication)

Combining PBHA storage partitions and asynchronous PBR for DR enables a 3-site replication solution for mission-critical workloads. HA focuses on preventing disruptions. DR prepares for worst-case scenarios.

You can manage HADR replication from the Storage partitions window in the management GUI.

Configuring 3-site with HADR

You can configure 3-site replication for a storage partition in any of the following ways:

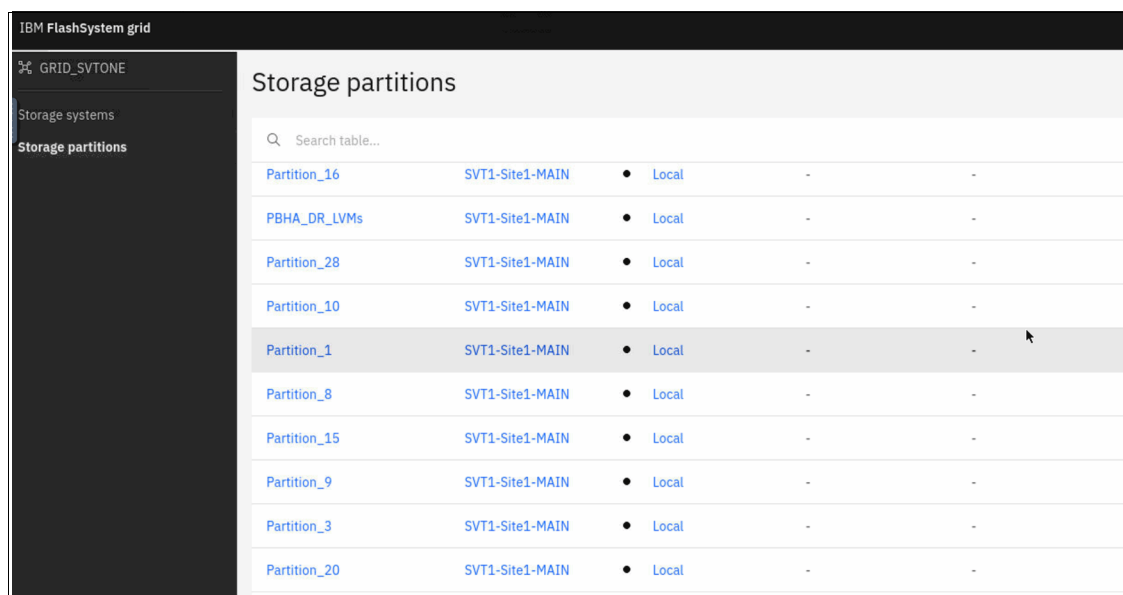
- ▶ Start with an existing 2-site HA configuration, then add replication for DR at a third site.
- ▶ Start with an existing 2-site configuration for DR by using storage partitions, then add HA to either partition.
- ▶ Configure 3-site replication for HADR in a single operation.

IBM Storage Virtualize provides a graphical wizard that guides you through creating 3-site replication (HADR).

To create 3-site replication, complete the following steps:

1. Select the partition from the site that you want to replicate in HADR (see Figure 4-30).

Note: In a 3-site HADR configuration, the target site for HA and the target site for DR are different physical locations. The site in Step 1 is the production site, where the application runs, and it must be part of HA.



Storage partitions				
Search table...				
Partition_16	SVT1-Site1-MAIN	Local	-	-
PBHA_DR_LVMs	SVT1-Site1-MAIN	Local	-	-
Partition_28	SVT1-Site1-MAIN	Local	-	-
Partition_10	SVT1-Site1-MAIN	Local	-	-
Partition_1	SVT1-Site1-MAIN	Local	-	-
Partition_8	SVT1-Site1-MAIN	Local	-	-
Partition_15	SVT1-Site1-MAIN	Local	-	-
Partition_9	SVT1-Site1-MAIN	Local	-	-
Partition_3	SVT1-Site1-MAIN	Local	-	-
Partition_20	SVT1-Site1-MAIN	Local	-	-

Figure 4-30 IBM FlashSystem grid Storage partitions view

2. Click **Configure replication** to start the wizard (see Figure 4-31).

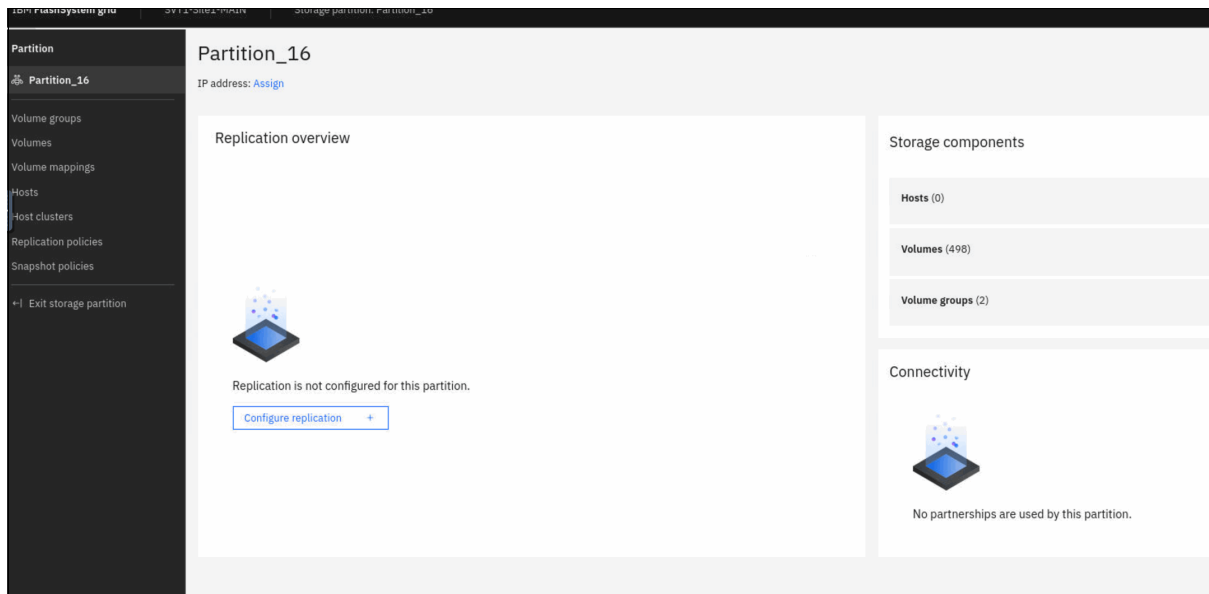


Figure 4-31 Inside the partition

3. Select **High availability and disaster recovery** as the replication type and click **Continue** (see Figure 4-32).

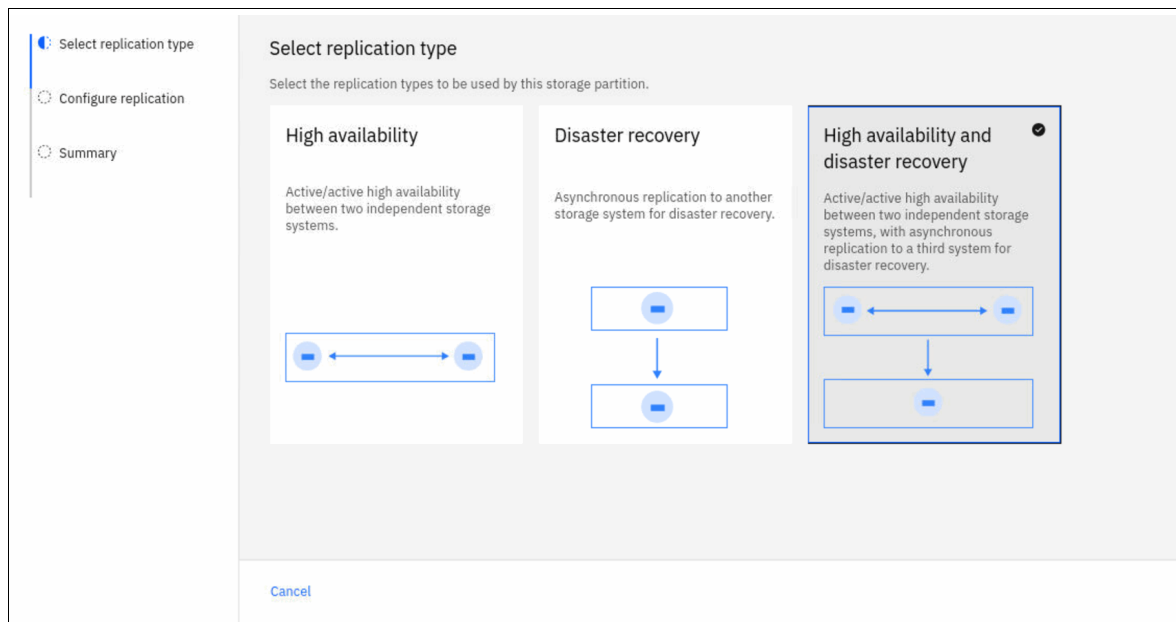


Figure 4-32 Selecting the type of replication

4. If partnerships do not exist, the wizard guides you through creating them, otherwise the wizard shows the next step. Click **Configure partnership** to begin (see Figure 4-33).

Figure 4-33 Creating partnerships

5. To set up your HA partnership, complete the following steps:
- Choose the connectivity type and partner system name from the drop-down list. You can create a new partnership or use an existing one.
 - Verify the system certificate.
 - Specify the bandwidth between your systems.

After you verify the details, click **Create** to finalize the partnership (see Figure 4-34).

Figure 4-34 Configure high availability partnership

- Repeat the process on the second system. After both partnerships are created and pools are linked, download the IP quorum application and verify that it is running (see Figure 4-35 and Figure 4-36).

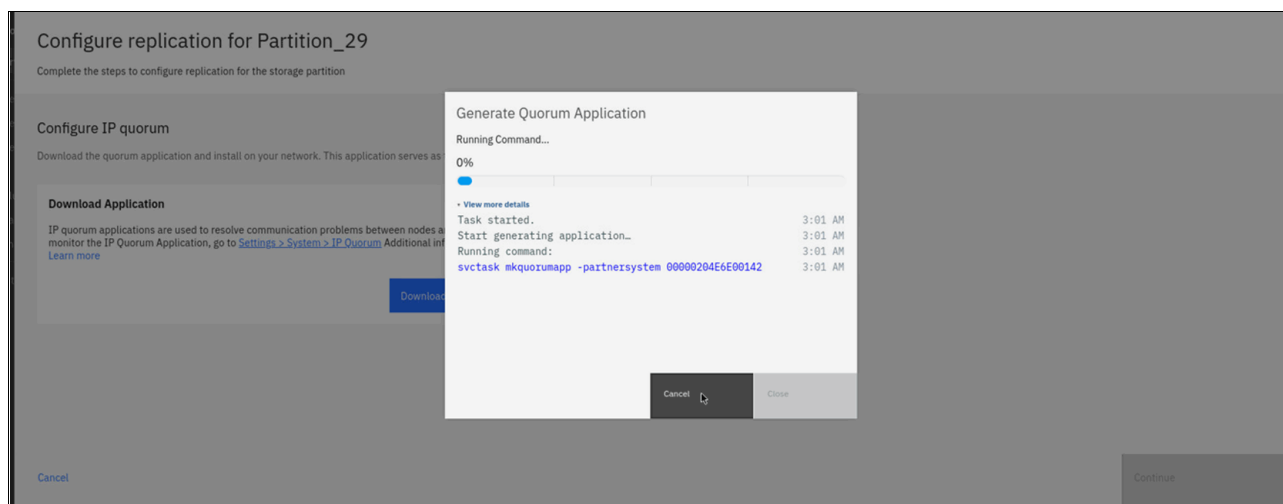


Figure 4-35 Creating the IP Quorum application

```
[21:19:00] Bantam-RHEL88:~ # java -jar ip_quorum.jar
=== IP quorum ===
Name set to null.
Discovering nodes for system: 172.168.13.9
Discovering nodes for system: 172.168.13.6
Successfully parsed the configuration, found 4 nodes.
App started
Trying to open socket
Trying to open socket
Trying to open socket
Trying to open socket
Handshaking
Handshaking
Handshaking
Handshaking
Creating UID
Waiting for UID
Waiting for UID
Waiting for UID
*Connecting with UID: 186916976721941
*Connecting with UID: 186916976721941
*Connecting with UID: 186916976721941
Connected to 172.168.13.7
Connected to 172.168.13.5
Connected to 172.168.13.4
Connected to 172.168.13.8
```

Figure 4-36 Example of an IP Quorum application running on a Red Hat Enterprise Linux host

Note: The server that hosts the quorum application must not be at Site 1 or Site 2. Use a separate isolated location.

- After you configure HA between Site 1 and Site 2, continue the HADR configuration. Create a partnership with the DR partner (Site 3). Click **Configure partnership** (see Figure 4-37).

Configure replication for Partition_16

Complete the steps to configure replication for the storage partition

Configure replication

Complete the actions below to configure high availability and disaster recovery for this partition.

Partition_16

SVT1-Site1-MAIN
Logged in here

High availability

SVT1-Site2-2ND

Partnership configured

Review linked portset

Pool links configured

IP quorum application active

Disaster recovery

DR Partner

Configure partnership

Link pools

Create partition

Cancel

Continue

Figure 4-37 DR partner configuration

- Create a partnership or select an existing partnership. Choose the connectivity type, partner system name, and the link bandwidth (see Figure 4-38).

Configure asynchronous partnership

Fill in the form fields to configure the partnership on this system.

Create new Use existing

Type

☒ Fibre Channel ☐ IP (long distances using TCP) ☐ IP (short distances using RDMA)

Partner system name

SVT1-Site3-AUX

Link specification

Enter the link bandwidth available between systems, in megabits per second (Mbps).

Link Bandwidth (Mbps)

Enter value

Cancel

Create

Figure 4-38 Configuring an asynchronous partnership

9. Verify the certificate. If it is correct, click **Create** (see Figure 4-39).

Configure asynchronous partnership
Fill in the form fields to configure the partnership on this system.

Certificate from 172.168.13.3

Subject Name	Issuer Name
Country: MX	Country: Country
Locality: GDL	Locality: Locality
Organization: IBM	Organization: International Business Machines Corporation
Organization: SVT	Organization: Spectrum Virtualize Root CA
Unit: virtualize	Unit: 78F15B9-1
Common Name: virtualize	Common Name: 78F15B9-1
Email Address:	Email Address:

[See fewer details](#)

Link specification
Enter the link bandwidth available between systems, in megabits per second (Mbps).
Link Bandwidth (Mbps): 32000

[Cancel](#) [Create](#)

Figure 4-39 Verifying the partner certificate

10. The wizard requests two partnerships. Follow the workflow to create both (see Figure 4-40).

Configure replication for Partition_29
Complete the steps to configure replication for the storage partition.

Steps: [Select replication type](#) | [Configure replication](#) | [Summary](#)

SVT1-Site1-MAIN (Logged in here)

Complete partnerships
You must configure partnerships between each system for replication. Clicking on a task below will open a new browser tab to begin creating the partnerships on that system. When the partnerships have been created, return to this screen to continue.

- [Configure 1 partnership on SVT1-Site2...](#)
- [Configure 2 partnerships on SVT1-Site3...](#)

[Close](#)

SVT1-Site3-AUX

[Cancel](#) [Continue](#)

Tasks on the right:
- Partnership configured
- Review linked portset
- Pool links configured
- IP quorum application downloaded

Buttons at the bottom right:
- Complete partnerships >
- Link pools >
- Create partition >

Figure 4-40 Creating asynchronous partnerships

Figure 4-41 shows that asynchronous partnerships were successfully created.

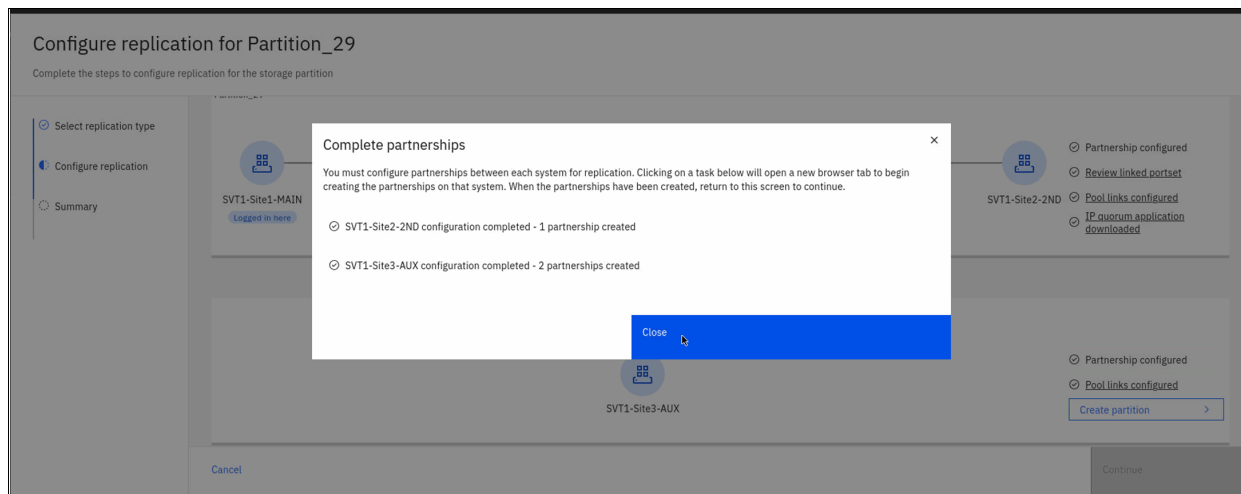


Figure 4-41 Asynchronous partnerships were successfully created

11. Click **Create partition** to create the partition at the DR site (see Figure 4-42.)

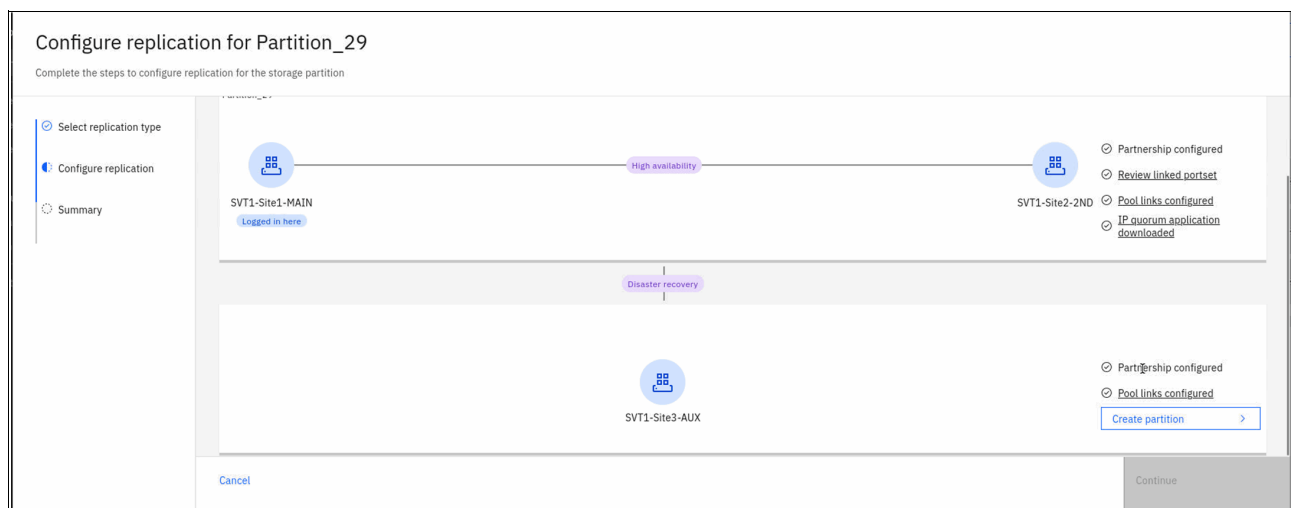


Figure 4-42 Creating the partition at the DR site

12.After you create and link the DR partition, the wizard displays a summary of the configuration (see Figure 4-43, Figure 4-44, and Figure 4-45 on page 99).

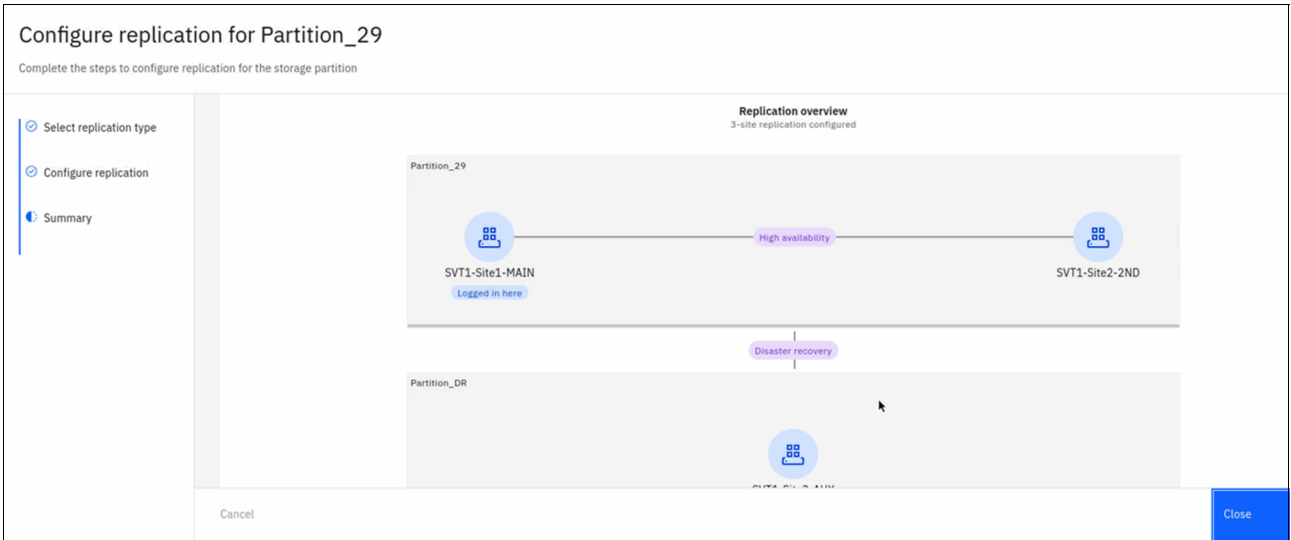


Figure 4-43 Replication overview 1

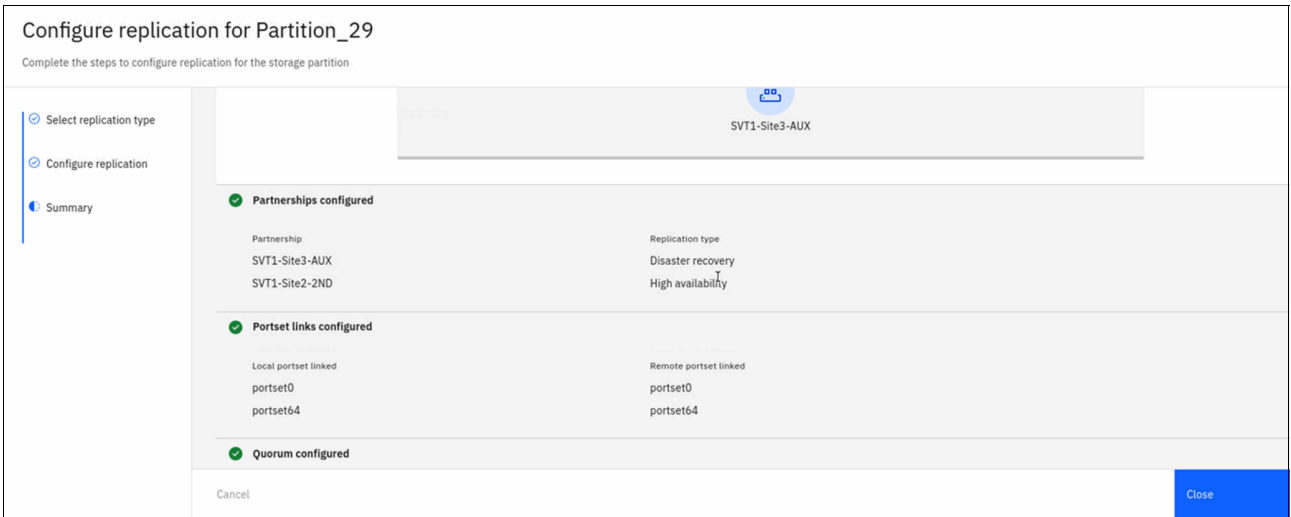


Figure 4-44 Replication overview 2

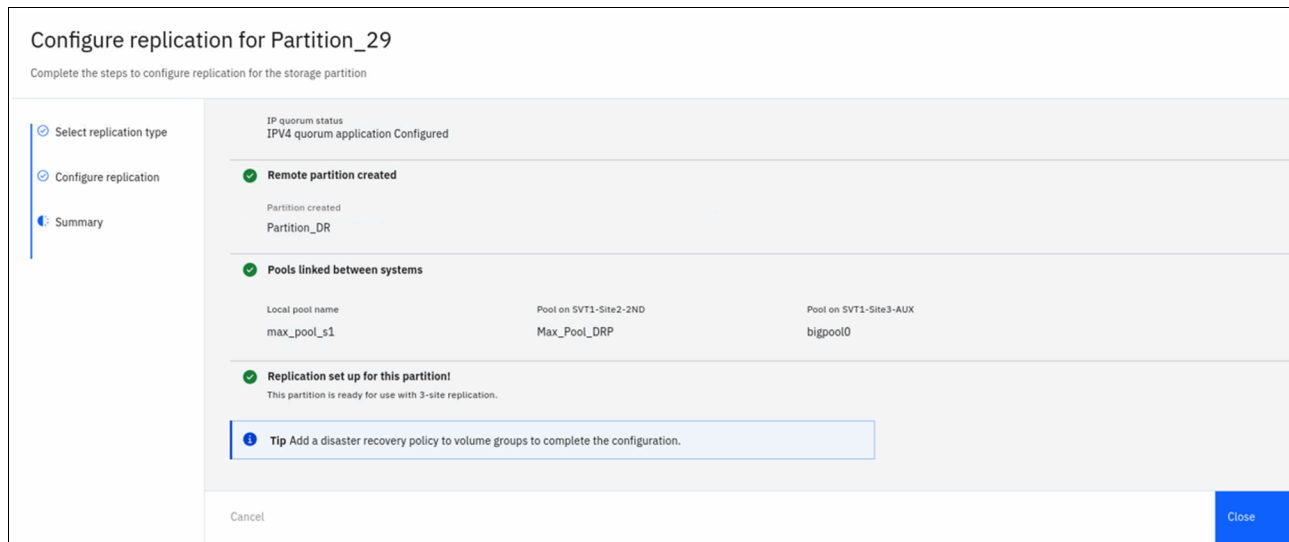


Figure 4-45 Replication overview 3

If something is missing, the replication overview shows what you need to address (see Figure 4-46).

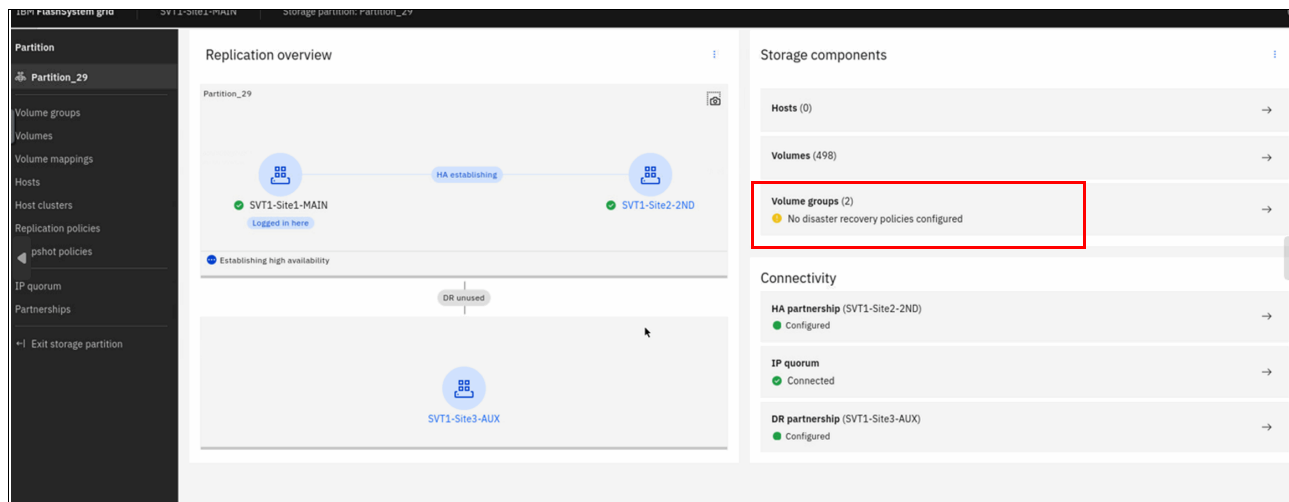


Figure 4-46 Replication overview in the partition GUI

13. Configure a DR policy by completing the following steps:

- a. Select the volume groups where you want to apply the replication policy (see Figure 4-47).

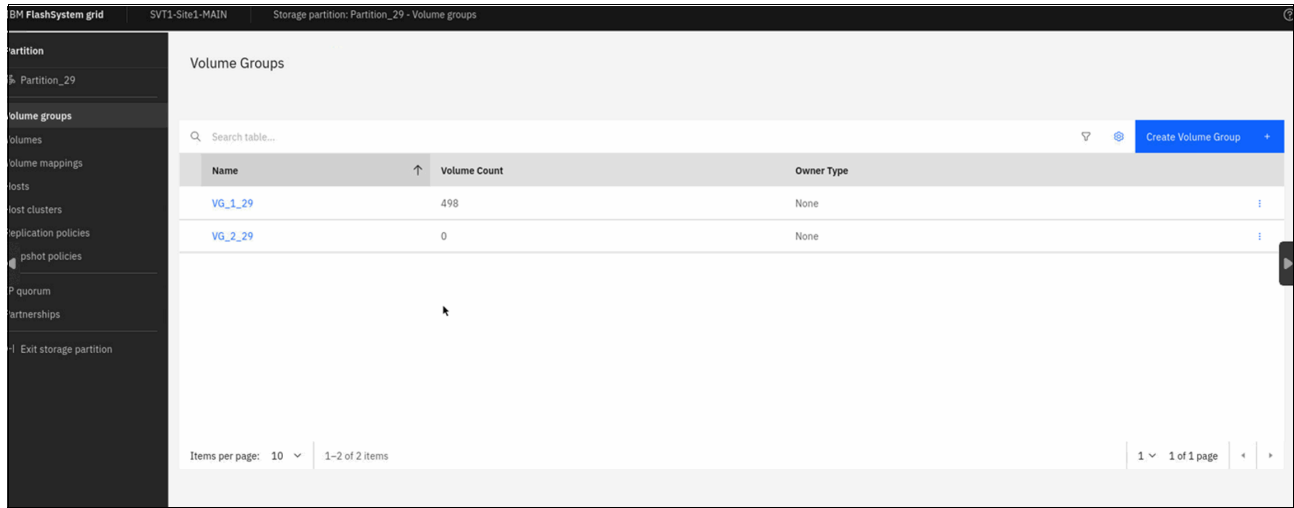


Figure 4-47 Selection of volume groups to add replication policies

- b. Add a DR policy to the first volume group (VG_1_29) and a DR policy to the second volume group (VG_2_29) (see Figure 4-48).

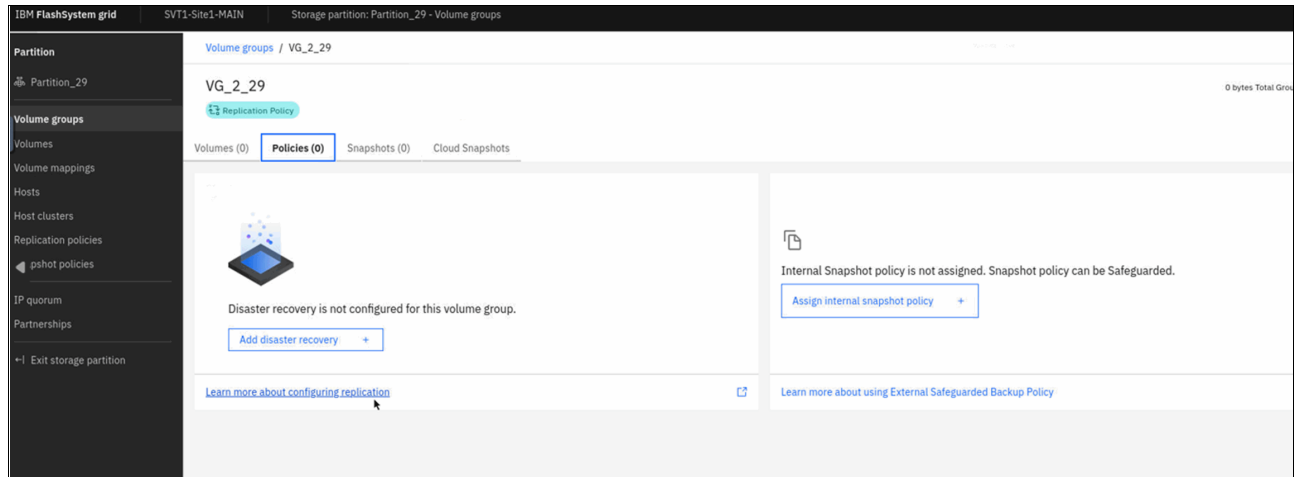


Figure 4-48 Adding a disaster recovery policy

- c. Configure the system to send an alert if the recovery copy is not updated for more than 5 minutes (see Figure 4-49 on page 101).

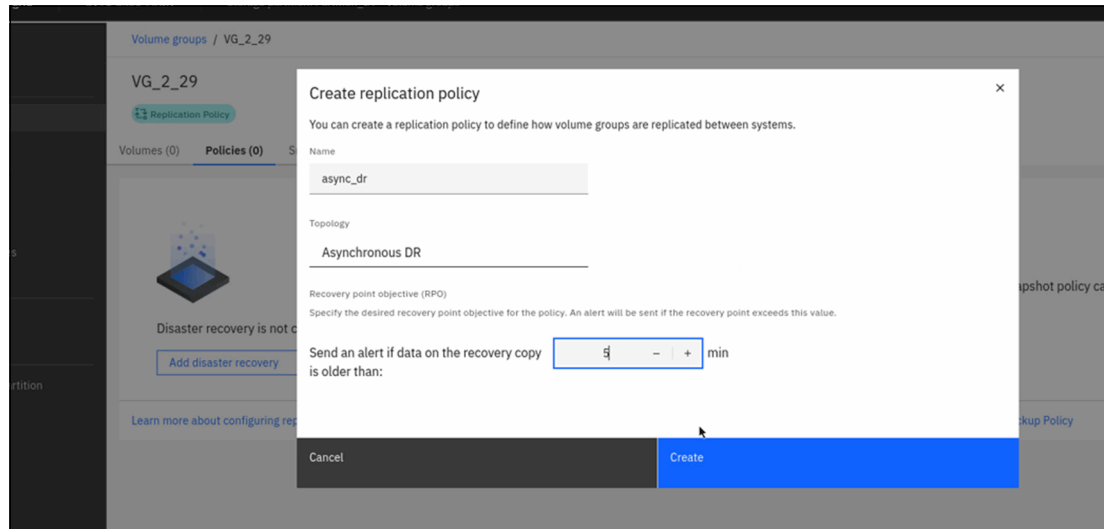


Figure 4-49 Creation of asynchronous replication policy

This setup for HADR with 3-site replication is complete.



Configuring and managing storage partition mobility

With the understanding of storage partition concepts introduced so far, this chapter describes how you migrate storage partitions across members in an IBM FlashSystem grid.

This chapter describes the following topics:

- ▶ “Understanding storage partition migration ” on page 104
- ▶ “Benefits of migrating a storage partition ” on page 105
- ▶ “Storage partition migration automation architecture” on page 105
- ▶ “Features and capabilities” on page 108
- ▶ “Planning and prerequisites” on page 108
- ▶ “Migration procedures” on page 110
- ▶ “Limitations and restrictions” on page 112

5.1 Understanding storage partition migration

Storage partitions contain the data and metadata that an application needs to run workloads. You can view a storage partition as a virtual storage controller that abstracts the storage configuration that is required to meet workloads objectives, such as:

- ▶ Volume sizes and provisioning settings
- ▶ Volume groups
- ▶ Host and host clusters configurations
- ▶ Volumes-to-host mappings
- ▶ Replication policies and settings
- ▶ Snapshots and Safeguarded Copy policies

In a typical storage environment without storage partitions, you must perform manual tasks to move storage resources, such as host objects and volume mappings between storage controllers. These tasks can include moving workloads to different controllers, configuring high availability and disaster recovery (HADR) policies, and creating snapshot policies on target systems.

In contrast, a storage partition serves as the unit of mobility, encapsulating the resources that are required to run a workload. You migrate the entire storage partition as a single entity. As Figure 5-1 illustrates, storage partition migration preserves DR relationships and snapshot policies automatically. It also maintains host access continuity, which enables non-disruptive migration of storage partitions from one grid member to another one.

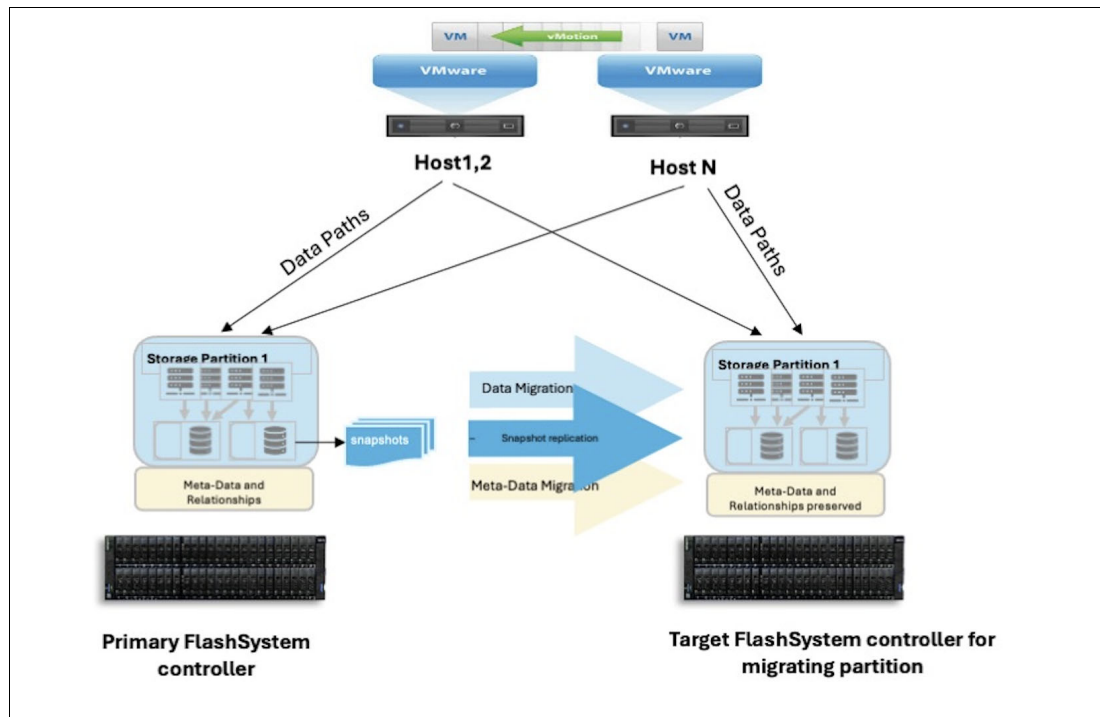


Figure 5-1 High-level view of a storage partition migration

5.2 Benefits of migrating a storage partition

Storage partition addresses the following business needs in a modern data center:

- ▶ Load-balancing among grid members to help maximize storage performance across the grid
- ▶ Movement of storage workloads during hardware repairs and upgrades
- ▶ Moving workloads to the most appropriate hardware based on business needs
- ▶ Data mobility that maintains application and workload continuity with uninterrupted operation

The next section describes the storage partition migration automation architecture. It focuses on simplicity and ease of use with minimum administrator interventions during migration.

5.3 Storage partition migration automation architecture

The storage partition migration automation aligns with the following design goals for modern data center administration and configuration:

- ▶ Minimal (near-zero) user interventions during the process
- ▶ An intuitive user interface (UI) to choose the target controller, with artificial intelligence (AI)-assisted advisories from IBM Storage Insights
- ▶ An IBM FlashSystem view for host servers that provide insights for storage discovery and I/O patterns during migration
- ▶ The ability to revert a migration and maintain full data availability
- ▶ Integration of the migration process into the IBM FlashSystem grid architecture, using observability and configuration controls
- ▶ Explicit confirmation for critical decisions, such as deleting the source copy of the data
- ▶ Uninterrupted asynchronous data replication, including 3-site data resiliency configurations

5.3.1 Behind the scenes for storage partition migrations

Figure 5-2 provides a detailed view of the mechanisms that are behind storage partition migrations.

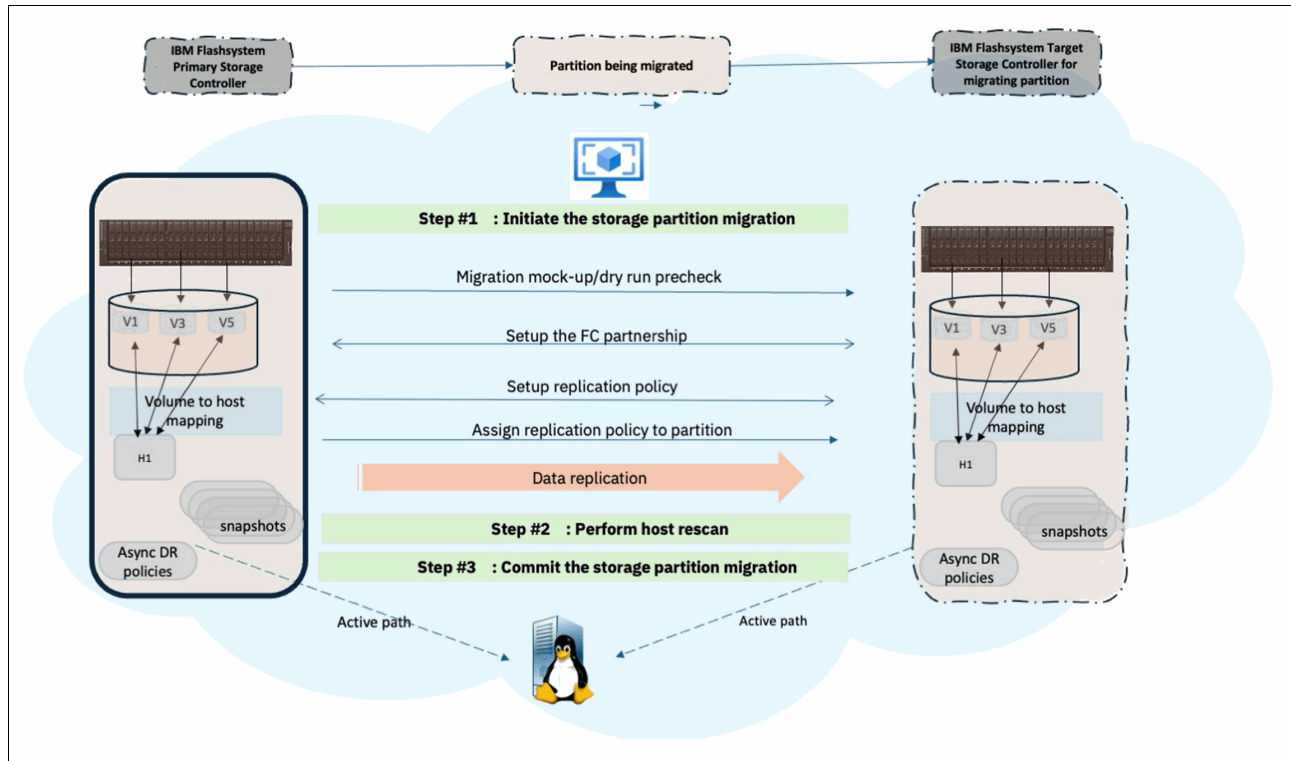


Figure 5-2 Detailed view of the mechanisms behind storage partition migrations

At a broad level, storage partition migration in an IBM FlashSystem grid occurs in three stages:

1. Target selection and migration initiation
2. Storage volume rescans at hosts (not required for **autostoragediscovery** capable hosts)
3. Commit to remove the source copy of the storage partition

Step 1: Target selection and migration initiation

Use the GUI guidance and IBM Storage Insights to identify an optimal target grid member for the storage partition.

After you select a suitable target system, the following storage operations are automated to deliver a simple migration experience:

1. Migration dry run. Performs version and capability checks to validate compatibility between the source and target systems.
2. Replica setup at target location
 - a. Linking default storage objects such as pools.
 - b. Creating replicas of essential storage objects, including volumes and host definitions.
 - c. Preserving volume-specific identifiers to help ensure consistency and avoid conflicts.
 - d. Replicating volume mappings to the appropriate hosts or host clusters.

3. Establishing inter-system connectivity
 - a. Configure Fibre Channel (FC) interconnects to support data replication.
 - b. Set up data resiliency policies. The policies include these items:
 - Data replication policies.
 - Snapshot policies.
4. Data synchronization. Initiates and monitors synchronization from source to target volumes to help ensure consistency and readiness for cutover

Step 2: Host rescan operations for storage detection

During data synchronization between source and target volumes, the target grid member initiates specific SCSI protocol sequences to signal readiness for active-active configurations. Because the replicated volumes retain the same unique identifiers as the source volumes, the host multipath software recognizes them as a single logical entity. The host then uses both the original and new paths for I/O operations.

To detect these new paths from the target storage partition, the host operating system must perform a storage rescan, either manually or automatically. ESX and IBM i typically perform automatic rescans at regular intervals by default. For other platforms, such as Red Hat Enterprise Linux (RHEL), enable automatic rescans by using an open-source utility or automation tool, such as Ansible. For more information, see this [GitHub page](#). IBM FlashSystem software automatically detects ESX operating systems and prepares them for storage rescan without extra configuration.

For environments where automatic rescans are enabled with third-party tools or open-source utilities, use the command-line interface (CLI) to update the **autostoragediscovery** attribute on the relevant host objects so the host detects new storage paths that are introduced during partition migration. For details, see the [IBM Documentation](#).

Step 3: Committing the migration operation

This step finalizes the storage partition migration. You provide explicit consent to remove the original copy of the storage partition from the source grid member. The target grid member becomes the primary storage system for the storage partition.

Before committing the removal, verify the following conditions:

- ▶ The host multipath software recognizes the target IBM FlashSystem paths as preferred and actively uses them for I/O.
- ▶ Application performance is stable, and I/O throughput and bandwidth from the target system meet expectations.
- ▶ Asynchronous DR operations continue from the target system, including enforcement of snapshot policies.
- ▶ No I/O traffic is routed to the source IBM FlashSystem. IBM FlashSystem software displays real-time insights during the commit event.
- ▶ The CLI (for example, **lshost <host_id>**) shows that host objects at the source IBM FlashSystem are inactive. For more information about the **lshost** command, see [IBM Documentation](#).

Most prerequisites are met through successful storage rescan operations and properly configured multipath policies on hosts that are mapped to storage partitions. These steps help host systems recognize and prioritize new paths from the target IBM FlashSystem, enabling a smooth and reliable partition migration.

5.4 Features and capabilities

The following features and capabilities provide an effective way to mobilize data within a modern data center:

- ▶ The solution provides non-disruptive data synchronization and storage partition migrations with an effective RTO of zero (meaning no downtime during the process).
- ▶ Non-disruptive asynchronous DR configurations become effective on the target storage system after storage partition migrations.
- ▶ Snapshot policies and safeguarded copy policies at the source storage systems are retained and effective post storage partition migrations.
- ▶ Use the GUI to choose the wanted grid members and committing the migration.
- ▶ Host I/O patterns information is available for making informed decisions regarding source copy deletions.
- ▶ Awareness of host storage rescan capabilities helps minimize user interventions during the migration process.
- ▶ With integration with IBM Storage Insights software, you can identify the most appropriate target storage systems based on AI analytics.

5.5 Planning and prerequisites

This section describes the planning and prerequisites for storage partition migration.

5.5.1 Connectivity requirements (FC, IP addresses, certificates, and NTP)

Table 5-1 shows the connectivity requirements for FC, IP addresses, certificates, and Network Time Protocol (NTP).

Table 5-1 Connectivity requirements (FC, IP, Certificates, and NTP)

Networking and communication technology	Requirements
FC connectivity	▶ Ensure that both source and target IBM Flash systems are connected through FC. ▶ Proper zoning must be configured to allow communication between systems and hosts.
IP address connectivity	Establish IP address connections between systems for management tasks, such as certificate exchange and system coordination.
Certificate	▶ Secure communication between systems requires certificate exchange. ▶ If the source and target systems are not part of the same IBM FlashSystem grid, a certificate exchange is mandatory to establish secure communication for storage partition migration.
NTP	Synchronize system clocks by using NTP to prevent time discrepancies that can affect migration operations.

5.5.2 IBM FlashSystem grid considerations

IBM FlashSystem grid provides a unified management and data migration framework, streamlining storage partition migrations between member systems with minimal configuration:

- ▶ Within the IBM FlashSystem grid
 - No certificate exchange required. Trust is implicitly established through grid membership.
 - Automatic system discovery. Member systems detect each other automatically.
 - Internally managed migration paths. All necessary connectivity and trust relationships are handled by the grid infrastructure.
- ▶ Outside the IBM FlashSystem grid
 - Manual certificate exchange is necessary to establish trust. For more information, see 5.5.1, “Connectivity requirements (FC, IP addresses, certificates, and NTP)” on page 108.
 - Explicit configuration of system trust and storage pool linking is required.
 - Host visibility. The target system must have visibility to the hosts originally served by the source system to help ensure seamless access post-migration.

5.5.3 System partnership requirements

To enable successful storage partition migration between IBM FlashSystem systems, the following partnership requirements must be met:

- ▶ Partnership establishment
 - When a migration is initiated, the source and target systems automatically attempt to establish an FC partnership.
 - This partnership is essential for enabling data replication and migration workflows.
 - Ensure that both systems have sufficient available partnership slots to support the new connection.
- ▶ Replication considerations

If the storage partition is configured for DR, verify either of the following options:

 - The target system and the DR-linked system already have an established partnership.
 - They are zoned so that the migration process can create one automatically.

5.5.4 Pool linking and replication dependencies

Here are the pool linking and replication dependencies:

- ▶ Pool linking
 - During migration, storage pool links define which pool on the target system will store the replicated volumes.
 - Establish appropriate pool links between the source and target systems before initiating the migration.

- ▶ Replication policies
 - If the target system is already configured for asynchronous replication with the source system, it cannot be used as a migration destination.
 - In such cases, administrators must use either of the following options:
 - Remove the existing replication configuration.
 - Select an alternative target system that is not involved in an active replication relationship with the source.

5.5.5 Host path validation and multipathing

To help ensure high availability (HA) and seamless storage access during partition migration, the following host-side configurations and validations are essential:

- ▶ Multipathing configuration
 - Enable multipathing on all hosts to provide redundancy and failover capability.
 - For Linux hosts, configure the `/etc/multipath.conf` file and manage the multipath daemon to apply and maintain changes.
- ▶ Path verification

Confirm that each host has multiple active paths to the storage systems. This configuration prevents single points of failure and helps ensure uninterrupted I/O during migration.
- ▶ Host automation
 - Automation tools that are based on RPM packages and Ansible playbooks are available to streamline storage path discovery and validation during migration.
 - These tools help reduce manual effort and ensure consistent configuration across large environments.

5.6 Migration procedures

Partition migration can be done by using the GUI or the CLI.

5.6.1 Using the GUI

You can use the management GUI to perform a storage partition migration. Figure 5-3 on page 111 shows the Storage partitions window of the IBM FlashSystem grid.

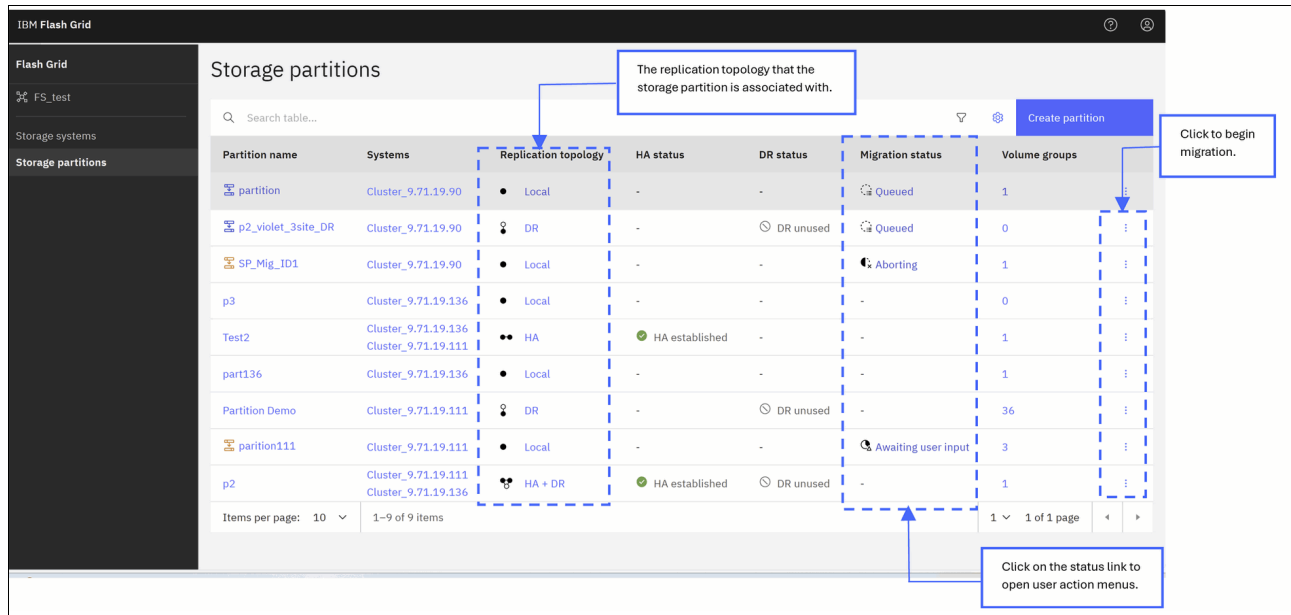


Figure 5-3 Storage partitions window of the IBM FlashSystem grid

To migrate a storage partition by using the management GUI, complete the following steps:

1. Click **Storage partitions** in the management GUI.
2. Select the storage partition to migrate.
3. Click **Migrate partition**.
4. Follow the prompts to select the target system and initiate the migration.
5. Monitor the migration status through the Migration status link, which provides guidance if user intervention is required.

Note: The Storage partitions window provides a comprehensive view of all partitions within the IBM FlashSystem grid. Key elements include a list of storage partitions with their status, and options to initiate migration, view details, and monitor statuses.

5.6.2 Using the CLI

To migrate a storage partition by using the CLI, use the following commands:

- ▶ The **chpartition** command modifies a storage partition, including initiating migration to a different system.
- ▶ The **lspartition** command lists storage partitions and provides detailed information about each partition.

Sample command usage

- ▶ To migrate a storage partition to a different system, run the following command:
`svctask chpartition -location <desired_system_name> <partition_name/id>`
- ▶ To list all storage partitions and the migration status, run the following command:
`svcinfolspartition`
- ▶ To monitor the migration status, use the **svcinfolspartition** command. The output indicates the current state of each partition, which helps administrators track progress and identify issues.

5.6.3 Volume discovery and path confirmation

After data migration (both source and target systems are in sync), ensure that hosts recognize the new paths to the migrated volumes. This process can involve rescanning the host's storage devices and verifying multipath configurations to confirm that all paths are active and correctly mapped.

5.7 Limitations and restrictions

There are some limitations and restrictions for partition migration:

- ▶ Storage partition migration is not supported for draft storage partitions or storage partitions that are configured for Policy-based High Availability (PBHA).
- ▶ Here are some protocol and topology constraints:
 - Migration is supported only with FC hosts and partnerships.
 - Ensure that the systems that are involved in the migration are within the same IBM FlashSystem grid or that the appropriate trust exchange and pool linking are done, as described in 5.5, “Planning and prerequisites” on page 108.
- ▶ Here are considerations for migration sequencing and queuing:
 - Only one storage partition can be migrated at a time from an IBM FlashSystem.
 - Subsequent migration requests are queued and automatically scheduled in the order that they were initiated after the previous migration completes.
- ▶ Here are some HA configurations and constraints:
 - For storage partitions that are configured with HA, migration can be performed only with the HA partner system.
 - Configure both the active and preferred management systems to support the migration process.



IBM FlashSystem grid monitoring with IBM Storage Insights

This chapter describes how to monitor IBM FlashSystem grid by using IBM Storage Insights.

IBM Storage Insights is a powerful cloud-based tool that effectively monitors the storage infrastructure. IBM Storage Insights provides an analytics layer for the IBM FlashSystem grid and offers valuable insights and recommendations to optimize storage operations by using artificial intelligence (AI).

When IBM FlashSystem grid was first released in release 8.7.0, IBM Storage Insights was the only user interface (UI)-based option for grid management and partition migration. Recent releases enable management and migration by using the command-line interface (CLI), GUI, and IBM Storage Insights to manage IBM FlashSystem grid and to monitor and migrate storage partitions.

This chapter explains how to properly configure IBM Storage Insights for IBM FlashSystem grid management. It provides step-by-step instructions for non-disruptive migration of storage partitions between systems by using IBM Storage Insights and introduces AI-driven advisories that IBM Storage Insights generates through continuous monitoring.

This chapter describes the following topics:

- ▶ “IBM FlashSystem grid observability” on page 114
- ▶ “Partition migration with IBM Storage Insights” on page 144

6.1 IBM FlashSystem grid observability

This section explains how to configure IBM Storage Insights to monitor an IBM FlashSystem grid effectively and displays the IBM FlashSystem grid views within IBM Storage Insights.

6.1.1 Introducing IBM Storage Insights

IBM Storage Insights is a key component of IBM storage management solutions. It enhances visibility across storage environments, reduces maintenance time, and optimizes performance through AI-driven recommendations.

IBM Storage Insights enables these capabilities:

- ▶ Monitor health, capacity, and performance events for all IBM block storage devices from a unified dashboard.
- ▶ Analyze historical data to identify trends in performance, capacity, and system health.
- ▶ Upload diagnostic logs to IBM Support for faster issue resolution.

Overall, IBM Storage Insights simplifies and streamlines storage management, enabling informed decisions and maintaining optimal system performance.

IBM Storage Insights is available in the following versions, each offering distinct capabilities:

- ▶ IBM Storage Insights (Trial Version)

This version provides essential monitoring and support integration capabilities. It is available at no charge for IBM block storage systems. Key features include basic health and performance monitoring, support log uploads, and a unified dashboard view.

- ▶ IBM Storage Insights Pro

The Pro Version adds proactive alerting, custom reporting, enhanced monitoring, predictive analytics, and long-term trend analysis. It also includes advanced performance metrics and extended data retention. The version that is available at no charge displays only the last 24 hours of performance data. The Pro Version retains data for up to one year and supports data export.

IBM Storage Insights Pro is a prerequisite for certain advanced features, including ransomware threat detection available with new FlashCore Module (FCM) drives.

Both versions support IBM FlashSystem grid and storage partition migration.

For a feature comparison, see [IBM Storage Insights vs IBM Storage Insights Pro](#).

Tip: For best results, use Chrome, Edge, or Safari to interact with IBM Storage Insights. Other browsers, such as Firefox, can affect the performance of the IBM Storage Insights UI.

Access requirements

Access to IBM Storage Insights requires an IBMid. If an IBMid is not available, create one by completing the [registration form](#).

The email address that is associated with the IBMid is used for support communication, including updates and announcements. Ensure that the email address is valid.

Getting started with IBM Storage Insights

To begin using IBM Storage Insights, complete the following steps:

1. Sign up for IBM Storage Insights by using the registration [link](#).
A welcome email is sent with the unique Uniform Resource Locator (URL) for the IBM Storage Insights instance.
2. Enable Call Home with cloud services.
3. Add the storage system. Connect IBM Storage Insights to the storage that requires monitoring. The dashboard for monitoring and managing the storage environment becomes available.

These steps are described in more detail in the following sections.

How to enable the 60-day Trial Version for IBM Storage Insights Pro

To activate the Trial Version of IBM Storage Insights Pro, complete the following steps:

1. Open the IBM Storage Insights dashboard.
2. In the top navigation bar, click **Try Pro for free**.
3. Review the information about the IBM Storage Insights Pro features.
4. Click **Start Trial** (see Figure 6-1).
5. Complete the activation process by following the on-screen instructions.

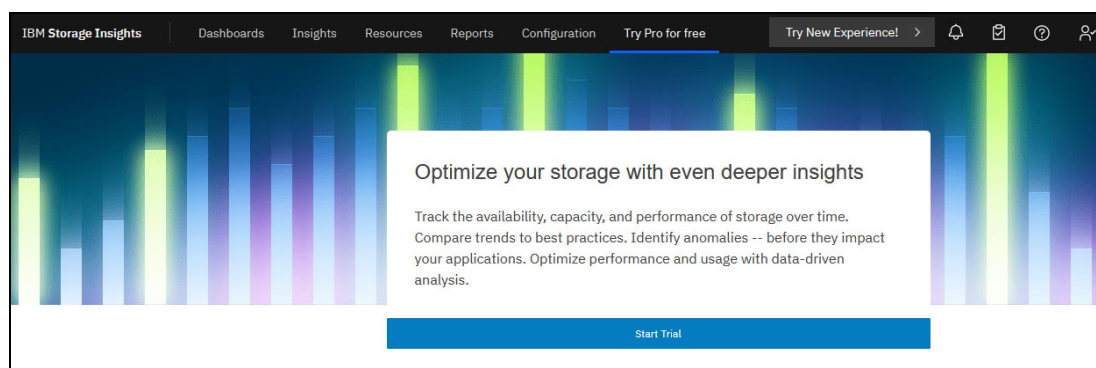


Figure 6-1 Starting the Trial Version of IBM Storage Insights Pro

Note: IBM also offers a 60-day trial of IBM Storage Insights Pro at no charge and can be started from the IBM Storage Insights dashboard. After the trial period ends, IBM Storage Insights automatically reverts to the Trial Version. The trial can be activated only once.

After the steps finish, the top navigation bar in the IBM Storage Insights instance displays IBM Storage Insights Pro instead of IBM Storage Insights. Figure 6-2 shows the updated navigation bar. Advanced functions become available.

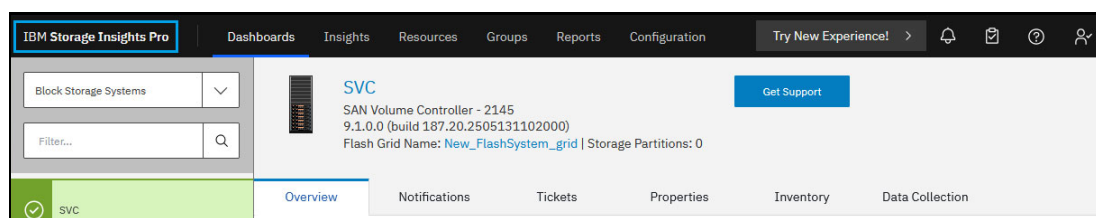


Figure 6-2 Top navigation bar displays IBM Storage Insights Pro

Useful reference resources

IBM Storage Insights is a cloud-based service. Communication is initiated one-way, encrypted, and compressed, and uses security-hardened endpoints.

The following resources provide additional information about IBM Storage Insights:

- ▶ [IBM Documentation for IBM Storage Insights](#)
- ▶ [IBM Storage Insights Fact Sheet](#)
- ▶ [IBM Storage Insights Getting Started Guide](#)
- ▶ [IBM Storage Insights Poster](#)
- ▶ [Frequently Asked Questions](#)
- ▶ [IBM Storage Insights Security Guide](#) provides a detailed description of the security aspects and data collection.
- ▶ *Maximize Your Storage Investment: Use IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, SG24-8584

6.1.2 How to set up a new IBM Storage Insights instance

Use the following steps to sign up for an IBM Storage Insights instance:

1. Go to the [registration page](#).
2. Click **IBM Storage Insights** (see Figure 6-3).

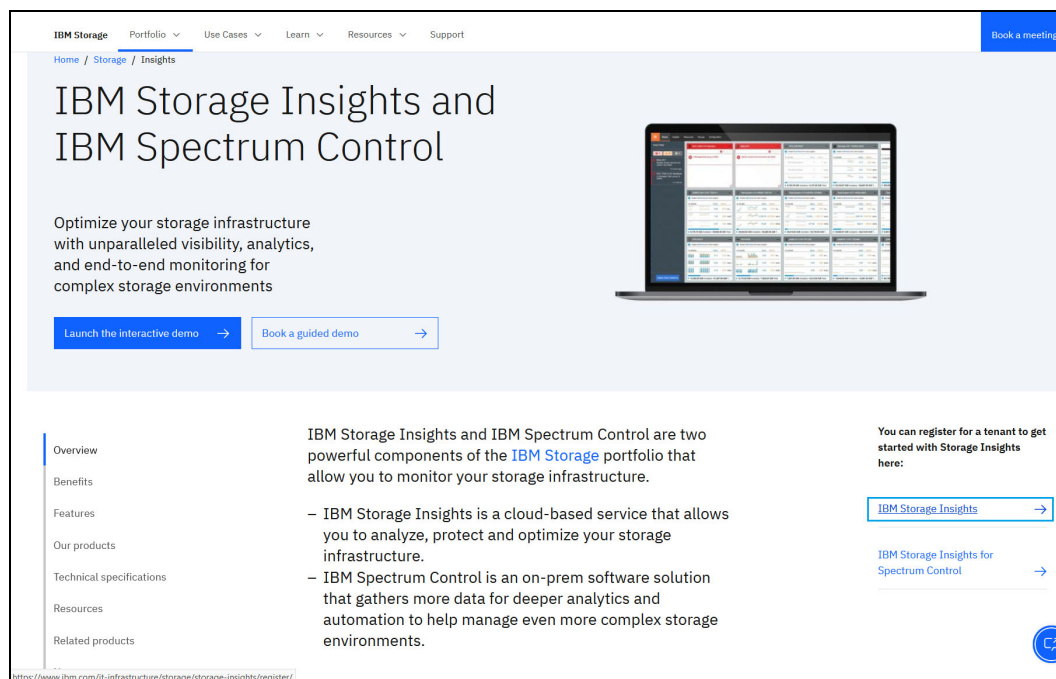


Figure 6-3 Registering for IBM Storage Insights

3. Sign in with your IBMId.
 - a. If you do not already have an account, click **Create an IBMId**.
 - b. The IBMId used for registration becomes the instance administrator and can invite more users and grant access.
 - c. Enter your IBMId and click **Continue** (see Figure 6-4).

IBM

Log in to IBM

IBMId

Continue →

☐ Remember me ⓘ

Don't have an account?

Create an IBMId →

Forgot IBMId? [Contact the IBMId help desk](#)

Contact Privacy Terms of use Accessibility Cookie preferences

Powered by IBM Security Veri

Figure 6-4 Log in with your IBMId or create a new ID

4. Assign a unique name for your IBM Storage Insights service.
 - a. Include your company name and other details such as geographic location or department.
 - b. Click **Submit**.
 - c. After successful registration, a confirmation appears.

Thanks for signing up for IBM Storage Insights. You'll receive an email when your account has been validated.
 - d. Check your inbox for a welcome email that contains the instance link.

- e. Click **Get Started!** to access your new instance.
- f. Bookmark the link for convenient access (see Figure 6-5).

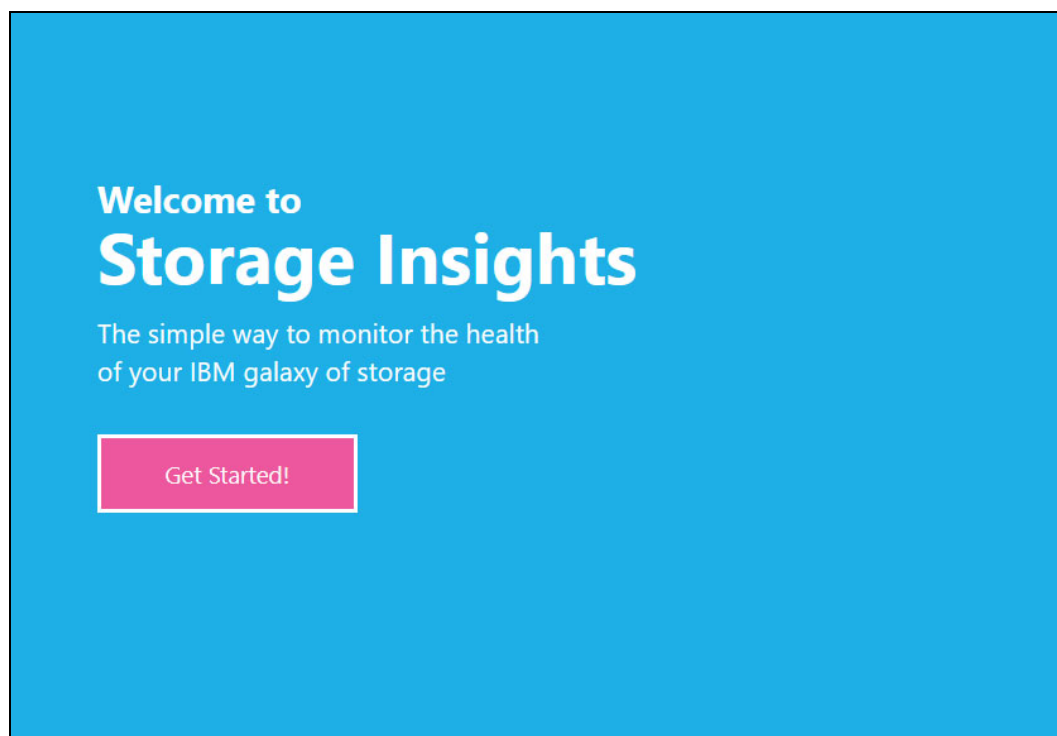


Figure 6-5 Welcome to IBM Storage Insights

Instance URL format

The URL for your instance follows a specific naming convention:

`https://insights.ibm.com/gui/xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx`

The final segment, `xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx`, is the IBM Storage Insights tenant ID. The IBM Storage Insights tenant ID is required to set up Call Home with cloud services, which is described in the next section.

6.1.3 Call Home with cloud services

IBM Storage Insights relies on Call Home with cloud services to communicate with the IBM FlashSystem or IBM SAN Volume Controller (SVC). For example, Call Home with cloud services can initiate a partition migration. To use IBM FlashSystem grid features within IBM Storage Insights, configure Call Home with cloud services as the data collection method.

Metadata that is collected through Call Home with cloud services includes the following items:

- ▶ Partition migration
- ▶ IBM FlashSystem grid
- ▶ Ransomware thread detection
- ▶ Workload anomaly detection
- ▶ Volume group metadata

To monitor the status of the storage devices effectively, it is essential to collect metadata. Starting with firmware Version 8.5.3, IBM Storage Virtualize supports CCH as a data collection method. With this approach, an external data collector is not required because the embedded collector in the IBM Storage Virtualize codebase handles the collection.

Note: A device can use either Call Home with cloud services or an external data collector, but not both simultaneously.

Requirements for using Call Home with cloud services

Call Home with cloud services requires at least 128 GB of memory for each node. The storage system must run IBM Storage Virtualize 8.5.3 or later.

Call Home with cloud services uses a REST API over an outbound secure HTTP connection (port 443) from IBM FlashSystem to IBM front-end servers.

The connection is one-way and inbound connections are not allowed. Transmitted data is encrypted and compressed. The front-end servers that are available at [IBM Support](#) are security-hardened systems. No personal, sensitive, or application data is collected.

Figure 6-6 shows the Call Home with cloud services connection without using a customer web proxy.

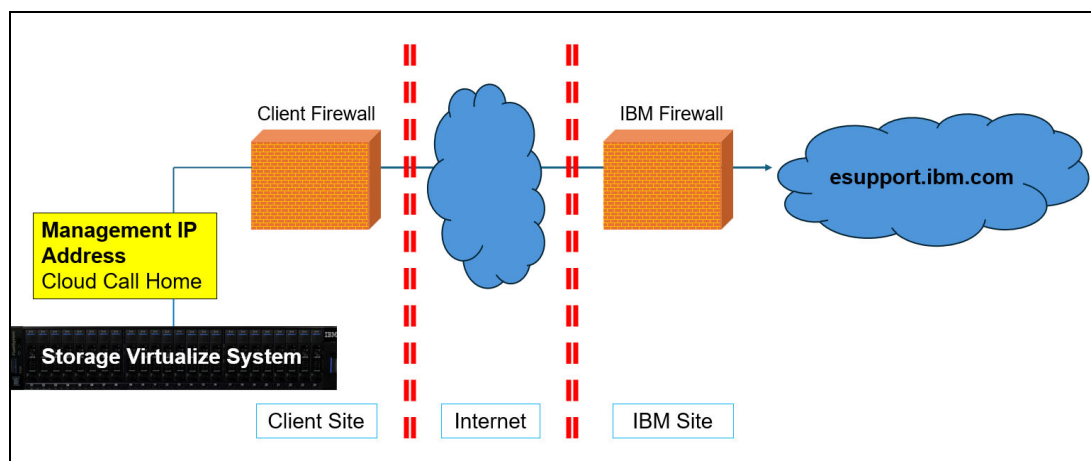


Figure 6-6 Call Home with cloud services connection overview

Table 6-1 lists required ports and connections for using Call Home with cloud services.

Table 6-1 TCP/IP requirements

Source	Protocol	Port	Target	Direction
Cluster IP address	HTTPS or TLS	443 TCP	esupport.ibm.com (DNS)	Outbound

In addition to the listed connections, the IBM Storage Virtualize system requires a Domain Name System (DNS) server to be configured.

Enabling Call Home with cloud services on the IBM FlashSystem

IBM Storage Virtualize systems can be integrated with IBM Storage Insights during initial system setup or later.

To enable Call Home with cloud services, use the IBM Storage Virtualize system management GUI to run the following steps:

1. Select **Settings** → **Support** → **Call Home**.
2. Click **Enable Call Home** to start the wizard (see Figure 6-7).

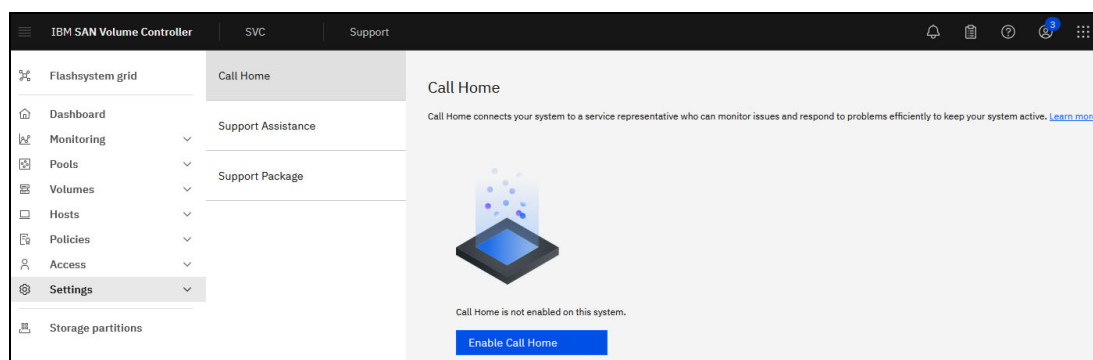


Figure 6-7 Enable Call Home with cloud services by using the management GUI

3. The wizard opens with a welcome message (see Figure 6-8). Click **Next**.

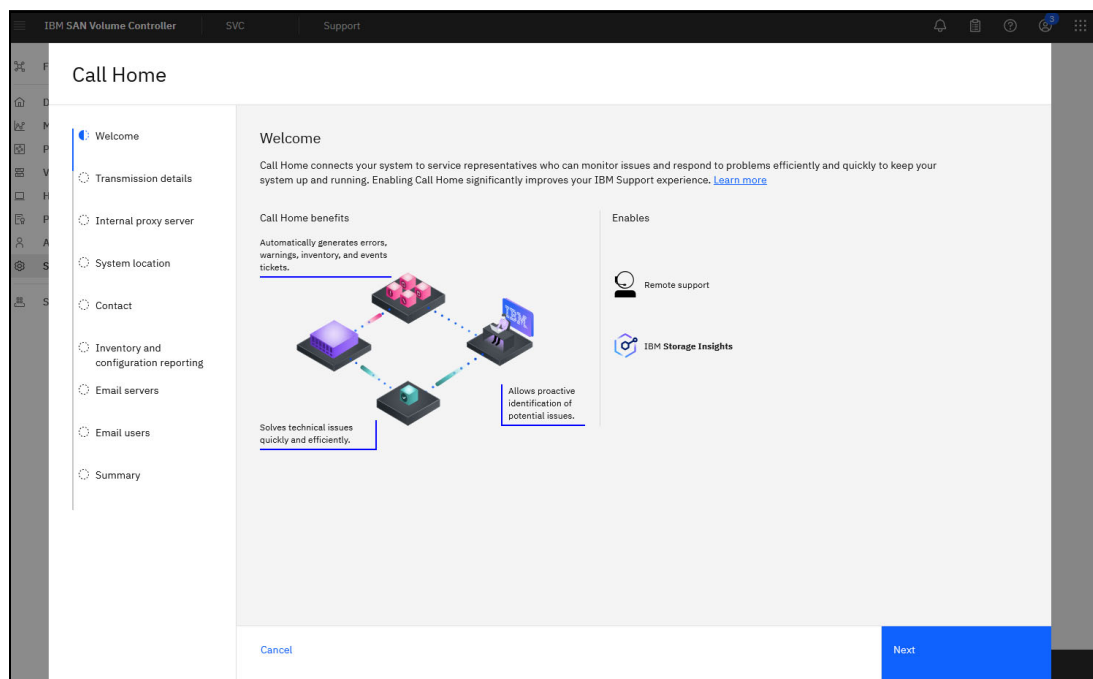


Figure 6-8 Call Home with cloud services wizard opens

4. Select the **Send using cloud services (recommended)** checkbox to choose Call Home with cloud services as the transmission type. Then click **Apply and next** (see Figure 6-9).

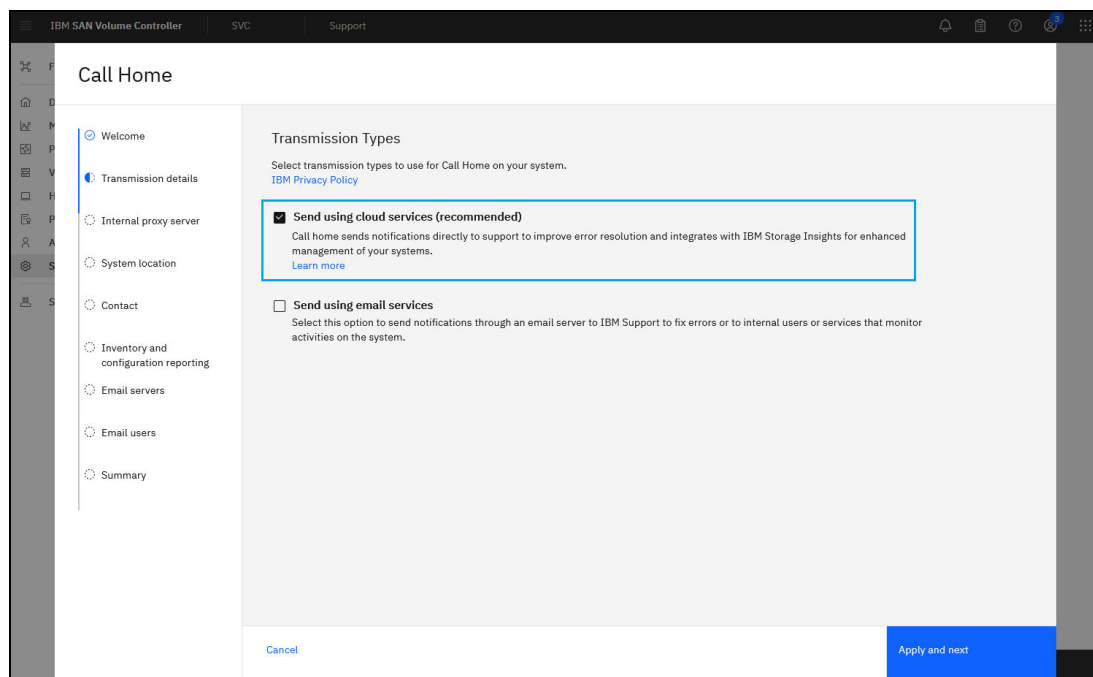


Figure 6-9 Selecting Call Home with cloud services as the transmission type

5. Review the “Call Home” Terms (see Figure 6-10). Read the terms, and accept by clicking **Accept and turn on Call Home**.

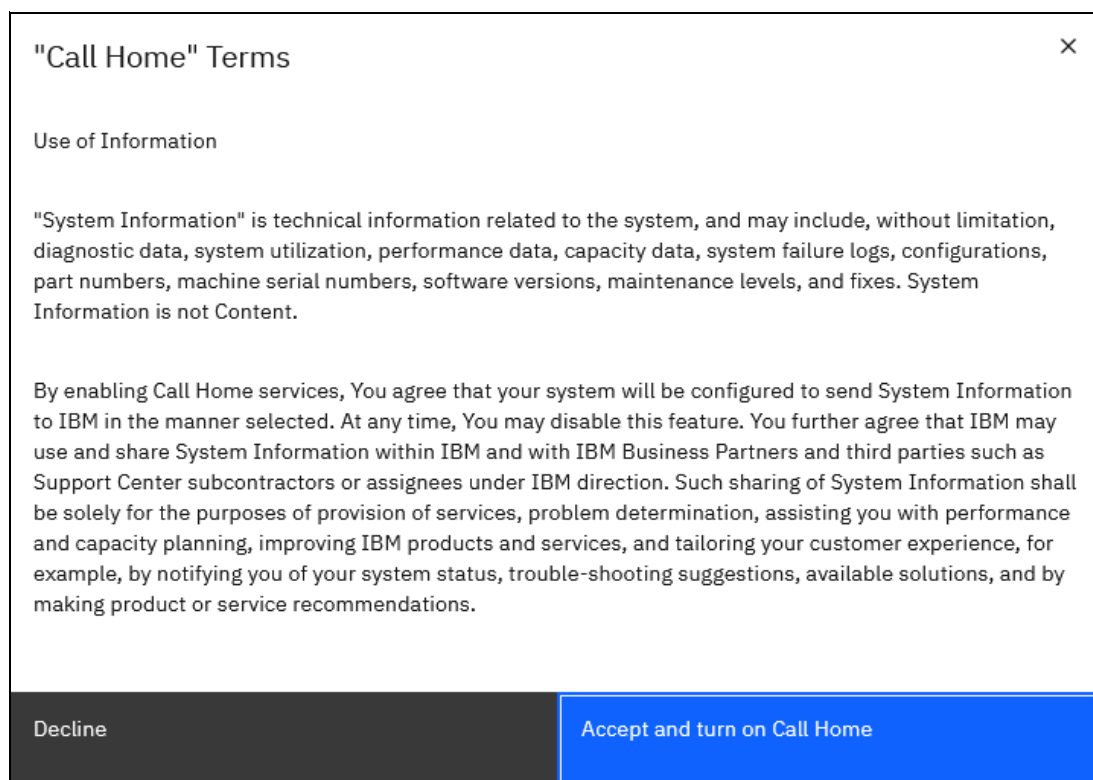


Figure 6-10 Call Home Terms

The terms explain which information the system periodically sends to IBM and how that information is used.

6. Configure a customer web proxy if your system does not have a direct connection to the internet.

If the system requires an internal web proxy, click the **Use proxy server to send data** and configure Open Proxy, Certificate, and Basic Authentication.

For each option, specify the proxy server URL and the related port. Select the option only when a proxy is required (Figure 6-11).

In this example, the option to use a proxy server is not selected.

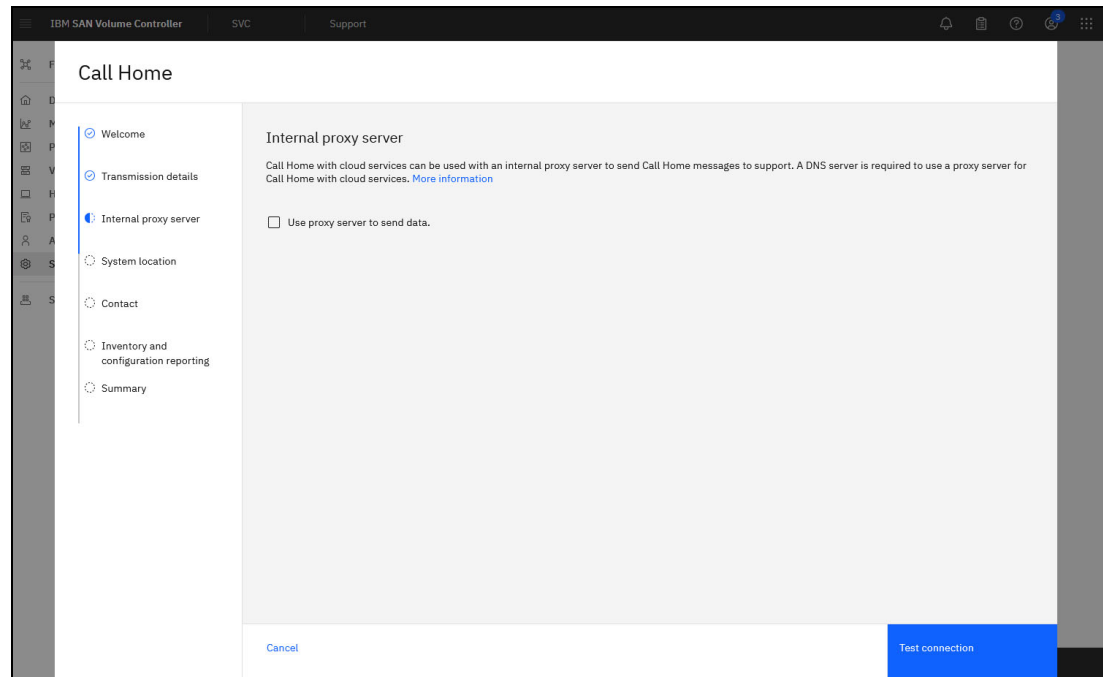


Figure 6-11 Internal proxy server configuration option

7. Click **Test connection**.

A connection test starts in a new window (see Figure 6-12).

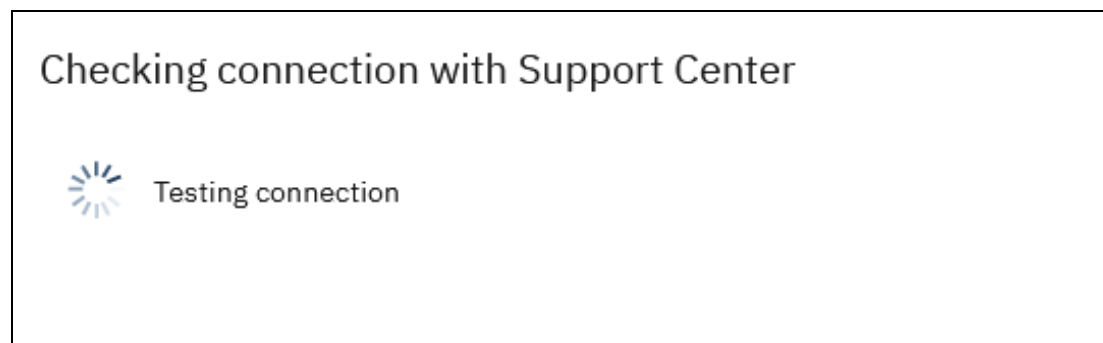


Figure 6-12 Connection test

8. If the connection is successful, click **Apply and next**.

9. Enter the system location details and click **Next** (Figure 6-13).

IBM SAN Volume Controller SVC Support

Call Home

- Welcome
- Transmission details
- Internal proxy server
- System location**
- Contact
- Inventory and configuration reporting
- Summary

System location

Enter the system location details where service parts need to be shipped.

Company name

System address

Organization that owns this system

Street address where this system is physically located

City

State or province

City where this system is physically located

2-3 uppercase characters representing US or Canadian provinces

Postal code

Country or region

Postal code where this system is physically located

Machine location

Additional details: eg floor, lab, etc

Cancel Next

Figure 6-13 System location configuration

10. Enter a contact name, email address, and phone number of the contact person in your organization. This information is required for support to make contact if any issues occur with the system.

- Use a valid email address. not a system address.
- Click **Apply and next** (Figure 6-14).

IBM SAN Volume Controller SVC Support

Call Home

- Welcome
- Transmission details
- Internal proxy server
- System location
- Contact**
- Inventory and configuration reporting
- Summary

Contact

Enter the contact information so that the support center can reach out to resolve system issues.

Enter business-to-business contact information
To comply with privacy regulations, personal contact information for individuals with your organization is not recommended.

Name

Email

Phone (Primary)

Phone (Secondary)

IBM may use my contact data to keep me informed of Storage related products, services, and offerings.
☐ Off

Cancel Apply and next

Figure 6-14 Contact information configuration

11. Configure more settings (see Figure 6-15):

- Inventory reporting (enabled by default). Sends a daily email containing hardware and configuration details.
- Configuration reporting (enabled by default). Recommended for support purposes, such as monitoring long-term corrected memory error statistics for DIMMs.
- Remove content that identifies system objects. Masks object names and IP addresses in inventory emails to protect privacy.

Click **Apply and next**.

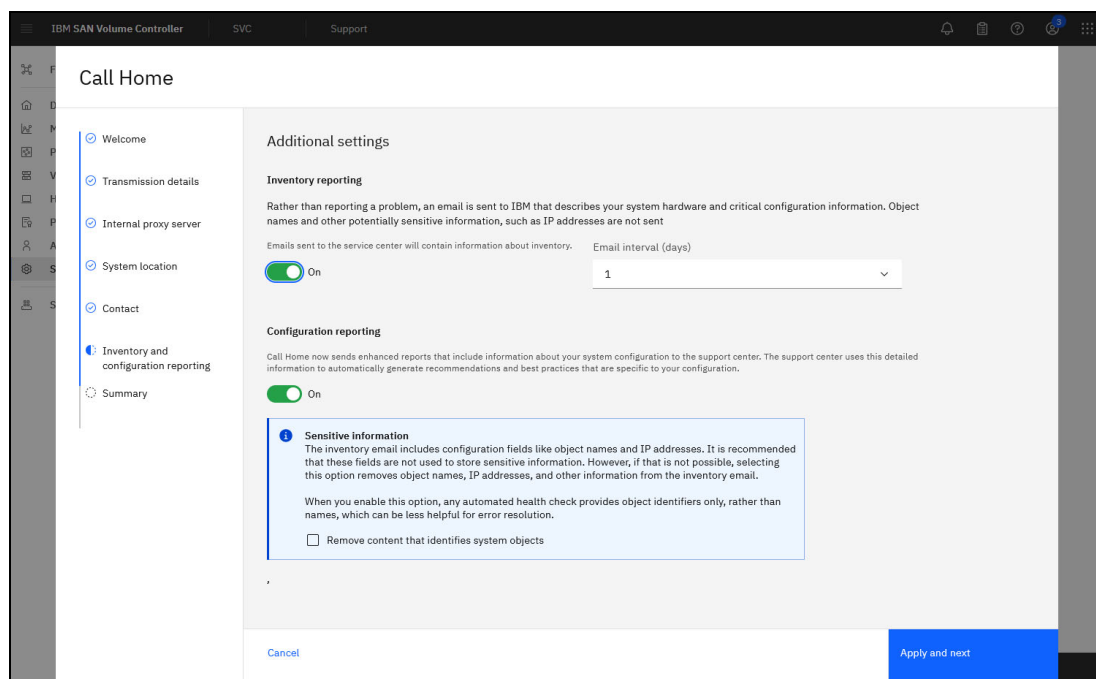


Figure 6-15 Configuring more settings

12.Review the summary and click **Finish** (see Figure 6-16).

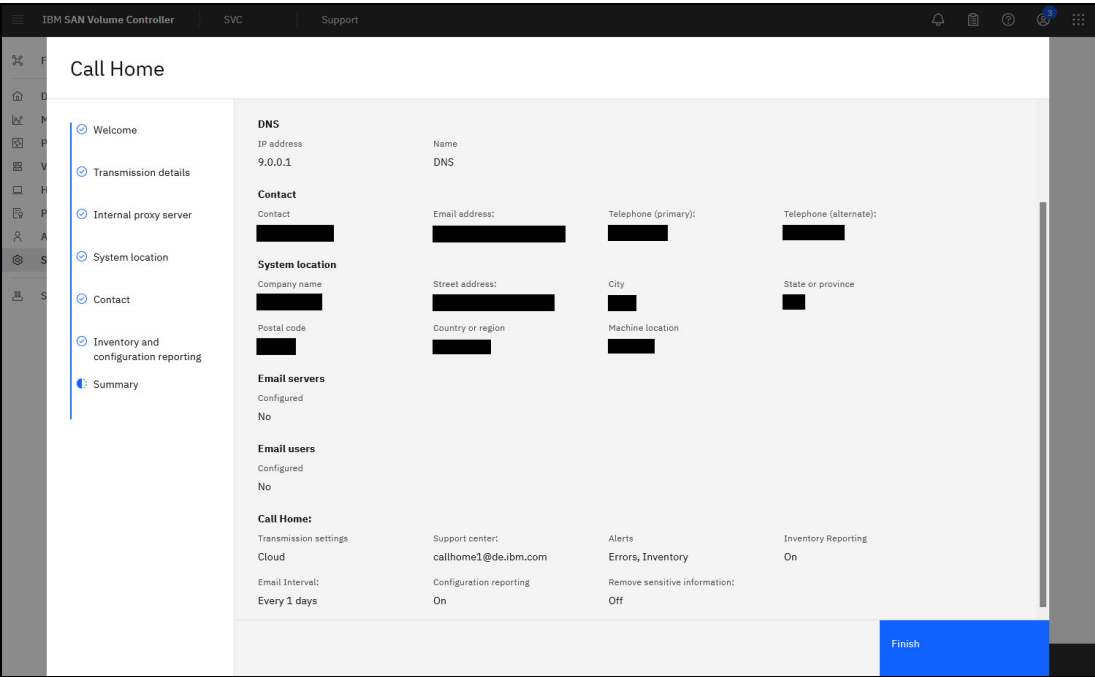


Figure 6-16 Summary information

13.The connection status shows active. IBM Storage Insights might still display “Not registered” (Figure 6-17).

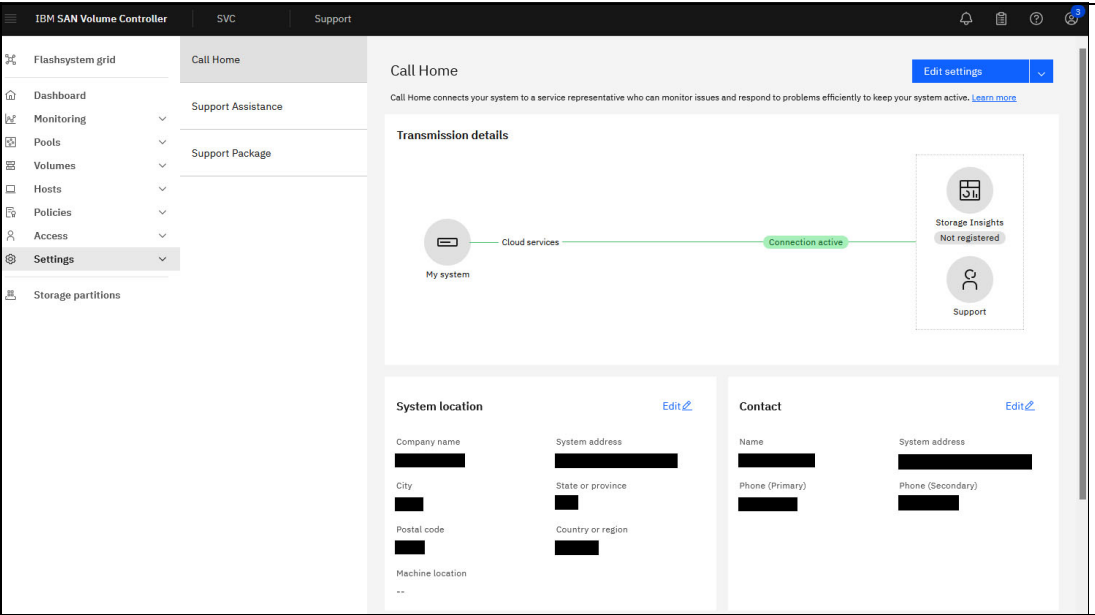


Figure 6-17 Call Home with cloud services is now enabled

Registering the System with the IBM Storage Insights (GUI)

To register the system with IBM Storage Insights by using the GUI, complete the following steps:

1. After Call Home with cloud services is enabled, connect the system to IBM Storage Insights by completing the following steps:
 - a. Select **Settings** → **Support** → **Call Home**:
 - b. Open the drop-down menu next to Edit settings.
 - c. Click **Add system to IBM Storage Insights** (Figure 6-18).

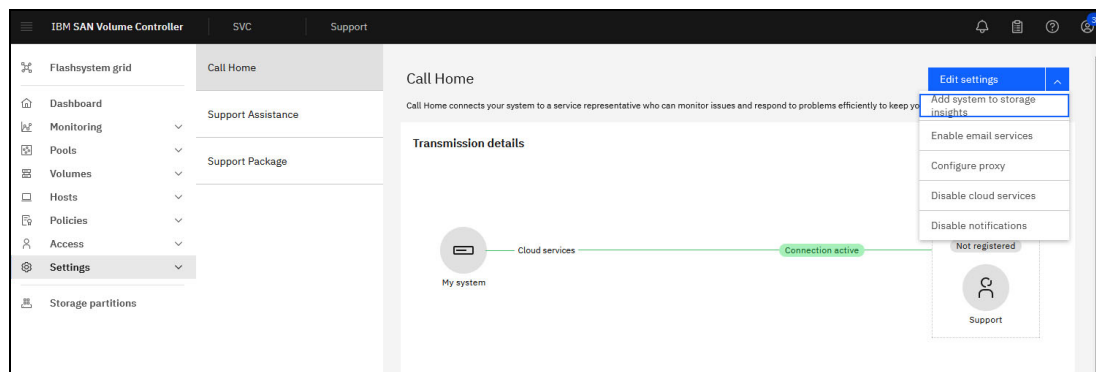


Figure 6-18 Adding a system to IBM Storage Insights

2. Enter the IBMid that is associated with the IBM Storage Insights tenant and click **Save** (Figure 6-19).

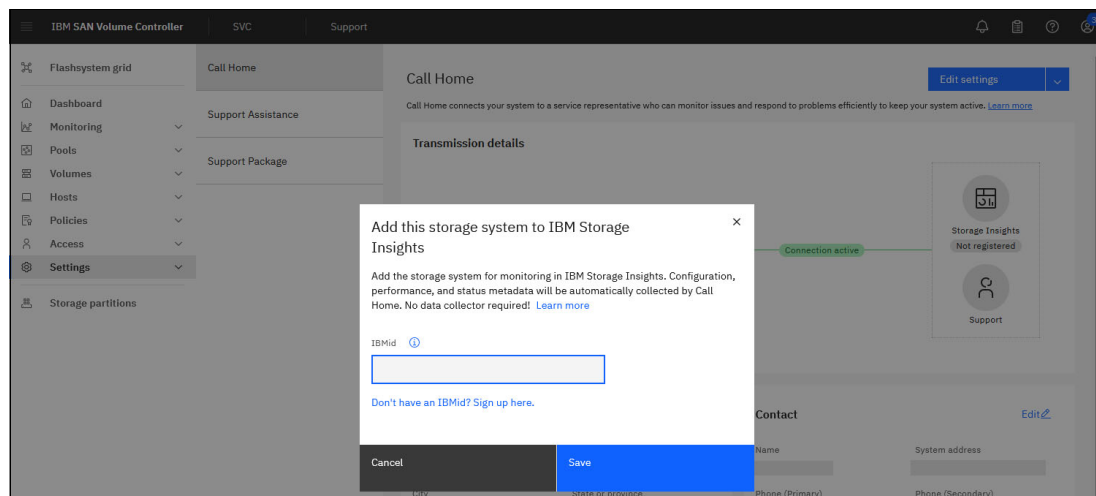


Figure 6-19 Entering your IBMid to integrate the system with IBM Storage Insights

3. Wait for the system to validate the IBMid.

4. If you have multiple instances of IBM Storage Insights, the GUI displays a list of available instances. Select the instance that you want to connect to your storage system.

When the process completes successfully, the GUI indicates that your system is registered with IBM Storage Insights.

Figure 6-20 shows the Call Home window with a status of Registered.

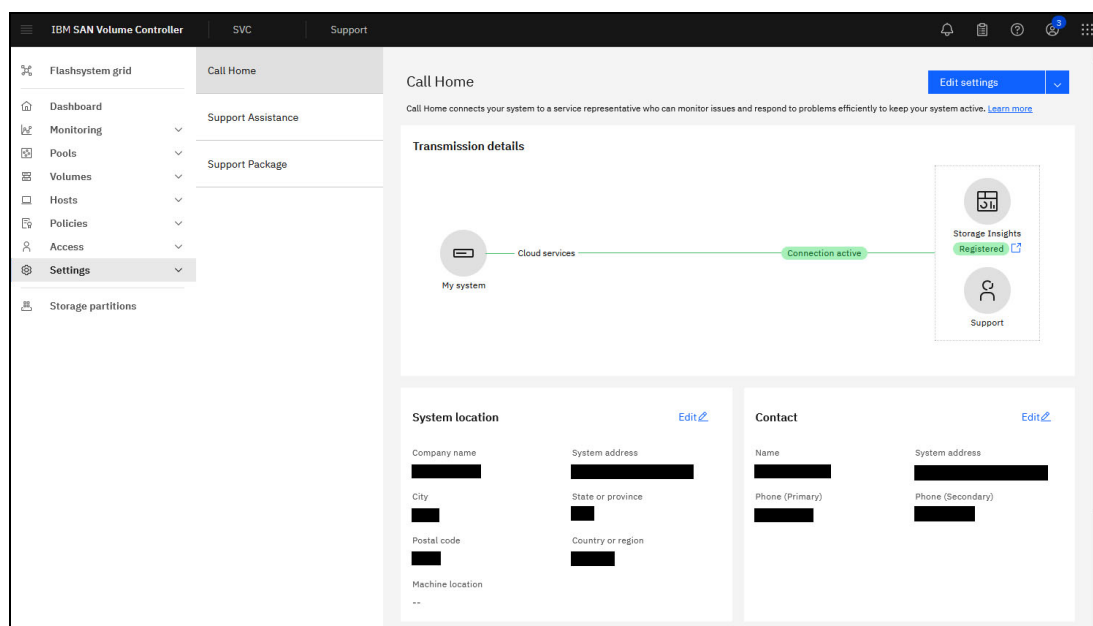


Figure 6-20 The system is now registered with IBM Storage Insights

Next to this status, a hyperlink directs you to your specific IBM Storage Insights instance. The next configuration steps must be completed in IBM Storage Insights.

Adding a storage system to IBM Storage Insights by using Call Home with cloud services

To add eligible IBM Storage Virtualize systems for monitoring through Call Home with cloud services, use either the classic view or the new IBM Storage Insights Overview window.

In the classic view, select **Configuration** → **Call Home Devices**. This window lists any pending devices that require approval. Approved Call Home devices are listed when Approved is selected.

Classic view

To add a storage device, complete the following steps:

- 1. Select **Configuration** → **Call Home Devices**. The action opens the Call Home Storage Systems window.
- 2. Approve pending devices by selecting the device under Pending Devices and clicking **Approve**. Figure 6-21 shows one pending device, which is an SVC.

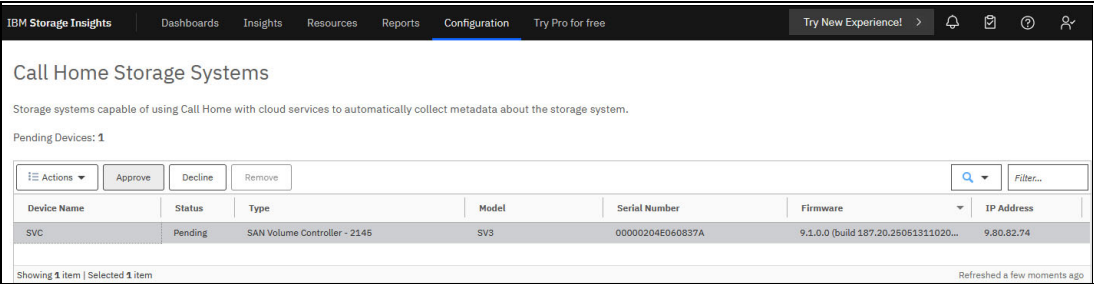


Figure 6-21 Approving a pending device

- 3. If IBM Storage Insights does not show any pending devices, see “Troubleshooting the IBM Storage Insights connection” on page 131.
- 4. Confirm approval by clicking **Approved** and confirming that your system is listed. The status in Figure 6-22 now shows Approved.

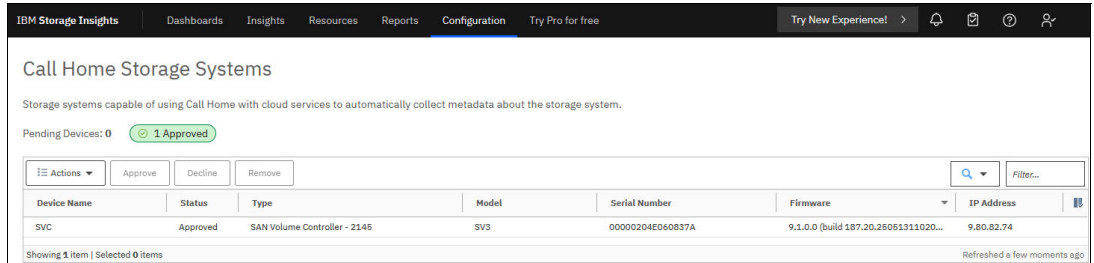


Figure 6-22 The new SVC system is now listed as an approved device

After approval, the IBM Storage Insights dashboard displays the system. The Data Collection Type field shows “Call Home with cloud services” (see Figure 6-23).

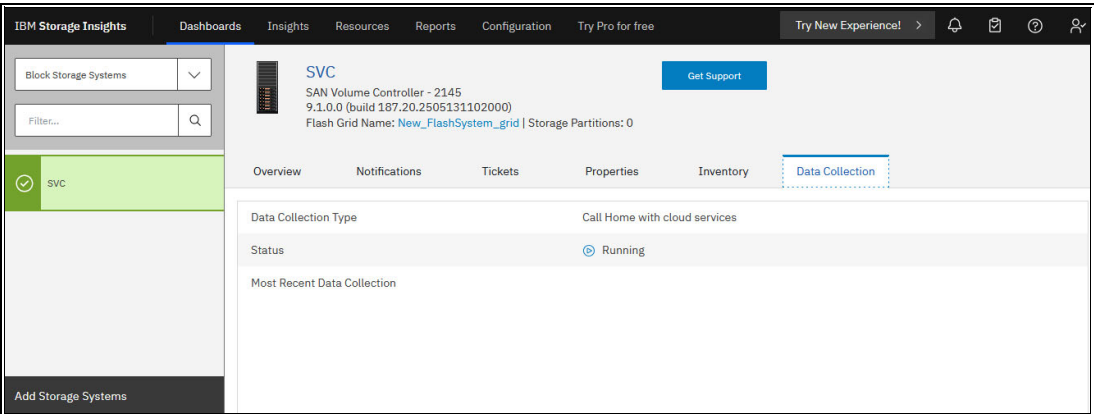


Figure 6-23 The SVC shows that the Data Collection Type is Call Home with cloud services

- 5. Repeat the procedure if you want to add more devices.

New experience (optional)

You can also use the new IBM Storage Insights Overview window to add devices.

To switch to the new view and respond to notifications, complete the following steps:

1. Click **Try New Experience!** in the navigation bar.
2. When a new Cloud Home with Cloud Services-enabled device is detected, the GUI displays a notification. Click **More details** (Figure 6-24).

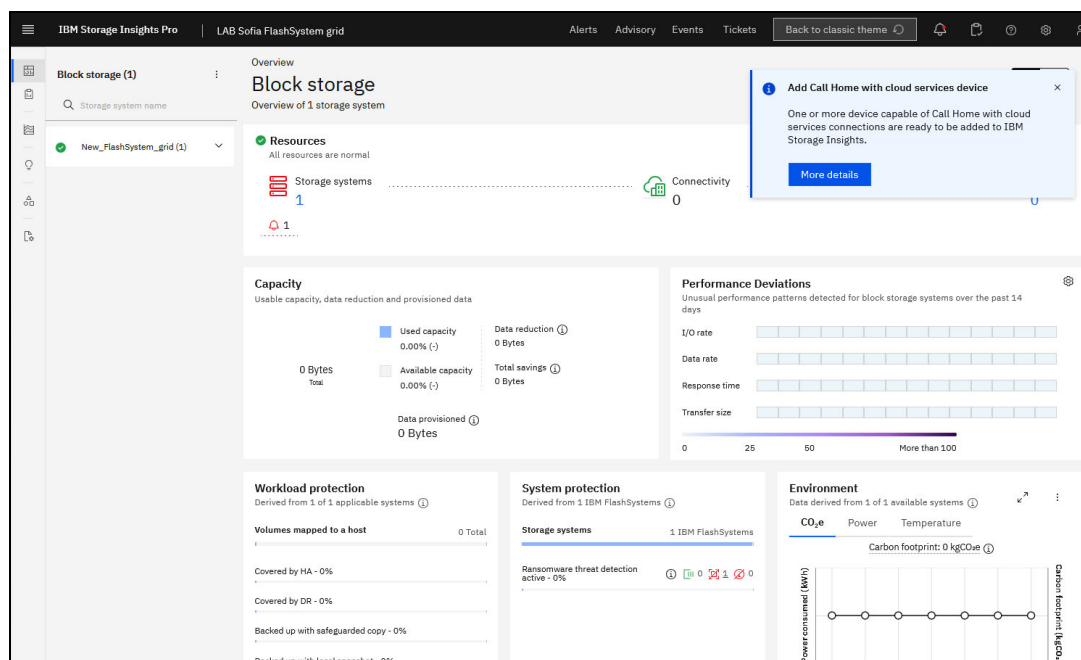


Figure 6-24 Notification to add a Call Home with cloud services device

3. Select the device from the list and click **Approve** (Figure 6-25). Approve multiple devices if needed.

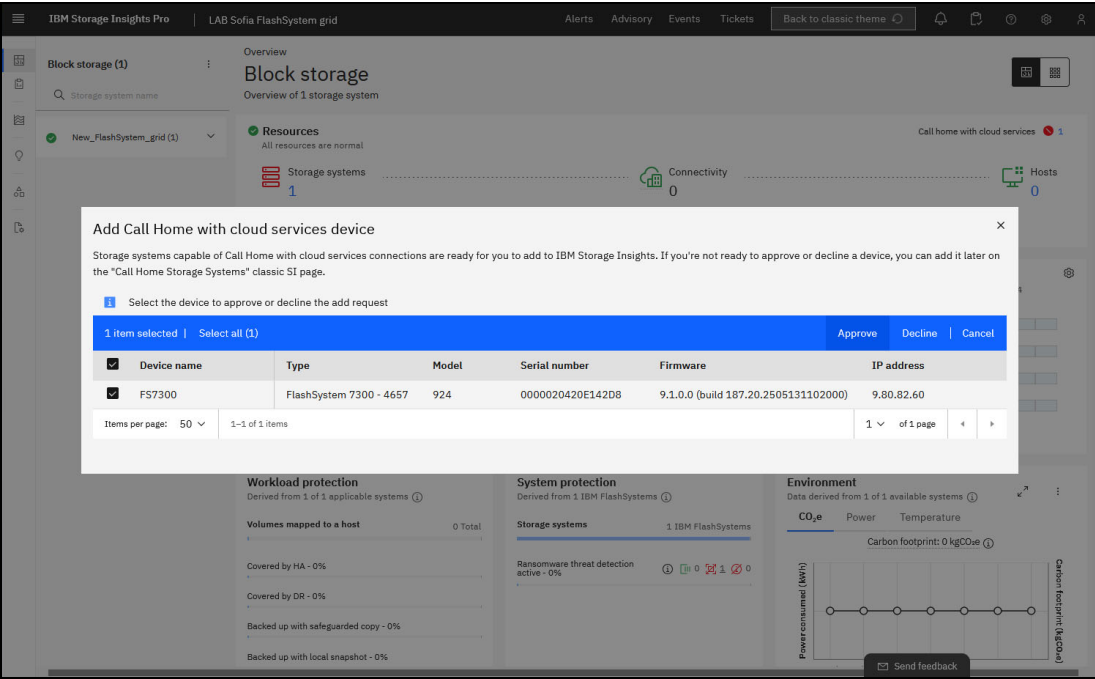


Figure 6-25 Selecting the new device to approve

IBM Storage Insights displays a notification to confirm that the action was successful (Figure 6-26).

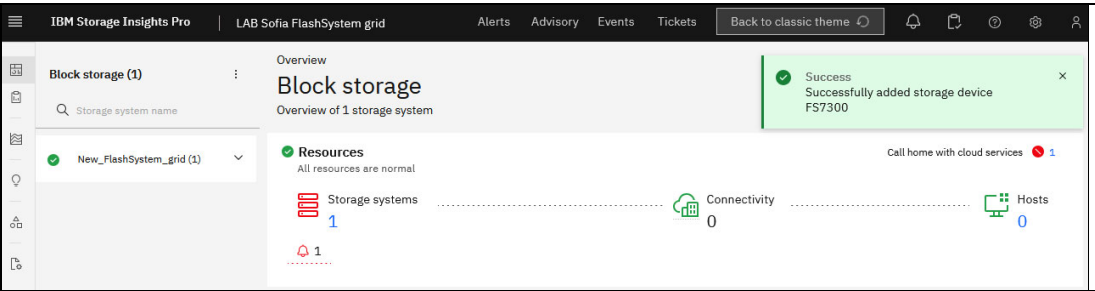


Figure 6-26 Successfully added the IBM FlashSystem 7300 storage device

4. After all devices are added, switch to the new IBM Storage Insights view by clicking **Try New Experience!** from the navigation bar.

After switching views, the Overview dashboard lists all storage systems. For an IBM FlashSystem grid, the dashboard also shows the grid name of the IBM FlashSystem grid with the number of systems in parentheses.

Figure 6-27 on page 131 shows an IBM FlashSystem grid that is named New_FlashSystem_grid, which consists of an IBM FlashSystem 7300, an IBM FlashSystem 9200, and an SVC.

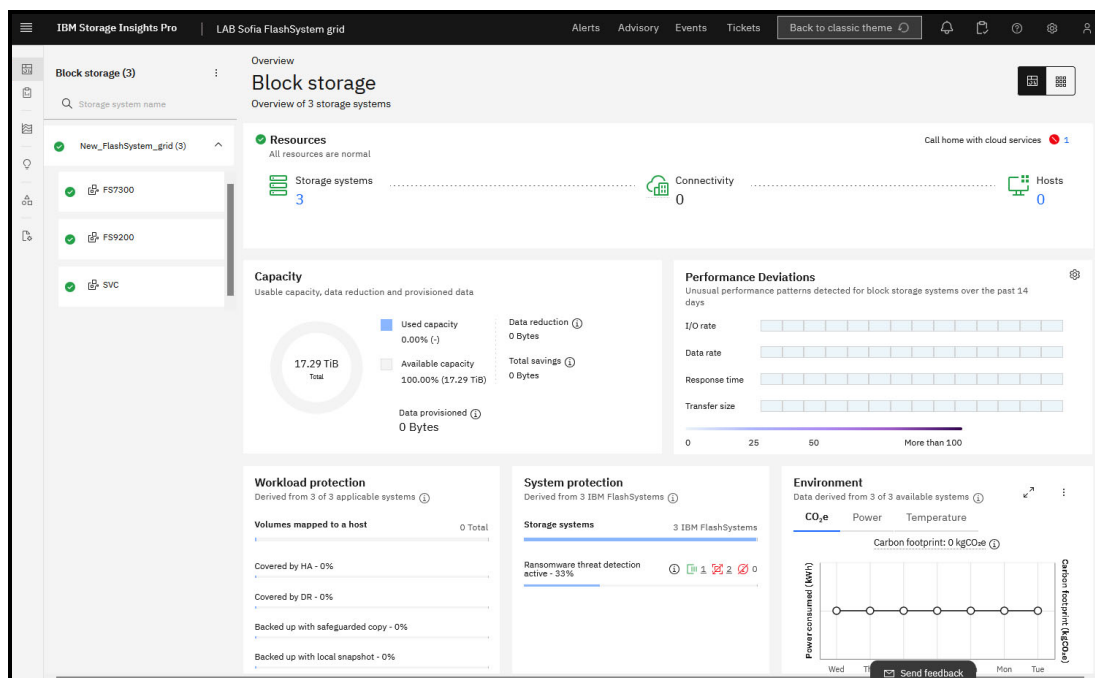


Figure 6-27 IBM Storage Insights Overview: An IBM FlashSystem grid with three devices

Troubleshooting the IBM Storage Insights connection

When you attempt to connect your system to a IBM Storage Insights instance by using the storage system management GUI, the system asks for an IBMid. Occasionally, the system might encounter issues verifying your IBM ID. You might see error messages such as Invalid IBMid or An error occurred when attempting to retrieve your IBM Storage Insights account.

The most effective way to resolve the issue is to configure the IBM Storage Insights tenant IBMid using the CLI.

To configure the tenant ID, complete the following steps:

1. Use SSH to connect to the cluster IP address of the storage system and sign in.
2. To configure the IBM Storage Insights tenant ID, run the following command:


```
> svctask chcloudcallhome -sitenantid xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx
```

Replace xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx with the final segment of your IBM Storage Insights URL, as described in 6.1.2, “How to set up a new IBM Storage Insights instance” on page 116).

3. Verify the configuration by running the following command:

```
svcinfolsccloudcallhome -delim :
```

This command displays the current configuration and confirms whether the IBM Storage Insights tenant ID was configured successfully.

Example 6-1 shows the expected output.

Example 6-1 Confirming the IBM Storage Insights tenant ID with the `lscloudcallhome` command

```
IBM_2145:SVC:superuser>svcinfolsccloudcallhome -delim :
status:enabled
connection:active
error_sequence_number:
last_success:250514082550
last_failure:
si_tenant_id:xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx
apikey_configured:no
```

If the system does not appear after one or two days, perform troubleshooting steps to verify the connection. IBM Storage Insights waits for an inventory report from the IBM Storage Virtualize system, which is sent once per day by default. To avoid waiting for the report automatic report, start an inventory report manually by using the CLI.

4. Send a one-time inventory report to the Support Center servers by running the following command:

```
svctask sendcloudcallhome -inventory
```

5. Check Pending Devices on the Call Home Storage Systems window. If the device does not appear after approximately 30 minutes, complete the following steps:

- a. Reset the tenant ID (required) by running the following command:

```
> svctask chcloudcallhome -clertenant
```

Wait about 5 minutes, and then reconfigure the tenant ID by running the following command:

```
> svctask chcloudcallhome -sitenantid xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx
```

- b. Disable and re-enable Cloud Call Home with cloud services (optional):

If you are using the GUI, complete the following steps:

- i. Select **Settings** → **Support** → **Call Home**.
- ii. Open the drop-down menu next to Edit settings and click **Disable cloud services**, as shown in Figure 6-28.

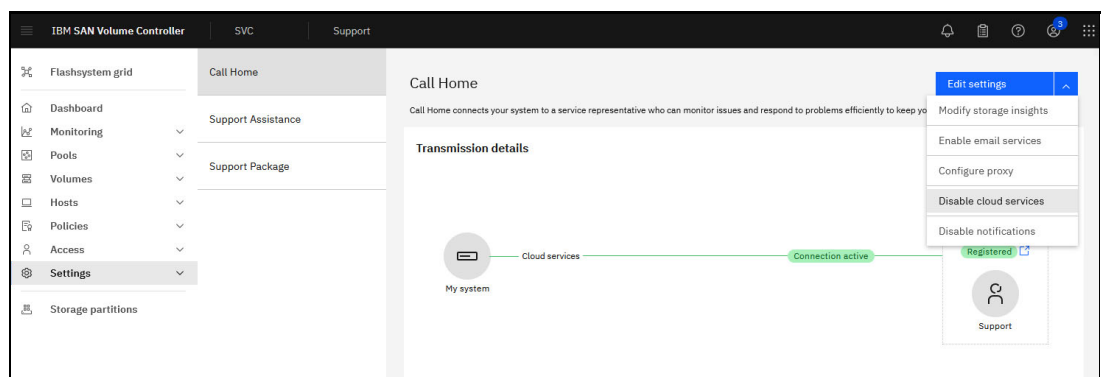


Figure 6-28 Disable Call Home with cloud services by using the GUI

- iii. Click **Enable Call Home**, and follow the wizard to complete the reconfiguration.

If you are using the CLI, complete the following steps:

- i. To disable the service, run the following CLI command:
`svctask chcloudcallhome -disable.`
- ii. Wait approximately 3 - 5 minutes.
- iii. To make sure that the service is running properly and to clear any temporary issues, re-enable the service run the following command:
`svctask chcloudcallhome -enable`

6. Test the connection by completing the following steps:

- a. Run the following CLI command:
`svctask sendcloudcallhome -connectiontest.`
- b. To check the result, run the following command:
`svcinfolsccloudcallhome -delim :`

This test indicates whether the connection was successful or failed.

Example 6-2 demonstrates how to perform a connection test and how to check the result. In this example, the connection was successful.

Example 6-2 Testing the Call Home with cloud services connection

```
IBM_2145:SVC:superuser>svctask sendcloudcallhome -connectiontest
IBM_2145:SVC:superuser>svcinfolsccloudcallhome -delim :
status:enabled
connection:active
error_sequence_number:
last_success:250514085550
last_failure:
si_tenant_id:xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx
apikey_configured:no
```

7. Verify that the system date and time are current and accurate by running the following command:

`svqueryclock`

8. Check for any network configuration issues that can affect Call Home connectivity. Run the following command from the CLI of the storage system:

`sainfo host -ip esupport.ibm.com`

Example 6-3 shows the expected output of the **sainfo host -ip esupport.ibm.com** command.

Example 6-3 Checking whether the DNS server can properly resolve esupport.ibm.com

```
IBM_2145:SVC:superuser>sainfo host -ip esupport.ibm.com

esupport.ibm.com has address 192.148.6.11
esupport.ibm.com has IPv6 address 2620:1f7:c010:1:1:1:1:11
```

9. As an alternative option, run a **curl** command from a host or laptop within the same subnet as the cluster. Check for a response to verify connectivity.

`curl https://esupport.ibm.com/connect/api/v1/callhome_event_service`

- 10.If your network uses an internal proxy server to manage connections between the system and IBM Support, ensure that the proxy server allows connections to the [IBM Support](#) DNS name.

Your IBM Storage Virtualize system can still fail to send inventory reports even if the firewall and proxy server are correctly configured. Sometimes, SSL certificate-related errors can occur when the proxy server intercepts traffic and behaves like a man-in-the-middle. This type of proxy, which is known as a *snooping proxy*, decrypts traffic to inspect its contents.

If your proxy server operates in this manner, you have the following options to resolve the issue:

- Exclude the URL [IBM Support](#) from snooping, which helps ensure that traffic to the destination remains encrypted and uninterrupted.
- Install the proxy server SSL certificate on the IBM Storage Virtualize system, which allows proper SSL certificate matching and successful communication.

- 11.To import a proxy certificate by using the GUI, complete the following steps:

- a. Select **Settings** → **Network** → **Internal proxy server** → **Certificate**.
- b. Upload the file or drag the file to the designated area.

This certificate allows the system to verify the authenticity of the internal proxy server.

- 12.To configure the proxy server certificate by using the CLI, complete the following steps:

- a. Transfer the certificate to the /upgrade directory by using Secure Copy (SCP), PuTTY Secure Copy Protocol (PSCP), or Secure File Transfer Protocol (SFTP). SFTP is the preferred method in Version 9.1.0 and later. For example, you can transfer the certificate into the /upgrade directory.
- b. Run the following command to configure the proxy server certificate:

```
svctask chproxy -sslcert /upgrade/proxy-server-certificate.pem
```

Use the **-sslcert** option to specify the path and file name of the proxy server certificate.

- c. To transfer the certificate by using SFTP, run the following command:

```
sftp -p22 <certificate-file> <username>@<system-ip-address>:<directory>
```

Example 6-4 shows how to use the **chproxy** command, assuming that the certificate file is already present on the system.

Example 6-4 How to specify a certificate for an already configured proxy server

```
> svctask chproxy -sslcert /upgrade/proxy-server-certificate.pem
```

For more information about the **chproxy** command, see [IBM Documentation](#).

6.1.4 IBM FlashSystem grid views in IBM Storage Insights

IBM Storage Insights provides a unified monitoring interface. Storage administrators can track capacity, performance, and health. It also displays storage partitions and monitor the status of any storage partition migration for systems within an IBM FlashSystem grid.

IBM FlashSystem grid in the Overview dashboard

Start from the Overview dashboard in the new IBM Storage Insights view. To switch from the classic view to the new experience, click **Try New Experience!** in the navigation bar.

To default to the new experience permanently, complete the following steps:

1. From the utility menu in the upper right of the navigation bar, select the user profile icon. It resembles a person.
2. In the drop-down menu, select the slider to switch the **Default to the new experience** selection from **Disabled** to **Enabled**.

Figure 6-29 shows how to change the default view.

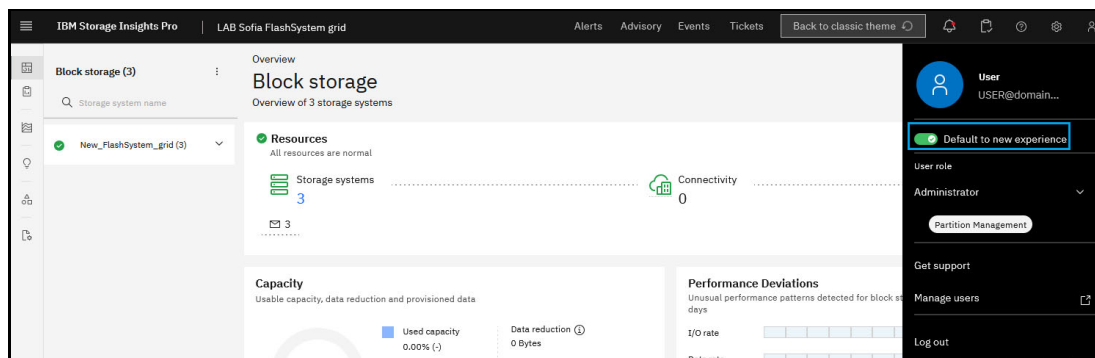


Figure 6-29 Changing the default IBM Storage Insights view to the new experience

Call Home with cloud services collects telemetry data and transmits it to IBM Cloud®. IBM Storage Insights analyzes historical patterns and converts metadata into recommendations. Performance metadata is sent automatically every five minutes. Inventory and capacity metadata is sent every 24 hours.

Identify an IBM FlashSystem grid by the chevron symbol next to the grid name, which expands to display all storage systems that belong to this IBM FlashSystem grid. The number in parentheses indicates the number of systems that are grouped within the grid.

Figure 6-30 shows an example view of an IBM FlashSystem grid within IBM Storage Insights Pro with the name New_FlashSystem_grid. It includes three systems in the grid. Any IBM FlashSystem grid-related records in the tabular view are highlighted in blue.

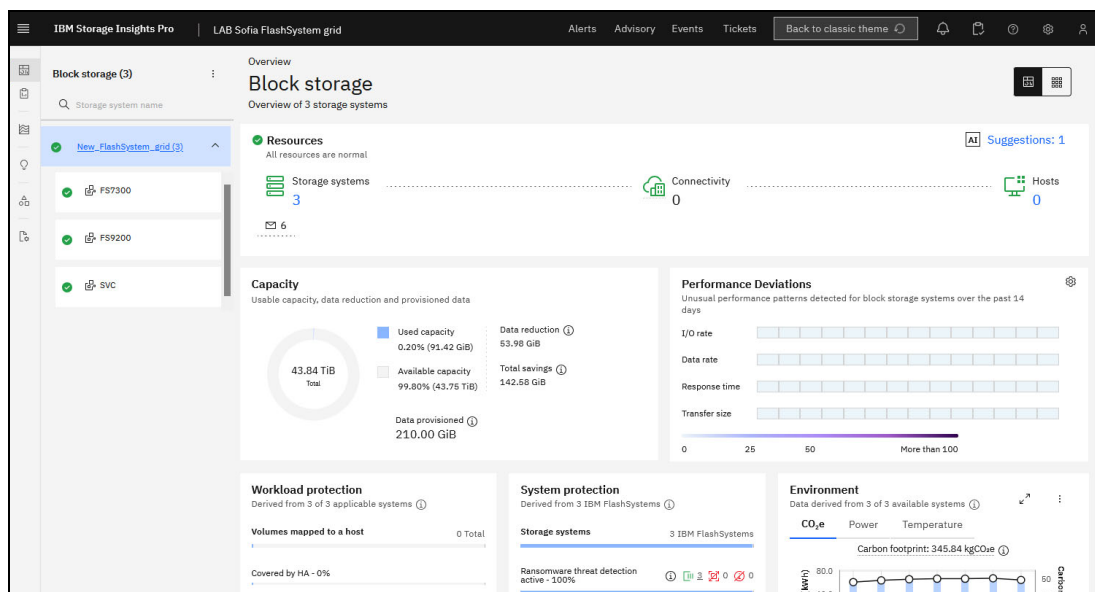


Figure 6-30 IBM FlashSystem grid information in the IBM Storage Insights Overview dashboard

To view AI Suggestions for IBM Storage Virtualize systems in an IBM FlashSystem grid, complete the following steps:

1. Select the IBM FlashSystem grid record in the list of storage systems.

The Overview window now shows a summary of the storage systems that are part of the IBM FlashSystem grid. AI Suggestions appears in the upper right (exclusive to IBM Storage Virtualize systems in the IBM FlashSystem grid). These suggestions can include recommendations for partition migration based on capacity forecasting and workload placement.

2. Click **Suggestions** to open a side pane with all the relevant AI suggestions. Click the arrow on a suggestion card to view details.

Figure 6-31 shows that the system warranty either expired or is about to expire soon. In addition to maintenance-related suggestions, it might also receive recommendations about capacity and performance.

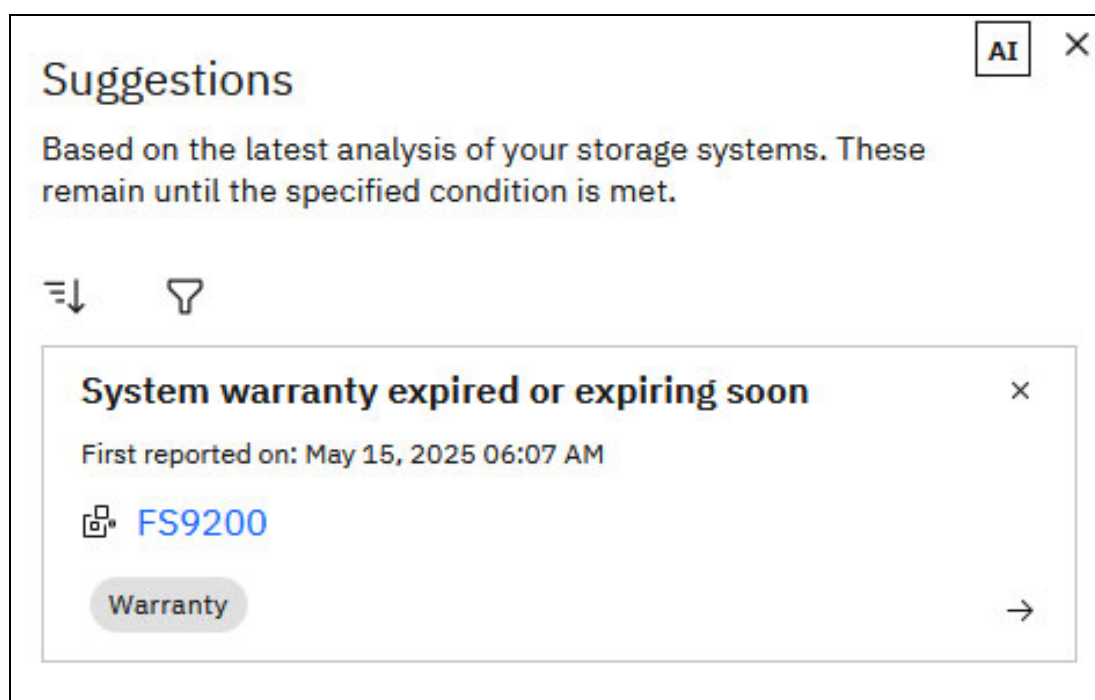


Figure 6-31 AI suggestion about an expiring system warranty of an IBM FlashSystem 9200 system

To view more details about a specific suggestion, click the arrow icon that is in the lower right of the suggestion card. Clicking the icon expands the card and displays additional information about the recommendation.

Figure 6-32 on page 137 shows the detailed view of the AI-powered recommendation for this condition.

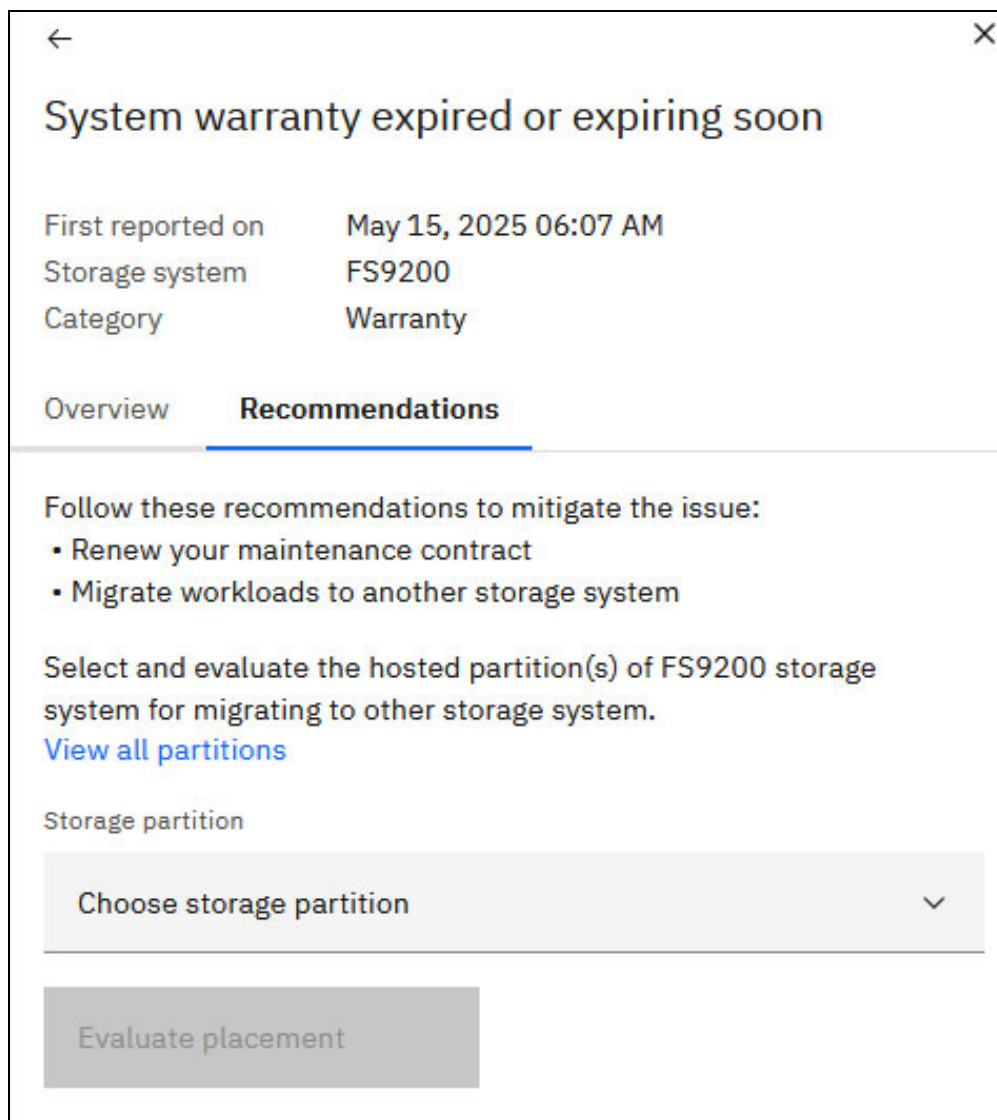


Figure 6-32 AI recommendations for checking the migration suitability of target systems

The window includes the following tabs:

- **Overview**
- **Recommendations**

One recommendation is to renew the maintenance contract. You can also migrate the workloads to a different storage system, for example, if you plan to decommission the system. AI recommendations help by presenting a list of storage partitions on this system that can be migrated.

You can select any partition in the drop-down menu and click **Evaluate placement** to see details about the migration suitability of the selected storage partition to available storage systems.

For more information about AI suggestions, see 6.2.4, “AI-based recommendations and System Health Advisories” on page 160.

IBM FlashSystem grid My World Navigation view

The IBM FlashSystem grid My World Navigation view provides an aggregated dashboard that shows systems and partitions that are grouped within an IBM FlashSystem grid. This view is accessible from the Overview dashboard.

Select a grid name in the Overview dashboard to open IBM FlashSystem grid My World Navigation, which displays detailed information about the selected grid.

Figure 6-33 shows the IBM FlashSystem grid that is listed within Storage systems in IBM Storage Insights Pro.

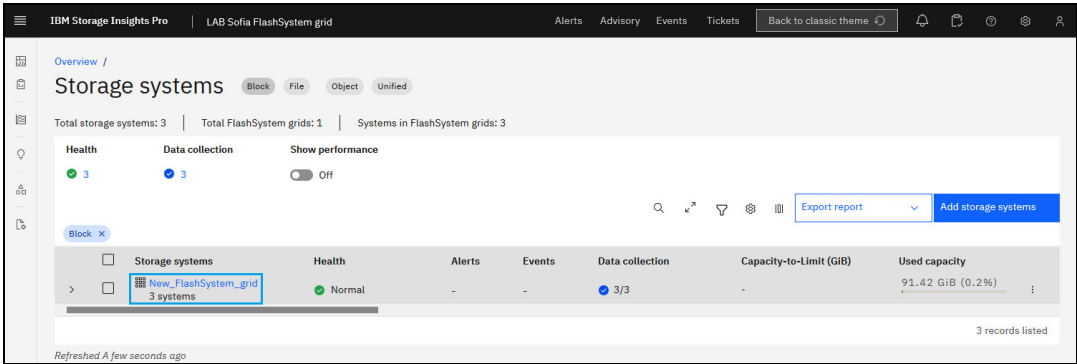


Figure 6-33 IBM FlashSystem grid listed under Storage systems within IBM Storage Insights Pro

Figure 6-34 shows the IBM FlashSystem grid My World Navigation Overview dashboard in IBM Storage Insights Pro. The example illustrates a partition migration in progress from an IBM FlashSystem 7300 to an IBM FlashSystem 9200.

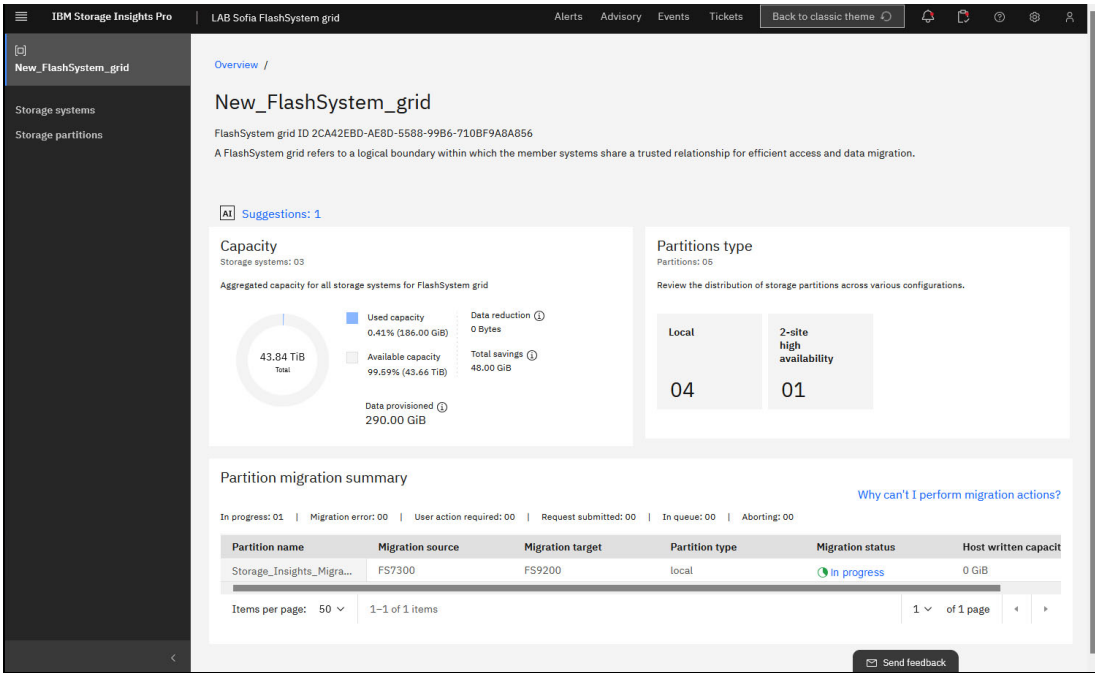


Figure 6-34 IBM FlashSystem grid: My World Navigation Overview dashboard

IBM Storage Insights for IBM FlashSystem grid presents a clean and structured interface to provide an overview of aggregated capacity, storage health, partition types, and partition migration status. The interface is organized into the following key sections:

- ▶ Navigation pane (left sidebar). The pane provides access to system views and configuration options.
- ▶ IBM FlashSystem grid Overview. This list includes aggregated grid information.
 - Capacity. The aggregated capacity of all grid systems.
 - Used capacity. The amount of data that is written to the storage arrays.
 - Available capacity. The remaining physical space, including unused and reclaimable capacity.
 - Data reduction. The savings from compression and deduplication.
 - Total savings. The sum of savings after data reduction and thin-provisioning.
 - Data provisioned. The total volume capacity, including capacity volumes that are not yet mapped to hosts.
 - Partition types.
 - Local partition. Available only on a specific storage system.
 - 2-site High Availability partition. Replicated across two sites.
 - Other types might also be displayed.
 - Partition migration summary. Details of ongoing migrations and partition types. The status of a migration can be one of the following states.
 - Request submitted. The system is about to start the migration. IBM Storage Insights sent a migration request to the storage system and is waiting for the device to accept the request.
 - User action required. The migration requires user intervention, which is necessary, for example, if a connection is lost, if the target system runs out of space, or if any other unexpected issue occurs.
 - In progress. The partition migration started and is in progress. The storage system accepted the migration request and is performing the migration activity.
 - Suspended. The migration is suspended by the user or because of one or more issues with the storage infrastructure.

Each section contributes to the monitoring experience and helps storage administrators make informed decisions about their IBM FlashSystem grid infrastructure.

Note: Migration state updates depend on the mini probe and full probe.

The left navigation pane is the main way to switch between different views and functions. The IBM FlashSystem grid Overview window is the main dashboard that displays high-level insights into storage capacity, partitions, and migration. It includes the following subsections:

- ▶ Storage system. A detailed view of the individual storage arrays in the IBM FlashSystem grid
- ▶ Storage partitions. A focused section for managing and analyzing the partitions that are configured within the IBM FlashSystem grid

Storage systems window

The Storage systems window contains details about the storage systems that are part of the IBM FlashSystem grid. It displays the name of each storage system and system properties such as firmware version, the Universally Unique Identifier (UUID) of the system, information about the system hardware and vendor, and the number of volume groups. In addition, the table displays capacity-related information such as *used capacity*.

The IBM FlashSystem grid window displays properties that help determine whether an active migration is in progress. This section also shows whether control access for IBM Storage Insights is enabled on the storage system. This setting is required to start a partition migration through IBM Storage Insights. Other IBM FlashSystem grid-related fields include the number of partitions and whether snapshots are enabled.

More columns can be displayed, and any columns can be hidden as needed.

Figure 6-35 shows the Storage systems window. All three storage systems that are part of the grid are listed: an SVC, an IBM FlashSystem 7300, and an IBM FlashSystem 9200. The figure shows a migration in progress for a partition with the name `Storage_Insights_Migration_Partition` from source system IBM FlashSystem 7300 to another system.

Storage systems	Partitions	Active migration	Volume groups	Snapshots	Control access
SVC	-	-	1	-	Yes
FS7300	04	Storage_Insights_	3	No	Yes
FS9200	02	-	1	No	Yes

Figure 6-35 Storage systems window

Table 6-2 describes each field that is available by default in the Storage systems window.

Table 6-2 Default columns in the Storage systems window

Column	Description
Storage systems	Displays the name of the storage system, for example SVC or IBM FlashSystem 7300. Select the name to open the storage system overview window, where it displays system-specific details such as hardware health, system capacity, performance, IP address, firmware version, and other information.
Partitions	Displays the total number of storage partitions allocated to the storage system.
Active migration	Indicates whether any active migration processes are running on the system. A storage system can have only one active migration at a time.
Volume groups	Displays the number of volume groups within the storage system. Select the number to view details about each volume group, including the volume group name and the number of volumes it contains.
Snapshots	Indicates whether snapshots are enabled for any of the volume groups (Yes/No).
Control access	Specifies whether the IBM Storage Insights Orchestrator permission is available. The expected value is Yes.
System UUID	Displays a unique identifier of the storage system to track it across environments.
Used capacity	Displays real-time storage consumption to help the storage administrator plan for capacity.
System type	Specifies the storage product, for example, SAN Volume Controller - 2145, IBM FlashSystem 7300 - 4657, IBM FlashSystem 9200 - 9848 or any other type of storage.
Vendor	Displays the provider of the storage system. In this case it is Although typically IBM, the IBM FlashSystem grid supports third-party vendor storage systems.
Firmware	An important parameter is to determine whether the firmware is up to date or outdated. Displays the firmware version and the build, such as Version 9.1.0.0 (build 187.20.2505131102000). Systems running firmware versions earlier than 8.7.0.0 do not display all details, and the storage administrator must upgrade.

Figure 6-36 shows the storage system overview window that is described in the Storage systems row in Table 6-2 on page 141. The figure shows the storage system overview window for an IBM FlashSystem 7300 system. One host in the system is offline.

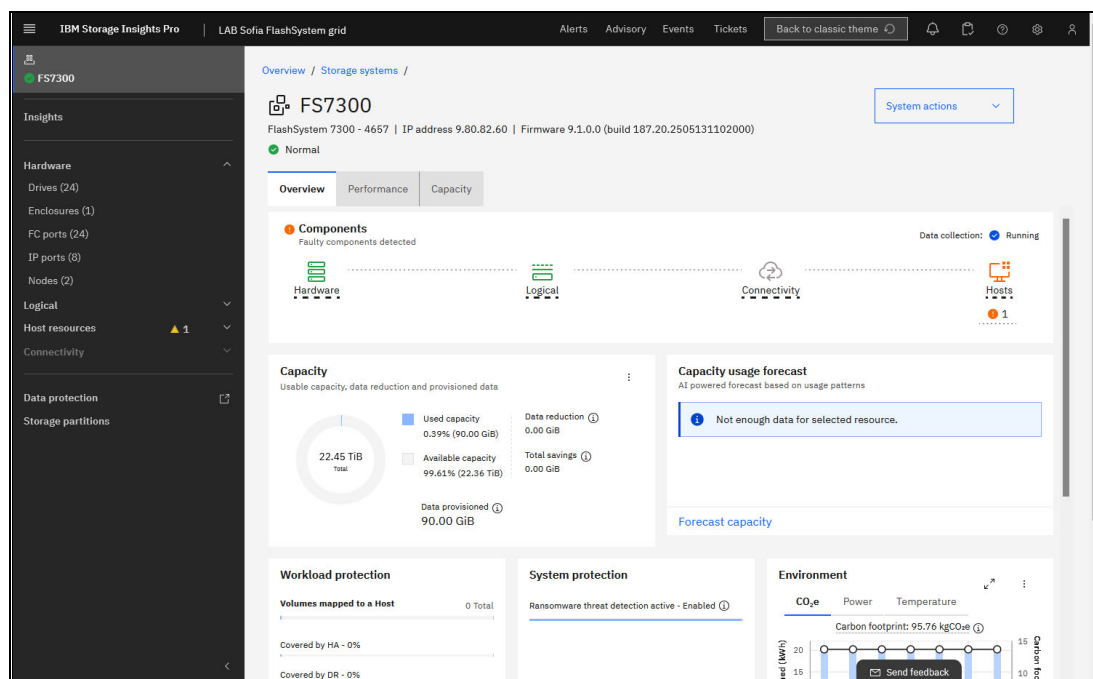


Figure 6-36 Storage system overview window of an IBM FlashSystem 7300 system

Storage partitions window

The Storage partitions window provides details about all partitions within a selected IBM FlashSystem grid storage system. It shows the partition name, partition type, and other associated information. It also helps administrators to track migration status. By default, the window displays five storage partitions. Scroll to view more storage partitions.

Figure 6-37 on page 143 displays the Storage partitions window for the IBM FlashSystem grid. There are five partitions present. Four of the partitions are local partitions, and you can initiate a partition migration for each partition by clicking **Migrate partition**. One of the storage partitions has a type of 2-site-ha. For this partition, the **Migrate partition** option is unavailable. Only one storage partition can be selected at a time to start a migration. The **Evaluate placement** option enables administrators to analyze the storage partition before migration.

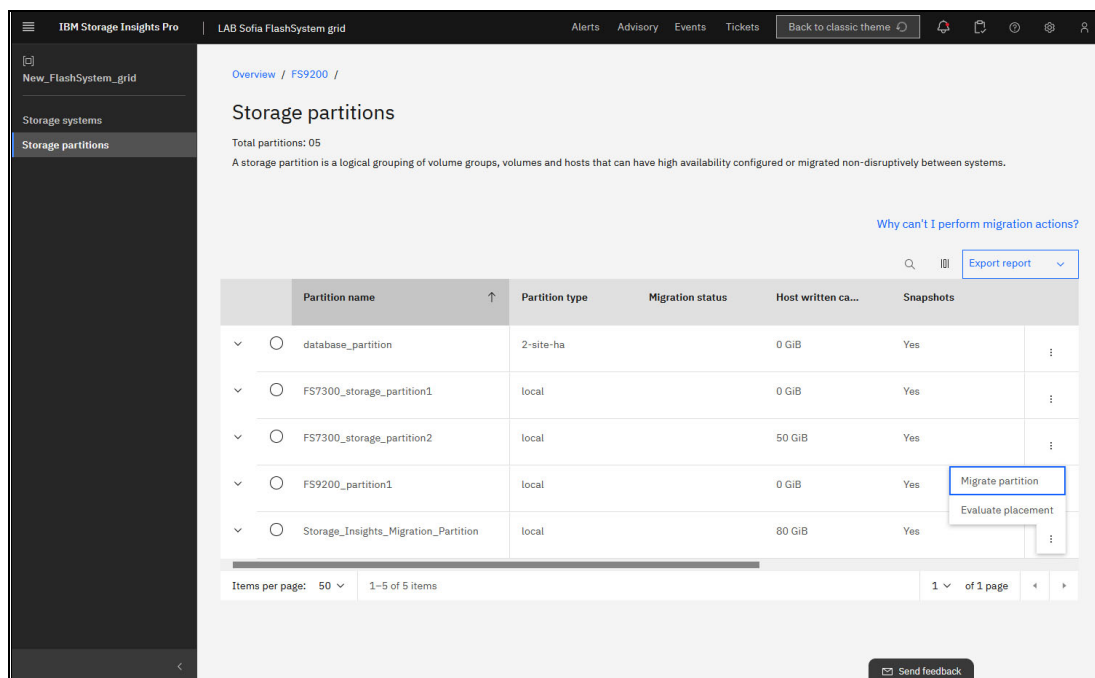


Figure 6-37 Storage partitions window

Table 6-3 describes each of the fields that are available by default in the Storage partitions window.

Table 6-3 Default columns in the Storage partitions window

Column	Description
Partition name	Displays the name of the storage partition, for example “database_partition” or “FS7300_storage_partition1”. If you click the name, the system displays a topological representation of the storage partition. The view includes replication details and connected hosts.
Partition type	Specifies the type of the storage partition, which can be Local, Disaster Recovery, 2-site High Availability, or 3-site High Availability.
Migration status	Indicates whether the partition is migrating (In progress) or queued for migration (for example, Request submitted or User action required). This field also indicates whether a migration error exists.
Host written capacity	Displays the total amount of data that the host wrote to all volumes in this storage partition.
Snapshots	Indicates whether snapshots are enabled for any of the volume groups that are associated with the partition (Yes/No).
Hosts	The number of hosts that are connected to the partition.
Volume groups	The number of volume groups that are assigned to the partition.
Last updated time	This field represents the timestamp of the most recent update to the storage partition. The timestamp determines whether the partition data is stale or whether it was updated recently.

Figure 6-38 shows the topological view for the storage partition `Storage_Insights_Migration_Partition`. This view shows more details, including host connectivity and the connected disaster recovery (DR). The green icon indicates that the system is functioning properly. Yellow icons indicate warnings.

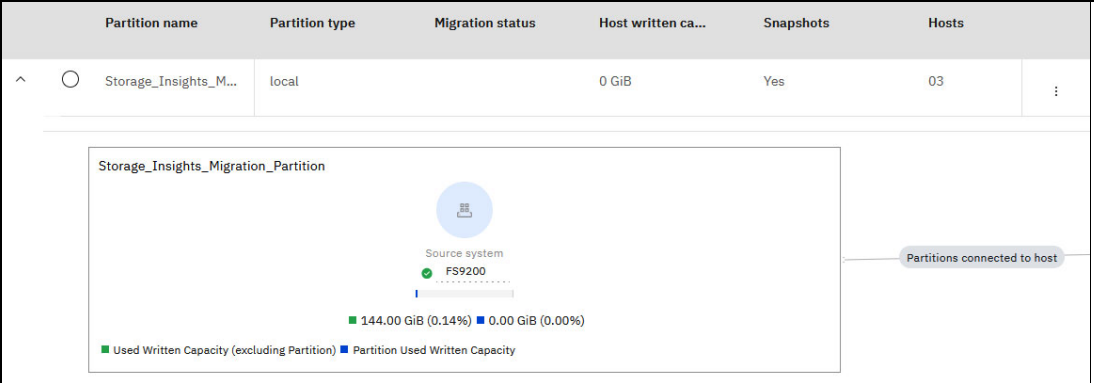


Figure 6-38 Storage partition topological view

Current limitations

At the time of writing, IBM Storage Insights does not support DR partitions or 3-site High Availability partitions.

Currently, IBM Storage Insights supports two main types of partitions for monitoring, which are local partitions and 2-site High Availability partitions. Local partitions are created and managed locally on a single IBM FlashSystem storage system. HA partitions span across multiple IBM FlashSystem storage systems to provide a highly available storage environment. In a 2-site HA setup, The volumes are mirrored between those two storage devices systems in a 2-site HA setup to help ensure redundancy.

Partition migration is supported only for local partitions from IBM Storage Insights.

It is not possible to initiate a new migration for a storage partition that is already in the process of migration. The ongoing migration must finish before starting another migration.

Note: An IBM Storage Virtualize system supports only one active storage partition migration at a time.

6.2 Partition migration with IBM Storage Insights

This section provides instructions for using IBM Storage Insights to perform partition migrations and outlines best practices for migration. IBM Storage Insights supports partition migration only for storage systems that are part of an IBM FlashSystem grid. The procedure requires connectivity between source and target systems. Systems in the grid must be zoned to help ensure mutual visibility.

A storage partition is a logical grouping of volume groups, volumes, host clusters, and hosts, typically for a specific application workload.

Partition migration relocates a partition from a source system to a target system and is non-disruptive within an IBM FlashSystem grid. Hosts must be zoned to all systems that contain the partition resources. If a workload is expected to move, ensure that host connectivity to the target system is established in advance.

Common use cases for migrating a partition include the following examples:

- Independent code update

If you plan to upgrade code, migrate the workload to another system, update the system to the new code level, then migrate the workload back to the original system. This approach prevents performance impact when a node is unavailable and provides flexibility to resolve hardware issues during the upgrade.

- Hardware replacement

If you plan to decommission an older system, migrate the workload to a newer system before you decommission the old system.

- Load-balancing (tiering within the same IBM FlashSystem grid)

If the workload requires better performance or a higher number of input or output operations per second (IOPS), migrate the partition to a different system with a more capable central processing unit (CPU) or higher performance flash drives to reduce latency.

If the application workload is not sensitive to response times, such as during backup operations, consider moving the workload to a lower-performance system, such as an IBM FlashSystem C200 that is suitable for bandwidth-intensive tasks. Although the system offers reduced response time performance, it is suitable for bandwidth-intensive workloads like backups.

- Capacity constraints

If the system is at risk of running out of space, migrate some workloads to a different system to free up capacity on the source system. The change can be temporary, for example, after adding FlashCore Modules (FCMs) to expand capacity and migrate back, or permanent.

For security and operational reasons, administrators primarily initiate partition migrations from the IBM FlashSystem grid dashboard. IBM Storage Insights adds value by continuously monitoring environments and providing AI-driven advisories to address potential issues proactively and reduce the risk of downtime.

Starting with Version 9.1.0, IBM Storage Insights provides AI-driven, real-time recommendations for migration targets on IBM Storage Virtualize systems. These recommendations help with selecting the most suitable target for a specific workload. They are based on a compatibility score that considers capacity, usage patterns, performance trends, and other metrics. Administrators can use AI-driven recommendations to help reduce human error when selecting a migration target. Administrators can switch between AI-driven target selection and manual selection within the grid GUI.

6.2.1 IBM Storage Insights control access and partition management permission

IBM Storage Insights uses a role-based access control (RBAC) model to manage permissions, including permissions for partition management. This model helps ensure that users have only the necessary access to view or configure storage resources and perform tasks such as partition migration.

IBM Storage Insights control access

By default, IBM FlashSystem provides IBM Storage Insights with monitoring access for telemetry data and support uploads through Call Home. To submit migration requests, IBM Storage Insights requires control access (*orchestrator permission*).

To determine whether IBM Storage Insights control access is enabled, use the IBM FlashSystem CLI and run the following command:

```
svcinfolssystem
```

Example 6-5 demonstrates how to verify whether IBM Storage Insights control access is enabled or disabled. In this example, the setting is disabled. The example also shows how to enable the setting by running the following command:

```
svctask chsystem -storageinsightscontrolaccess yes
```

Example 6-5 Managing IBM Storage Insights control access from the IBM FlashSystem CLI

```
IBM_FlashSystem:FS9200:superuser>svcinfolssystem | grep insights
storage_insights_control_access no
IBM_FlashSystem:FS9200:superuser>svctask chsystem -storageinsightscontrolaccess
yes
IBM_FlashSystem:FS9200:superuser>svcinfolssystem | grep insights
storage_insights_control_access yes
```

After running the **chsystem** command to enable IBM Storage Insights control access, IBM FlashSystem allows IBM Storage Insights to submit migration requests to the storage system.

Partition management permission

To perform partition migration and management, the IBM Storage Insights administrator or monitor user needs Partition Management permission. By default, this option is disabled and must be enabled manually per user.

To grant more privileges, such as Partition Management, complete the following steps from the User Access Management window as an administrator:

1. Log in to IBM Storage Insights by using the administrator credentials.
2. Select **Configuration** → **User Access Management**, as shown in Figure 6-39.

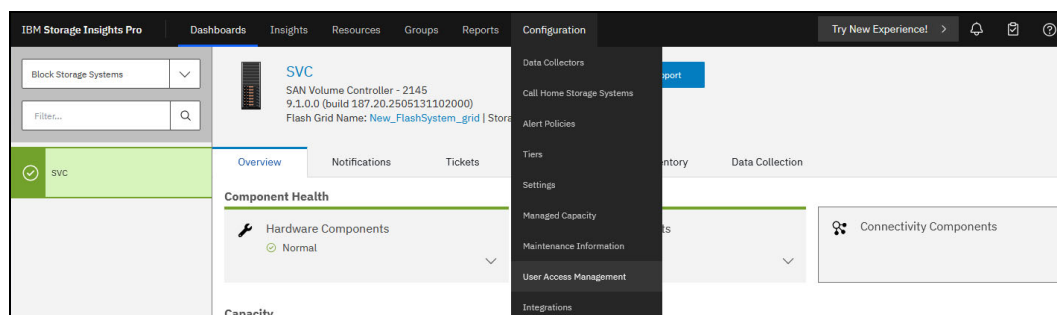


Figure 6-39 Selecting User Access Management from the Configuration drop-down menu

- Find the user for whom you want to grant the additional permission. Click the menu with the three vertical dots next to the user, and then click **Manage additional privileges**, as shown in Figure 6-40.

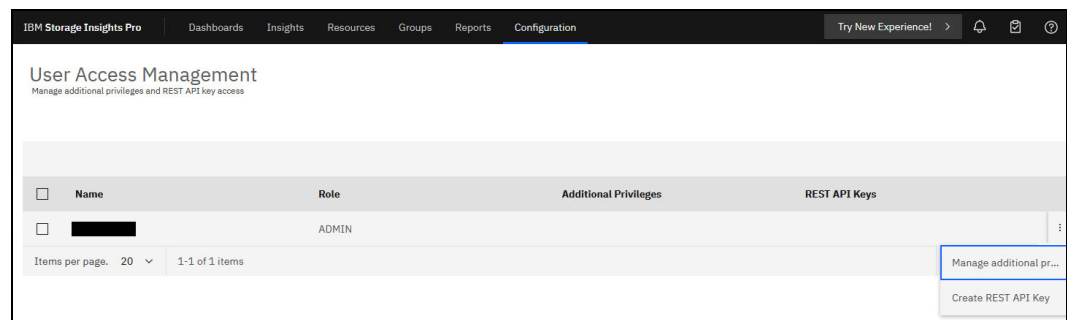


Figure 6-40 Manage additional privileges window: Specific user

- Select the **Partition Management** checkbox to enable the option, as shown in Figure 6-41.

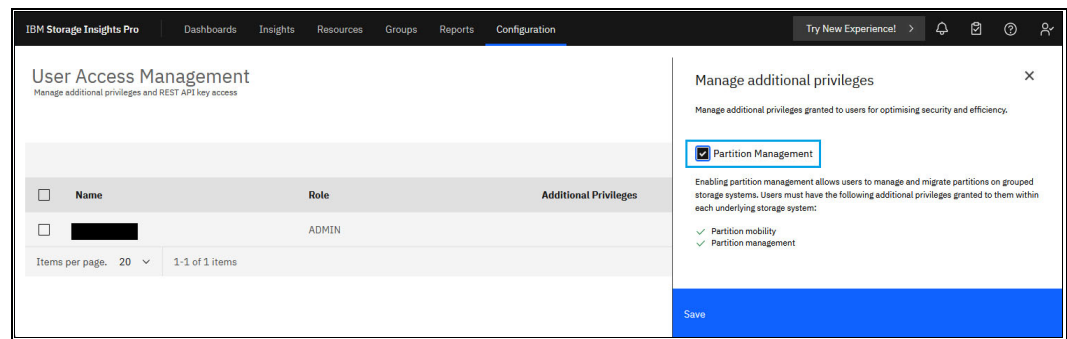


Figure 6-41 Enabling Partition Management

- Click **Save**.

As shown in Figure 6-42, the **User Access Management** window now displays Partition Management as an additional privilege next to the username.

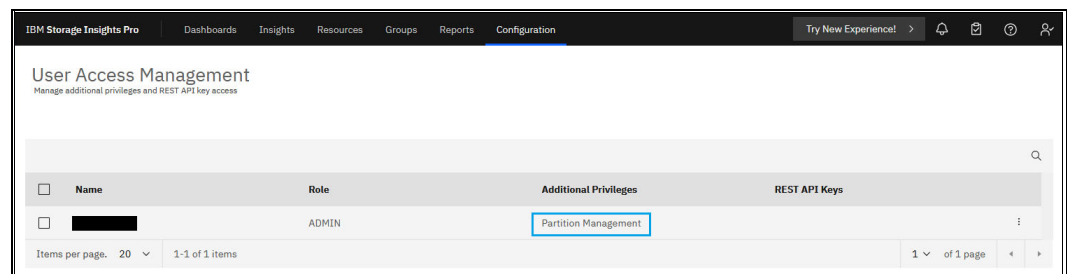


Figure 6-42 The Partition Management permission was granted to the user

Now that you know how to obtain the required permission before initiating a migration, review how the process works. Notifications and tasks in IBM Storage Insights are also available for tracking the migration.

Before you start a migration, ensure that you have the required permissions and that control access is enabled on the source and target systems. Verify that the systems meet the version requirements for IBM Storage Insights-initiated migrations.

Note: IBM Storage Insights provides the only UI for initiating and monitoring partition migration of storage partitions (supported from IBM Storage Virtualize 8.7.0 onward). Automatic host migration support with storage partitions was introduced in IBM FlashSystem 8.7.2.

The phases of a storage partition migration that is initiated from IBM Storage Insights in an IBM FlashSystem grid include the following steps:

1. Initiate a migration.
2. Monitor the migration.
3. Use control commands for switch primary, host rescan, and commit.
4. View notifications and tasks in IBM Storage Insights for migration tracking.
5. Review dump files and errors during a migration.
6. Address monitoring challenges across multiple systems.
7. Apply recommended best practices from IBM Storage Insights.

6.2.2 Initiating a migration by using IBM Storage Insights

The migration process begins by initiating the migration from IBM Storage Insights. This section describes the steps that are required to start the migration and select the appropriate target system.

Note: Run AI Advisory before starting the migration. This tool helps assess whether the selected target system is the most appropriate option based on historical data from the partition and system. For more information, see 6.2.4, “AI-based recommendations and System Health Advisories” on page 160.

To initiate a migration in IBM Storage Insights, complete the following steps:

1. In the Storage partitions window, click the overflow icon and click **Migrate partition** (Figure 6-43 on page 149).

Note: If the **Migrate partition** option is unavailable, verify that the Partition Management permission is granted to the user and that control access is enabled on the source and target storage systems. If the **Migrate partition** option remains unavailable, the action might be unsupported. For example, migrating a DR partition or a 3-site HA partition by using IBM Storage Insights is not possible.

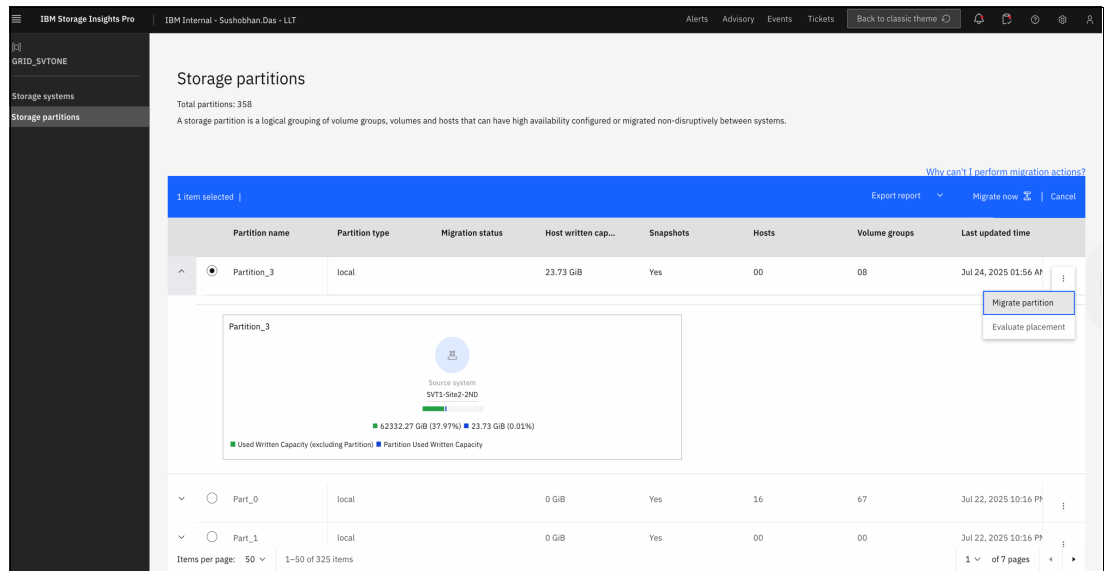


Figure 6-43 Clicking Migration Partition to initiate migration

2. The Migration overview window opens (Figure 6-44). Click **Next**.

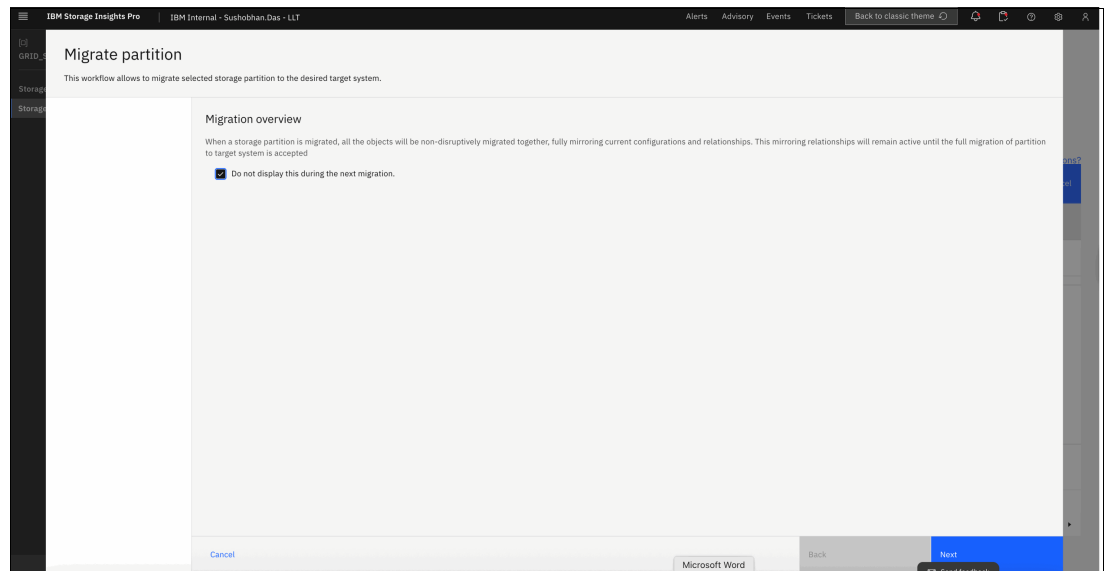


Figure 6-44 Migration overview before starting the migration

3. Select a target system for the partition to be migrated (Figure 6-45).

Note: Only systems with control access that is granted at the IBM FlashSystem level appear in the view.

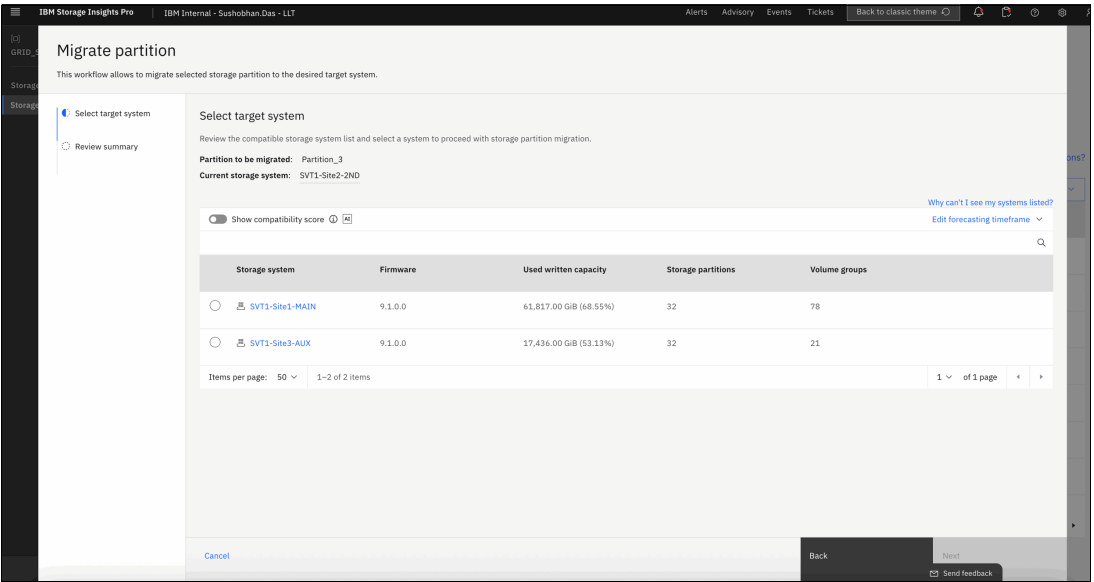


Figure 6-45 Selecting a target system for the partition

4. After the target system is selected, click **Next** to view an overview of the migration plan (Figure 6-46).

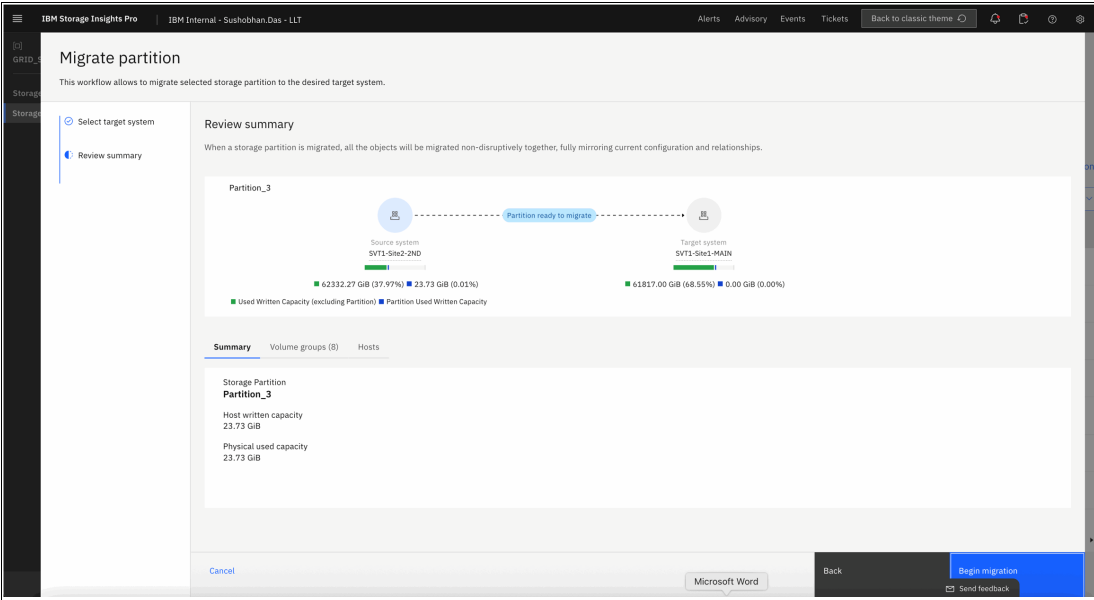


Figure 6-46 Reviewing the migration

5. The review window (Figure 6-46) shows the migration plan as a topology and the status as Partition ready to migrate. After you confirm the plan, click **Begin migration**.

6.2.3 Monitoring the migration

IBM Storage Insights creates a task to start migration to the target system, but does not perform the migration action. Instead, it creates the command and queues it for the storage system to process. The storage system runs the command only if all parameters, the source partition, and the target system are in a valid state when the storage device processes the command.

Because data synchronization between IBM Storage Insights and the device occurs at intervals (typically every five minutes), expect a processing delay of approximately five minutes before the storage device processes any action (Figure 6-47).

The screenshot shows the IBM Storage Insights interface. On the left is a sidebar with navigation options: GRID_SVTONE, Storage systems, and Storage partitions (selected). The main content area is titled 'Storage partitions' and shows 'Total partitions: 358'. A notification box in the top right corner states: 'Migration task created successfully. Storage partition Partition_3 will be migrated from sv15-s1e4 to SVT1-S1e2-WA0N. This process can take a while to complete and its progress can be monitored as a background task during this interval.' Below the notification is a table of storage partitions. The table has columns: Partition name, Partition type, Migration status, Host written cap..., Snapshots, Hosts, Volume groups, and Last updated time. The first row shows 'Partition_3' with a status of 'Request submitted'. Other rows show partitions like 'Part_0', 'Part_1', 'Part_10', 'Part_11', 'Part_12', 'Part_13', and 'Part_14' with various capacities and update times. At the bottom of the table, it says 'Items per page: 50' and '1-50 of 325 items'.

Partition name	Partition type	Migration status	Host written cap...	Snapshots	Hosts	Volume groups	Last updated time
Partition_3	local	Request submitted	23.73 GiB	Yes	00	08	Jul 24, 2025 01:56 AM
Part_0	local		0 GiB	Yes	16	67	Jul 22, 2025 10:16 PM
Part_1	local		0 GiB	Yes	00	00	Jul 22, 2025 10:16 PM
Part_10	local		0 GiB	Yes	00	00	Jul 22, 2025 10:16 PM
Part_11	local		0 GiB	Yes	00	00	Jul 22, 2025 10:16 PM
Part_12	local		0 GiB	Yes	00	02	Jul 22, 2025 10:16 PM
Part_13	local		0 GiB	Yes	00	00	Jul 22, 2025 10:16 PM
Part_14	local		0 GiB	Yes	00	02	Jul 22, 2025 10:16 PM

Figure 6-47 Partition migration task created notification

Notes:

- If a connection loss occurs between the storage device and IBM Call Home server, the command that IBM Storage Insights initiates remains in the submitted state. If the device reconnects within one hour, the command is processed. If the connection is not restored within that time, the command task times out and the migration action is marked as failed in the Task view of IBM Storage Insights.
- If a notification is missed and information about the request and response is needed, download the audit log in IBM Storage Insights by selecting **Settings** → **Export Audit Log**, and select **Review the request and response for the action that is taken by IBM Storage Insights related to partition migration**.

To monitor the progress and timestamps of the migration, click **Migration Status**, where “Request submitted” is displayed (Figure 6-48).

Note: Pay close attention to the Updates window. This section tracks the progress of the migration, including timestamps and any changes in status.

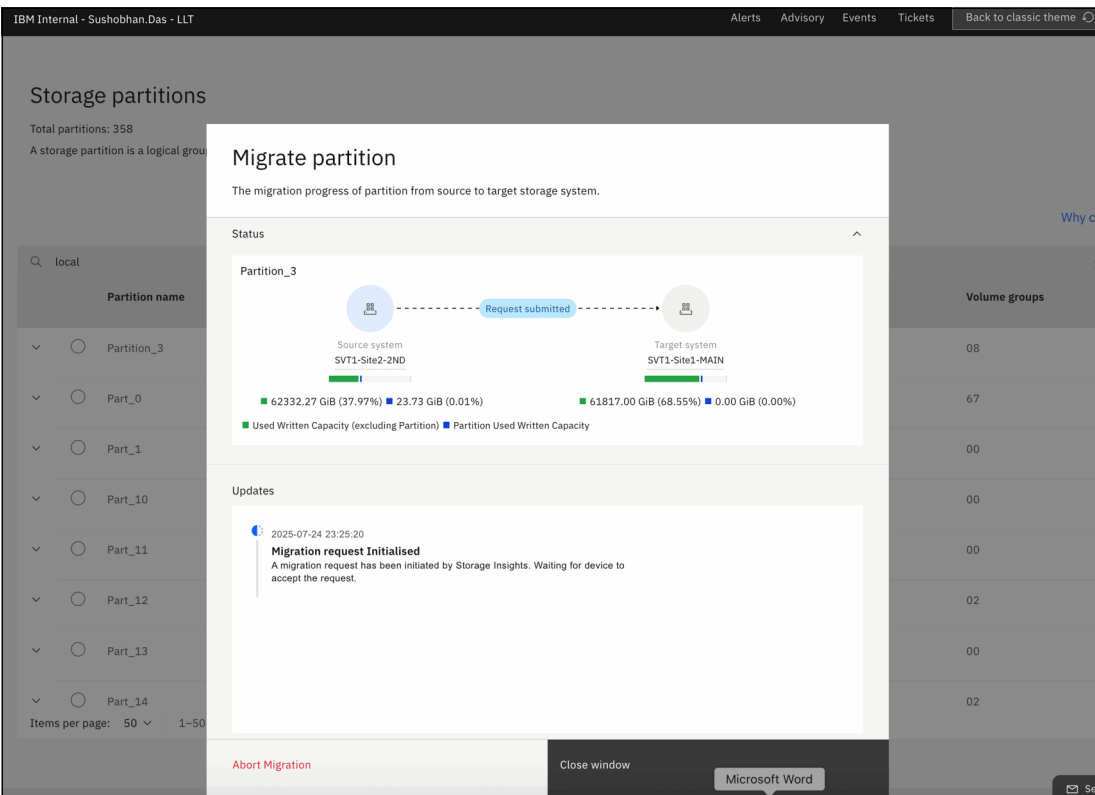


Figure 6-48 Migration status

When device accepts the migration command, the migration status changes from Request submitted to “In progress”, as shown in (Figure 6-49 on page 153).

Note: The migration status that is displayed in IBM Storage Insights is based on the status of the partition object on the storage device. Updates depend on the probe processing intervals within IBM Storage Insights.

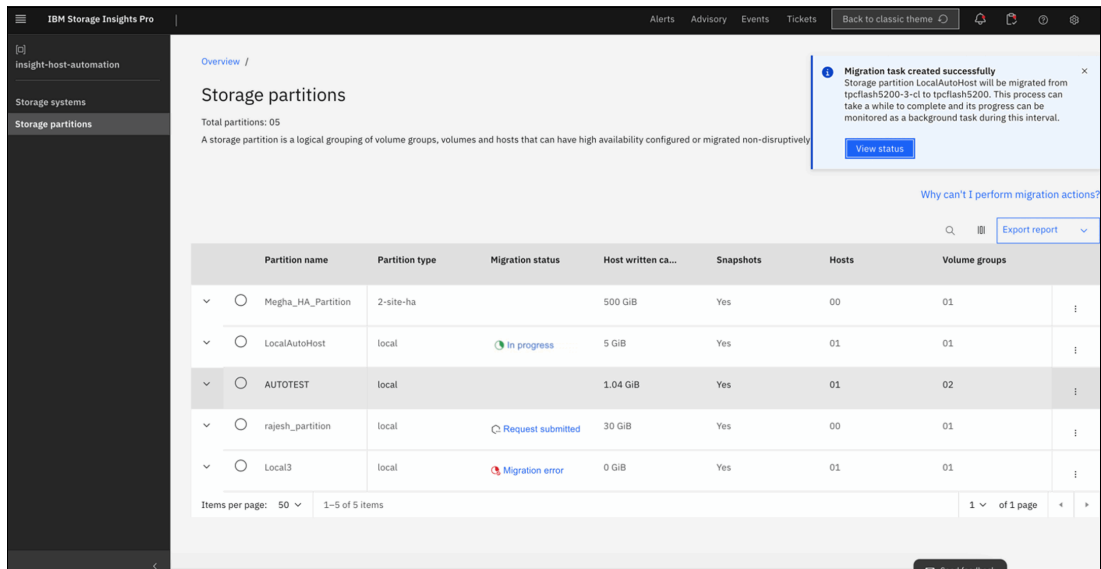


Figure 6-49 Migration in progress

Selecting the migration status displays detailed information about the migration, including timestamps and any changes in status, as shown in Figure 6-50.

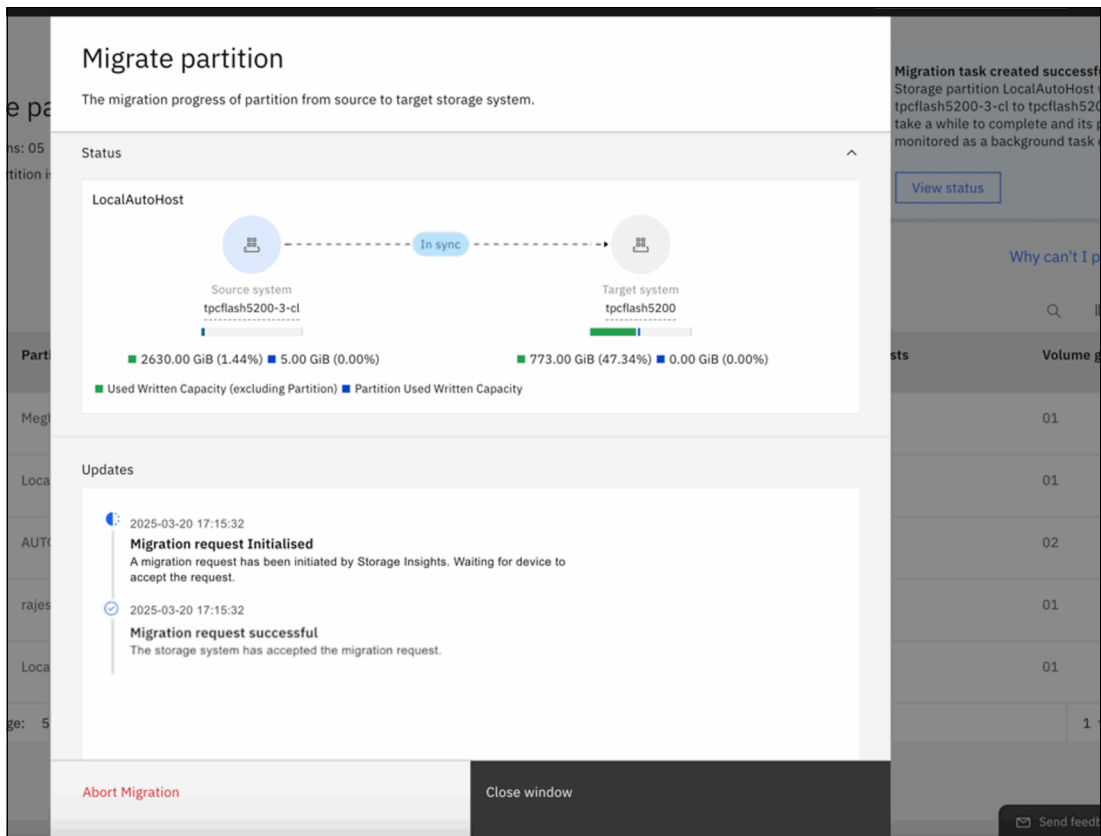


Figure 6-50 Migration details

Migration can also be monitored by selecting **Overview** → **Migration Status** on the IBM FlashSystem grid overview window.

Control commands for switch primary

Storage Partition Migration in an IBM FlashSystem grid is a multi-step process that might require user intervention to proceed. A migration configuration requirement or a blocking error can trigger these actions.

To investigate and complete the migration, complete the following steps:

1. Check the migration status and open the status window.

When user intervention is required, the migration status changes to User action required, as shown in Figure 6-51.

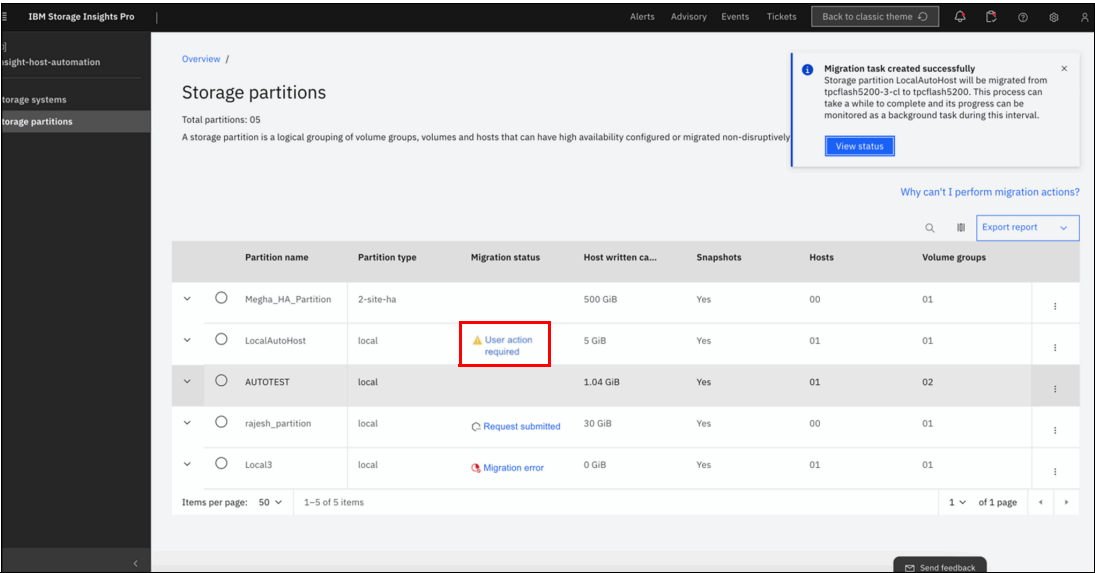


Figure 6-51 User action required

2. Click **View status** to investigate the required user action.

The system opens a window that displays detailed updates, including the specific cause for the User action required status, as shown in Figure 6-52.

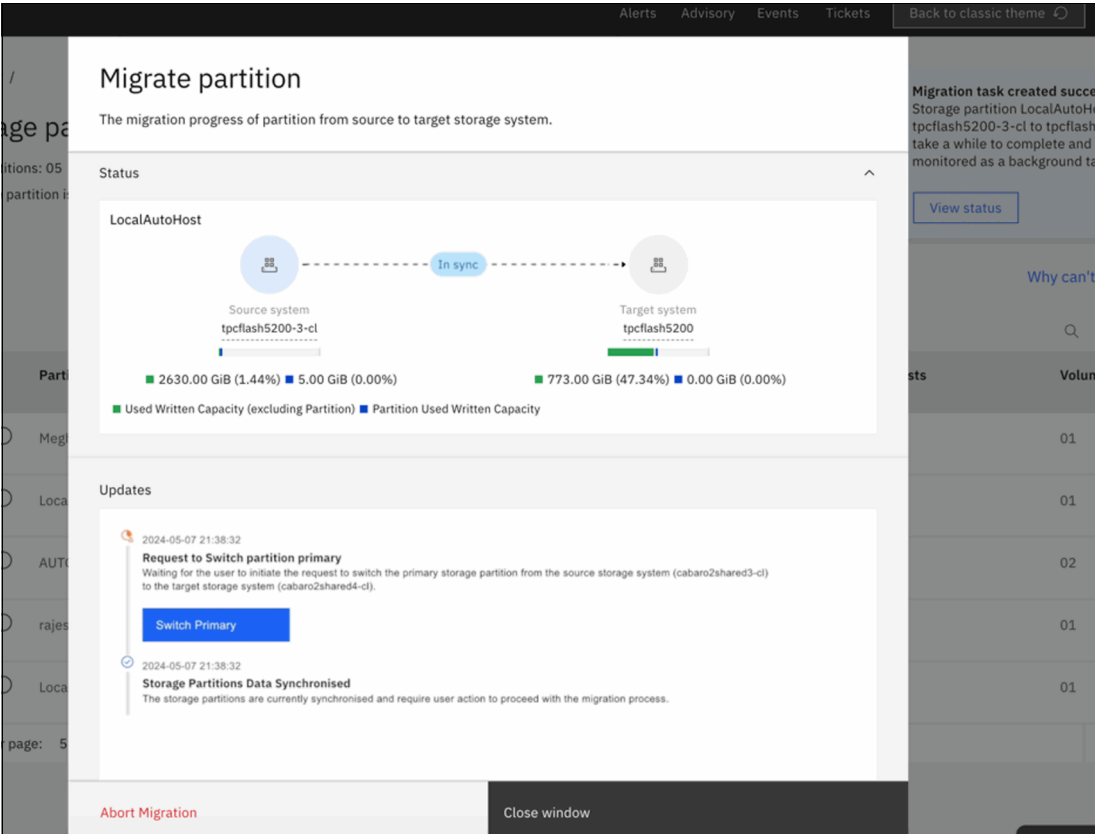


Figure 6-52 Switch primary

3. Switch the primary storage partition.

Review the updates to confirm that the system is awaiting a command to switch the primary storage partition from the source to the target, as shown in Figure 6-53. Then, click **Switch primary**. A confirmation notification appears in the upper right of the window.

Notes:

- ▶ Before IBM Storage Virtualize 8.7.2, the first migration action was Switch primary. This user action indicated that the data on both sites was synchronized. The user was required to ensure that host connections from the target storage system to partition-related volumes were present before switching the Partition Management system from source to target. This approach helps ensure that switching does not interrupt I/O if sessions on the target are missing.
- ▶ For IBM Storage Virtualize 8.7.2 and later, the switch of the active management system from source to target happens automatically without user intervention. When an active session exists on the target, the target storage system raises a user action to confirm that the rescan is complete.

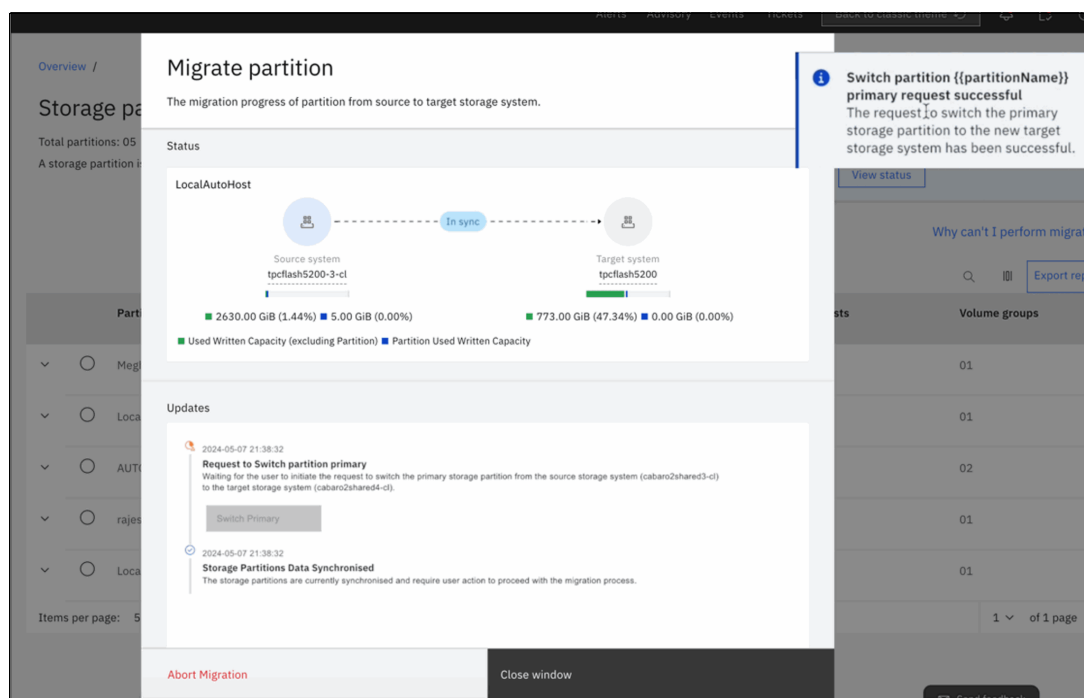


Figure 6-53 Switch primary window

4. Complete the migration by committing the changes.

Any command that starts during migration follows the same mechanism. The storage device accepts the command, and the appropriate status appears in IBM Storage Insights after the probe is processed. After the management system in the storage partition switches from source to target, the migration continues with the next steps. As the migration nears completion, the status changes back to User action required.

5. Check the status again, and then click **Commit** to finalize the migration, as shown in Figure 6-54.

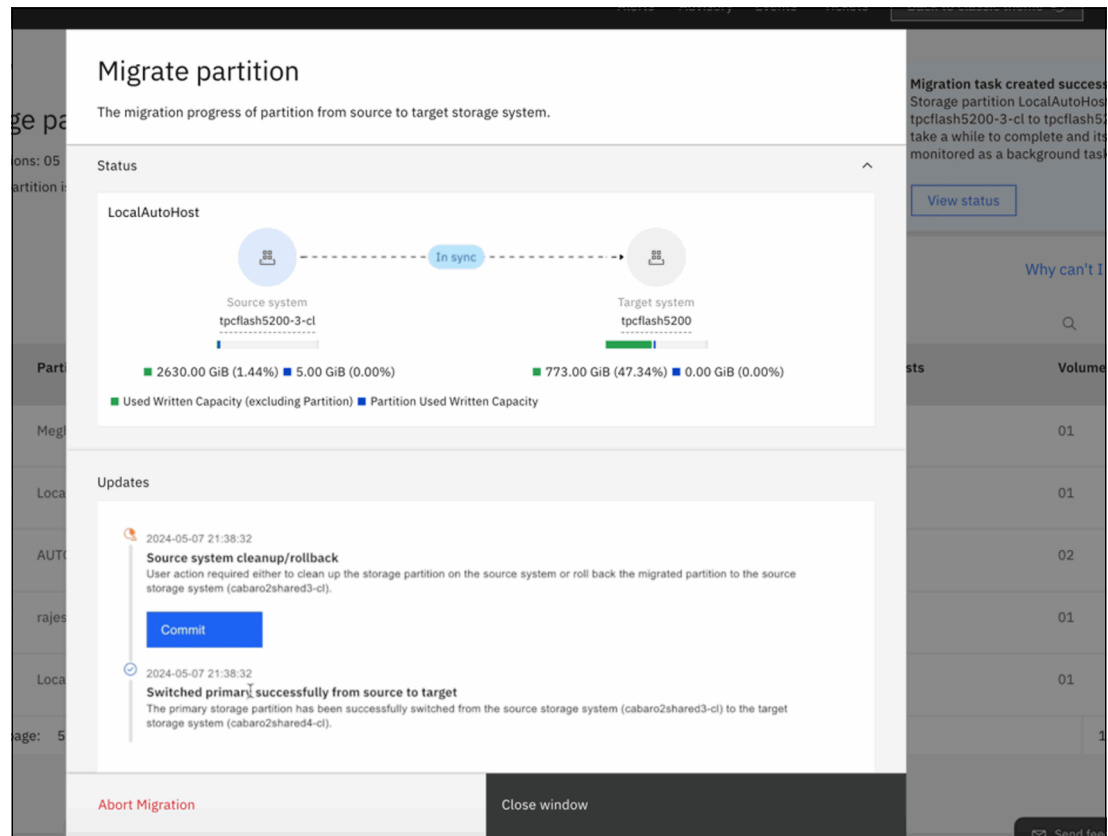


Figure 6-54 Committing the migration

6. Confirm the committed action and close the window.

A confirmation notification acknowledges the commit action, which cleans up the source partition to finalize the migration. Review the confirmation notification, as shown in Figure 6-55. Close the window.

Note: The cleanup process might take some time, especially if snapshots exist for any of the volume groups within the storage partition. These snapshots cannot be automatically deleted during the cleanup phase. In such cases, both the source and target objects within the storage partition remain present. Wait for the cleanup to finish, or manually delete any snapshots that are not safeguarded to allow the migration to complete.

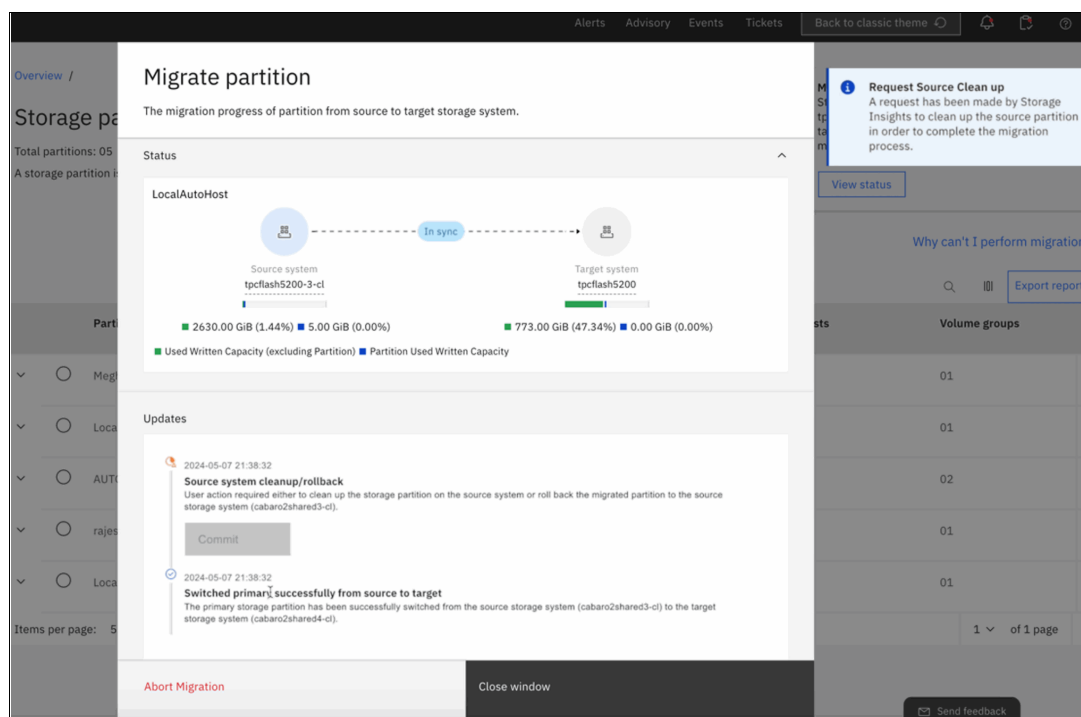


Figure 6-55 Committing the migration: Cleanup message

7. The migration completes successfully after the cleanup finishes. Click **View status** again to see the updates, as shown in Figure 6-56.

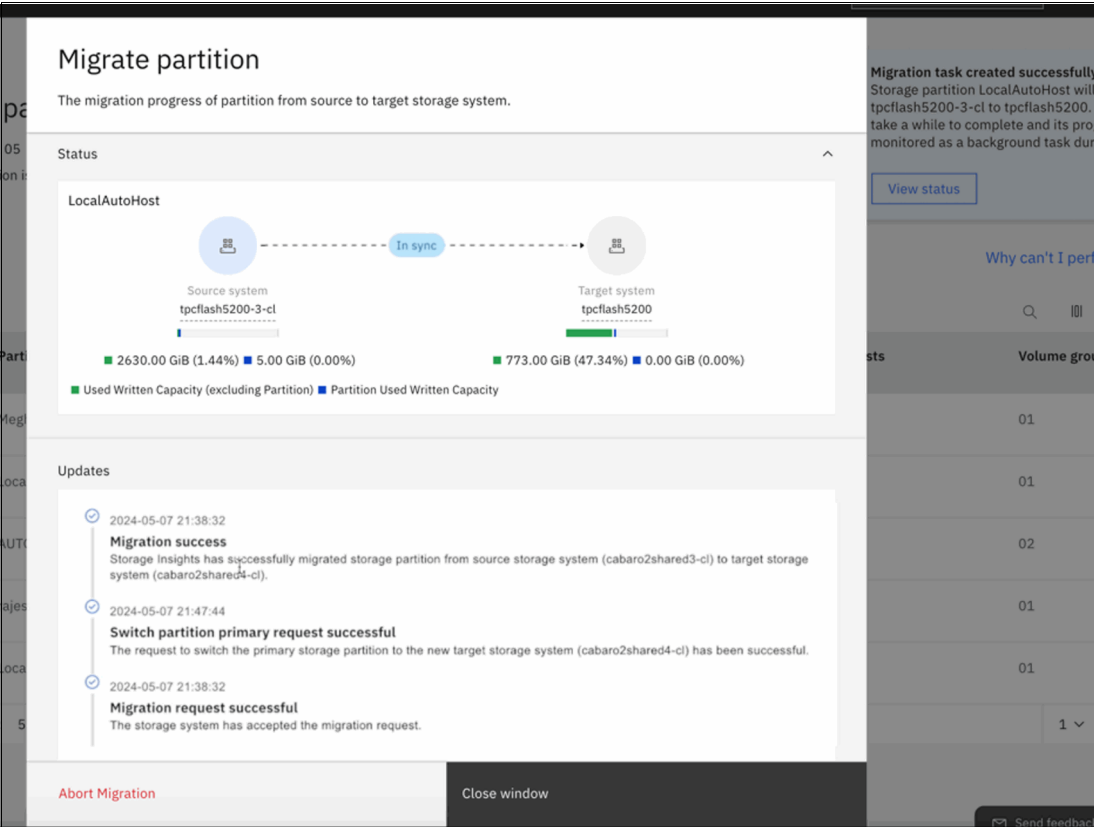


Figure 6-56 Migration successful

The partition list window now shows no status because the migration completed successfully (see Figure 6-57).

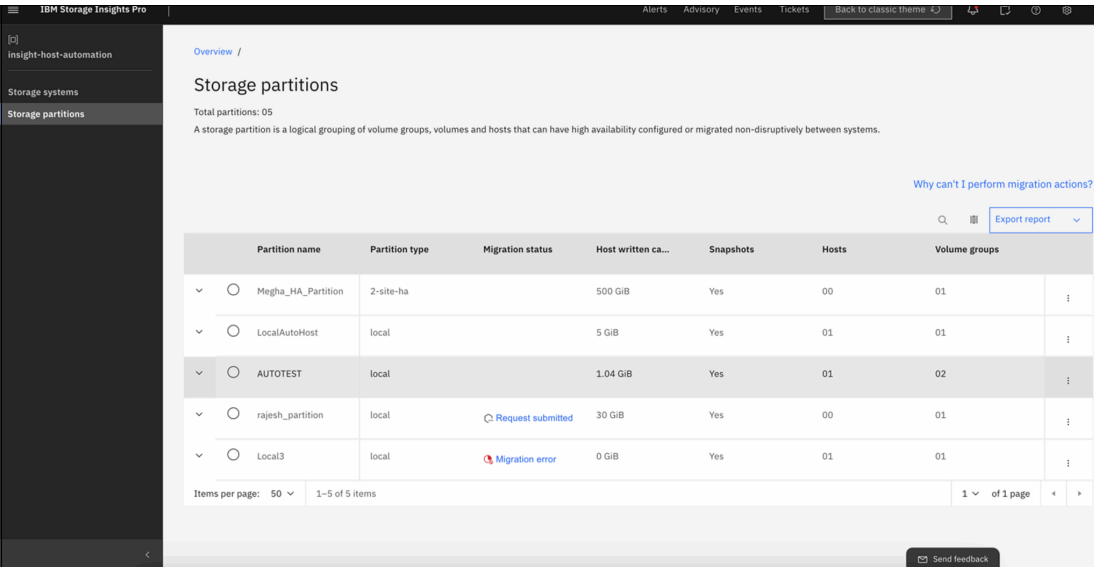


Figure 6-57 Partition list window

6.2.4 AI-based recommendations and System Health Advisories

IBM Storage Insights provides advisories that are designed to help users identify issues, understand their impact, and take informed actions. These advisories include health checks that monitor the overall condition of storage systems and offer actionable guidance.

Similarly, AI-driven suggestions help with operational decisions, such as storage partition migration, based on real-time analysis of system capacity, maintenance schedules, and performance metrics. These AI-driven suggestions apply only to block storage systems within an IBM FlashSystem grid.

This section focuses on the System Health Advisory capabilities for IBM FlashSystem grids.

Conditions of AI suggestions

AI suggestions are generated as part of the post-processing that occurs after daily inventory data collection is completed. Currently, IBM Storage Insights supports AI suggestion for these conditions:

- ▶ **Warranty and Maintenance.** An AI suggestion (System warranty expired or expiring soon) is displayed for a block storage system that is part of an IBM Storage FlashSystem grid when the warranty or maintenance is scheduled to expire within the next 30 days.
- ▶ **Capacity.** An AI suggestion (Insufficient storage capacity to sustain growth) is displayed for a block storage system that is part of an IBM FlashSystem grid when the forecasted used capacity is expected to reach or exceed 80% of the total capacity within the next 30 days.
- ▶ **Performance.** An AI suggestion (High CPU usage detected) is generated for a block storage system when it is part of an IBM FlashSystem grid and the 95th percentile of CPU usage is greater than equal to 75% during the past 2 days.

For block storage systems that experience a ransomware threat, AI suggestions are not generated, and any existing AI suggestions are removed. A ransomware threat is a critical alert that must be resolved first, as it can affect the data points used for to evaluate AI suggestion criteria.

After each probe completion, AI suggestions are revalidated. If the condition is no longer met, the suggestion is deleted.

Accessing AI suggestion

The AI suggestions are accessible in the following windows:

- Overview window (L2 window) for block storage systems (see Figure 6-58). The AI suggestion count shows the total number of current AI suggestions that are generated for block storage systems under this tenant.

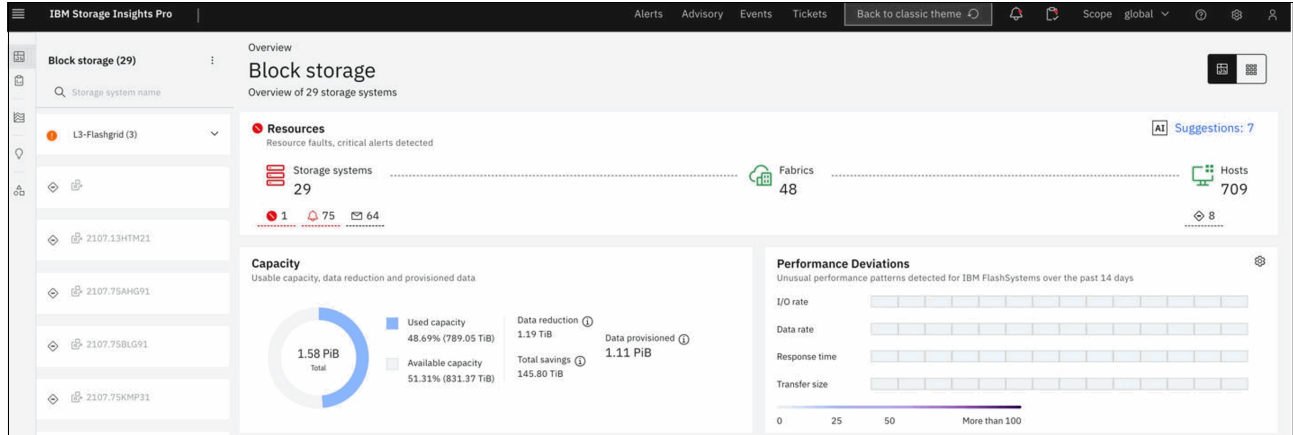


Figure 6-58 Block storage overview AI suggestion

- Overview window of the IBM FlashSystem grid (see Figure 6-59). The AI suggestion count shows the total number of current AI suggestions that are generated for block storage systems under this IBM FlashSystem grid.

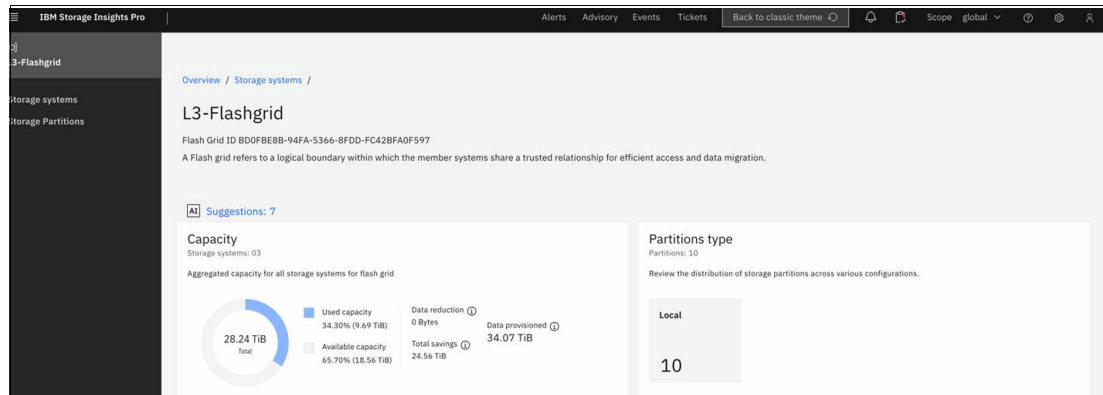


Figure 6-59 IBM FlashSystem grid overview AI suggestion

- Details window of a block storage system (see Figure 6-60). The AI suggestion count shows the total number of current AI suggestions that are generated for this block storage system.

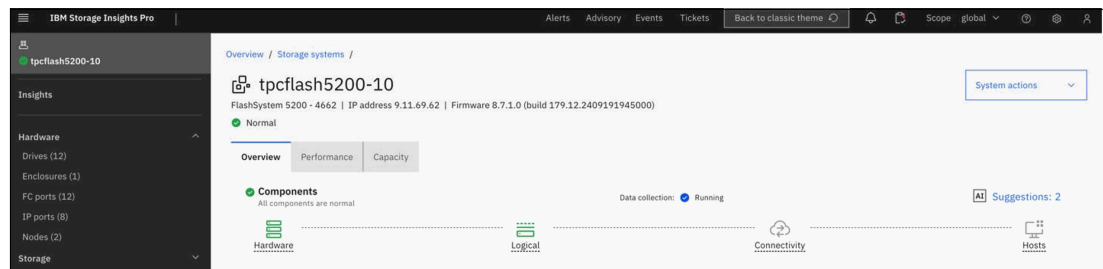


Figure 6-60 IBM FlashSystem storage overview AI suggestion

Summary view of AI suggestions

IBM Storage Insights provides a centralized summary view of AI suggestions to help users quickly assess the health and operational status of their block storage systems within an IBM FlashSystem grid. These AI-driven insights are generated daily based from inventory data and highlight key areas that require attention, such as warranty status, capacity trends, and performance metrics.

This summary enables IT administrators to prioritize actions, streamline maintenance, and proactively address potential issues before they impact system availability or performance.

Accessing the AI suggestion count from any location opens the right pane, which displays a summary of the AI suggestions, as shown in Figure 6-61.

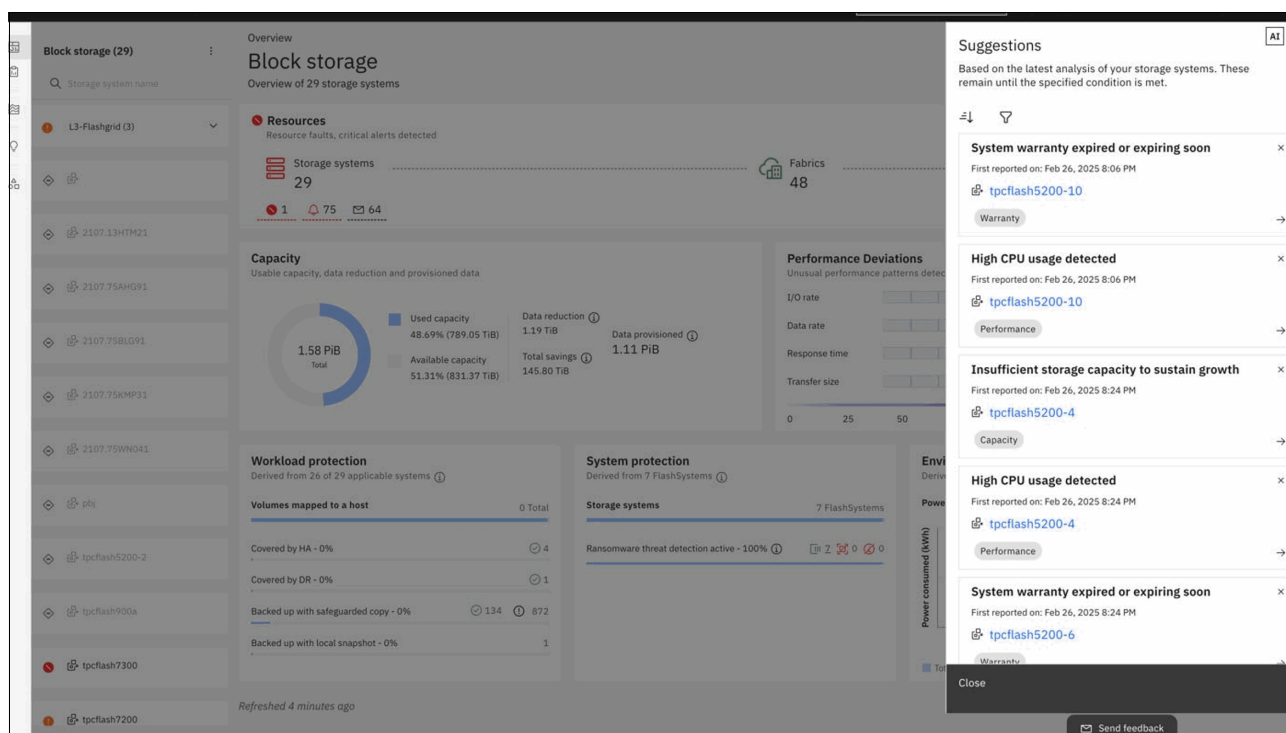


Figure 6-61 Summary view of AI suggestions

The AI suggestion list summary view is available from the individual IBM FlashSystem details window, as shown in Figure 6-62.

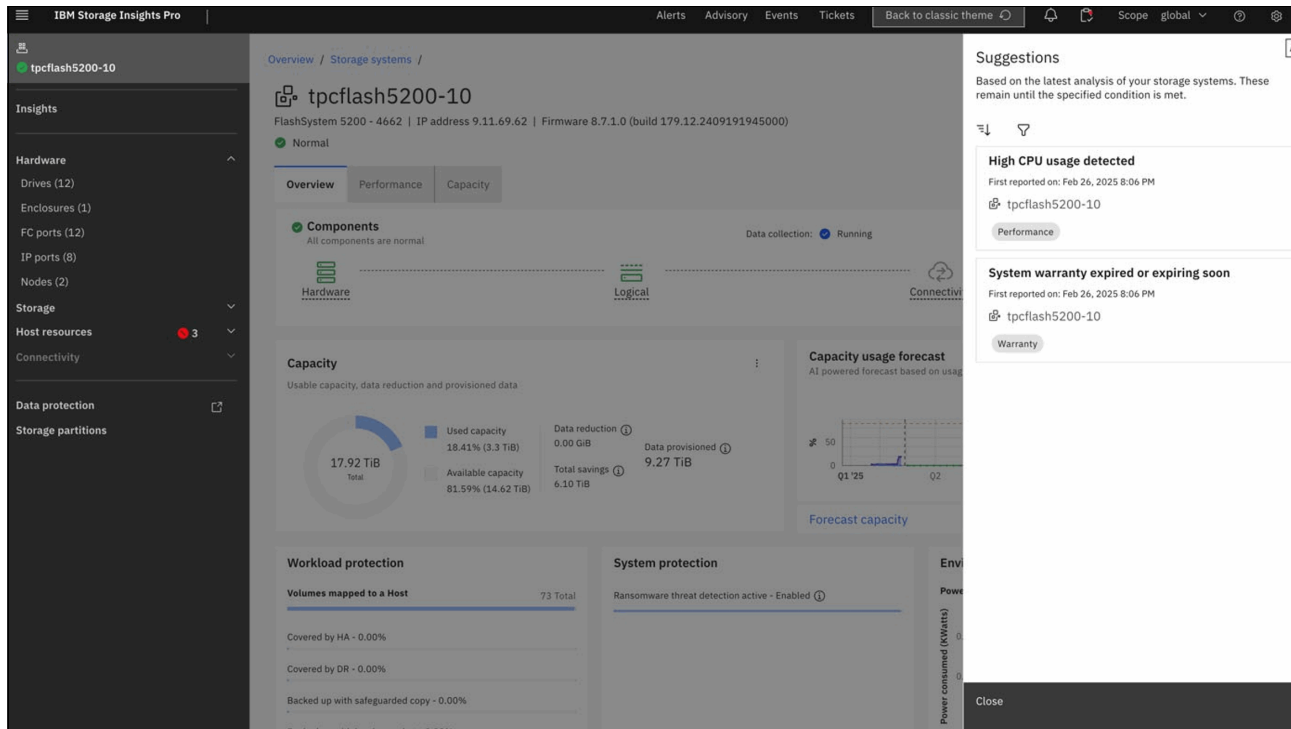


Figure 6-62 Summary view of AI suggestion for individual IBM FlashSystem units

The summary list view enables users to sort AI suggestions by creation date and category. This function helps users organize and prioritize suggestions based on when they were generated or the type of condition that they address, such as warranty, capacity, or performance.

Detail view of AI suggestions

Opening an AI suggestion displays its detailed view. The **Overview** tab analyzes the AI suggestion and includes a link for deeper understanding of the condition. The **Recommendation** tab lists recommendations relevant to the AI suggestion and provides links to available actions.

Maintenance AI suggestions

Figure 6-63 shows an example of an AI suggestion for a storage system with maintenance that expired or is scheduled to expire within 30 days.

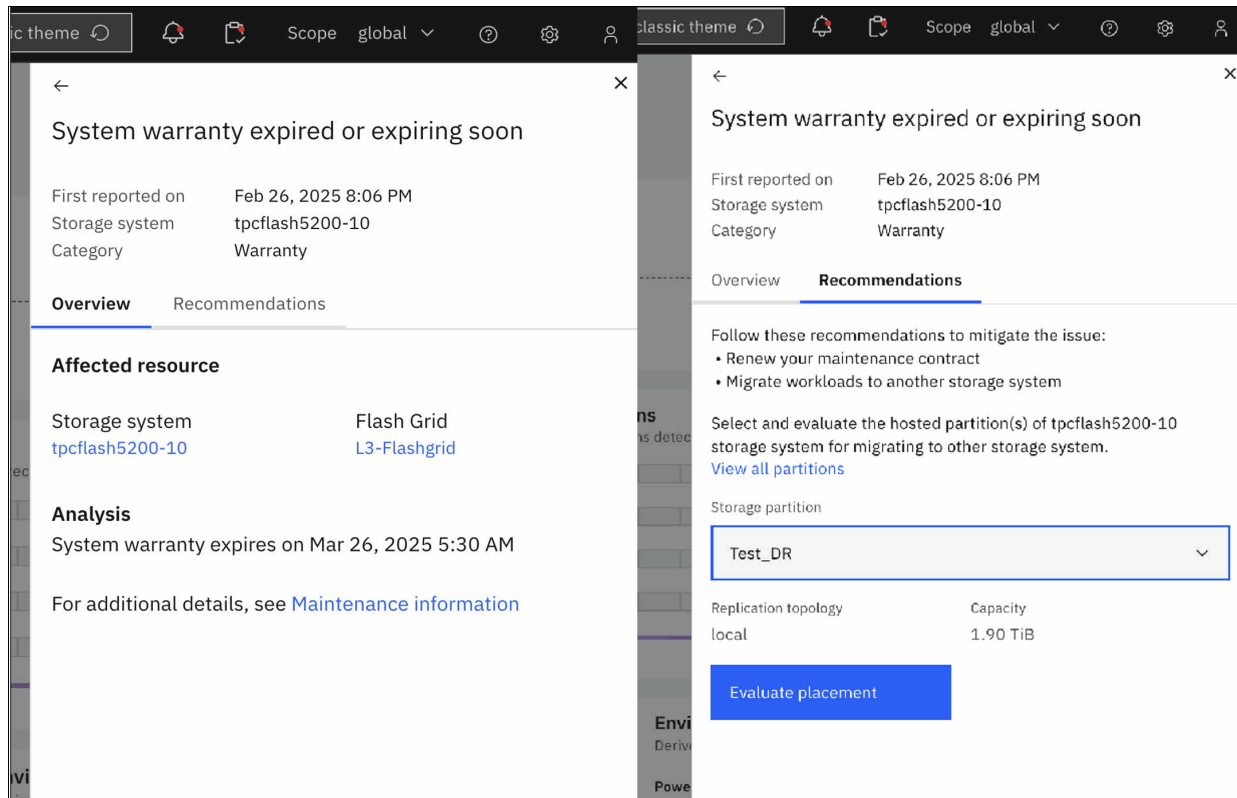


Figure 6-63 Detail view of a maintenance AI suggestion

AI-generated recommendations offer multiple ways to address identified issues. For example, if a system's maintenance contract is expiring, these primary options are available:

- ▶ Renew the maintenance contract to continue using the current storage system with valid IBM Support.
- ▶ Migrate workloads to another storage system that has an active maintenance contract to help ensure continued support and security for critical workloads.

To assist with workload migration decisions, the Evaluate Placement feature in IBM Storage Insights can be used to assess the suitability of alternative systems based on capacity, performance, and support status.

Capacity AI suggestions

Figure 6-64 shows an example of an AI suggestion for a storage system with forecasted capacity that is expected to reach 80% within the next 30 days.

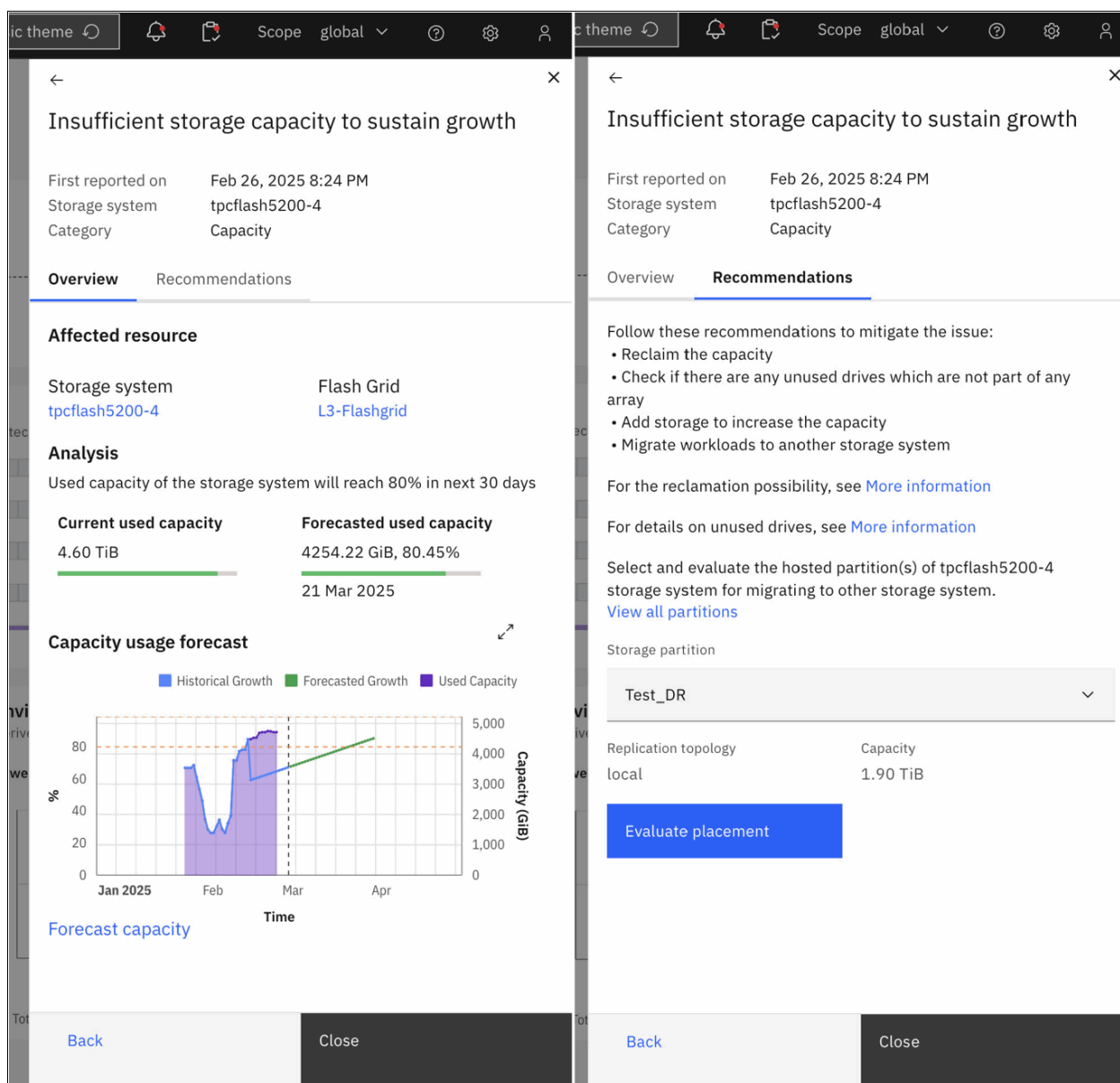


Figure 6-64 Detail view of a capacity AI suggestion

AI-generated recommendations provide several approaches to resolve capacity-related issues. The following actions can help resolve the issues:

- ▶ Use the Reclamation feature to identify and recover reclaimable volumes and unused storage.
- ▶ Add more storage to increase overall system capacity.
- ▶ Migrate workloads to another storage system with greater available capacity to help ensure continued performance and security for critical workloads.

To support migration decisions, users can use the Evaluate Placement feature in IBM Storage Insights. This tool helps assess alternative systems based on capacity, performance, and support status, enabling informed and efficient workload placement.

Performance AI suggestions

Figure 6-65 is an example of an AI suggestion that appears when CPU usage reaches 75% for the 95th percentile of sample data that is collected over the past 2 days.

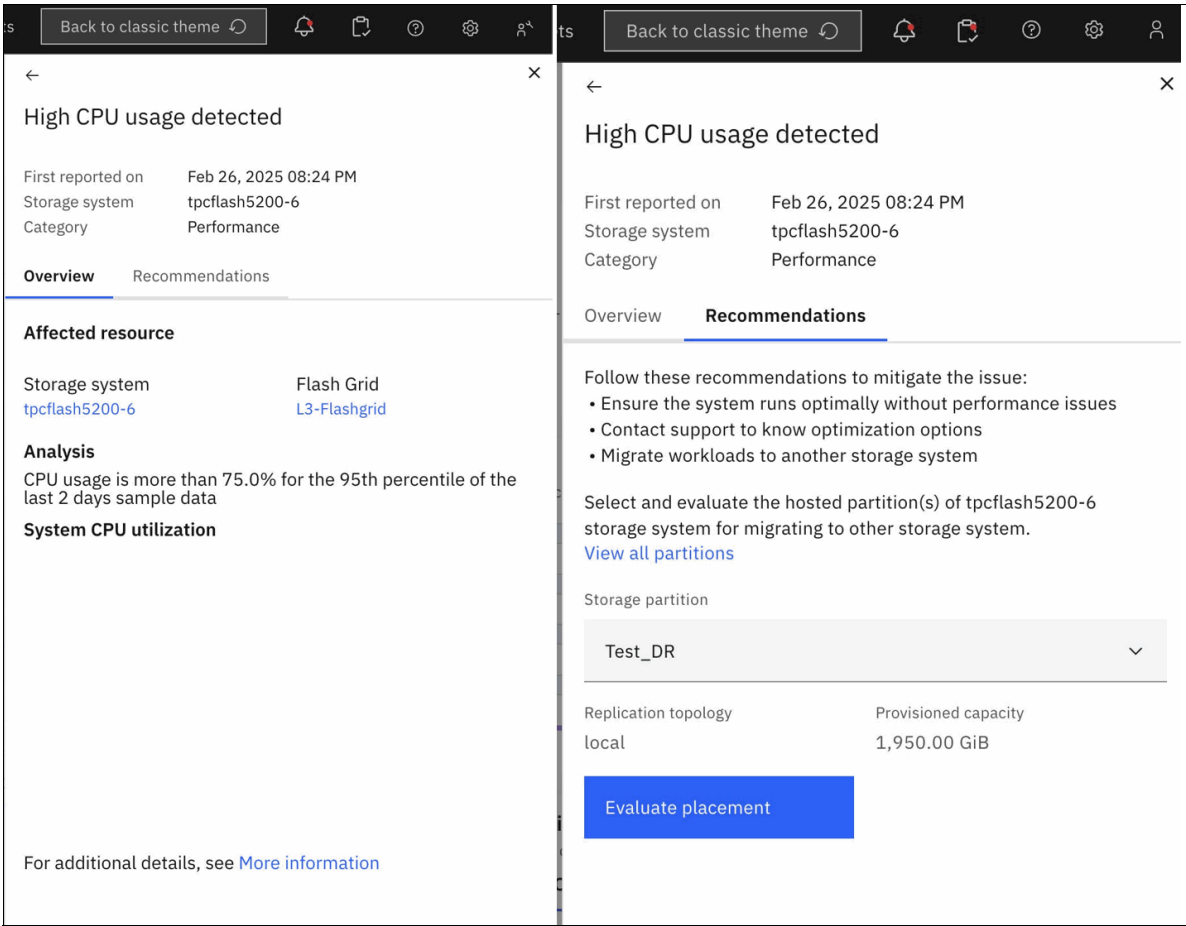


Figure 6-65 Detail view of a performance AI suggestion

Recommendations offer various ways to resolve the issue. Users must implement appropriate solutions to help ensure that the system operates without performance problems. Alternatively, users can migrate workloads from the current storage system to another system with greater processing power to safeguard the workloads. To assist with this decision, users can use the Evaluate Placement feature in IBM Storage Insights.

6.2.5 Placement Advisory for Storage Partitions

IBM Storage Insights supports storage partition migration between storage systems within an IBM FlashSystem grid. A *storage partition* is a logical grouping of resources that represents a workload. Migrating a storage partition typically requires careful planning to help ensure that the workload remains uninterrupted.

One of the key advantages of using storage partitions is that IBM FlashSystem migration technology supports live workload migration with zero application downtime. Although this capability is powerful, it also introduces complexity, particularly in selecting a suitable target system that can sustain the workload over time without performance degradation.

To address this need, IBM Storage Insights offers the Placement Advisory for Storage Partitions. As explained earlier, this feature helps administrators plan migrations by analyzing historical data and forecasting future growth to ensure that the selected target system is both compatible and sustainable.

When an administrator initiates a migration from within IBM Storage Insights, the system provides AI-recommended target storage systems. These recommendations include detailed explainability, similar to what was described in the previous section. As shown in Figure 6-66, administrators can enable Compatibility Score to view AI-based recommendations and make informed decisions about the target system for migration.

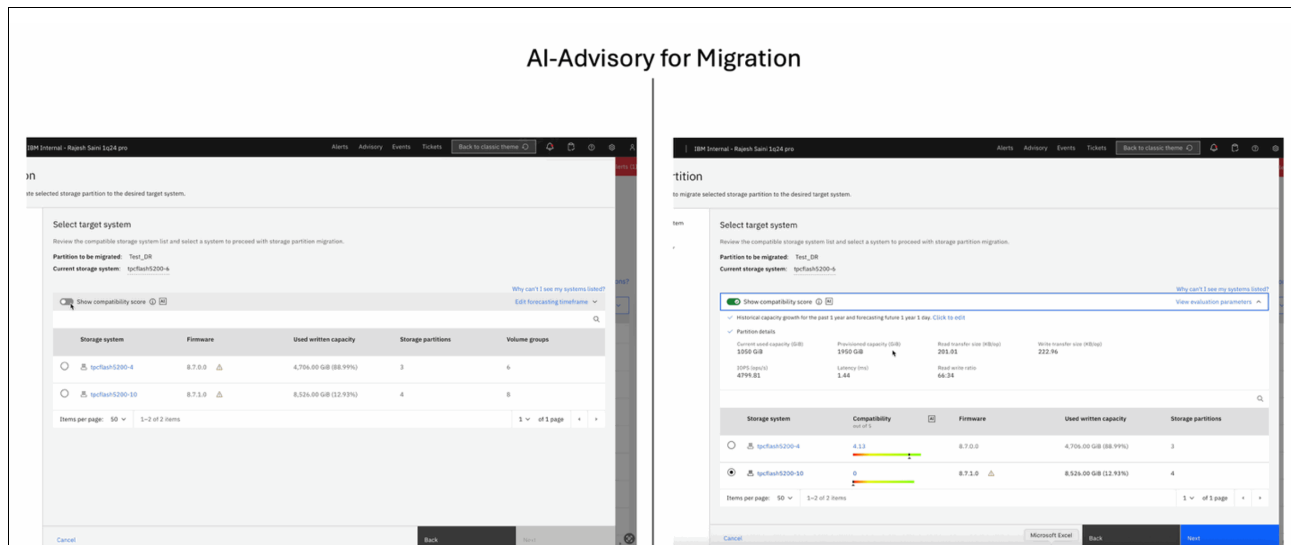


Figure 6-66 Migration partition with AI Advisory

Note: IBM Storage Insights also enables AI-advisory recommendations for selecting a target storage system when an IBM FlashSystem partition migration scenario is triggered from the IBM FlashSystem UI. This capability is available starting with IBM Storage Virtualize 9.1.0.

Apart from the user-initiated workload advisory, IBM Storage Insights also monitors resources within the IBM FlashSystem grid and provides recommendations to users based on the system performance and storage needs. These recommendations apply to resources that are part of the IBM FlashSystem grid. The advisories appear on the Overview Dashboard and IBM FlashSystem Grid Dashboard, as shown in Figure 6-67.

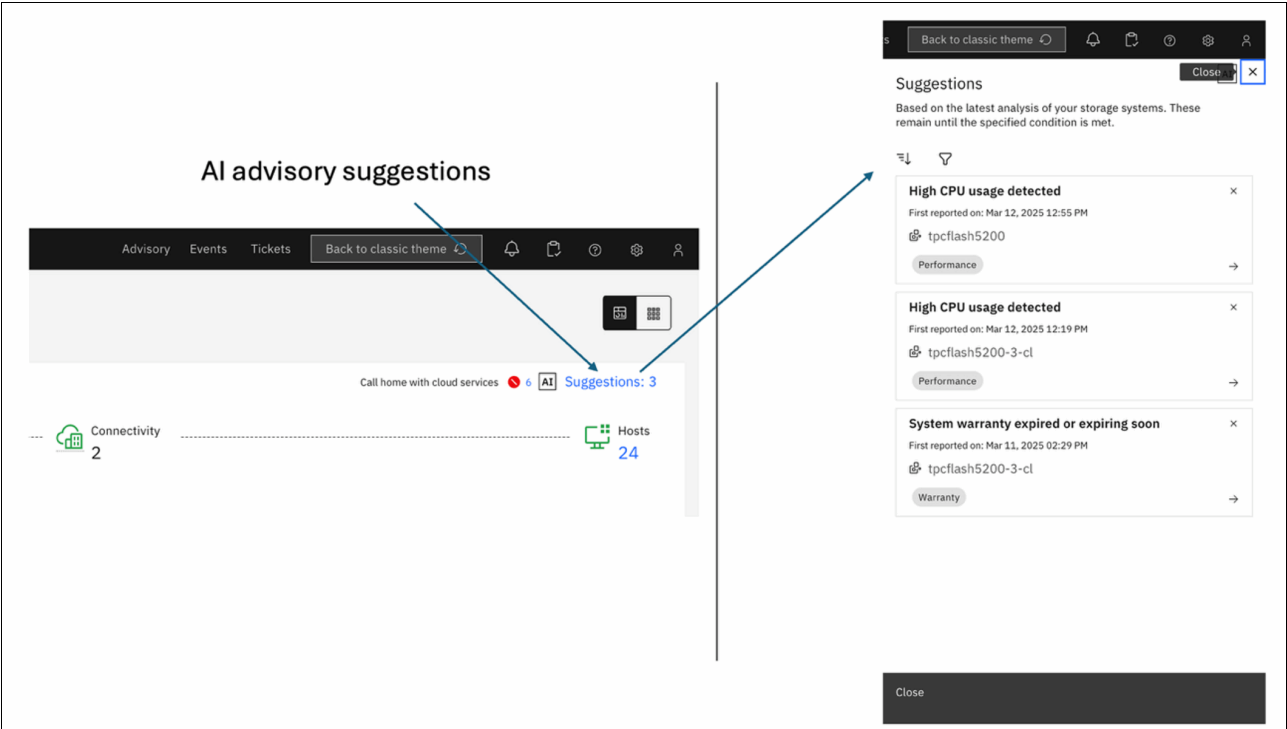


Figure 6-67 AI-Advisory suggestions on overview dashboard

For more information about AI-Advisory recommendations, see [IBM Storage Insights AI-Workload Advisory](#).

6.2.6 AI-Advisory integration with IBM FlashSystem Element Manager

The AI-Advisory capability of IBM Storage Insights is also available within the IBM FlashSystem Element Manager, enabled through integration with the IBM Storage Insights REST API. The IBM FlashSystem Element Manager UI now includes an option to display system-level AI advisories.

This view is available only for systems that are part of an IBM FlashSystem grid, and it is accessible exclusively through the IBM FlashSystem grid UI console within Element Manager.

As shown in Figure 6-68 on page 169, this integration allows administrators to view AI-driven insights directly within the system management interface, enhancing visibility and decision making at the system level.

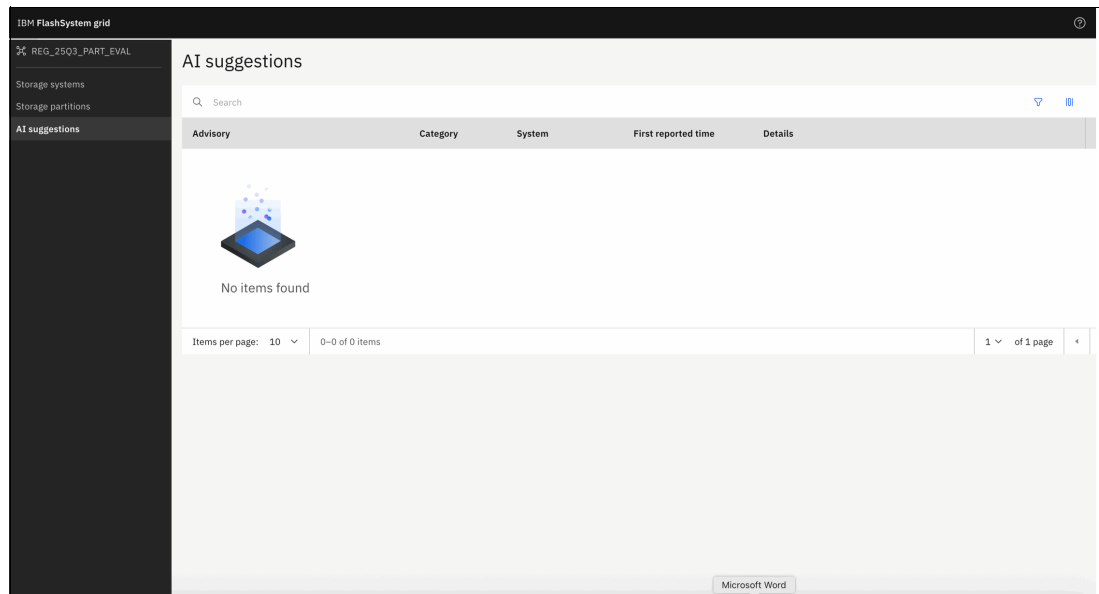


Figure 6-68 IBM FlashSystem grid AI Suggestions view

The IBM FlashSystem grid, which is powered by the IBM Storage Insights REST API infrastructure, enables users to perform partition placement advisories for any partition within the grid, as shown in Figure 6-69.

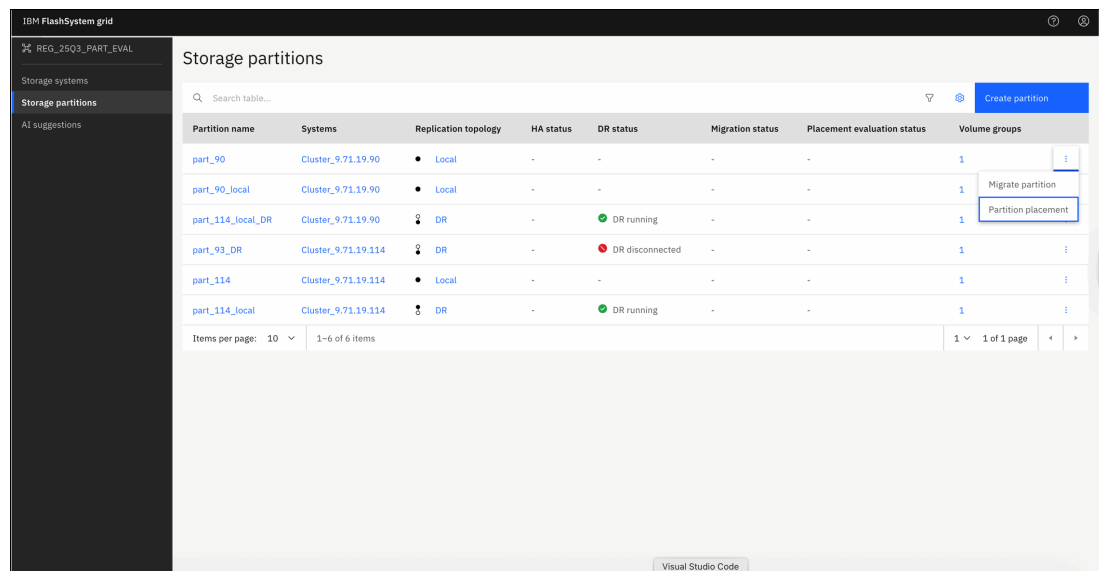


Figure 6-69 IBM FlashSystem grid storage partition evaluation for placement

Note: This feature is available only for partitions whose associated systems are registered under an IBM Storage Insights Pro tenant.

When a user initiates an evaluation request, the IBM FlashSystem UI starts the IBM Storage Insights REST API to determine the most suitable target storage system for migration. This evaluation is based on historical performance, capacity trends, and system health data collected by IBM Storage Insights.

This capability allows administrators to pre-assess migration options independently, without immediately initiating the migration process. It helps ensure that the selected target system can support the workload effectively over time, minimizing risk and maximizing performance continuity.

The migration evaluation process can be time-consuming. Several factors influence the process, including the number of systems that are added to the IBM FlashSystem grid and how many of those systems are visible to IBM Storage Insights. Only systems that are visible to IBM Storage Insights have the necessary historical data that is needed for accurate trend prediction and for evaluating potential target systems for partition migration.

When a user clicks **Partition Placement**, it initiates a flow in which a request is sent to IBM Storage Insights to evaluate the best-fit target system for migration. This process is shown in Figure 6-70.

This evaluation can be performed independently of the migration. It allows administrators to assess the suitability of target systems in advance by using historical performance and capacity data that IBM Storage Insights collects.

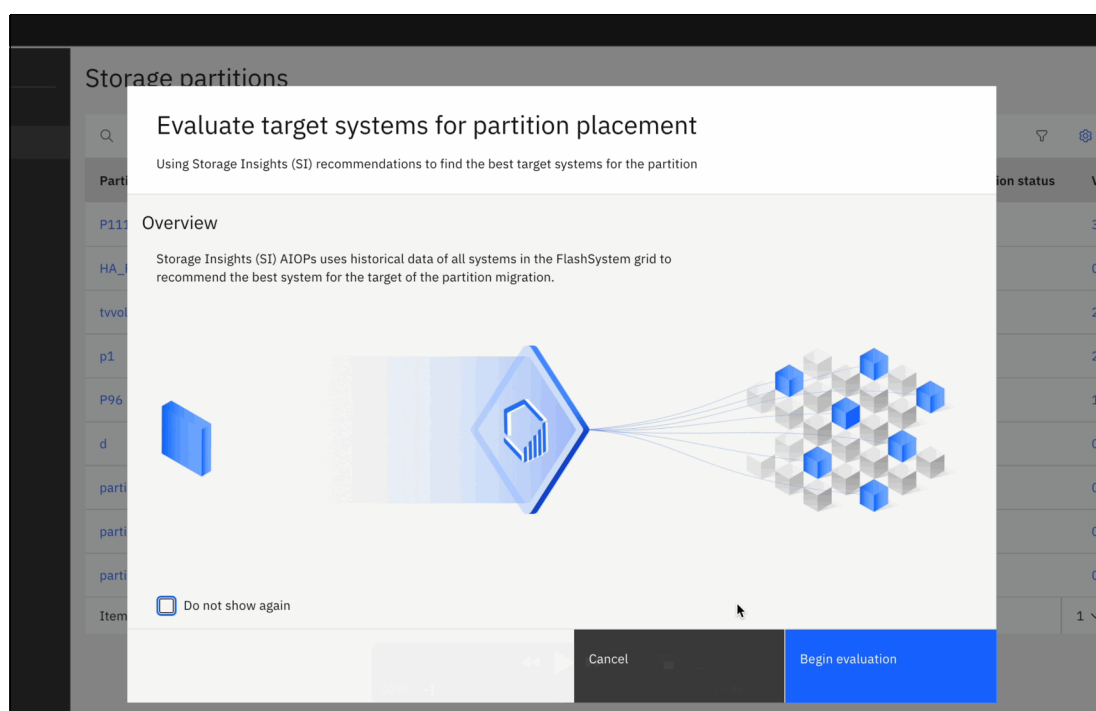


Figure 6-70 Evaluating a partition placement flow in an IBM FlashSystem grid

API key requirement for partition placement evaluation

As described earlier, the IBM FlashSystem grid connects with IBM Storage Insight REST API. Users must provide the API key if one is not added to IBM FlashSystem. The API key from IBM Storage Insights can be generated for partition migration and provided as shown in Figure 6-71 on page 171.

Note: For more information about IBM Storage Insights API key creation, see [IBM Documentation](#).

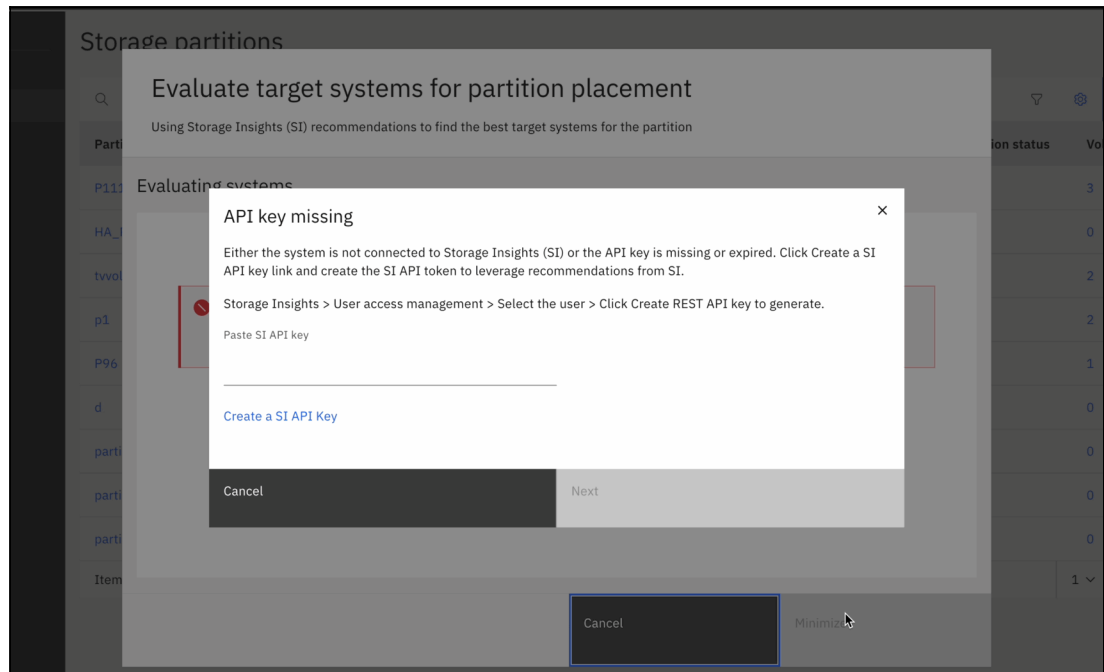


Figure 6-71 API key that is provided for an advisory on placement

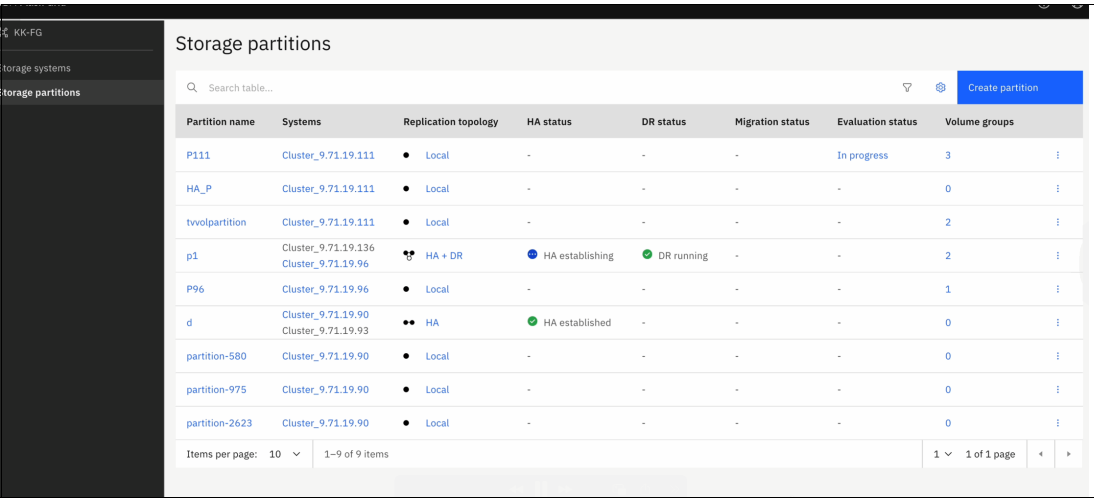
Partition placement evaluation task

When a partition placement advisory starts, IBM Storage Insights creates a task to perform the evaluation. This task uses historical data that IBM Storage Insights collects to analyze the partition's trends in capacity, performance, and partition limits, such as the number of volumes, volume groups, and partitions on the system.

The evaluation considers both the current system that hosts the partition and all potential target systems that meet the following criteria:

- ▶ They are part of the IBM FlashSystem grid.
- ▶ They are added to IBM Storage Insights through the Call Home interface for monitoring.
- ▶ They have sufficient historical data available for trend analysis.

During this process, users can track the evaluation progress directly within the IBM FlashSystem Element Manager UI, as shown in Figure 6-72.



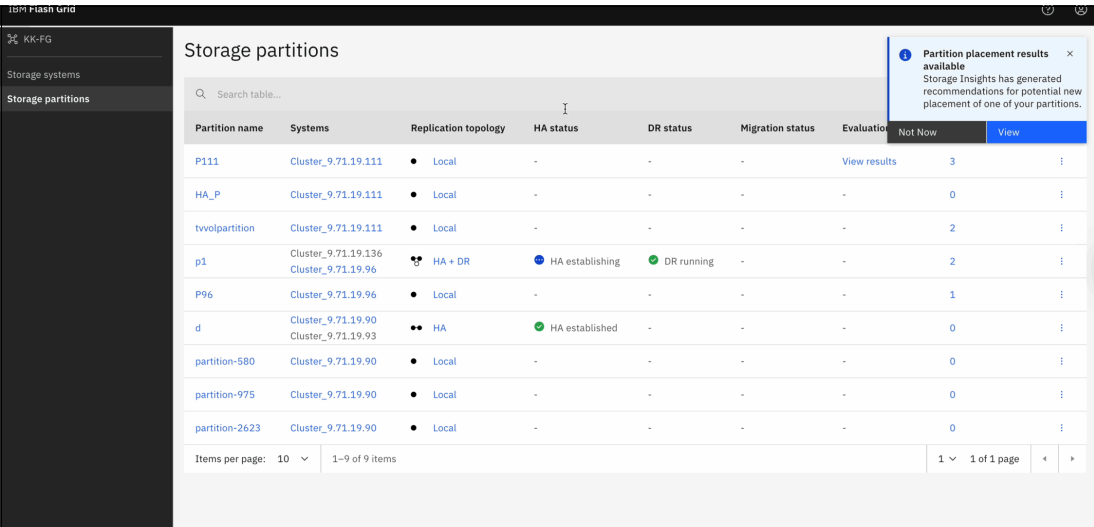
Partition name	Systems	Replication topology	HA status	DR status	Migration status	Evaluation status	Volume groups
P111	Cluster_9.71.19.111	Local	-	-	-	In progress	3
HA_P	Cluster_9.71.19.111	Local	-	-	-	-	0
tvvolpartition	Cluster_9.71.19.111	Local	-	-	-	-	2
p1	Cluster_9.71.19.136 Cluster_9.71.19.96	HA + DR	HA establishing	DR running	-	-	2
P96	Cluster_9.71.19.96	Local	-	-	-	-	1
d	Cluster_9.71.19.90 Cluster_9.71.19.93	HA	HA established	-	-	-	0
partition-580	Cluster_9.71.19.90	Local	-	-	-	-	0
partition-975	Cluster_9.71.19.90	Local	-	-	-	-	0
partition-2623	Cluster_9.71.19.90	Local	-	-	-	-	0

Figure 6-72 Tracking the progress of a partition placement evaluation

Tracking a migration evaluation status

IBM FlashSystem uses the same REST API framework to track the status of a migration evaluation request that is sent to IBM Storage Insights. Both the IBM FlashSystem storage system and IBM Storage Insights must maintain a healthy Call Home connection. If connectivity is disrupted, the REST API cannot retrieve accurate status updates.

When IBM Storage Insights completes the partition placement evaluation by using historical partition data, the results return to the IBM FlashSystem grid UI, and a notification appears, as shown in Figure 6-73.



Partition name	Systems	Replication topology	HA status	DR status	Migration status	Evaluation	Volume groups
P111	Cluster_9.71.19.111	Local	-	-	-	View results	3
HA_P	Cluster_9.71.19.111	Local	-	-	-	-	0
tvvolpartition	Cluster_9.71.19.111	Local	-	-	-	-	2
p1	Cluster_9.71.19.136 Cluster_9.71.19.96	HA + DR	HA establishing	DR running	-	-	2
P96	Cluster_9.71.19.96	Local	-	-	-	-	1
d	Cluster_9.71.19.90 Cluster_9.71.19.93	HA	HA established	-	-	-	0
partition-580	Cluster_9.71.19.90	Local	-	-	-	-	0
partition-975	Cluster_9.71.19.90	Local	-	-	-	-	0
partition-2623	Cluster_9.71.19.90	Local	-	-	-	-	0

Figure 6-73 Notification on IBM FlashSystem about AI evaluation results being ready

To view the results of a partition placement advisory, click **View Results**. A dialog box appears to display the systems that were evaluated, along with their respective compatibility scores. These scores are calculated based on multiple factors relevant to workload placement.

As shown in Figure 6-74, the evaluation results provide a clear comparison of potential target systems. For systems that are not part of IBM Storage Insights, the UI indicates that results are unavailable and offers an option to add the system to IBM Storage Insights. Adding the system enables more accurate evaluations in future placement advisories by using historical performance and capacity data.

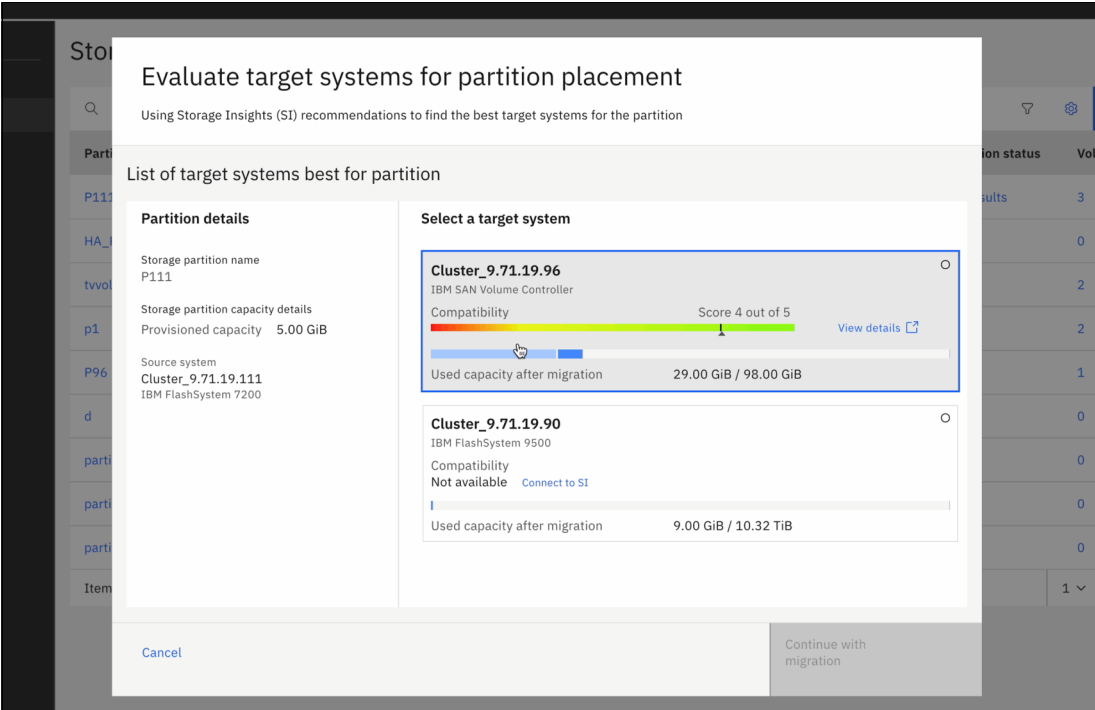


Figure 6-74 Result evaluation

After reviewing the evaluation results, proceed with the migration by selecting a recommended target system and then click **Continue with Migration**. This action starts the migration workflow in the IBM FlashSystem grid UI with the selected target system prefilled based on the evaluation.

To explore more details about the compatibility score, use the link that is provided in the evaluation dialog to access in-depth insights. As shown in Figure 6-74 on page 173, this link opens a detailed view of the evaluation criteria. The corresponding results in IBM Storage Insights are shown in Figure 6-75. The figure provides a detailed breakdown of the factors that influence the recommendation.

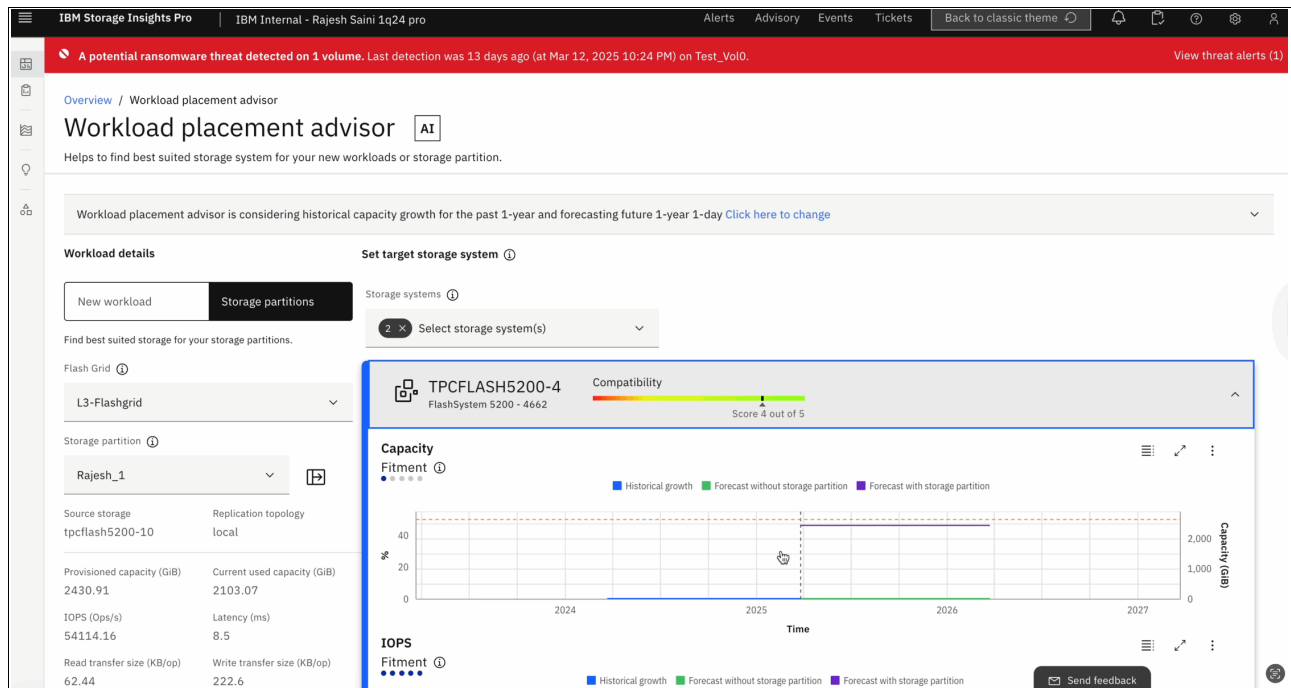


Figure 6-75 IBM Storage Insights details view of a partition evaluation

The integration of IBM Storage Insights AI Advisory capabilities into the IBM FlashSystem grid Element Manager UI empowers users to manage their entire storage grid with AI-driven insights from a single, unified interface. Although IBM Storage Insights provides detailed recommendations and analytics, this integration improves the experience by enabling quick access, evaluation, and interaction with AI advisories without exiting the IBM FlashSystem management environment.

This seamless integration streamlines decision making and enables administrators to respond to system needs efficiently and take advantage of the insights that are provided by IBM Storage Insights.

Abbreviations and acronyms

AI	artificial intelligence	TCT	Transparent Cloud Tiering
CA	certificate authority	UI	user Interface
CLI	command-line interface	URI	Uniform Resource Identifier
CN	common name	URL	Uniform Resource Locator
CPU	central processing unit	UUID	Universally Unique Identifier
CRLF	Carriage Return Line Feed	WWPN	worldwide port name
CSR	Certificate Signing Request		
DMP	Device Management Page		
DNS	Domain Name System		
DR	disaster recovery		
DRP	Data Reduction Pools		
ESCC	EMEA Storage Competence Center		
FC	Fibre Channel		
FCM	FlashCore Module		
FDMI	Fabric Device Management Interface		
FQDN	fully qualified domain name		
HA	high availability		
HADR	high availability and disaster recovery		
HSE	high-speed Ethernet		
IBM	International Business Machines Corporation		
LUN	logical unit number		
NAS	network-attached storage		
NDVM	Non-Disruptive Volume Movement		
NTP	Network Time Protocol		
OU	organizational unit		
PBHA	Policy-based High Availability		
PBR	Policy-based Replication		
PSCP	PuTTY Secure Copy Protocol		
PTR	Pointer		
RBAC	role-based access control		
RHEL	Red Hat Enterprise Linux		
RPO	Recovery Point Objective		
RTO	Recovery Time Objective		
SAN	storage-attached network		
SCP	Secure Copy		
SFTP	Secure File Transfer Protocol		
STSM	Senior Technical Staff Member		
SVC	IBM SAN Volume Controller		

Related publications

The publications that are listed in this section are considered suitable for a more detailed description of the topics that are covered in this book.

IBM Redbooks

The following IBM Redbooks publications provide more information about the topics in this document. Some publications that are referenced in this list might be available in softcopy only.

- ▶ *Ensuring Business Continuity: A Practical Guide to Policy-Based Replication and Policy-Based High Availability for IBM Storage Virtualize Systems*, SG24-8569
- ▶ *Maximize Your Storage Investment: Use IBM Storage Insights for Observability and Artificial Intelligence for IT Operations*, SG24-8584

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Online resources

The following websites are also relevant as further information sources:

- ▶ IBM Storage Insights documentation
<https://www.ibm.com/docs/en/storage-insights?topic=overview-storage-insights>
- ▶ IBM Storage Virtualize 9.1 documentation
<https://www.ibm.com/docs/en/flashsystem-9x00/9.1.0>

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